The Igusa Square Root

Borcherds Denominators, Protected Hall Theory, Chiral Geometry, and the $K3 \times E$ Realization

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Abstract

Let S be a smooth projective K3 surface and E an elliptic curve. Reduced curve counting on $S \times E$, in classes primitive and nonzero in the K3 factor, has generating series $-\chi_{10}{}^{-1}$ in the normalization used here, where χ_{10} is the monic Igusa cusp form. Its square root D_5 is an automorphic section of weight five with an order-two character. A geometric realization asks for extension operations on states, their parity, an orientation line, and a comparison of its section with D_5.

The automorphic construction begins with the weak Jacobi form $\phi_{0,1}$, the type-II Lorentzian root lattice, its positive roots, and its Weyl vector. The Borcherds–Gritsenko–Nikulin construction supplies a Lie superalgebra $\mathfrak{g}_{\Delta_5}$ and the identity $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$. Proper-height Chevalley chains give its homological denominator. A diagonal operator built from signed multiplicities gives the root Fredholm product on an absolute-convergence chamber; its Weyl monomial and half-coordinate pullback give the automorphic denominator. The root Lie superalgebra carries the bracket, while the minimal virtual superspace carries the signed product.

The compact comparison begins with sheaves, extensions, and flags on $S \times E$. It requires reduced critical coefficients and orientation comparisons on those correspondences, then a root-graded map preserving the Hall operations. Identifying a Pfaffian with the automorphic section also requires a map of lines and their transport. The reduced counting identity supplies the scalar value to recover.

The comparison problem

An extension $0 \to A \to F \to B \to 0$ remembers the relative elliptic position of its endpoints. After translation quotient, A and B must therefore live on the same elliptic torsor. For E-stacks M, N, the relevant product is

$$[(M \times N)/E_{\mathrm{diag}}] \simeq [M/E] \times_{BE} [N/E],$$

where BE is the stack of elliptic torsors. The ordinary product of the two quotients forgets the identification of their torsors. For $M = N = E$, the relative product is E, with coordinate $y - x$, whereas the ordinary product is a point.

On the full coherent-sheaf stack, the middle-term map of an extension correspondence is proper on each fixed Hilbert-polynomial component. Its fibre is the appropriate Quot space. Iterated correspondences are full flag stacks, and the zero sheaf gives the unit over BE. A critical cohomological realization must provide the coefficient maps and orientation transport on these same flags. The primitive comparison begins with geometric classes satisfying the root presentation. The resulting root-to-primitive map must preserve brackets and both parity dimensions; identifying it as an equivalence also requires the generation and injectivity statements. A separate map of square-root lines must preserve the orientation character.

The later algebraic and field-theoretic comparisons explain which operations such maps must retain. The relative Rees and oscillator calculations supply finite examples. Their compact Hall interpretation requires a map from geometric states to the specified representations. A Hochschild centre, a monoidal categorical centre, and a trace on a representation each have their own carrier and comparison map. The geometric realization keeps these structures until scalar evaluation.

Notation and normalization

The ground field has characteristic zero unless stated otherwise. Small perfect categories and presentable categories of modules are distinguished where they enter. Each construction specifies its tensor product and the colimits it preserves. Positive-weight, proper-height, formal-parameter and pro-cohomological completions retain their respective quotient systems; they are not identified without a comparison. The Igusa normalization is

$$\Delta_5^{\theta} = 64 D_5, \qquad \chi_{10} = D_5{}^2,$$

with D_5 monic. Physical charge is additive; Gram degree is carried through its polarization cocycle. Bar, factorization, and Hopf coproducts retain distinct notation and distinct geometric meanings.

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Book I

Denominators and geometric realizations

Chapter 1

From an automorphic product to operations

A protected index counts states with signs. Its coefficients do not specify the states, their extension operations or the line in which an operator determinant takes values. The Igusa product makes these questions concrete: one automorphic function has a root-algebra construction and a reduced curve-counting interpretation. A geometric comparison must connect their objects and operations before evaluating that function.

Use $Z = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right) \in \mathbb{H}_2$, with $q = e^{2\pi i\tau}$, $r = e^{2\pi iz}$ and $s = e^{2\pi i\sigma}$. Write

$$\phi_{0,1}(\tau, z) = \sum_{n\geq 0,\, \ell\in\mathbb{Z}} f(n,\ell) q^n r^\ell, \qquad f(0,0) = 10,$$

for the normalized weak Jacobi form of weight zero and index one. The K3 elliptic genus is $2\phi_{0,1}$. In the cusp convention specified below, its multiplicative lift is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,\ell,m)>0} (1 - q^n r^\ell s^m)^{f(nm,\ell)}, \qquad \Delta_5^\theta = 64 D_5, \qquad \chi_{10} = {D_5}^2.$$

The order on (n, ℓ, m), the theta phase and the coefficient ring of this product will be fixed before its construction. The monic section D_5 has weight five and an order-two modular character ν_5.

The Gritsenko–Nikulin presentation supplies a Lie superalgebra $\mathfrak{g}_{\Delta_5}$, with positive half $\mathfrak{n}_{\Delta_5,+}$, for which

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z).$$

The product records signed root dimensions. The presentation supplies both parity dimensions and the bracket. In particular, the first interior root has dimensions $(29 \mid 93)$; its signed dimension -64 is not its ordinary dimension.

1.1 Algebraic and analytic constructions

A proper positive height makes the bounded root-degree pieces finite. Complete by total root height, retaining the specified coproduct and completed tensor product. The ordinary Chevalley chain complex of $\mathfrak{n}_{\Delta_5,+}$ then has Euler supercharacter, with Weyl vector ρ,

$$P = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_{\Delta_5}\alpha}, \qquad e^{-\rho} P = \operatorname{den}(\mathfrak{g}_{\Delta_5}).$$

Its enveloping algebra has PBW supercharacter P^{-1}. The ordinary bar–Chevalley comparison, the current envelope on a curve and an associative envelope on Ran space have different carriers; their maps and tensor products are specified separately.

Definition 1.1 (Root constructions)**.** Let Real_{Δ_5} denote the following collection associated with the fixed minimal Igusa presentation: its positive height-complete enveloping algebra and Chevalley coalgebra;

its level-zero current envelope; its quantized root datum with a specified Cartan and Borel completion; and its diagonal Fredholm model on a declared root Hilbert superspace and convergence domain. An operation-preserving comparison specifies a map on every component it compares.

Theorem 3.6 constructs the primitive map from simple-root representatives satisfying the relations. Surjectivity and equality of the two parity-refined PBW series make that map an isomorphism. It then induces the ordinary enveloping, Chevalley and current maps. A Hall-category comparison, a quantum-double pairing and a Fredholm intertwiner each require additional structure.

The analytic root operator K_Z acts on a positive root vector by $e^{-\alpha(Z)}$. On its trace-class domain,

$$\operatorname{sdet}_F(1 - K_Z) = P = e^{\rho} D_5(2Z).$$

An oriented skew doubling realizes this determinant as a Pfaffian ratio. This constructs an operator model of the scalar product. A global geometric family and its orientation transport are further comparisons.

1.2 The compact geometric problem

Let $X = S \times E$, where S is a smooth projective K3 surface and E is an elliptic curve. In the primitive nonzero K3-class sector and the reduced incidence convention, curve counting gives

$$\mathcal{Z}_X^{\mathrm{red}} = -\chi_{10}{}^{-1} = -\big(\jmath^* \operatorname{den}(\mathfrak{g}_{\Delta_5})\big)^{-2}, \qquad \jmath(Z) = Z/2.$$

The negative sign and the half-coordinate pullback are part of this identity. The two copies of the root Fock space reproduce the inverse square after the Weyl normalization; an identification with geometric states is additional.

Physical charge is additive, whereas its three Gram invariants are quadratic. For $c = (Q, P) \in \widetilde{H}(S, \mathbb{Z})^{\oplus 2}$, write $\Pi(c) = (Q^2/2, Q \cdot P, P^2/2)$ and put

$$B(c, d) = \Pi(c + d) - \Pi(c) - \Pi(d).$$

The group law

$$(c, \lambda)(d, \mu) = (c + d, \lambda + \mu + B(c, d))$$

makes $c \mapsto (c, \Pi(c))$ additive. This split central extension retains the cross term; projecting away that term does not make the original Gram map additive.

Write $\mathcal{L}$ for the weight-one modular line.

Definition 1.2 (Geometric realization datum)**.** A compact realization datum consists of reduced oriented coefficient complexes on object, extension and flag moduli of X; their additive charge grading and polarized Gram refinement; point-supported and wrapped elliptic sectors with their mixed correspondences; primitive representatives and an integration map to the root presentation; a specified coproduct and full Hopf pairing when a double is compared; and a map of orientation lines to the automorphic line $\mathcal{L}^5 \otimes \nu_5$, preserving their sections.

These moduli share an elliptic torsor after translation quotient. The coefficient maps must therefore act on the common-torsor extension and flag diagrams. The geometric primitive map satisfies the same presentation criterion as the algebraic map above. Its relations, generation and both parity dimensions must be proved on the actual classes. The coproduct, chiral collision operations, orientation transport and state comparison retain their own equations. Their existence is the compact realization problem.

Book II

Root algebras, extension geometry and operators

Chapter 2

Lattices and automorphic denominators

The denominator records signed dimensions. A bracket requires the labelled generators and relations behind those dimensions. The lattice fixes their degrees, the automorphic product fixes the chamber correction, and that correction specifies the minimal Borcherds presentation.

2.1 The Lorentzian root lattice

Let

$$L = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus 2\mathbb{Z}f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

Here U is the unimodular hyperbolic plane, and $U(4)$ has its pairing multiplied by four. The Gram-index map is

$$\alpha : \Gamma_{\text{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \qquad \alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2}.$$

Then

$$(\alpha(n, \ell, m), \alpha(n', \ell', m')) = -2\big(2nm' + 2n'm - \ell\ell'\big),$$

and

$$(\alpha(n, \ell, m), \alpha(n, \ell, m)) = -2(4nm - \ell^2).$$

2.1.1 *Real simple roots and Weyl vector*

Put

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Proposition 2.1 (Type-II Cartan matrix)**.** *The Gram matrix of the real simple roots is*

$$\big((\delta_i, \delta_j)\big)_{i,j=1}^{3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. □

The Weyl vector is

$$\rho = f_2 - \frac{1}{2} f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

Use formal root-torus monomials

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \qquad e^{-\rho} = Q^{1/2} R^{1/2} S^{1/2}.$$

The substitution relating this torus to the denominator period coordinate will be specified below.

2.2 The root reflection matrices

In the basis (f_2, f_3, f_{-2}), the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \qquad S_i^{\mathsf{T}} G S_i = G.$$

2.3 The protected Jacobi input

2.3.1 The K3 elliptic genus

Write $q = e^{2\pi i \tau}$ and $r = e^{2\pi i z}$. Let S be a K3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N} = (4, 4)$ sigma model is the elliptic genus

$$Z_{K3}(\tau, z) = 2\phi_{0,1}(\tau, z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0} \sum_{\ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau, z) = 4 \sum_{a=2}^{4} \frac{\vartheta_a(\tau, z)^2}{\vartheta_a(\tau, 0)^2} \in J^{\mathrm{w}}_{0,1}$$

[33]. The coefficient depends on the discriminant $4n - \ell^2$ and on ℓ modulo 2.

Theorem 2.2 (Exact low Fourier coefficients). *The first four q-rows are*

$$\begin{aligned}\phi_{0,1} = {} & r^{-1} + 10 + r \\ & + q\left(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2\right) \\ & + q^2\left(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3\right) \\ & + q^3\left(-64r^{-3} + 808r^{-2} - 2752r^{-1} + 4016 \right. \\ & \qquad \left. - 2752r + 808r^2 - 64r^3\right) + O(q^4).\end{aligned}$$

In particular,

$$f(0,0) = 10, \qquad \frac{f(0,0)}{2} = 5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t = q^{1/8}$ and $x = r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x. The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Section 2.4. □

2.3.2 *The virtual half-index*

For every (n, ℓ) choose the minimal super vector space $U_{n,\ell} = \mathbb{C}^{\max(f(n,\ell),0)|\max(-f(n,\ell),0)}$. It satisfies

$$\operatorname{sdim} U_{n,\ell} = f(n,\ell).$$

The integer $f(n,\ell)$ is its superdimension. In the ordinary Grothendieck group of finite-dimensional super vector spaces, the even and odd dimensions remain separate. The signed integer is their image under superdimension. The determinant depends only on that image. Adding an even–odd pair preserves the determinant, but changes the vector space. The minimal choice makes every zero coefficient contribute the zero space.

Lemma 2.3 (Fock character and Berezinian). *Let $V = \bigoplus_\gamma V_\gamma$ be a super vector space graded by a pointed positive lattice cone, with a proper positive height and finite-dimensional homogeneous pieces and $V_0 = 0$. In the total-height completion,*

$$\operatorname{sch}\left(\operatorname{Sym}(V_{\bar 0}) \otimes \bigwedge(V_{\bar 1})\right) = \prod_\gamma (1 - x^\gamma)^{-\operatorname{sdim} V_\gamma},$$

and

$$\operatorname{sdet}_V(1 - x) = \prod_\gamma (1 - x^\gamma)^{\operatorname{sdim} V_\gamma}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^\gamma)^{-1}$. An odd one-dimensional summand contributes $1 - x^\gamma$. Multiplication over a homogeneous basis proves both identities. □

2.4 Exact theta-series calculation

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\begin{aligned}\vartheta_2(\tau, z) &= \sum_{n\in\mathbb{Z}} t^{4n^2+4n+1} x^{2n+1}, \\ \vartheta_3(\tau, z) &= \sum_{n\in\mathbb{Z}} t^{4n^2} x^{2n}, \\ \vartheta_4(\tau, z) &= \sum_{n\in\mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.\end{aligned}$$

Cancel the common lowest power of t in each quotient before formal division in $\mathbb{Q}[x^{\pm 1}][[t]]$. At $q = 0$, the second theta quotient contributes $(r + 2 + r^{-1})/4$ and the other two contribute one each. Multiplication by four gives $r^{-1} + 10 + r$. Successive convolution and inversion through t^{24} give the next three rows of Theorem 2.2. Fractional powers cancel before conversion back to (q, r).

2.5 The Borcherds product

2.5.1 *The effective chamber*

For $Z = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right) \in \mathbb{H}_2$, put $s = e^{2\pi i \sigma}$. Define

$$\Gamma_{\mathrm{eff}} = \{(n, \ell, m) \in \mathbb{Z}^3 : m > 0,\ n \geq 0\} \cup \{(n, \ell, 0) : n > 0\} \cup \{(0, \ell, 0) : \ell < 0\}.$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber. For $\gamma = (n, \ell, m)$ put

$$V_\gamma = U_{nm,\ell}, \qquad \operatorname{sdim} V_\gamma = f(nm, \ell).$$

Definition 2.4 (Monic virtual determinant)**.** The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \operatorname{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

Theorem 2.5 (Borcherds–Gritsenko–Nikulin product)**.** *The preceding product converges normally sufficiently far inside the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5. The theta-normalized form is*

$$\Delta_5^\theta = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0, 0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [8, 9, 44, 46]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. □

2.5.2 *The theta constant normalization*

For $a, b \in \{0, 1\}^2$, fix the series

$$\vartheta[a, b](Z, 0) = \sum_{u \in \mathbb{Z}^2} \exp\big(\pi i \big((u + a/2)^t Z (u + a/2) + b^t u\big)\big).$$

Let $\mathfrak{E}$ be the ten pairs with $a^t b$ even, and set

$$\Delta_5^\theta(Z) = \prod_{(a,b) \in \mathfrak{E}} \vartheta[a, b](Z, 0).$$

This is the Gritsenko–Nikulin phase convention. Replacing the last term by $b^t(u + a/2)$ multiplies the ten-factor product by -1: only $a = b = (1, 1)$ contributes a nontrivial product phase.

Proposition 2.6 (Ten-factor leading coefficient)**.** *In the stated convention,*

$$[q^{1/2} s^{1/2}] \Delta_5^\theta = 64(r^{1/2} - r^{-1/2}), \qquad [q^{1/2} r^{1/2} s^{1/2}] \Delta_5^\theta = 64.$$

Proof. The four factors with $a = (0,0)$ start with 1. The two with $a = (1,0)$ start with $2q^{1/8}$, and the two with $a = (0,1)$ start with $2s^{1/8}$. The remaining pair has leading terms

$$2q^{1/8}s^{1/8}(r^{1/4} + r^{-1/4}), \qquad 2q^{1/8}s^{1/8}(r^{1/4} - r^{-1/4}).$$

Their product retains both elliptic monomials. Multiplication by the other eight factors gives the formula. In particular the normalization $\Delta_5^{\theta} = 64D_5$ fixes one coefficient and does not discard the second term of the boundary coefficient. □

Proposition 2.7 (Boundary factor of the Borcherds product)**.** *The $m = 0$ part of the monic product is*

$$\prod_{n>0,\ell\in\mathbb{Z}} (1 - q^n r^\ell)^{f(0,\ell)} \prod_{\ell<0} (1 - r^\ell)^{f(0,\ell)} = (1 - r^{-1}) \prod_{n\geq 1} (1 - q^n r^{-1})(1 - q^n)^{10}(1 - q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2}r^{1/2}(1 - r^{-1}) \prod_{n\geq 1} (1 - q^n r^{-1})(1 - q^n)^{10}(1 - q^n r) = \eta(\tau)^9 \vartheta_{11}(\tau, z),$$

where the odd theta function is fixed by the exact convention

$$\vartheta_{11}(\tau, z) = -q^{1/8}r^{-1/2} \prod_{n\geq 1} (1 - q^{n-1}r)(1 - q^n r^{-1})(1 - q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0,-1) = 1$, $f(0,0) = 10$, and $f(0,1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24} \prod_{n\geq 1}(1 - q^n)$ give the second, including its leading sign and monomial. □

Corollary 2.8 (The scalar square)**.** *The monic weight-ten form is*

$$\chi_{10} = {D_5}^2 = 2^{-12}{\Delta_5^{\theta}}^2.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^{\theta} = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. □

2.6 Divisor, character, and the meaning of the square root

Let H_1 be the image of $z = 0$ in the genus-two modular stack. Its inverse image in $\mathbb{H}_2$ is the orbit of that hypersurface.

Theorem 2.9 (Diagonal divisor)**.** *On $\mathbb{H}_2$, the divisor of D_5 is the orbit of $z = 0$, with multiplicity one. Equivalently its divisor on the modular stack is H_1. The form transforms by the Maass character ν_5 of order two, and $\chi_{10} = {D_5}^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [19, 52, 53]. □

Proposition 2.10 (Linear and slash reflection signs)**.** *For the real reflections, the linear tube action has*

$$D_5(s_{\delta_i} Z) = -D_5(Z).$$

For the block-diagonal symplectic lifts $g_i = \operatorname{diag}(U_i^{-t}, U_i)$, with

$$U_1 = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the Maass character satisfies $\nu_5(g_i) = 1$. Thus the homogeneous linear reflection character is the sign character of W, while these weight-five slash actions have value $+1$.

Proof. The maps $Z \mapsto U_i^{-t} Z U_i^{-1}$ give the three tube reflections. Every U_i has determinant -1, and each g_i reduces to the identity modulo two. The Maass character is the sign of the permutation of the six odd characteristics, so $\nu_5(g_i) = 1$. The automorphy factor is $j(g_i, Z) = \det U_i = -1$. Hence $D_5(g_i Z) = \nu_5(g_i) j(g_i, Z)^5 D_5(Z) = -D_5(Z)$. Products of these reflections give the sign character in the linear action. Dividing by j^5 gives the stated slash action. □

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borcherds algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

2.6.1 *Automorphic section, orientation line, and Pfaffian operator*

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\text{automorphic section} \longrightarrow \text{determinant-line identification}$$
$$\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.}$$

A geometric comparison must therefore provide its determinant line, a coherent square root, and a section-preserving map to the modular line. An operator model also specifies its domain and regularization; a differential-operator model specifies its symbol and analytic conditions. Chapter 6 constructs a skew Fredholm model of the product and retains the global geometric comparison as a separate problem.

2.7 The minimal simple-root presentation

The real roots form a basis of L_{II}. Write

$$\alpha = b_1\delta_1 + b_2\delta_2 + b_3\delta_3, \qquad h(\alpha) = b_1 + b_2 + b_3.$$

For $\alpha = \alpha(n, \ell, m)$, this is

$$(b_1, b_2, b_3) = (n, m, n + m - \ell), \qquad h(\alpha) = 2n + 2m - \ell.$$

The active product support lies in $Q_+ = \sum_i \mathbb{N}\delta_i$. Indeed, weak index-one Jacobi coefficients vanish when $4nm - \ell^2 < -1$. For $n, m \geq 0$ with $n + m > 0$, this implies $\ell \leq n + m$, except possibly $n = m$, where integrality gives the same inequality. The boundary factor has $(n, \ell, m) = (0, -1, 0)$. Every bounded-height region is finite.

Let W be the real reflection group and put

$$C = \{a = \textstyle\sum b_i\delta_i : b_i \geq 0,\ (a, \delta_i) \leq 0 \text{ for all } i\}.$$

Its generating rays are $a_{12} = \delta_1 + \delta_2$, $a_{13} = \delta_1 + \delta_3$, and $a_{23} = \delta_2 + \delta_3$. To see this, write

$$a = t_{12}a_{12} + t_{13}a_{13} + t_{23}a_{23}, \qquad t_{ij} = \tfrac{1}{2}(b_i + b_j - b_k).$$

The chamber inequalities say exactly $t_{ij} \geq 0$, and

$$(a, a) = -8(t_{12}t_{13} + t_{12}t_{23} + t_{13}t_{23}).$$

Thus the only null rays are the three displayed ones; all other nonzero chamber vectors are timelike.

In root-torus notation the chamber Fourier expansion has the form

$$D_5(Z_{\mathrm{root}}) = \sum_{w\in W} \det(w) e^{-w\rho}\left(1 - \sum_{0\neq a\in C\cap L_{II}} m(a)e^{-wa}\right).$$

Here $m(a) = -c_{\rho+a}$, where $c_{\rho+a}$ is the coefficient of that monomial in the monic theta product. The reflection law makes the Fourier coefficients alternating; wall coefficients vanish, and each remaining orbit has a unique chamber representative. This gives the displayed expansion [44, Theorem 2.3]. Along each primitive null ray,

$$1 - \sum_{n\geq 1} m(na)t^n = \prod_{n\geq 1}(1-t^n)^9.$$

The exponent nine follows from the boundary input $\eta^9\vartheta_{11}$, with the other rays obtained by chamber permutations.

Definition 2.11 (Minimal imaginary generators)**.** At a timelike chamber vector a, take $\max(m(a),0)$ even and $\max(-m(a),0)$ odd generators. At every nonzero multiple of each primitive null ray, take nine even generators and no odd generator. Together with the three even real generators, these form a labelled set I_Δ, with degree α_i and parity $p(i)$.

The null-ray Euler exponents determine its simple multiplicities; the coefficients $m(na)$ are not themselves those multiplicities. At a timelike degree the signed simple multiplicity likewise need not equal the signed dimension of the whole root space, which also contains brackets.

Definition 2.12 (The Igusa Lie superalgebra)**.** Take the even Cartan space $\mathfrak{h} = L_{II}\otimes\mathbb{C}$ and generators e_i, f_i of parity $p(i)$ and degrees $\alpha_i, -\alpha_i$. Identify $h_i = \alpha_i \in \mathfrak{h}$ and impose

$$[h,h'] = 0, \quad [h,e_i] = (h,\alpha_i)e_i, \quad [h,f_i] = -(h,\alpha_i)f_i, \quad [e_i,f_j] = \delta_{ij}h_i.$$

For real $i\neq j$, impose

$$(\operatorname{ad} e_i)^{1-(\alpha_i,\alpha_j)}e_j = 0, \qquad (\operatorname{ad} f_i)^{1-(\alpha_i,\alpha_j)}f_j = 0.$$

For $(\alpha_i,\alpha_j) = 0$, impose $[e_i,e_j] = [f_i,f_j] = 0$. These are super brackets, with $[x,y] = -(-1)^{|x||y|}[y,x]$. Denote this presentation by $\mathfrak{g}_{\Delta_5}$, its positive half generated by the e_i by $\mathfrak{n}_{\Delta_5,+}$, and its negative half generated by the f_i by $\mathfrak{n}_{\Delta_5,+}{}^-$.

Theorem 2.13 (Root supermultiplicities and denominator)**.** *This presentation has triangular decomposition* $\mathfrak{g}_{\Delta_5} = \mathfrak{n}_{\Delta_5,+}{}^- \oplus \mathfrak{h}\oplus\mathfrak{n}_{\Delta_5,+}$*, with finite even and odd root dimensions. On the active positive support,*

$$\operatorname{sdim}\mathfrak{g}_{\Delta_5\,\alpha(n,\ell,m)} = f(nm,\ell).$$

In the Gritsenko–Nikulin period convention,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6}\Delta_5^{\theta}(2Z).$$

Proof. Distinct simple vectors have nonpositive pairing: the real pairings are -2, the mixed pairings are the chamber inequalities, and two future nonspacelike vectors pair nonpositively. Thus the symmetrizable Borcherds presentation theorem applies. Its free labelled Cartan reduces by the central kernel of $h_i\mapsto\alpha_i$, giving the stated Cartan

and triangular decomposition. In a bounded positive degree only finitely many labelled generators and tensor words can occur, so PBW gives finite root dimensions.

The Weyl correction of this minimal superdatum is the chamber Fourier correction above. For the null rays the pairwise orthogonal even generators supply the Euler product of exponent nine. Timelike generators supply their signed coefficients. The Gritsenko–Nikulin denominator theorem therefore identifies the root product with the monic Igusa product [44–46]. The change to its period coordinate is $Z_{\mathrm{root}} = 2Z$. □

For example, at $\delta_1 + \delta_2 + \delta_3$ there are 29 even and 93 odd vectors. Its even part consists of the two real multilinear brackets and 27 brackets between a null-ray simple generator and the remaining real generator. Its 93 odd vectors are the timelike simple generators, since the correction coefficient there is -93. Thus the signed dimension is -64, while the total dimension is 122.

Chapter 3

Positive completions and current constructions

Positive height controls products and their limits. It gives a finite calculation in each root degree, but distinguishes completion of an enveloping algebra from the enveloping algebras of Lie quotients. The ordinary bar comparison begins in that setting; current fields and Ran envelopes then add their coefficient categories.

3.1 Positive height and the ordinary bar

Let $Q_+ = \mathbb{N}\partial_1 + \mathbb{N}\partial_2 + \mathbb{N}\partial_3$, with height equal to the sum of the three coefficients. Every bounded-height set of labelled simple generators is finite. The free tensor algebra on these generators therefore has finite-dimensional homogeneous pieces, as do its quotient $U(\mathfrak{n}_{\Delta_5,+})$ and the embedded Lie superalgebra $\mathfrak{n}_{\Delta_5,+}$.

For a positively root-graded algebra A, write

$$A_{\le H} = A \Big/ \bigoplus_{h(\gamma)>H} A_\gamma, \qquad \widehat{A}_h = \varprojlim_H A_{\le H} = \prod_{\gamma\in Q_+} A_\gamma.$$

The ideal is defined by total root height, including the height of a product. The completed tensor product uses the same total height in both factors. For $A = U(\mathfrak{n}_{\Delta_5,+})$, the primitive coproduct $x \mapsto x\otimes 1 + 1\otimes x$ extends continuously to that tensor product. At finite height its target is the quotient of $A\otimes A$ by total height, not the untruncated tensor product of two height quotients.

There is also a finite-dimensional nilpotent Lie quotient

$$\mathfrak{n}_H = \mathfrak{n}_{\Delta_5,+} \Big/ \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta_5\alpha}.$$

It is useful for calculations in degrees at most H, but $U(\mathfrak{n}_H)$ still contains arbitrarily long products of low-height generators. Thus $\varprojlim_H U(\mathfrak{n}_H)$ and $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$ are different constructions. For the one-dimensional even abelian Lie algebra $\mathbb{C}x$ of height one, the former limit is $\mathbb{C}[x]$, whereas the latter is $\mathbb{C}[[x]]$.

Let $B_{\mathrm{word}}A$ be the normalized coaugmented ordinary bar coalgebra of an augmented algebra A with $A_0 = \mathbb{C}$. Write $\overline{A}$ for its augmentation ideal. Its underlying graded object is $T^c(s\overline{A})$, with deconcatenation and the differential from adjacent multiplication. Here s raises homological degree by one; intrinsic parity is recorded separately. Complete this coalgebra by the total height of its inputs, and use the same completion for the Chevalley chains $C_*(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Sym}(s\mathfrak{n}_{\Delta_5,+})$.

Theorem 3.1 (Positive-height bar–Chevalley comparison)**.** *Suspended antisymmetrization gives a natural quasi-isomorphism of height-complete dg coalgebras*

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) \longrightarrow \widehat{B}_{\mathrm{word},h}U(\mathfrak{n}_{\Delta_5,+}).$$

In every fixed root degree it is already obtained from any $\mathfrak{n}_H$ containing that degree. Its Euler supercharacter is

$$\chi_{\mathrm{root}}\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\,\alpha}}.$$

Proof. The map is

$$sx_1\cdots sx_r \longmapsto \sum_{\sigma\in S_r}\varepsilon(\sigma;sx_1,\dots,sx_r)[x_{\sigma(1)}|\cdots|x_{\sigma(r)}].$$

Here ε is the permutation sign in the symmetry of complexes of super vector spaces: homological degree and intrinsic parity each contribute their Koszul sign. There is no factor $1/r!$: the unshuffle coproduct on the source agrees with deconcatenation of the permutation sum. The binary differential agrees because the commutator in the enveloping algebra is the given bracket. PBW reduces the filtered comparison to the Koszul resolution of the super-symmetric algebra. In each fixed root degree all inputs have bounded positive height, and the filtration and chain groups are finite. The ordinary Koszul contraction therefore proves the quasi-isomorphism there. Products of complexes of vector spaces preserve these quasi-isomorphisms, giving the completed map.

An intrinsically even root vector becomes an odd Chevalley generator and contributes $1-e^{-\alpha}$ to the Euler supercharacter. An intrinsically odd root vector contributes its reciprocal. The parity sign and the homological Euler sign together give the displayed exponent. Positive height makes each coefficient finite. □

Proposition 3.2 (The bar of the Lie-quotient tower)**.** *There is also a natural quasi-isomorphism of inverse-limit complexes*

$$\varprojlim_H C_*(\mathfrak{n}_H) \longrightarrow \varprojlim_H B_{\mathrm{word}}U(\mathfrak{n}_H).$$

In each homological degree the left side is $\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+})$. The natural coefficient map from the right side to $\widehat{B}_{\mathrm{word},h}U(\mathfrak{n}_{\Delta_5,+})$ is a quasi-isomorphism as well. These comparisons do not identify $\varprojlim_H U(\mathfrak{n}_H)$ with $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$.

Proof. At each H, ordinary antisymmetrization is a quasi-isomorphism by PBW, and the comparisons commute with the quotient maps. The bonding maps on Chevalley and bar chain groups are surjective: they are induced by surjections of Lie algebras, enveloping algebras and augmentation ideals. The same is true on the mapping cones K_H of the comparisons. Consequently the Roos sequence

$$0\longrightarrow\varprojlim_H K_H\longrightarrow\prod_H K_H\xrightarrow{1-\mathrm{shift}}\prod_H K_H\longrightarrow 0$$

is degreewise exact. Products of complexes of vector spaces preserve acyclicity. Since every K_H is acyclic, its inverse limit is acyclic, proving the first assertion.

In Chevalley degree r, a monomial with inputs of height at most H has total height at most rH. Conversely, a monomial of total height at most H has all inputs of height at most H. These two filtrations are therefore cofinal in each chain degree, identifying the completed Chevalley complexes. Every fixed root coefficient of the bar tower also stabilizes, giving the stated coefficient map. Its composite with antisymmetrization is the map of Theorem 3.1, so it is a quasi-isomorphism. The example $\mathbb{C}[x]$ versus $\mathbb{C}[[x]]$ above still distinguishes the two enveloping constructions. □

Write Z_{root} for the root-torus coordinate with

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \qquad e^{-\rho} = Q^{1/2}R^{1/2}S^{1/2}.$$

The root-torus product is $D_5(Z_{\mathrm{root}})$. The Gritsenko–Nikulin period convention substitutes $Z_{\mathrm{root}} = 2Z$. Consequently

$$e^{-\rho}\chi_{\mathrm{root}}\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z).$$

This is an ordinary word-bar identity. An associative algebra on Ran space and a chiral field realization require their own coefficient categories and comparison maps.

Proposition 3.3 (The two-sided root product)**.** *The Lie bracket has no jointly continuous extension to the coordinatewise product of all nonzero root spaces with discrete Cartan factor.*

Proof. Choose one even simple pair e_n, f_n at each degree $na, -na$ on a primitive chamber null ray. The presentation gives $[e_n, f_m] = \delta_{nm} na$. Both sequences of partial sums converge coordinatewise, but their brackets have Cartan component $N(N+1)a/2$, which is not eventually constant. Continuity would force convergence in the discrete Cartan space. □

The positive completion with a discrete finite-support negative half does have a continuous bracket. Every mixed output coefficient then receives only finitely many terms. A completion of a full quantum double needs an analogous specified domain; root-degree preservation alone does not provide one.

3.2 Currents and their associative recipient

On a coordinate disc, the level-zero current Lie conformal superalgebra is

$$\mathrm{Cur}(\mathfrak{n}_H) = \mathbb{C}[\partial] \otimes \mathfrak{n}_H, \qquad [a_\lambda b] = [a, b].$$

Its coordinate-free input is the induced right differential module $N_H = \mathfrak{n}_H \otimes D_E$, with its Lie-* bracket. Let $V_H = U^{\mathrm{ch}}(N_H)$ be its BD current envelope [6]. In a coordinate, its vacuum module is spanned by ordered negative modes $a(-j)$, $j \geq 1$.

Proposition 3.4 (Current character)**.** *At fixed root degree and energy, the quotient system $(V_H)_H$ stabilizes. Its coefficientwise supercharacter is*

$$\prod_{j \geq 1} \prod_{\alpha > 0} (1 - t^j e^{-\alpha})^{-\operatorname{sdim} \mathfrak{g}_{\Delta_S \alpha}}.$$

The current generators are primitive for the level-zero current coproduct. The coefficientwise completion does not by itself assert uniform pole bounds for all completed states.

Proof. PBW identifies the vacuum space with the super-symmetric algebra on $\mathfrak{n}_H \otimes t^{-1}\mathbb{C}[t^{-1}]$. The mode $a(-j)$ has energy j; an even mode gives a symmetric factor and an odd mode an exterior factor. At fixed root height and energy only finitely many modes and products contribute. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$, and the quotient maps preserve it. Lower truncation for arbitrary completed states is a separate continuity condition on fields. □

An associative bar needs an associative algebra and its tensor. Use complexes of super vector spaces, retaining intrinsic parity separately from cochain degree. For nonempty finite sets and surjections, form

$$\mathcal{R}_E = \varprojlim_I D\text{-Mod}(E^I; \mathrm{sVect}_{\mathbb{C}}), \qquad \mathcal{R}_E^+ = D(\mathrm{sVect}_{\mathbb{C}}) \times \mathcal{R}_E,$$

using exceptional pullback along the diagonals. For an ordered partition $I = I_1 \sqcup I_2$, let j_{I_1,I_2} be the locus where the two blocks are disjoint. The chiral convolution is

$$(F \star G)_I = \bigoplus_{I = I_1 \sqcup I_2} j_{I_1,I_2,*}\, j^!_{I_1,I_2}(F_{I_1} \boxtimes G_{I_2}).$$

Its unit K has only the empty coefficient $K_\varnothing = \mathbb{C}$. A positive object has zero empty coefficient. The BD vacuum is the unit map $\omega_E \to V_H$. It is different from this empty-component unit.

The underlying chiral Lie algebra of V_H gives the positive main-diagonal Lie object $\mathcal{L}_H = \iota_* V_H$, with $(\iota_* V_H)_I = \delta_{I,*} V_H$. Define its augmented associative envelope $\mathcal{A}_H = U_\star(\mathcal{L}_H)$.

Proposition 3.5 (Bar of the Ran envelope). *In* $(\mathcal{R}^+_E, \star, K)$,

$$B_\star \mathcal{A}_H \simeq \mathrm{CE}_\star(\mathcal{L}_H), \qquad \Omega_\star \,\mathrm{CE}_\star(\mathcal{L}_H) \simeq \mathcal{A}_H.$$

The Chevalley coalgebra satisfies factorization.

Proof. A positive convolution word of length r vanishes on a configuration of cardinality less than r. Bar and cobar filtrations are therefore finite on each configuration. Flattening a bar–cobar word gives one two-term contractible gap complex at every internal cut; only the empty and one-input terms survive. This proves the associative unit and counit equivalences.

Suspended symmetrization gives the envelope comparison, with the same permutation sum as in Theorem 3.1. Its PBW filtration reduces to the super-symmetric Koszul complex; the Euler contraction divided by positive polynomial degree is valid in characteristic zero. The finite configuration filtration proves the equivalence. Finally, the main-diagonal support of $\mathcal{L}_H$ implies factorization of its chiral Chevalley coalgebra [36, Theorems 5.1.1–5.2.1]. □

The last cited equivalence is Lie–commutative. The associative assertion also uses the envelope and the bar–cobar argument. The objects N_H, V_H, $\mathcal{L}_H$ and $\mathcal{A}_H$ have different carriers. In particular, this proposition does not identify $\mathcal{A}_H$ with the BD field algebra V_H, or construct a second independently defined field dual. Such a comparison must preserve its specified supported operations and units.

3.3 The primitive comparison and its extensions

Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra with finite-dimensional homogeneous pieces and a proper positive height. A primitive realization of the Igusa presentation is a parity-preserving homogeneous map from its labelled simple-root spaces to $\mathfrak{p}$, satisfying the defining relations.

Theorem 3.6 (Universal presentation map). *Let* $\mathfrak{p}$ *be a positive-root-graded Lie superalgebra over* $\mathbb{C}$*, with finite-dimensional homogeneous pieces. Suppose a homogeneous parity-preserving map from every simple-root generating space of* $\mathfrak{n}_{\Delta_s,+}$ *to* $\mathfrak{p}$ *satisfies the defining Borcherds–Serre relations. It extends uniquely to a graded Lie-superalgebra map*

$$f : \mathfrak{n}_{\Delta_s,+} \longrightarrow \mathfrak{p}.$$

It induces continuous maps on height completions, enveloping algebras, Chevalley chains, and level-zero currents. The map f *is an isomorphism if and only if it is surjective and the root-graded PBW series agree in each parity.*

Proof. Extend the chosen maps to the free Lie superalgebra on the simple-root spaces. The relation hypothesis makes the defining ideal lie in its kernel, giving f. The generators prove uniqueness. On enveloping algebras the map sends $x_1 \cdots x_j$ to $f(x_1) \cdots f(x_j)$. On Chevalley chains it is the graded symmetric map on the shifted generators. It commutes with the Chevalley differential because f preserves brackets. The current map is $\partial^j x \mapsto \partial^j f(x)$, preserving $[x_\lambda y] = [x, y]$. Root degree preservation gives compatible maps at each height and hence on the stated limits.

If f is surjective, $U(f)$ is surjective. Proper positive height makes each root-graded PBW component finite-dimensional. Equality of its even and odd dimensions makes $U(f)$ bijective in every component. PBW embeds each Lie superalgebra into its enveloping algebra, so f is injective. The converse follows by applying the constructions to its inverse. □

A geometric comparison also requires functors on the actual Hall and critical categories, a compatible coproduct and Hopf pairing, and maps of determinant lines and operator domains. These categories and lines are not the domain of the primitive Lie map. The additional maps must be constructed and checked against that map. They are not consequences of its universal property.

3.4 The local–wrapped root algebra

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m. Set

$$\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)=0}} \mathfrak{g}_{\Delta_5\alpha}, \qquad \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)>0}} \mathfrak{g}_{\Delta_5\alpha}.$$

Proposition 3.7 (Semidirect local–wrapped decomposition)**.** *The space $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. □

Write $\mathbb{U}_{\Delta_5} = \widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$, and retain the current system $(V_H)_H$ of Section 3.2. The adjoint action of $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ on $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ extends to derivations of its enveloping algebra. Let $\#$ denote the resulting smash product, completed by total root height.

Theorem 3.8 (Local–wrapped enveloping algebra)**.** *There are natural isomorphisms*

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}),$$

and

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) \cong \widehat{\mathrm{Tot}}_h C_*(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}; C_*(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})).$$

The primitive Lie superalgebra of the completed enveloping algebra is $\widehat{\mathfrak{n}_{\Delta_5,+}}_h$. The Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_5(2Z)$.

Proof. PBW orders wrapped generators before local generators. Moving a local generator past a wrapped one adds their adjoint bracket, which is exactly the smash-product relation. All relations preserve total height, so the isomorphism extends to the completion. Splitting a Chevalley word into its two types gives the Hochschild–Serre bicomplex: its differentials are the two internal brackets and the adjoint action. This proves the second isomorphism.

In characteristic zero, the primitives of a super-symmetric Hopf algebra are its degree-one generators. Apply PBW to the highest polynomial degree of a primitive in each finite root-degree piece. All higher degrees vanish. This identifies the primitives before and after completion. The denominator identity gives the character. □

These operations define the algebraic semidirect-product example of a local–wrapped theory. Extending the local Lie action to chiral currents acting on geometric wrapped objects requires supported coefficient maps. The adjoint action alone does not specify those maps.

3.5 The colored elliptic Ran space

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \coprod_{m\geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The names local and wrapped refer here to the degree colors $m = 0$ and $m > 0$; the underlying current operations remain supported on configurations of points.

Let

$$\mathfrak{n}_{\Delta_S,+} = \bigoplus_{m\geq 0} \mathfrak{n}_{\Delta_S,+}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta_S,+}[m], \mathfrak{n}_{\Delta_S,+}[n]] \subset \mathfrak{n}_{\Delta_S,+}[m + n].$$

Construction 3.9 (Elliptic-degree factorization coalgebras)**.** For each H, take

$$\mathcal{F}_H = \mathrm{CE}_\star(\mathcal{L}_H), \qquad \mathcal{L}_H = \iota_* V_H.$$

Every current state has an elliptic degree, obtained by adding the root degrees of its negative modes. Give each Chevalley word the sum of these degrees and place its degree-m coefficient on $\mathrm{Ran}(E)_m$. The empty coefficient remains the unit K defined in Section 3.2.

Theorem 3.10 (Elliptic-degree factorization)**.** *The coalgebras $\mathcal{F}_H$ factorize compatibly with the elliptic-degree grading and the quotient system in H. Disjoint-union structure maps add the degree labels. Under the Lie–commutative Koszul equivalence, their derived Lie primitives are $\mathcal{L}_H$. The energy-one current states recover $\mathfrak{n}_H$, with its bracket given by the zeroth product.*

Proof. The current bracket and the Chevalley differential preserve root degree. The factorization theorem of Proposition 3.5 therefore respects its elliptic component. Its Lie inverse recovers $\mathcal{L}_H$. In a coordinate, PBW identifies the energy-one subspace of V_H with the states $a(-1)|0\rangle$, for $a \in \mathfrak{n}_H$. Their zeroth product is $[a, b](-1)|0\rangle$. All constructions commute with the root-graded quotient maps. □

The degree-zero sector acts on every positive-degree sector through the current operations. This is a grading of a point-supported current construction. A compact sheaf projecting onto the whole elliptic curve has different support. Its comparison must map the common-torsor extension correspondences to these current operations, preserving both support and charge.

3.6 Cotangent fields and finite BV models

Choose a holomorphic volume form Ω_E on E. For each finite-height Lie algebra $\mathfrak{n}_H$, the local dg Lie algebra on $E \times \mathbb{R}$ is

$$\mathfrak{L}_H = \Omega^{0,*}(E)\widehat{\otimes}\Omega^*(\mathbb{R}) \otimes \mathfrak{n}_H, \qquad d = \bar\partial + d_{\mathbb{R}}.$$

The tensor product uses smooth local sections. With cohomological shifts, the cotangent fields are

$$\mathcal{E}_H = \big(\Omega^{0,*}(E)\widehat{\otimes}\Omega^*(\mathbb{R})\big) \otimes \big(\mathfrak{n}_H[1] \oplus \mathfrak{n}_H^\vee\big).$$

The first summand is A and the second is B. Integration against Ω_E makes their pairing have degree -1, so this is the odd cotangent theory of $\mathfrak{L}_H[1]$. Its action, on compactly supported fields, is

$$S_H(A,B) = \int_{E\times\mathbb{R}} \Omega_E \wedge \left\langle B, dA + \tfrac{1}{2}[A,A] \right\rangle.$$

Proposition 3.11 (Classical cotangent action)**.** *This action satisfies the classical master equation. Its classical local observables form a factorization algebra. The same construction on* $\mathbb{R}^2_{\mathrm{top}} \times E$ *uses*

$$\mathfrak{L}_H^{(2)} = \Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(E) \otimes \mathfrak{n}_H$$

with fields in $\Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(E) \otimes (\mathfrak{n}_H[1] \oplus \mathfrak{n}_H^\vee[1])$ *and action*

$$S_H^{(2)} = \int_{\mathbb{R}^2\times E} \Omega_E \wedge \langle B, (d+\bar\partial)A + \tfrac{1}{2}[A,A]\rangle.$$

Proof. The vector field $Q(A) = dA + [A,A]/2$ squares to zero because $d^2 = 0$, d is a derivation, and the bracket satisfies Jacobi. Its cotangent lift is Hamiltonian with Hamiltonian S_H. Thus $\{S_H, S_H\} = 0$. Restriction of fields and extension of compactly supported test functions give the local observable maps; disjoint supports give their factorization products. The Dolbeault–de Rham complex is elliptic; its local descent gives Weiss descent for the classical cotangent observables [21, 22]. The same argument applies with the second topological direction. □

There is a separate finite-dimensional quantum calculation. Put $M_H = \mathfrak{n}_H[1]$, and take its odd cotangent space $T^*[-1]M_H$. In linear coordinates c^a and their cotangent coordinates b_a, let Q be the Chevalley vector field and set

$$S_H^{\mathrm{fin}} = \sum_a b_a Q(c^a).$$

Use the constant Berezin volume to define the BV Laplacian Δ.

Proposition 3.12 (Finite cotangent quantum master equation)**.** *The finite action satisfies*

$$\tfrac{1}{2}\{S_H^{\mathrm{fin}}, S_H^{\mathrm{fin}}\} + \hbar\Delta S_H^{\mathrm{fin}} = 0.$$

Proof. The classical term vanishes by $Q^2 = 0$. The BV term is the divergence of Q; its linear coefficients are the modular supertrace $x \mapsto \mathrm{str}(\mathrm{ad}\, x)$. For an even homogeneous x, the adjoint action strictly raises positive root height and is upper triangular with zero diagonal. For an odd x, it is an odd endomorphism and has zero supertrace. Linearity proves that the entire divergence vanishes. □

A reduction of the local field theory must specify a contraction of its field complex and the induced pairing and operations. Reducing only a topological interval leaves the Dolbeault fields on E. Even their harmonic complex contains $H^{0,*}(E) = \mathbb{C} \oplus \mathbb{C}\eta$, with η in degree one. It is not the single copy $\mathfrak{n}_H[1]$ used in the finite proposition. A quantum pushforward must in addition control its measure, boundary conditions and loop terms.

The algebraic boundary model can be specified independently: quantize the linear Lie–Poisson space by $U(\mathfrak{n}_H)$ and take its two augmentation modules. Their derived tensor product is

$$\mathbb{C} \otimes^{\mathbb{L}}_{U(\mathfrak{n}_H)} \mathbb{C} \simeq C_*(\mathfrak{n}_H).$$

The bar–Chevalley map proves this statement. The holomorphic current model is V_H. Identifying these algebraic objects with boundary observables and an interval amplitude of the field theory requires boundary quantization and sewing maps. At infinite height the appropriate total-height completion must also be preserved; the limit of the uncompleted $U(\mathfrak{n}_H)$ is different. Dualizing the Lie quotient maps reverses their direction on the B-fields, so a field-theoretic limit also requires compatible pushforward maps.

Chapter 4

Extensions, charges and quiver Hall models

A geometric Hall product remembers extensions and their relative elliptic position. Its first construction is a correspondence over the stack of common translation torsors. Quiver models provide explicit algebraic examples, while critical coefficients and their orientation lines impose additional equations on the same flags.

4.1 Extensions over the elliptic classifying stack

Work on locally noetherian complex schemes with the fppf topology. Let BE be the stack of right E-torsors. For a left E-stack M, an object of $[M/E](T)$ is a right torsor $P \to T$ and an equivariant map $P \to M$. Forgetting the map gives $[M/E] \to BE$. All products of stacks below are two-fibre products.

Proposition 4.1 (The common translation torsor)**.** *There is a natural equivalence*

$$[(M \times N)/E_{\mathrm{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

The equivalences for any finite number of factors commute with regrouping and insertion of the terminal E-stack.

Proof. An object on the right is $(P, m), (Q, n)$ together with a torsor isomorphism $\alpha : P \to Q$. Send it to $(P, m, n\alpha)$. Conversely, use a common torsor twice and its identity isomorphism. A pair of torsor maps a, b on the right satisfies $b\alpha = \alpha' a$, so b is determined by a. The same construction identifies the maps and their descent isomorphisms. It commutes with base change. For several factors, every regrouping gives one torsor with the same list of equivariant maps, proving all stated coherences. □

For $M = N = E$ with translation, the diagonal quotient is E, with coordinate $y - x$. Both separate quotients are points. Their fibre product over BE retains the relative translation, while their ordinary product loses it.

Let $\mathfrak{M}$ be the stack of coherent sheaves on $X = S \times E$, with families flat over the parameter scheme, and put $\mathfrak{Q} = [\mathfrak{M}/E]$. An object of $\mathfrak{Q}(T)$ is a torsor $P \to T$ and a flat sheaf on $S \times P$. Fix a polarization and write $\mathfrak{Q}_\alpha$ for a Hilbert-polynomial component. Translation preserves these polynomials. Let $\mathfrak{H}_{\alpha,\beta}$ classify short exact sequences $0 \to A \to F \to B \to 0$ on a common $S \times P$, with all terms flat over T and polynomials α, $\alpha + \beta$, β. Its span over BE is

$$\mathfrak{Q}_\alpha \times_{BE} \mathfrak{Q}_\beta \xleftarrow{p} \mathfrak{H}_{\alpha,\beta} \xrightarrow{q} \mathfrak{Q}_{\alpha+\beta}.$$

The category of spans of fppf stacks over BE has composition by fibre product and tensor product $\times_{BE}$. Its unit object is the identity map of BE.

Theorem 4.2 (Geometric Hall correspondence)**.** *For fixed α, β, q is representable by algebraic spaces and proper. The spans form a coherently associative unital Hall object in the category of spans over BE. Its unit is the zero-sheaf span $BE \leftarrow BE \rightarrow \mathfrak{Q}$.*

Proof. Over a fixed (P, F), the fibre of q classifies quotients $F \twoheadrightarrow B$ of polynomial β. Its kernel is flat, because F and B are flat. An automorphism fixing the quotient and F is the identity. Trivializing P fppf-locally identifies this fibre with the fixed-polynomial Quot scheme, projective over the parameter scheme by Grothendieck's theorem [47, Theorem 3.2]. The quotient functor descends to an algebraic space, and properness descends with it.

Both composites of three Hall spans classify $0 \subset A \subset D \subset F$ on the same torsor. One bracketing concatenates inclusions. The other starts with $A \subset F$ and a subobject $B \subset F/A$, and sets $D = F \times_{F/A} B$. These constructions are inverse on objects and isomorphisms. They preserve flatness and commute with base change. For longer flags all regroupings identify the same subobjects and quotients. Their comparison maps agree, proving the pentagon and higher coherences. Finally $\mathfrak{Q}_0 = BE$. The sequences with zero subobject or zero quotient give the unit and its coherences. □

Properness here concerns the full coherent-sheaf stack on a fixed pair of Hilbert-polynomial components. A stability restriction can replace its fibre by an open part of a Quot scheme, which need not be proper. A bounded charge window also needs closure under the extensions and flags used in its operations. A reduced critical-coefficient realization must supply pullback, pushforward, and orientation comparisons on these same flags. Its unit has stabilizer E. A coefficient theory for this quotient must include that nonaffine stabilizer, or work relative to BE, on the unit and every flag.

4.2 Additive physical charge and quadratic automorphic degree

Let

$$\Lambda = \widetilde{H}(S, \mathbb{Z})$$

be the Mukai lattice, with pairing

$$(r, c, s) \cdot (r', c', s') = c \cdot c' - rs' - r's$$

and signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\text{phys}} = \Lambda \oplus \Lambda, \qquad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\text{ind}}.$$

Proposition 4.3 (Polarization identity)**.** *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = \big(Q_1 \cdot Q_2, Q_1 \cdot P_2 + Q_2 \cdot P_1, P_1 \cdot P_2 \big).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B. □

4.2.1 *The polarized central extension*

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\mathrm{ind}} \longrightarrow \widetilde{\Gamma}_\Pi \longrightarrow \Gamma_{\mathrm{phys}} \longrightarrow 0, \qquad \widetilde{\Gamma}_\Pi = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = \big(c + d, \lambda + \mu + B(c, d)\big).$$

The cocycle identity makes this multiplication associative. In fact $(c, \lambda) \mapsto (c, \lambda - \Pi(c))$ identifies it with the direct-product group, so the extension is split. The polarization identity makes the graph map

$$j_\Pi : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_\Pi, \qquad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $A = \bigoplus_{c \in \Gamma_{\mathrm{phys}}} A_c$ be a charge-graded associative algebra, and let $\Bbbk[\Gamma_{\mathrm{ind}}]$ have monomials u^λ. Define the polarized central-extension algebra

$$\mathcal{R}_\Pi(A) = A \otimes \Bbbk[\Gamma_{\mathrm{ind}}]$$

with product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d \, u^{\lambda + \mu + B(c,d)}.$$

Give $a_c u^\lambda$ degree $(c, \lambda) \in \widetilde{\Gamma}_\Pi$.

Theorem 4.4 (Graph-multiplicative polarization)**.** *The product $\star$ is associative and $\widetilde{\Gamma}_\Pi$-graded. The map*

$$\iota_\Pi : A \longrightarrow \mathcal{R}_\Pi(A), \qquad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_\Pi(c)$.

Proof. The two parenthesizations differ by

$$B(c, d) + B(c + d, e) - B(d, e) - B(c, d + e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_\Pi(a_c) \star \iota_\Pi(b_d) = a_c b_d \, u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. □

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

4.3 Point-local and wrapped sectors

Let $p_E : S \times E \to E$ be projection.

Lemma 4.5 (Positive elliptic degree is wrapped)**.** *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E. Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C, is all of E. □

Objects whose support has finite image in E give the point-local sector. Positive elliptic degree gives support over the whole curve.

Definition 4.6 (Mixed geometric operations)**.** Fix a symmetric monoidal coefficient theory on the relevant configuration, object and flag stacks, with its pullbacks and pushforwards. A local–wrapped datum consists of a local factorization object $\mathcal{F}_{\mathrm{loc}}$, wrapped coefficient objects W_m for $m > 0$, ordered products

$$W_{m_1} \otimes \cdots \otimes W_{m_r} \longrightarrow W_{m_1+\cdots+m_r},$$

and mixed maps from local insertions and wrapped inputs to wrapped outputs on their specified incidence diagrams. These maps must commute with disjoint local factorization, regrouping of wrapped extensions, substitution of mixed operations, and insertion of the unit. A geometric realization supplies the diagrams and coefficient maps, not only the resulting grading.

Write $\mathsf{Ch}_E(k)$ for a chosen local chiral operation object and $\mathsf{Ass}(r)$ for the ordered associative operation object. Mixed substitution is additional data. The expression $\mathsf{Ch}_E(k) \otimes \mathsf{Ass}(r)$ alone does not define it: substituting several wrapped products requires a rule assigning the local insertions to their factors. One can form the free colored operad on local, wrapped and mixed generators and then impose the identities of a supplied geometric datum. Identifying its operation objects with the simple tensor expression requires a proved normal-form theorem and the required distribution maps.

There is a concrete algebraic example. Take $L = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$, $W = U(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$, and let L act by adjoint derivations. For homogeneous $x \in L$ and $u, v \in W$,

$$x \cdot (uv) = (x \cdot u)v + (-1)^{|x||u|}u(x \cdot v), \qquad x \cdot 1 = 0.$$

Jacobi makes this a Lie action. It extends to a module-algebra action of $U(L)$, whose primitive coproduct is the distribution rule in the displayed formula. Thus the colored operad generated by a Lie bracket, an associative product and a derivation action has this local–wrapped algebra. The total-height completion preserves it. Chiral currents on a curve and compact wrapped coefficients need their supported analogues of these maps.

For $S \times E$, common-torsor extension and mixed-support flag correspondences provide the geometric diagrams to compare. A critical coefficient system satisfying base change and the mixed substitution equations would realize the preceding datum. Properness of a middle-term map alone supplies none of its critical pullback or collision maps.

4.3.1 Two product conventions for the Gram graph

The positive cocycle in $\mathcal{R}_\Pi(A)$ makes $a_c \mapsto a_c u^{\Pi(c)}$ a homomorphism. In the ordinary untwisted product on $A \otimes \mathbb{C}[\Gamma_{\mathrm{ind}}]$, the same graph monomials instead satisfy

$$(a_c u^{\Pi(c)})(b_d u^{\Pi(d)}) = u^{-B(c,d)}\, a_c b_d u^{\Pi(c+d)}.$$

This follows by subtracting the polarization identity. The negative cocycle therefore describes multiplication in graph coordinates for the untwisted algebra. The positive cocycle describes the changed multiplication that makes the graph map multiplicative. Both retain additive physical charge; neither makes the quadratic Gram map additive in its original target Γ_{ind}.

4.4 Finite simple-root windows

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borcherds superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Use the positive root height $h(i) = h(\alpha_i)$, whose sublevel sets of labelled generators are finite. Put

$$I_H = \{i \in I_\Delta : h(i) \le H\}.$$

The restriction of the invariant form gives a finite symmetric Borcherds–Cartan matrix

$$\begin{aligned} A_H &= (a_{ij})_{i,j\in I_H}, \\ a_{ii} &= 2 && \text{for real } i, \\ a_{ii} &\le 0 && \text{for imaginary } i, \\ a_{ij} &\le 0 && \text{for } i \ne j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 4.7 (The Igusa Borcherds quiver)**.** Order I_H. For $i < j$ place $-a_{ij}$ arrows from i to j. At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $Q_{\Delta,H}$.

Proposition 4.8 (Euler form)**.** *For dimension vectors use the Euler form* $\langle d, e\rangle = \sum_i d_i e_i - \sum_{a:i\to j} d_i e_j$. *Its symmetrization on the standard vertex basis is* A_H*:*

$$\langle e_i, e_j\rangle + \langle e_j, e_i\rangle = a_{ij}.$$

Proof. For $i \ne j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij}. At i the value is $2 - 2g_i = a_{ii}$. □

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

4.5 Hall realization of the positive algebra

In this Hall construction q is a prime power. Let $\mathcal{E}_H = \mathrm{Rep}_{\mathbb{F}_q}(Q_{\Delta,H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N\rangle} \sum_{[L]} |\{N' \subset L : N' \cong N,\ L/N' \cong M\}|\,[L], \qquad v^2 = q.$$

The class $[0]$ is the unit: the required subobject is unique on either side. Associativity follows by counting flags $N_1 \subset N_2 \subset L$ in two orders. The twist factors agree because the Euler form is bilinear:

$$\langle \alpha, \beta\rangle + \langle \alpha + \beta, \gamma\rangle = \langle \beta, \gamma\rangle + \langle \alpha, \beta + \gamma\rangle.$$

Thus this normalization defines the unital twisted Hall algebra on isomorphism classes.

Theorem 4.9 (Finite-window composition algebra)**.** *At each vertex i of $Q_{\Delta,H}$, choose the one-dimensional representation S_i supported at i, with every loop acting by zero. Use the generic composition algebra generated by these classes across finite fields, with the twisted subobject-count multiplication above. After extending coefficients from $\mathbb{Z}[v, v^{-1}]$ to $\mathbb{C}(v)$, this is the positive quantum generalized Kac–Moody algebra for the even Borcherds–Cartan matrix A_H. The parameter in Kang–Schiffmann is $v_{\mathrm{KS}} = v^{-1}$. Their specified semisimple perverse-sheaf construction on nilpotent representation varieties gives the corresponding canonical basis.*

Proof. This is the selected-simple composition construction of Kang–Schiffmann, Section 2.3 and Theorem 2.2, with one selected zero-loop simple at each vertex. Their geometric realization uses its split graded Grothendieck group over $\mathbb{Z}[v_{\mathrm{KS}}, v_{\mathrm{KS}}^{-1}]$ (Theorem 4.1), with the stated inversion of parameter [55, 56]. The selected geometric basis objects can be semisimple, as they are for a one-loop vertex; they are not all the individual simple perverse sheaves on the representation stack. The theorem does not use every simple object of the full representation category. A vertex with c loops has q^c one-dimensional simple representations over $\mathbb{F}_q$, whereas only the chosen zero-loop representation is the generator here. □

A semi-derived Hall realization of a full quantum group additionally specifies two-periodic complexes, the invertible acyclic classes and Cartan reduction [3, 69]. Its coproduct, pairing and completion are separate from the positive composition algebra used here.

4.5.1 Parity and the quantized superdatum

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. It is additive on the free lattice of labelled simple generators. Projection to the Lorentzian degree can identify contributions of different parity. Both parity dimensions must therefore be retained in each projected root space. Hong's construction applied to this Borcherds superdatum gives the positive part

$$U_{q,H}^{+}$$

and the full quantized Borcherds superalgebra $U_{q,H}$.

Theorem 4.10 (Ordinary and super presentations)**.** *The ordinary quiver Hall presentation and Hong's quantum superalgebra use the same specified Cartan matrix and labelled simple generators. Their relation ideals and parity conventions are different inputs. Equality of their Cartan gradings alone does not identify their PBW characters.*

Proof. The two presentations are those of the preceding Hall construction and Hong's superalgebra construction. The difference already occurs for a single isotropic generator. The enveloping algebra of a one-dimensional even abelian Lie algebra is $\mathbb{C}[x]$. Declaring its generator odd only as a graded vector space leaves a nonzero vector x^2 in degree two. The enveloping algebra of a one-dimensional odd abelian Lie superalgebra is $\bigwedge(\xi)$, since its defining relation is $2\xi^2 = [\xi, \xi] = 0$. Its degree-two component is zero. Thus parity assignment to an ordinary PBW basis does not produce the super relation ideal or its character. A comparison must respect these relations degree by degree. □

4.6 The real-root seed window

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$A_{\mathrm{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 4.11 (Real-root Hall seed)**.** *The symmetrized Euler form of Q_{re} is A_{re}. The generic twisted Hall composition algebra over $\mathbb{C}(v)$ generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re}.*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2. The Ringel–Green theorem identifies the generic composition algebra with the positive quantum algebra for this symmetric Cartan matrix [43, 80]. □

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

4.6.1 The first Ginzburg differential

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \to j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \to i$ of degree -1 and loops t_i of degree -2. The canonical three-Calabi–Yau completion has

$$da_{ij}^{(r)} = 0, \qquad d(a_{ij}^{(r)})^* = 0,$$

and

$$dt_i = \sum_{a : s(a) = i} a a^* - \sum_{a : t(a) = i} a^* a. \tag{4.1}$$

The identity $d^2 = 0$ follows immediately. Formula (4.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 4.12 (Vertex one)**.** At vertex 1,

$$dt_1 = \sum_{r=1}^{2} \Big(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \Big).$$

At vertex 3,

$$dt_3 = -\sum_{r=1}^{2} \Big((a_{13}^{(r)})^* a_{13}^{(r)} + (a_{23}^{(r)})^* a_{23}^{(r)} \Big).$$

The sum over vertices is $\sum_a (aa^* - a^* a)$. It vanishes in the cyclic quotient and after matrix trace, but generally not in the path algebra.

Proof. The outgoing arrows at vertex 1 have targets 2 and 3. The incoming arrows at vertex 3 have sources 1 and 2. The displayed formulas follow from these incidence lists. Each arrow contributes one commutator to the sum over vertices. The trace of a commutator vanishes, giving the asserted trace identity. □

4.7 The three-Calabi–Yau completion

Let $kQ_{\Delta,H}$ be the path algebra. Its algebraic three-Calabi–Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Write J for the ideal generated by all arrows in its graded quiver. Its completed version is the separate topological dg algebra $\widehat{\Gamma}_{\Delta,H} = \varprojlim_n \Gamma_{\Delta,H}/J^n$. The algebraic Calabi–Yau theorem is used for $\Gamma_{\Delta,H}$; a topological Calabi–Yau assertion for $\widehat{\Gamma}_{\Delta,H}$ requires its completed bimodule category. This is the Ginzburg model with zero potential: a degree-minus-one reverse arrow for every arrow and a degree-minus-two loop at each vertex. The loop differential is the signed sum of incident arrow–reverse-arrow products.

Theorem 4.13 (Borcherds–Ginzburg projections)**.** *For each finite quiver $Q_{\Delta,H}$, its undeformed three-Calabi–Yau completion $\Gamma_{\Delta,H}$ is the Ginzburg dg algebra with zero potential. If $H \leq H'$ and $Q_{\Delta,H}$ is the full subquiver on a subset of vertices, there is a surjective dg algebra map*

$$\pi_{H'H} : \Gamma_{\Delta,H'} \longrightarrow \Gamma_{\Delta,H}.$$

It kills the added vertices, every arrow incident to them, the corresponding reverse arrows, and the added degree-minus-two loops. These maps compose strictly and extend continuously to the completed algebras.

Proof. The finite Calabi–Yau assertion is the Ginzburg model of Keller's completion theorem [38, 57]. We concatenate paths from left to right, so aa^* is based at the source of a. Original and reverse arrows have zero differential. For the degree-minus-two loop t_i, the differential is

$$dt_i = \sum_{s(a)=i} aa^* - \sum_{t(a)=i} a^*a.$$

The projection deletes exactly the terms involving an added vertex, leaving the same sum for the full subquiver. It therefore commutes with the differential. It preserves path length and extends to the completed path algebra. Successive projections kill the union of their omitted vertices and arrows, proving strict composition. □

The evident inclusion of graded quivers does not give the reverse dg map. A single old vertex has loop differential zero. Adding an arrow $a : i \to j$ changes that differential to aa^* at i. The map fixing the old loop therefore fails to commute with the differential.

Write $\Gamma_\Delta = (\Gamma_{\Delta,H}, \pi_{H'H})_H$ for this pro-system of dg algebras. It is separate from the ordinary Hall embeddings obtained by extension by zero on representations of a full subquiver. A direct system of Calabi–Yau dg categories compatible with those Hall embeddings requires additional dg functors. It does not follow from the inclusion of the displayed Ginzburg generators.

The quantum group retains its separately specified Borel completion. Its transition maps must preserve the Cartan generators, coproduct, and pairing. Quotienting the positive nilpotent Lie algebra by height and truncating a full quantum group are different operations.

4.8 Hall states and protected superstates

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the composition-algebra construction. The full equivariant derived category is a larger carrier. Already for one real vertex and dimension one, its quotient is $B\mathbb{G}_m$, with $H^*(B\mathbb{G}_m;\mathbb{Q}) = \mathbb{Q}[c_1]$, where $|c_1| = 2$. Its total equivariant cohomology is infinite-dimensional. Finite root multiplicities therefore refer to the specified composition algebra, not to total equivariant cohomology of this stack.

Definition 4.14 (Igusa Hall category)**.** For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

Use bounded equivariant constructible complexes, with coefficients $\overline{\mathbb{Q}}_\ell$ away from the field characteristic. The direct sum over dimension vectors has convolution $q_! p^*$ along the extension stack, with endpoints on the left and middle term on the right. Flag base change proves associativity. Denote this monoidal category by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_\Delta = \varinjlim_H \mathcal{B}_{\Delta,H}.$$

Independently, let $\mathfrak{n}_{I_H}$ be the positive Lie superalgebra generated by the selected labelled simple roots, with their Borcherds relations. Its one-particle space is

$$\mathcal{H}_{\mathrm{ip},H} = \mathfrak{n}_{I_H} = \bigoplus_{\alpha>0} (\mathfrak{n}_{I_H})_\alpha.$$

It generally has infinitely many root degrees. The bound H restricts the simple generators; it is different from the nilpotent quotient $\mathfrak{n}_H$, which discards all root spaces above height H. The corresponding multiparticle space is the PBW superspace of $U(\mathfrak{n}_{I_H})$.

Theorem 4.15 (Hall and protected-state comparison)**.** *The composition subalgebra in the finite-window Hall theorem is the ordinary positive quantum generalized Kac–Moody algebra. This statement does not identify the Grothendieck group of the full category $\mathcal{B}_{\Delta,H}$ with that composition algebra. The protected one-particle superspace has signed root multiplicities*

$$\mathrm{sdim}\, \mathcal{H}_{\mathrm{ip},H,\alpha} = \mathrm{sdim}(\mathfrak{n}_{I_H})_\alpha.$$

Taking the union of labelled windows and then completing by total root height gives the PBW character

$$\mathrm{sch\,Fock}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0} (1 - e^{-\alpha})^{-\mathrm{sdim}\, \mathfrak{g}_{\Delta_5,\alpha}},$$

and its Weyl-shifted inverse is $D_5(2Z)$. The composition canonical basis and the protected root presentation use the specified simple-root indexing and Cartan grading; the parity refinement belongs to the Borcherds superdatum.

Proof. The composition-algebra assertion is the finite-window theorem. The protected assertion is PBW for the Borcherds Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. □

4.9 Extension determinants and critical orientations

Let C be a three-Calabi–Yau dg category over $\mathbb{C}$, with its chosen trace. For families of objects A, B over a base T, let $C(A,B)$ denote the relative derived Hom complex on T. Suppose these complexes are perfect, and that the trace gives base-change-compatible duality

$$C(B,A) \simeq C(A,B)^\vee[-3].$$

Put $L(A,B) = \det C(A,B)$ and $D(A) = L(A,A)$. The determinant convention is $\det C = \otimes_i (\det C^i)^{(-1)^i}$, with symmetry sign specified by $\chi(C) \bmod 2$.

Theorem 4.16 (Extension determinant)**.** *For an exact triangle $A \to F \to B \to A[1]$, there is a base-change-compatible isomorphism*

$$D(F) \simeq D(A) \otimes D(B) \otimes L(A,B)^{\otimes 2}.$$

The determinant comparisons obtained by refining a flag agree, with the stated determinant symmetry signs.

Proof. Apply derived Hom to the triangle in each variable. Determinant additivity gives

$$D(F) \simeq L(B,A) \otimes D(A) \otimes D(B) \otimes L(A,B).$$

Dualization and the odd shift each invert the determinant line. Thus Calabi–Yau duality identifies $L(B,A)$ with $L(A,B)$. Reorder by determinant symmetry to obtain the formula. For a full flag, all iterated decompositions identify with the same ordered tensor product of $C(A_i, A_j)$, for its successive quotients A_i. Determinant additivity for refined filtrations proves coherence. Each operation commutes with base change. □

A square-root orientation consists of lines o_A and isomorphisms $o_A^{\otimes 2} \simeq D(A)$. Extension compatibility requires maps

$$o_A \otimes o_B \otimes L(A,B) \longrightarrow o_F$$

whose squares are the determinant comparisons. Choosing coordinates on a smooth critical chart trivializes its ambient canonical line. Hall convolution requires transition maps compatible with the cross line and the zero object. On three-step flags, the two orientation maps can differ by an element of μ_2, even when their squares agree. That sign must be trivialized coherently.

Example 4.17 (A nontrivial cross line)**.** Let X be a smooth projective Calabi–Yau threefold, with a chosen volume form, and let $x \in X$. On $X \times T$, where $T = \mathbb{P}^1$, take

$$A = \mathcal{O}_X \boxtimes \mathcal{O}_T(1), \qquad B = \mathcal{O}_x \boxtimes \mathcal{O}_T.$$

Then $C(A,B) = \mathcal{O}_T(-1)[0]$, and relative Serre duality gives $C(B,A) = \mathcal{O}_T(1)[-3]$. Thus $L(A,B) = L(B,A) = \mathcal{O}_T(-1)$. The self-Hom complexes of A and B have constant determinant lines, since the factors $\mathcal{O}_T(1)$ cancel in the first self-Hom. Nevertheless

$$D(A \oplus B) = D(A) \otimes D(B) \otimes \mathcal{O}_T(-2).$$

The orientation cross factor is $\mathcal{O}_T(-1)$. A sign change cannot replace this line of nonzero degree.

Corollary 4.18 (Oriented critical coefficients)**.** *Suppose the chosen critical charts have square-root orientation maps on overlaps and extension flags, with unit and flag coherences. Suppose their vanishing-cycle comparison maps preserve these data. Then the local vanishing-cycle complexes descend to the specified oriented coefficient object. Their Hall convolution is associative whenever its pullback and pushforward maps are defined and satisfy base change on the flag diagrams.*

Proof. The overlap cocycle identifies the local coefficient complexes by descent. On a three-step flag, orientation coherence identifies the two Thom–Sebastiani coefficient maps. Proper or compact-support base change, on the stipulated domain, identifies the two composites of convolution. The zero flag gives the unit equation. □

For the quiver charts, the remaining construction is an equivariant system of square roots and coefficient maps on the actual representation and extension stacks, compatible with changes of height. For compact $S \times E$, the reduced complexes and the common elliptic torsor must also be retained. A coordinate volume on an affine chart specifies neither descent system by itself.

Chapter 5

The quantum root datum

The labelled presentation also admits a quantum Hopf algebra. Its Cartan lattice, parity signs and Borel completion are part of the construction. The canonical pairing produces a root tensor; a separate Cartan weight operator turns it into the intertwiner on the specified representation domain.

5.1 The integral colored datum

Retain the labelled simple-root set I_Δ, including its multiplicities and parity $p(i)$. For a finite sublevel I_H, use the free lattice

$$Q_H = \bigoplus_{i \in I_H} \mathbb{Z}\epsilon_i, \qquad \pi(\epsilon_i) = \alpha_i \in L_{II}, \qquad a_{ij} = (\alpha_i, \alpha_j).$$

The sign bicharacter is $\theta(\epsilon_i, \epsilon_j) = (-1)^{p(i)p(j)}$. It lives on the labelled lattice. Since π can identify opposite parities, it need not descend to L_{II}.

The matrix A_H is integral and symmetric, with real diagonal two, imaginary diagonal nonpositive and even, and off-diagonal entries nonpositive. Odd simple generators are imaginary. These are the hypotheses for the colored quantum Borcherds construction [51, Section 2]. Work over $\mathbb{C}(q)$, where q is indeterminate, so no root-of-unity specialization is implied.

For the extended Cartan, take independent h_i, d_i with

$$(h_i, h_j) = a_{ij}, \qquad (h_i, d_j) = \delta_{ij}, \qquad (d_i, d_j) = 0.$$

Its form is nondegenerate. The simple root as a Cartan weight is $x \mapsto (h_i, x)$; the d_i separate repeated labelled roots. The derived quantum algebra uses only the torus generated by the h_i, together with the positive and negative generators.

5.2 The reduced Cartan and coproduct

The derived algebra is generated by E_i, F_i, K_h, with $h \in \bigoplus_i \mathbb{Z}h_i$, $|E_i| = |F_i| = p(i)$, and

$$K_h K_{h'} = K_{h+h'}, \qquad K_h E_i K_h^{-1} = q^{(h,h_i)} E_i, \qquad K_h F_i K_h^{-1} = q^{-(h,h_i)} F_i.$$

Its mixed relation is

$$E_i F_j - (-1)^{p(i)p(j)} F_j E_i = \delta_{ij} \frac{K_{h_i} - K_{h_i}^{-1}}{q - q^{-1}}.$$

For real $i \neq j$, the positive Serre relation is

$$\sum_{r=0}^{1-a_{ij}} (-1)^r \begin{bmatrix} 1 - a_{ij} \\ r \end{bmatrix}_q E_i^{1-a_{ij}-r} E_j E_i^r = 0,$$

with the corresponding relation for F. If $a_{ij} = 0$, the two generators supercommute. Here $[n]_q = (q^n - q^{-n})/(q - q^{-1})$, and the bracket denotes the associated Gaussian binomial coefficient. All tensor products of superalgebras use the Koszul multiplication sign.

The coproduct and counit are

$$\begin{aligned} \Delta E_i &= E_i \otimes 1 + K_{h_i} \otimes E_i, & \epsilon(E_i) &= 0, \\ \Delta F_i &= F_i \otimes K_{h_i}^{-1} + 1 \otimes F_i, & \epsilon(F_i) &= 0, \\ \Delta K_h &= K_h \otimes K_h, & \epsilon(K_h) &= 1. \end{aligned}$$

The antipode is $S(E_i) = -K_{h_i}^{-1} E_i$, $S(F_i) = -F_i K_{h_i}$, and $S(K_h) = K_{-h}$.

Theorem 5.1 (Quantization with the Lorentzian Cartan)**.** *Impose $K_h = 1$ for*

$$h \in \ker\left(\bigoplus_{i \in I_H} \mathbb{Z} h_i \longrightarrow L_{II},\ h_i \longmapsto \alpha_i \right).$$

This is a central Hopf quotient. Once I_H contains the three real generators, its torus is $\mathbb{C}(q)[L_{II}]$, and it has triangular decomposition

$$U_{q,H}^{\mathrm{red}} \cong U_{q,H}^{-} \otimes \mathbb{C}(q)[L_{II}] \otimes U_{q,H}^{+}.$$

The positive and negative labelled PBW dimensions and parities agree with those of the classical Borcherds presentation. The inclusions of finite labelled sets give a quantum algebra $U_q^{\mathrm{red}}(\mathfrak{g}_{\Delta_\varsigma})$ with these properties.

Proof. Hong's presentation and triangular decomposition apply to the extended datum [51, Propositions 2.2–2.3 and 3.1]. A Cartan vector in the displayed kernel pairs to zero with every h_j. Its torus element is therefore central. Moreover

$$\Delta(K_h - 1) = (K_h - 1) \otimes K_h + 1 \otimes (K_h - 1),$$

so these relations generate a Hopf ideal. In the triangular decomposition the quotient acts only on the middle group algebra. Its quotient is the group algebra of the image lattice, which is L_{II} because the real roots form a basis.

The labelled positive and negative relations remain unchanged. Hong's character theorem identifies their graded dimensions with the classical ones. Relations involving an added generator have positive degree in that generator, so they impose no new relation on an older positive or negative subalgebra. Triangular decomposition then gives the injective transition maps. In a fixed positive Lorentzian degree, proper height permits only finitely many labelled words. The finite character identity therefore survives both the union and the projection to that degree, in each parity. □

The full quantum algebra has not been completed by taking a product of all positive and negative root spaces. The discontinuity in Proposition 3.3 persists as a warning against that topology. The Borel completion used for the canonical tensor has a different definition.

5.3 The canonical tensor and the weight operator

For a finite labelled set, let U be the extended quantum algebra. If $\beta \in Q_H^+$, let

$$U^{+,\beta} = \bigoplus_{\gamma \not\leq \beta} U_\gamma^+, \qquad \widehat{U} = \varprojlim_{\beta} U / U\, U^{+,\beta}.$$

The order is coordinatewise on the labelled positive lattice. These are left-ideal neighbourhoods, not two-sided root-height quotients. Hong constructs the algebra structure and its tensor completion in this topology [51, (7.2)].

The nondegenerate Borel pairing gives a canonical element $C_\beta \in U_\beta^+ \otimes U_{-\beta}^-$. Put $h_\beta = \sum_i \beta_i h_i$ and $K_\beta = K_{h_\beta}$. The completed element is

$$C = \sum_{\beta \in Q_H^+} \theta(\beta,\beta) q^{(h_\beta, h_\beta)} (K_\beta^{-1} \otimes K_\beta) C_\beta.$$

It satisfies a twisted intertwining identity. The twist Φ fixes pairs of torus elements and has

$$\Phi(E_i \otimes 1) = E_i \otimes K_i, \quad \Phi(1 \otimes E_i) = K_i \otimes E_i,$$
$$\Phi(F_i \otimes 1) = F_i \otimes K_i^{-1}, \quad \Phi(1 \otimes F_i) = K_i^{-1} \otimes F_i.$$

Here $K_i = K_{h_i}$. On weight modules for which the weight pairing is defined, set

$$Z(v_\lambda \otimes w_\mu) = q^{(\lambda,\mu)} v_\lambda \otimes w_\mu.$$

Adjoin the required powers of q if the chosen weight lattice has rational pairing. Restrict to tensor products on which the series for C and its inverse are defined. For example, a highest-weight bound on the first factor makes each vector receive only finitely many positive-root terms.

Theorem 5.2 (The quantum R operator)**.** *On this weight-module domain,*

$$R = Z^{-1} C$$

is invertible and satisfies $R\Delta(a) = \Delta^{\mathrm{op}}(a) R$ and the Yang–Baxter equation. The opposite coproduct uses the super flip. These conclusions hold for the full labelled system on modules satisfying the same finiteness conditions.

Proof. Nondegeneracy of the Borel pairing is [51, Theorem 4.6]. The canonical tensor satisfies $C\Delta(a) = \Phi(\Delta^{\mathrm{op}} a) C$ and the two compatible coproduct identities of [51, Propositions 7.4–7.5 and Theorem 7.6]. Conjugation by Z realizes Φ. Multiplying by Z^{-1} gives the ordinary intertwining identity. The two coproduct identities give the Yang–Baxter equation, as in [51, Theorem 7.7]. The finiteness hypothesis permits these identities to act on the stated tensor products. In the full system, every contributing finite set of labelled degrees lies in a finite window, where the same identities hold. □

The Cartan factor is necessary even for one real root. On the standard two-dimensional $U_q(\mathfrak{sl}_2)$-module, choose v_+, v_- with $Kv_\pm = q^{\pm 1} v_\pm$, $Ev_- = v_+$, $Fv_+ = v_-$, and $Ev_+ = Fv_- = 0$. Then

$$C = 1 - (q - q^{-1}) E \otimes F, \qquad Z = \mathrm{diag}(q^{1/2}, q^{-1/2}, q^{-1/2}, q^{1/2}).$$

With the coproduct above, $Z^{-1}C$ intertwines it with its opposite; C alone does not. This example also fixes the sign of the root term. The reduced Cartan quotient has the Hopf algebra constructed above. Its braided representation theory must retain weight pairings and a domain on which the same operator descends.

A geometric Hall comparison requires a coproduct, Hopf pairing and module functor compatible with this quantization. A current or fusion construction requires a further comparison of its supported operations with the stated weight-module braiding. Neither follows from equality of denominator characters.

Chapter 6

Fredholm determinants and compact comparisons

A formal determinant becomes an analytic operator determinant only when both parity dimensions are summable. The minimal simple-root presentation gives such a bound through an absolute theta estimate and the free tensor algebra. This constructs the root Fredholm model before asking for its compact geometric realization.

6.1 Finite determinants

For a positive height bound H, set

$$\mathcal{V}_H = \bigoplus_{h(\alpha)\le H} \mathfrak{g}_{\Delta_5\alpha}, \qquad \mathscr{D}_H|_{\mathfrak{g}_{\Delta_5\alpha}} = (1 - e^{-\alpha})\,\mathrm{id}\,.$$

All root spaces here are positive. Their two parity dimensions are those of the fixed minimal presentation.

Proposition 6.1 (Finite Berezinian and formal limit)**.** *In the total-height completion,*

$$\mathrm{Ber}(\mathscr{D}_H) = \prod_{h(\alpha)\le H} (1 - e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}, \qquad e^{-\rho}\lim_H \mathrm{Ber}(\mathscr{D}_H) = D_5(2Z).$$

Pullback by $\jmath(Z) = Z/2$ sends the last expression to $D_5(Z)$, whose square is $\chi_{10}(Z)$.

Proof. On a finite super vector space, the Berezinian is the even determinant divided by the odd determinant. This gives the product. Each formal coefficient uses only finitely many positive-height roots, so it stabilizes as H increases. The denominator theorem supplies the Weyl prefactor and the coordinate substitution. □

6.2 An absolute bound for both root parities

Let $M(t)$ count the labelled simple generators by their ordinary dimension, with even and odd generators both counted positively. For the minimal presentation,

$$M(t) = 3t + \frac{27t^2}{1 - t^2} + \sum_{\substack{a\in C\cap L_{II}\\ (a,a)<0}} |m(a)|\,t^{h(a)}.$$

The second term counts nine even generators at every multiple of each of the three height-two null rays.

Lemma 6.2 (Theta majorant)**.** *The series $M(t)$ converges for $0 < t < 1$, and $M(t) \to 0$ as $t \to 0^+$.*

Proof. Specialize the root-torus variables to

$$Q = S = t^2, \qquad R = t^{-1}.$$

A monomial of root degree $\alpha(n, \ell, m)$ has exponent $2n - \ell + 2m = h(\alpha)$. A term of a genus-two theta constant, indexed by $x, y \in \mathbb{Z} + \{0, 1/2\}$, has absolute value

$$t^{x^2+y^2-xy}, \qquad x^2 + y^2 - xy = \tfrac{1}{2}((x - y)^2 + x^2 + y^2) \ge \tfrac{1}{2}(x^2 + y^2).$$

Each absolute theta series is therefore Gaussian and converges. The product of the ten absolute series bounds the sum of absolute Fourier coefficients of Δ_5^{θ}, including possible cancellations in its actual coefficients. Since $m(a) = -c_{\rho+a}$, the last sum in $M(t)$ is bounded by this product divided by $64t^{h(\rho)}$, where $h(\rho) = 3/2$. This proves convergence at every fixed $0 < t < 1$. All terms of M have positive integral height. Dominated convergence using any fixed $t_* < 1$ then proves $M(t) \to 0$. □

Theorem 6.3 (Full root-dimension bound). *There exist $C_0, C_1 > 0$ such that*

$$\dim \mathfrak{g}_{\Delta_5 \alpha} \le C_0 \exp(C_1 h(\alpha)) \tag{6.1}$$

for every positive root. More strongly, for any $t_0 \in (0,1)$ with $M(t_0) < 1$,

$$\sum_{\alpha>0} \dim \mathfrak{g}_{\Delta_5 \alpha}\, t_0^{h(\alpha)} \le \frac{1}{1 - M(t_0)} - 1.$$

Both parity dimensions are included in this estimate.

Proof. Let V be the superspace of labelled simple generators. The tensor algebra has ordinary dimension series

$$\sum_{r\ge 0} \mathrm{Hilb}(V^{\otimes r}; t_0) = \sum_{r\ge 0} M(t_0)^r = \frac{1}{1 - M(t_0)}.$$

It surjects onto $U(\mathfrak{n}_{\Delta_5,+})$. PBW embeds $\mathfrak{n}_{\Delta_5,+}$ in that enveloping algebra. All maps preserve height and parity, so dimensions can only decrease. Removing the unit gives the stated sum. Each summand is bounded by its right side, giving the exponential estimate with $C_1 = -\log t_0$. The lemma ensures that such t_0 exists. □

A signed multiplicity bound alone would not prove this theorem. At the first timelike root it bounds 64, whereas the full root space has dimension 122. The tensor-algebra argument starts from the specified simple generators and controls all their brackets.

6.3 The two root Hilbert spaces

Choose a homogeneous orthonormal basis in every root space. Define

$$\mathcal{H}_\Delta = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta_5 \alpha}.$$

There is also a minimal virtual space

$$V_\alpha^{\mathrm{vir}} = \mathbb{C}^{\max(\dim \mathfrak{g}_{\Delta_5 \alpha}, 0) \mid \max(-\dim \mathfrak{g}_{\Delta_5 \alpha}, 0)}, \qquad \mathcal{H}_\Delta^{\mathrm{vir}} = \widehat{\bigoplus}_{\alpha>0} V_\alpha^{\mathrm{vir}}.$$

The hats here mean Hilbert direct sums. The first carrier retains both root parities; the second retains only their difference and has no supplied root bracket.

On the denominator period domain, put $Z_{\mathrm{root}} = 2Z$. Thus $Q = e^{4\pi i \tau}$, $R = e^{4\pi i z}$, $S = e^{4\pi i \sigma}$, and use the additive exponent

$$\alpha(Z) = -4\pi i(n\tau + \ell z + m\sigma), \qquad e^{-\alpha(Z)} = Q^n R^\ell S^m.$$

Let $\mathfrak{C}_{\mathrm{abs}}$ be the domain where $\Re \partial_i(Z) > 0$ for all three real simple roots and

$$\sum_{\alpha>0} \dim \mathfrak{g}_{\Delta_5 \alpha}\, |e^{-\alpha(Z)}| < \infty.$$

Theorem 6.4 (Nonempty trace-class domain)**.** *For every direction Z_0 with $c = \min_i \Re\delta_i(Z_0) > 0$, the ray TZ_0 lies in $\mathfrak{C}_{\mathrm{abs}}$ for all sufficiently large $T > 0$. On both Hilbert superspaces the operator*

$$K_Z|_\alpha = e^{-\alpha(Z)}\,\mathrm{id}$$

is trace class in each parity. The family is holomorphic in trace norm on the interior of its convergence domain.

Proof. For $\alpha = \sum_i b_i\delta_i$, with $b_i \geq 0$, $\Re\alpha(TZ_0) \geq Tc\,h(\alpha)$. Choose T so that $e^{-Tc} \leq t_0$ in the preceding theorem. Its summable bound dominates the trace norm of K_Z. The virtual space has no larger dimension in any degree, so the same estimate applies there. At any interior point of the convergence domain, choose a nearby convergent point with all three real simple exponents smaller. Its absolute series dominates a neighbourhood. The diagonal holomorphic entries then give a trace-norm holomorphic family. □

6.4 Skew doubling and the Fredholm product

Put $D_Z = 1 - K_Z$. The real parts are positive on the chosen domain, so no eigenvalue of K_Z is one. Its even and odd Fredholm determinants are therefore nonzero. On either root Hilbert superspace form, separately in each parity,

$$\mathbb{A}_{Z,\tilde{p}} = \begin{pmatrix} 0 & D_{Z,\tilde{p}} \\ -D_{Z,\tilde{p}}^{\mathrm{t}} & 0 \end{pmatrix}.$$

The transpose uses the chosen basis and its dual. Order the finite root truncations by height and then by a fixed order within each degree. Normalize each Pfaffian by its value at $K = 0$.

Theorem 6.5 (Oriented Fredholm realization)**.** *The normalized finite Pfaffians converge, and their even-to-odd ratio satisfies*

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha>0}(1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_{\frac{1}{2}}\alpha}}.$$

For both the full and the minimal virtual root carrier,

$$e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_{\frac{1}{2}}(2Z)$$

in the Gritsenko–Nikulin period convention.

Proof. For a finite square matrix D, the Pfaffian of $\left(\begin{smallmatrix} 0 & D \\ -D^t & 0 \end{smallmatrix}\right)$ is $(-1)^{N(N-1)/2}\det D$. Normalizing at $D = 1$ removes that sign. Hence the paritywise normalized Pfaffians equal their determinants at every finite height. The truncations of K_Z converge in trace norm, so determinant continuity gives the Fredholm limit. The odd determinant is nonzero on the domain, allowing the ratio. Both carriers have the same superdimension in every degree, so they give the same product. The denominator identity supplies its Weyl shift. □

This is a skew operator model with a specified polarization and relative Pfaffian normalization. It does not supply a first-order differential operator or its principal symbol. A compact geometric family requires its own domain, symbol, line comparison and orientation transport, together with an intertwiner to the root model.

6.5 What the automorphic character determines

Proposition 6.6 (Information retained by the product)**.** *With the cusp and automorphy law fixed, the normalized product records its signed exponents, Weyl prefactor, divisor and modular multiplier. It does not determine a Lie bracket, the two separate parity dimensions, extension correspondences, or an orientation comparison with a geometric determinant line.*

Proof. The recorded data are the exponents, prefactor and transformation law. To see the lost information concretely, put even vectors x, y in degree one and z in degree two. The abelian bracket and the Heisenberg bracket $[x, y] = z$ give nonisomorphic Lie algebras with the same PBW character

$$\frac{1}{(1-t)^2(1-t^2)}.$$

An added even–odd pair leaves every signed exponent unchanged. Neither character calculation specifies an extension correspondence. For the line ambiguity, the two square-root pairs on $\mathbb{C}^\times$, with square isomorphisms given by 1 and by the coordinate t, have the same square line. They are not isomorphic as root pairs, since no unit of $\mathbb{C}[t, t^{-1}]$ has square t. A comparison must retain these structures separately. □

Recognition Theorem 6.7 (Primitive recognition from characters)**.** *Let $\mathfrak{p}$ satisfy the proper-height finiteness hypotheses of Theorem 3.6, and let $\Psi : \mathfrak{n}_{\Delta_5,+} \twoheadrightarrow \mathfrak{p}$ be a graded, parity-preserving Lie map. It is an isomorphism if the two parity-refined PBW series agree. More generally, for a surjection between positive Lie superalgebras whose root spaces have prescribed pure parity, equality of their signed PBW characters also suffices. Their induced height completions, Chevalley complexes and current envelopes are then isomorphic.*

Proof. The first assertion is the universal presentation criterion. For the second, the logarithm of the PBW product determines signed primitive dimensions inductively in height: contributions from proper multiples use smaller degrees and can be subtracted. Pure parity recovers ordinary dimensions from these signed numbers. Degreewise surjectivity and finite dimension give injectivity. All the other maps are induced functorially. □

The pure-parity criterion does not apply to the first timelike Igusa root, where both parities occur. Its comparison uses both parity dimensions.

6.5.1 The automorphic line and its reflection action

The scalar product, with its Weyl prefactor, continues to a section of $\mathcal{L}^5 \otimes \nu_5$, where $\mathcal{L}$ is the weight-one modular line. This constructs its automorphic transition functions. Operator transition maps are additional: scalar determinant equality does not identify Hilbert spaces, operator domains or their skew families.

Proposition 2.10 specifies the linear and weight-five slash actions separately.

An orientation comparison with a geometric skew family must specify which of these actions it intertwines, together with the weight-five line. The monodromy of a root line is not determined solely by the vanishing order of one scalar section.

6.6 Reduced curve counting on $K3 \times E$

Let S be a smooth projective K3 surface and E an elliptic curve. For a nonzero effective primitive class $\beta_h \in H_2(S, \mathbb{Z})$, with $\beta_h^2 = 2h - 2$, choose $\beta_h^\vee \in H^2(S, \mathbb{Q})$ satisfying $\int_{\beta_h} \beta_h^\vee = 1$. Define

$$N_{g,h,d} = \int_{[\overline{M}_{g,1}(S\times E,(\beta_h,d))]^{\mathrm{red}}} \mathrm{ev}_1^*(\beta_h^\vee \otimes [\mathrm{pt}]_E).$$

The bullet permits disconnected domains with no collapsed connected components; $2g-2$ denotes minus their Euler characteristic. The insertion fixes the reduced incidence convention. Reduced deformation invariance makes the result depend only on g, h, d. As h varies, use any pairs (S, β_h) with the stated square. Put

$$\mathcal{Z}(u,q,s) = \sum_{g,h,d\geq 0} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 6.8 (Primitive Igusa cusp-form theorem)**.** *The primitive reduced series is the Laurent expansion in u of*

$$\mathcal{Z}(u,q,s) = -\chi_{10}(q,r,s)^{-1}\big|_{r=e^{iu}}.$$

For fixed q, s degrees, this is the expansion of the corresponding rational function of r at $r=1$. Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12}\Delta_s^{\theta-2}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [75]; their normalization is the monic Borcherds product χ_{10}. The second formula is the identity $\chi_{10} = 2^{-12}\Delta_s^{\theta^2}$. □

The theorem identifies the numerical generating series in its stated primitive sector. The Borcherds root algebra supplies a separate algebraic character to compare with that series.

Corollary 6.9 (Exact character comparison)**.** *Let $\jmath : \mathbb{H}_2 \to \mathbb{H}_2$ be the half-coordinate map $\jmath(Z) = Z/2$, and let*

$$\mathrm{Fock}(\mathfrak{n}_{\Delta_s,+}) = \mathrm{Sym}(\mathfrak{n}_{\Delta_s,+_{\bar{0}}}) \otimes \bigwedge(\mathfrak{n}_{\Delta_s,+_{\bar{1}}}).$$

Then

$$\mathcal{Z}^{\mathrm{red}}_{S\times E} = -\jmath^*\Big(e^{2\rho}\,\mathrm{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_s,+})^{\otimes 2}\big)\Big) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha>0}(1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_s}\alpha}.$$

The denominator theorem gives $\mathrm{den}(\mathfrak{g}_{\Delta_s}) = e^{-\rho}P = D_s(2Z)$, while PBW gives $\mathrm{sch}\,\mathrm{Fock}(\mathfrak{n}_{\Delta_s,+}) = P^{-1}$. Hence

$$e^{2\rho}\,\mathrm{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_s,+})^{\otimes 2}\big) = \mathrm{den}(\mathfrak{g}_{\Delta_s})^{-2}.$$

Pullback by $\jmath$ sends $\mathrm{den}(\mathfrak{g}_{\Delta_s})$ to $D_s(Z)$, whose square is χ_{10}. The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. □

The equality compares the completed root-Fock supercharacter with the numerical reduced generating series. Its proof uses the denominator identity and the half-coordinate pullback. A map from compact geometric states to this Fock space, preserving operators and their trace, is an additional comparison.

6.7 The eight diagonal-divisor forms

Cléry and Gritsenko classify eight genus-two diagonal-divisor forms [19]. The table records their groups, weights and character orders. Half-integral weights are interpreted on the corresponding multiplier covers. These eight rows are not a list of cyclic CHL orders. The reflective presentations in [40–42, 44–46] have their own specified Jacobi inputs, lattices and chambers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_0^{(2)}(2)$	3	2
∇_2	$\Gamma_0^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Proposition 6.10 (Product models and presentation refinements)**.** *Fix one row with an integral-exponent product*

$$F(Z) = e^{-\rho_F(Z)} \prod_{\gamma>0} (1 - e^{-\gamma(Z)})^{c_\gamma}$$

in its declared cusp coordinates and proper positive cone. Its minimal virtual root superspace has superdimension c_γ and gives this formal determinant after the Weyl prefactor. On any domain where $\sum_\gamma |c_\gamma e^{-\gamma(Z)}| < \infty$ and the factors are nonzero, it also gives a Fredholm Berezinian. If a symmetrizable Borcherds superpresentation for the same product is supplied, the quiver, Calabi–Yau and quantum constructions above apply to that datum at their respective stated domains.

Proof. Choose the minimal even or odd vector space according to the sign of c_γ. Proper height proves formal local finiteness; the summability assumption is exactly the trace-norm condition for its diagonal operator. Their determinants give the displayed product. A supplied presentation specifies the labelled Cartan matrix and parity, which are the inputs to the later constructions. The ordinary quiver Hall algebra uses the even relation ideal; the quantum superalgebra uses the supplied parity relations. The shared matrix alone does not identify those two algebras. □

The Igusa row has the full root-space summability proof given above. For another row, an absolute coefficient estimate must be checked against its actual product or simple presentation. Neither a classification by divisor nor the equality of one scalar product constructs its root bracket, quantum pairing or compact Hall map.

6.8 The collection of comparison targets

Definition 6.11 (Igusa comparison targets)**.** Associated with the fixed minimal presentation are

$$\mathfrak{K}_\Delta = \Big(\Gamma_\Delta, \mathcal{B}_\Delta, \mathfrak{g}_{\Delta_5}, (V_H, \mathcal{A}_H, \mathcal{F}_H)_H, U_q^{\mathrm{red}}(\mathfrak{g}_{\Delta_5}), \mathbb{A}_Z\Big).$$

These are respectively the pro-system of finite Calabi–Yau completions, the ordinary quiver Hall categories, the root Lie superalgebra, the current and Ran constructions, the quantum Hopf superalgebra, and the oriented skew family on the root Hilbert space. The finite BV cotangent models and the classical local BF theories provide separate field-theoretic comparison targets.

Theorem 6.12 (Scalar invariants of the root constructions)**.** *Put*

$$P = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}}.$$

The ordinary Chevalley coalgebra has Euler supercharacter P, the positive PBW/Fock superspace has supercharacter P^{-1}, and the skew Fredholm model has super-Pfaffian P on its convergence domain. Consequently

$$e^{-\rho} P = \operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z), \qquad e^{\rho} \operatorname{sch} \operatorname{Fock}(\mathfrak{n}_{\Delta_5,+}) = \operatorname{den}(\mathfrak{g}_{\Delta_5})^{-1},$$

and in the primitive reduced incidence convention,

$$\mathcal{Z}^{\mathrm{red}}_{S\times E} = -\big(j^* \operatorname{den}(\mathfrak{g}_{\Delta_5})\big)^{-2} = -\chi_{10}{}^{-1}.$$

Proof. The ordinary bar–Chevalley comparison gives the first character; PBW gives its reciprocal. The Fredholm theorem gives the same signed product on its trace-class domain. The denominator theorem and the primitive reduced curve-counting theorem then give the remaining identities with their stated Weyl and coordinate factors. □

The constructions retain distinct operations. The minimal root datum supplies a bracket and both parities. The quiver supplies extension correspondences and a separate ordinary composition algebra. The current envelope supplies local fields, and its associative Ran envelope supplies a bar in a specified tensor category. Hong's Borel pairing supplies an R operator on its weight-module domain. The skew doubling supplies a Pfaffian ratio, while the automorphic form supplies a global modular section. These scalar identities preserve the common denominator without identifying these carriers.

For a compact realization, the first maps are the reduced critical coefficient maps on common-torsor extensions and full flags. They must preserve the additive physical charge and its polarized Gram refinement. Primitive representatives must satisfy the simple-root relations and generate with the required parity dimensions, so that Theorem 3.6 applies. A comparison of doubles additionally preserves a coproduct and full Hopf pairing. A local–wrapped comparison preserves the supported mixed operations. Finally an operator and state comparison must identify the relevant orientation lines, sections, domains and sewing maps. The classical cotangent action, the finite quantum master equation and the reduced scalar formula leave these geometric constructions as separate mathematical problems.

Book III

Ordered chiral geometry

Chapter 7

Collision, order, and coefficient maps

Two observables on $X \times \mathbb{R}$, with X a smooth complex curve, can approach one another in two different directions. Collision in X probes singular coefficients along a diagonal. Composition along the oriented real line retains the order of the operators. A third operation, exchanging the operators along paths in a plane, requires transport beyond the two compositions. The distinction is already visible for two observables; it persists in their bar constructions and in the categories on which they act.

Principle 7.1 (Two directions of locality). A factorization coefficient on X describes separation and collision of holomorphic supports. An internal associative multiplication describes composition in an independent oriented direction. A braid action is additional transport data, subject to its compatibility with those operations.

The geometric sources of the operations are

geometry	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered composition
oriented plane	braid transport
circle	Hochschild chains
bordered or stable curves	sewing with its specified contractions.

A comparison between rows must give a map preserving the operations in question.

7.1 Coefficients on configurations

Let $\Delta : X \hookrightarrow X^2$ be the diagonal and $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$ its complement. A Beilinson–Drinfeld chiral Lie operation on a right D-module $\mathfrak{g}$ is

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g},$$

with skew-symmetry and the chiral Jacobi identity [6]. The module, the singular coefficient extension, and the operation are separate data. For the pole filtration on $\mathcal{O}_{X^2}(*\Delta)$,

$$\mathrm{gr}^F_m \mathcal{O}_{X^2}(*\Delta) \cong \mathcal{O}_\Delta(m\Delta) \cong N^{\otimes m}_{\Delta/X^2}, \qquad m \geq 1.$$

This is the normal symbol of a principal part of order m. It follows from the quotient of consecutive pole-order sheaves. A residue for a meromorphic one-form recovers its first principal-part coefficient; higher poles require their higher normal symbols. A coordinate change can mix lower coefficients while preserving the filtration.

Fix a stable presentable category $\mathcal{D}_X$ of coefficients on the unital Ran space $\mathrm{Ran}^+(X)$, whose points include the empty configuration. Specify its restriction and collision functors and their coherent base-change maps. We use the pointwise tensor $\otimes^!$ and the chiral convolution with chart formula

$$(F \otimes^{\mathrm{ch}} G)_I = \bigoplus_{I=I_1 \sqcup I_2} j_{I_1,I_2,*}\, j^!_{I_1,I_2}(F_{I_1} \boxtimes G_{I_2}).$$

Here j_{I_1,I_2} includes the locus where the two blocks are disjoint; empty blocks are allowed. The direct image is the $*$ extension appearing in the singular chiral coefficient model, with its stipulated transition maps. Its replacement by extension by zero would be a different coefficient construction. The pointwise unit is denoted $\mathbf{1}_\perp$; the convolution unit $\mathbf{1}_{\mathrm{ch}}$ is supported at the empty configuration. No identification of these units is part of the notation. We also write $\star^{\mathrm{ch}}$ for $\otimes^{\mathrm{ch}}$. Preservation of a realization or limit is a hypothesis where that construction uses it.

A factorization object consists of coefficients F_I on the finite configuration charts, collision maps, and coherent equivalences

$$j^! F_I \simeq \bigboxtimes_{a=1}^{r} F_{I_a} \qquad (I = I_1 \sqcup \cdots \sqcup I_r)$$

on the corresponding disjoint locus, together with its empty-configuration datum. Write $\mathrm{Fact}_D(X)$ for the resulting category in the fixed enhancement. When that enhancement realizes factorization objects as chiral commutative coalgebras, this notation refers to the factorizing subcategory of $\mathrm{CoCAlg}_{\mathrm{ch}}(\mathcal{D}_X)$. The Lie-to-commutative-coalgebra equivalence of chiral Koszul duality has this Lie-theoretic scope [36, 50]; an associative duality requires its own functors.

Proposition 7.2 (Pointwise tensor of factorization objects)**.** *Suppose the pointwise tensor is compatible with the specified collision functors and external products. Then it restricts to a symmetric monoidal structure on* $\mathrm{Fact}_D(X)$.

Proof. For two blocks, the comparison on the disjoint locus is

$$\begin{aligned} j^!(F_I \otimes^! G_I) &\simeq j^! F_I \otimes^! j^! G_I \\ &\simeq (F_{I_1} \boxtimes F_{I_2}) \otimes^! (G_{I_1} \boxtimes G_{I_2}) \\ &\simeq (F_{I_1} \otimes^! G_{I_1}) \boxtimes (F_{I_2} \otimes^! G_{I_2}). \end{aligned}$$

The same reordering gives the comparison for any finite number of blocks. Refinement, permutation and collision coherence follow from those for the inputs and the assumed tensor compatibilities. The specified pointwise unit has the corresponding factorization structure. □

7.2 The two associative structures

Definition 7.3 (Transversely ordered chiral algebra)**.** An ordered chiral algebra is an augmented associative algebra

$$A^\perp \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{Fact}_D(X), \otimes^!).$$

Thus its multiplication and unit are morphisms of factorization objects. On the chart I the multiplication is $m_I : A_I \otimes^! A_I \to A_I$. For a binary decomposition its compatibility with separation is

$$\Delta_{I_1,I_2} m_I = (m_{I_1} \boxtimes m_{I_2})\sigma(\Delta_{I_1,I_2} \otimes^! \Delta_{I_1,I_2}),$$

where σ interchanges the two middle tensor factors with their graded sign. All higher associativity homotopies, units, and collision maps obey the same compatibility.

Proposition 7.4 (Locality recognition)**.** *An ordered chiral algebra is equivalently a family of associative algebra objects A_I whose units and products commute coherently with the transition maps and factorization equivalences.*

Proof. A morphism in $\mathrm{Fact}_D(X)$ is a componentwise morphism preserving these data. Apply this description to every operation of the associative operad. The displayed binary equation is its multiplication condition; the coherent higher equations give the remaining algebra structure. In a strict dg model these reduce to associativity, units, and the stated compatibility diagrams. □

Definition 7.5 (Associative chiral-convolution algebra)**.** An associative chiral-convolution algebra is an augmented object

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{D}_X, \otimes^{\mathrm{ch}}).$$

Its commutator is a Lie algebra object for the same convolution tensor. A chiral Lie algebra $\mathfrak{g}^{\mathrm{ch}}$ has this latter Lie type.

The maps $m^{\perp} : A \otimes^{!} A \to A$ and $m^{\mathrm{ch}} : A \otimes^{\mathrm{ch}} A \to A$ have different domains. Neither definition contains a comparison between them. Even the elementary operation of adjoining a unit must specify its tensor. In a stable monoidal dg category whose tensor preserves sums, a nonunital associative algebra (M, μ) has the augmented unitalization $\mathbf{1} \oplus M$ with multiplication

$$(a, m)(b, n) = (ab, an + bm + \mu(m, n)).$$

It is square-zero precisely when $\mu = 0$. Associativity follows by expanding this formula and using associativity of μ. This construction applies separately to either chosen tensor when its hypotheses hold. It does not make an ambient direct sum a factorization object, nor does it identify the two different unitalizations on a common carrier.

7.3 Common decompositions and collision

For a finite set I, write $\mathrm{Dec}(I)$ for its ordered decompositions $I = A \sqcup B$. In a linear species model suppose

$$\mathcal{M}_I = \bigoplus_{d_1, d_2 \in \mathrm{Dec}(I)} \mathcal{M}_{d_1, d_2}.$$

There is a projection e_I onto $\mathcal{M}^{\mathrm{al}}_I = \bigoplus_d \mathcal{M}_{d,d}$. If a_I denotes the diagonal of the discrete decomposition index, this is extension by zero after restriction, $a_{I,!} a^{!}_I$.

Proposition 7.6 (Finite alignment and its descent condition)**.** *The projections e_I are idempotent and natural under bijections. On three decomposition indices the two iterated alignments are the projection onto $d_1 = d_2 = d_3$. They descend to a coefficient diagram with collision transitions if those transitions intertwine the projections coherently.*

Proof. On each direct summand e_I is either zero or the identity. Bijections preserve equality of decomposition indices, and both triple composites retain exactly the common diagonal. For a transition T_{IJ} the required equation is

$$T_{IJ} e_I \simeq e_J T_{IJ}.$$

Together with its coherence under composition, this equation is exactly what makes the chartwise endomorphisms an endomorphism of the full diagram. Idempotence then passes to that diagram and, in an idempotent-complete setting, its image defines the aligned subobject. □

The collision condition is substantive. The surjection $\{1, 2\} \to \{*\}$ does not carry the decomposition $(\{1\}, \{2\})$ to a disjoint decomposition: its two images overlap. Consequently the finite discrete indexing argument alone does not supply a Ran transition or its base-change square. A particular geometric model may supply the required comparison; the failure of this argument is no obstruction to every possible mixed distributive law.

A *mixed associative chiral datum* is a common factorization carrier with both augmented associative products, a specified comparison between their mixed composites on a stated correspondence, and coherent unit and iteration diagrams. When the correspondence is the aligned image above, its descent condition is part of the datum. To extend an aligned comparison to the entire mixed object one must also define it on the complementary sector. Equality of the two finite triple projectors proves only the alignment identity, not these additional laws.

7.4 From an oriented line to an internal algebra

Let $\mathcal{F}$ be a product-generated factorization algebra on $X^{\mathrm{an}} \times \mathbb{R}$ that is holomorphic along X and locally constant along the real direction. Its value on an interval, considered as an object of the chosen holomorphic factorization category, has an E_1 structure: ordered disjoint intervals give its products, and each order chamber is contractible. Factorization in X commutes with these products because the product embeddings commute.

To obtain an object of $\mathrm{Fact}_D(X)$ one further needs a specified monoidal comparison from that holomorphic coefficient theory to the chosen D-module theory, preserving the units, collision maps and relevant colimits. Under those hypotheses the comparison transports the internal associative algebra. A regular-holonomic Riemann–Hilbert equivalence can be used only on coefficient objects within its domain and with these compatibilities established. It does not algebraize an arbitrary holomorphic factorization algebra.

7.5 Block-ordered operations

The preceding definition is intrinsic. A geometric model arises after one chooses an infinity-operad Ch_X representing the selected chiral factorization theory. The Boardman–Vogt tensor product

$$\mathsf{OrdCh}_X = \mathsf{E}_1 \otimes \mathsf{Ch}_X$$

then represents compatible ordered and chiral operations.

Theorem 7.7 (Operadic exponential law)**.** *Let P and Q be infinity-operads whose Boardman–Vogt tensor product exists, and let $\mathcal{V}$ be a compatible presentable symmetric monoidal infinity-category. Then*

$$\mathrm{Alg}_{P\otimes Q}(\mathcal{V}) \simeq \mathrm{Alg}_P\big(\mathrm{Alg}_Q(\mathcal{V})\big).$$

In particular, after a chiral operad Ch_X has been constructed, the tensor product $\mathsf{E}_1 \otimes \mathsf{Ch}_X$ classifies E_1-algebra objects in Ch_X-algebras.

Proof. The tensor product of infinity-operads is characterized by the displayed universal property [70]. The second statement is the specialization $P = \mathsf{E}_1$ and $Q = \mathsf{Ch}_X$. This formal equivalence applies to each geometric model of Ch_X after that model and its operadic tensor product have been constructed. □

A profile is an ordered list $(I_1, \dots, I_r)$ of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \cdots \times X^{I_r}$. Chiral morphisms refine and collide labels inside one block. Transverse morphisms merge adjacent blocks. The tensor-product operad imposes their mixed squares.

7.6 Mixed forests

Suppose a cellular resolution of the chosen mixed operad is given, whose cells are levelled forests with transverse and chiral internal edges and specified coefficient complexes. A Boardman–Vogt model can be used when its cells and coefficient maps have been constructed in that geometric theory. Set

$$\mathrm{or}(F) = \det(E_{\perp}(F)) \otimes \det(E_{\mathrm{ch}}(F)).$$

The candidate differential is the internal coefficient differential plus one-edge contractions.

Theorem 7.8 (Two-color forest differential)**.** *Let C_* be the chain complex freely generated by oriented two-color forests, with one determinant-line generator for every transverse edge and one for every chiral edge. Suppose contraction of an edge is functorial on the coefficient system and every codimension-two contraction square commutes before orientations are inserted. With the product orientation, the transverse and chiral contraction operators satisfy*

$$d_{\perp}^2 = 0, \qquad d_{\mathrm{ch}}^2 = 0, \qquad d_{\perp} d_{\mathrm{ch}} + d_{\mathrm{ch}} d_{\perp} = 0.$$

Thus $d = d_{\mathrm{int}} + d_{\perp} + d_{\mathrm{ch}}$ squares to zero whenever the internal differential anticommutes with both contraction operators.

Proof. Every term of $d_{\perp}^2$ contracts the same pair of transverse edges in two orders. The underlying coefficient maps agree by functoriality, while exchanging the two transverse determinant generators reverses the sign. The terms cancel in pairs. The same argument proves $d_{\mathrm{ch}}^2 = 0$. For one edge of each color the coefficient maps again form a commuting square and exchanging the two determinant generators contributes the Koszul sign -1, proving the mixed anticommutator identity. The internal terms vanish by the chain-map assumption. □

The sign calculation proves that this candidate is a complex under its stated hypotheses. Identifying it with a resolution of the geometric operad still requires the cellular comparison. The intrinsic definition of an ordered chiral algebra is independent of that choice.

Chapter 8

Bars, completions, and module equivalence

The bar remembers the multiplication used to form it. We first distinguish the three operadic types, then construct the transverse bar in the factorization category. Positive weights provide a separate setting in which the complete tensor cobar and its module functors can be proved to reconstruct.

8.1 Three bar constructions

We distinguish three bars by their monoidal product and algebraic operad. The two associative augmentations are to their respective tensor units. Use cohomological degrees: $sM = M[1]$ lowers the degree of an element by one. In a chosen dg model, $\bar{A}$ is the reduced augmentation ideal and $T^c_{\otimes}(M) = \bigoplus_{n\geq 0} M^{\otimes n}$ is the tensor coalgebra with deconcatenation; its empty tensor is the coaugmentation.

Definition 8.1 (Transverse, associative-chiral, and Lie-chiral bars)**.** For the two augmented associative objects and a chiral Lie algebra, define

$$\mathrm{Bar}^{\perp}(A^{\perp}) := \mathbf{1} \otimes^{\mathbb{L},!}_{A^{\perp}} \mathbf{1},$$

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}}) := \mathbf{1} \star^{\mathrm{ch}\,\mathbb{L}}_{A^{\mathrm{ch}}_{\mathsf{Ass}}} \mathbf{1},$$

and

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}}) := C^{\mathrm{ch}}_*(\mathfrak{g}^{\mathrm{ch}}) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}[1])$$

with its chiral Chevalley differential. The first bar is formed with $\otimes^!$, the second with $\star^{\mathrm{ch}}$, and the third is the Lie-to-commutative-coalgebra Koszul functor.

Theorem 8.2 (Typed bar theorem)**.** *Choose a symmetric monoidal dg coefficient model in which the indicated tensors compute derived tensors and direct-sum normalization computes realization. The three underlying bars then have the following chain models.*

(i) $\mathrm{Bar}^{\perp}(A^{\perp})$ *is the tensor coalgebra* $T^c_{\otimes^!}(s\bar{A})$ *with coderivation induced by* $m^{\perp}$.

(ii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}})$ *is the tensor coalgebra* $T^c_{\star^{\mathrm{ch}}}(s\bar{A})$ *with coderivation induced by* m^{ch}.

(iii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}})$ *is the chiral symmetric coalgebra with differential induced by the chiral Lie bracket.*

Each differential squares to zero by the defining relation of its own operad. The outputs live in different coalgebra categories and are compared by a displayed monoidal or PBW morphism.

Proof. Normalize the two simplicial two-sided bars in their respective monoidal categories. Degeneracies remove units, leaving the two tensor coalgebras, and faces contract adjacent factors using the relevant multiplication. The Lie formula is the ordinary operadic bar of the Lie operad internal to chiral convolution. Associativity and Jacobi are exactly the quadratic identities that make the corresponding coderivations square to zero. □

Definition 8.3 (PBW-admissible chiral Lie algebra)**.** A chiral Lie algebra $\mathfrak{g}^{\mathrm{ch}}$ is PBW-admissible when its associative enveloping algebra $U_{\star^{\mathrm{ch}}}(\mathfrak{g})$ for chiral convolution has a complete separated PBW filtration, the canonical map

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}_{\mathrm{PBW}}\, U_{\star^{\mathrm{ch}}}(\mathfrak{g})$$

is an equivalence, and the bar filtration is strongly convergent in the chosen nilcomplete Ran category.

Theorem 8.4 (Filtered chiral PBW comparison)**.** *Let $\mathfrak{g}$ be a chiral Lie algebra in a complete filtered chiral category. Assume:*

(i) *its associative enveloping algebra $U_{\star^{\mathrm{ch}}}(\mathfrak{g})$ for chiral convolution exists;*

(ii) *the PBW filtration is exhaustive, complete, and separated;*

(iii) *the chiral PBW morphism*

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}\, U_{\star^{\mathrm{ch}}}(\mathfrak{g})$$

is an isomorphism;

(iv) *the induced filtrations on the two bar complexes are strongly convergent.*

Then antisymmetrization induces a filtered quasi-isomorphism

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\big(U_{\star^{\mathrm{ch}}}(\mathfrak{g})\big) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}).$$

Proof. The associated graded of the associative chiral bar is the Koszul bar of $\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g})$. The usual Koszul antisymmetrization identifies this coalgebra with the chiral Chevalley coalgebra. Hypotheses (2)–(4) give a complete, separated, strongly convergent comparison spectral sequence. Since its associated-graded map is a quasi-isomorphism, the filtered map is a quasi-isomorphism. This is an associative-enveloping-to-Chevalley comparison under the stated PBW and convergence hypotheses. It does not identify arbitrary associative chiral algebras with chiral Lie algebras. □

Definition 8.5 (Bar-admissible mixed datum)**.** A mixed associative chiral datum is bar-admissible when its aligned distributive law acts on the full bisimplicial two-sided bar and both geometric realizations commute with the tensor products occurring in that bisimplicial object.

Theorem 8.6 (Mixed Fubini theorem)**.** *Let A be a mixed algebra in a presentable stable category. Assume the mixed distributive datum extends to the bisimplicial two-sided bar, and that both monoidal products preserve its geometric realizations separately. Then the two iterated realizations and the diagonal realization are canonically equivalent:*

$$\Big|\, p \mapsto \big|q \mapsto B_{p,q}(A)\big|_{\mathrm{ch}}\Big|_{\perp} \simeq \big|\mathrm{diag}\, B_{\bullet,\bullet}(A)\big| \simeq \big|\, q \mapsto \big|p \mapsto B_{p,q}(A)\big|_{\perp}\big|_{\mathrm{ch}}.$$

Proof. The distributive datum makes the two simplicial directions commute, so $B_{\bullet,\bullet}(A)$ is a genuine bisimplicial object. Geometric realization is a colimit. Since the two tensor products preserve the relevant colimits, Fubini for colimits identifies both iterated realizations with the realization of the diagonal simplicial object. □

The notation used in the remainder of this part is

$$\mathrm{Bar}^{\mathrm{ord}}_{X} = \mathrm{Bar}^{\perp}.$$

It always refers to transverse order.

8.2 Realization and factorization

Factorization compares the coefficient at a disjoint configuration with an external tensor product of its coefficients at the separate configurations. This condition is nonlinear. For example, suppose those coefficients are k at one point and at two distinct points, where k is a field. The componentwise sum of two copies has coefficient k^2 at both configurations. Factorization would instead require $k^2 \otimes_k k^2 \cong k^4$ at two points. A componentwise shift also fails: its two-point coefficient is $k[1]$, whereas the tensor product of its singleton coefficients is $k[2]$. Thus an ambient stable coefficient category does not make its factorization objects closed under componentwise sums or shifts.

Geometric realization has a different colimit property. A small indexing ∞-category J is *sifted* when it is nonempty and its diagonal $J \to J \times J$ is cofinal. Here cofinal means that, for every diagram D on $J \times J$, the canonical map $\operatorname{colim}_J (D \circ \Delta) \to \operatorname{colim}_{J \times J} D$ is an equivalence whenever these colimits exist. Iteration gives the same property for $J \to J^m$, for every finite $m \geq 1$. The simplex category with arrows reversed is sifted, so simplicial geometric realization is a sifted colimit.

Proposition 8.7. *Fix coefficient categories on configuration charts, with specified restriction and collision functors and their coherent identifications. Suppose the coefficient categories admit J-indexed colimits and these functors preserve them. Suppose external tensor products preserve colimits separately in each variable. Let $(F_j)_{j \in J}$ be a diagram of factorization objects and morphisms preserving their factorization equivalences. If J is sifted, the componentwise colimit is a factorization object.*

Proof. Write $F_I = \operatorname{colim}_{j \in J} (F_j)_I$ on a chart indexed by I. The collision maps and their coherences pass to the colimit by hypothesis. For a decomposition $I = I_1 \sqcup \cdots \sqcup I_m$, let u denote restriction to its disjointness locus. Restriction, factorization, cofinality, and separate preservation give

$$\begin{aligned} u F_I &\simeq \operatorname*{colim}_{j \in J} u(F_j)_I \\ &\simeq \operatorname*{colim}_{j \in J} \boxtimes_{a=1}^{m} (F_j)_{I_a} \\ &\simeq \operatorname*{colim}_{(j_1, \ldots, j_m) \in J^m} \boxtimes_{a=1}^{m} (F_{j_a})_{I_a} \simeq \boxtimes_{a=1}^{m} F_{I_a}. \end{aligned}$$

Each comparison is natural in the diagram and the decomposition. Refinements and permutations therefore preserve the given coherence diagrams. If an empty-configuration unit is specified, its constant diagram also has that unit as colimit, since a sifted category has contractible groupoid completion. □

For a simplicial two-sided bar, this proposition applies when every simplicial term and every face and degeneracy are already in the specified factorization category. It also requires the actual restriction and collision functors to preserve realization. Normalization may then be computed in an ambient stable category. Its individual shifted summands need not be factorization objects. A coproduct on the normalized complex requires its own compatibility with the factorization equivalences.

8.3 The internal ordered bar

Let A be an augmented ordered chiral algebra. Assume the restriction and collision functors preserve geometric realizations, and the external and pointwise tensor products preserve them separately. The two-sided simplicial bar has

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Every face and degeneracy is a factorization morphism by the algebra and augmentation axioms. The sifted-colimit proposition therefore constructs its realization $\operatorname{Bar}^{\mathrm{ord}}_X(A)$ as a factorization object.

Theorem 8.8 (Normalized bar). *Choose a symmetric monoidal dg model for the ambient coefficients in which the displayed tensors compute derived tensors and direct-sum normalization computes realization. Write $\bar{A}$ for the augmentation ideal in that model. The underlying bar complex is*

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \big(T^c(s\bar{A}), d_{\mathrm{int}} + b_m\big).$$

Here s decreases cohomological degree by one. The direct sum and its individual shifted summands are formed in the ambient coefficient category. Deconcatenation makes this complex a conilpotent coassociative coalgebra there:

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

Proof. Normalization removes words containing a unit and leaves the shifted augmentation ideal in every slot. The faces give the internal differential and adjacent multiplication. The simplicial identities give a square-zero differential. Two successive cuts enumerate the same three subwords in either order. An internal differential or multiplication lies on one side of any surviving cut, with the tensor sign of its prefix. Thus the differential is a coderivation. Every element has finite word support, so sufficiently many reduced cuts give zero. □

8.4 The bar comparison on separated configurations

The normalized word decomposition is not itself a decomposition in $\mathrm{Fact}_D(X)$. Its compatibility with factorization is expressed by a map between whole bar complexes. For augmented dg algebras P, Q in the chosen model, define

$$\mathrm{sh} : BP \otimes BQ \longrightarrow B(P \otimes Q)$$

by shuffling the factors $p_i \otimes 1$ and $1 \otimes q_j$, keeping their respective orders. The sign is the Koszul sign for the permutation of the suspended factors. For degree-zero factors,

$$\mathrm{sh}([p] \otimes [q]) = [p \otimes 1|1 \otimes q] - [1 \otimes q|p \otimes 1].$$

Proposition 8.9 (Shuffle and factorization). *The shuffle is a natural quasi-isomorphism and a coalgebra map, where its source has the tensor-product coalgebra structure. For the geometric coefficient models above, suppose these maps commute with the specified restriction and collision functors. Then they give factorization equivalences for the whole bar and make deconcatenation a morphism of factorization objects in the derived category.*

Proof. In the bar differential of a shuffle, a merge of an entry from P with one from Q occurs twice with opposite suspended signs. The equality $(p \otimes 1)(1 \otimes q) = (-1)^{|p||q|}(1 \otimes q)(p \otimes 1)$ cancels these terms. Merges within one factor give the differential of that factor. Internal differentials commute with the same signed permutation. Thus the shuffle is a cochain map.

A cut of a shuffled word determines one cut in each input word, a shuffle of their prefixes, and a shuffle of their suffixes. Conversely these four choices determine exactly that cut shuffle. The signs on both sides come from the same permutation of suspended factors. This proves the coalgebra identity with the tensor-product coproduct.

To verify the quasi-isomorphism, form the bisimplicial object $B_p(k, P, k) \otimes B_q(k, Q, k)$, with k denoting the tensor unit. Its iterated realization is $BP \otimes BQ$. Its diagonal is the simplicial bar of $P \otimes Q$. Cofinality of the simplicial diagonal identifies these realizations. On normalized chains the map from the iterated realization to the diagonal is the shuffle just defined: the monotone paths from $(0, 0)$ to (p, q) triangulate the product of its two simplices. Each path inserts precisely the degeneracies of its shuffle. Adjacent paths have opposite orientations on a shared face, so their signed sum has exactly the product boundary. This is the realization comparison, hence an equivalence.

For $I = I_1 \sqcup I_2$ on the disjointness locus, the given algebra factorization identifies A_I with $A_{I_1} \boxtimes A_{I_2}$. The shuffle therefore gives the comparison $BA_{I_1} \boxtimes BA_{I_2} \to BA_I$. Its inverse in the derived category is the bar factorization map. The coalgebra identity just proved gives compatibility with deconcatenation. For several blocks, iterated shuffles enumerate the same ordered multishuffles, with the same Koszul signs. This proves refinement coherence. Naturality and the stated functor compatibility give the collision coherence. □

The restriction and collision compatibility in this proposition is a hypothesis on the chosen geometric model. It does not follow from naming an ambient stable category. Conilpotence here refers to finite normalized words in that model. No word-length completion is implicit.

8.5 Three coproducts

Three coproducts occur on different carriers:

$$\Delta_{\mathrm{fact}} : \text{separation of chiral support},$$
$$\Delta_{\mathrm{dec}} : \text{cutting an ordered bar word},$$
$$\Delta_{\mathrm{Hopf}} : \text{fusion of representations}.$$

The first two are present on the transverse bar under the preceding geometric compatibility hypotheses. The third is additional bialgebra data on an acting algebra, with an antipode when a Hopf algebra is intended.

Proposition 8.10 (Coproduct separation)**.** *Factorization separation is carried by every ordered coefficient. Deconcatenation is generated functorially by the bar construction. An algebra-compatible coproduct is supplied by an independent bialgebra structure. A coalgebra comparison morphism relates the Hopf coproduct to either geometric coproduct whenever a theory identifies their carriers.*

Proof. Factorization separation is part of the coefficient object, and deconcatenation is the canonical cut map on tensor words. A Hopf coproduct is an algebra map to the specified tensor square, with counit and antipode identities; a completion is used only when supplied. Its source, target, and defining universal property differ from the first two, so a relation among them is exactly the data of a comparison morphism. □

8.6 Interval factorization homology

For a right A-module R and a left A-module L, factorization homology of an interval with endpoint labels gives

$$\int_{([0,1],\{0,1\})} (A; R, L) \simeq R \otimes^{\mathbb{L}}_{A} L.$$

This statement holds internally in every presentable monoidal category supporting factorization homology [4, 70]. Taking $R = L = \mathbf{1}$ identifies the interval amplitude with the ordered bar.

8.7 Cobar and twisting cochains

Let C be a coaugmented conilpotent differential graded coalgebra in the ambient coefficient model, and write $\bar{C}$ for its coaugmentation coideal. The ordinary, direct-sum tensor cobar algebra is

$$\mathrm{Cobar}^{\mathrm{ord}}_{X}(C) = \big(T(s^{-1}\bar{C}), d_{\mathrm{int}} + d_{\Delta}\big).$$

A twisting cochain $\tau : C \to A$ has degree 1, vanishes on the coaugmentation, and has zero composite with the augmentation of A. It satisfies

$$d\tau + \tau d + m(\tau \otimes \tau)\bar{\Delta} = 0.$$

The universal twisting cochains are

$$\tau_A : \mathrm{Bar}_X^{\mathrm{ord}}(A) \to A, \qquad \tau_C : C \to \mathrm{Cobar}_X^{\mathrm{ord}}(C).$$

Theorem 8.11 (Bar–cobar adjunction)**.** *For ordinary augmented dg algebras and coaugmented conilpotent dg coalgebras in the specified coefficient model, tensor cobar is left adjoint to tensor bar. Their mapping set is the set of twisting cochains. An internal adjunction on factorization objects additionally requires both constructions to produce factorization objects and their twisting maps to preserve the geometric data.*

Proof. An algebra map from the tensor algebra $T(s^{-1}\bar{C})$ is determined by its restriction to $s^{-1}\bar{C}$. Compatibility with the cobar differential is the Maurer–Cartan equation above. The same equation describes coalgebra maps from C to the cofree conilpotent coalgebra $T^c(s\bar{A})$. □

8.8 Complete bar–cobar reconstruction

Ordinary tensor cobar uses finite words. Completion changes that carrier. Let k be a commutative ring and $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ an augmented algebra in degree zero, with $A_0 = k$, augmentation given by projection onto A_0, and multiplication adding weights. Put $D = \Omega B(A_{\mathrm{alg}})$ using ordinary direct-sum tensors, and let D_w be its weight-w subcomplex. The complete algebra and its complete resolution are

$$\widehat{A} = \prod_{w \geq 0} A_w, \qquad \widehat{D} = \prod_{w \geq 0} D_w.$$

Products of complexes are taken degreewise, with the weight topology.

Theorem 8.12 (Complete reconstruction)**.** *The componentwise counit $\widehat{D} \to \widehat{A}$ is a quasi-isomorphism of complete differential graded algebras.*

Proof. A weight-w block word has at most w positive-weight entries. Filter by their number. Multiplication lowers this number and splitting preserves it. At a fixed list of $N \geq 2$ entries, the splitting complex is the tensor product of the coefficient module with $N - 1$ copies of $[k \xrightarrow{1} k]$ in degrees $-1, 0$. Changing the first outer cut to an inner gap contracts it. The other differential terms cancel by their prefix signs. This identity survives tensoring with any coefficient module. Subtracting a primitive at the largest entry number terminates, so negative cohomology vanishes. In degree zero, the boundaries impose $c(a)c(b) = c(ab)$, leaving exactly A_w. Taking the degreewise product preserves kernels and images of these differentials, and therefore preserves the computed cohomology. □

For $k[x]$ with weight one on x, the ordinary counit recovers $k[x]$, whereas this weight completion recovers $k[[x]]$. A completed geometric bar–cobar equivalence needs a specified completed tensor product, coalgebra category, and comparison maps compatible with collision and inverse limits. An operadic nilcompletion or a boundary double centralizer is not identified with the ordinary tensor counit by this weight calculation. The coalgebra-unit and geometric duality statements require their own convergence and comparison arguments.

8.9 Quadratic recognition

Over a field k, let V have weight one and let $A = T(V)/(R)$ with $R \subset V \otimes V$. Its quadratic dual coalgebra $A^{\mathrm{q!}}$ is cogenerated by sV with quadratic component $s^2 R$. The canonical map $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ induces the twisting cochain whose left and right twisted tensor complexes are the Koszul complexes.

Theorem 8.13 (Quadratic Koszul criterion)**.** *Assume positive weight and degreewise finite duals. The following conditions are equivalent:*

(i) $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ *is a quasi-isomorphism;*

(ii) $\mathrm{Cobar}(A^{\mathrm{q!}}) \to A$ *is a quasi-isomorphism;*

(iii) *the left and right Koszul complexes have homology concentrated in weight zero;*

(iv) $\mathrm{Tor}_i^A(\mathbf{1}, \mathbf{1})$ *is concentrated in internal weight* i*.*

Proof. The universal twisting morphism $A^{\mathrm{q!}} \to A$ is Koszul exactly when the associated twisted tensor products resolve the unit. The fundamental theorem of twisting morphisms identifies this condition with the bar and cobar quasi-isomorphisms. Weight purity is the homological form of the same statement [67, 78]. □

8.10 Modules, comodules, and Morita theory

The ordinary tensor counit reconstructs an algebra. Reconstruction of its modules requires the unit and counit of the twisted module functors as well. The following weight conditions make these maps finite in every component.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an associative algebra in cohomological degree zero. Assume $A_0 = k$, multiplication adds weights, and augmentation projects onto A_0. Put $C_{\mathrm{alg}} = B(A_{\mathrm{alg}})$, with its bar weight grading, and set

$$A = \prod_{w \geq 0} A_w, \qquad C = \prod_{w \geq 0} (C_{\mathrm{alg}})_w.$$

Products are degreewise in complexes. The universal twisting map $\tau : C \to A$ has degree one, sends $[a]$ to a, and vanishes on the empty word and all longer words. The bar differential has the earlier convention $b_2[a|b] = [ab]$.

A complete weight-graded left A-module here means a product $M = \prod_{w \geq w_0} M_w$ of complexes, for some integer w_0, with differential preserving weight and actions $A_u \otimes M_v \to M_{u+v}$. A complete weight-graded left C-comodule means such a product N with a counital coassociative coaction $\rho : N \to C \widehat{\otimes} N$ preserving weight and differential. The completed tensor product is defined componentwise by

$$(C \widehat{\otimes} N)_w = \bigoplus_{u+v=w} (C_{\mathrm{alg}})_u \otimes N_v.$$

This sum is finite since $u \geq 0$ and $v \geq w_0$. Each reduced coaction lowers the weight of its final N-component. It therefore vanishes after more than $w - w_0$ iterations on N_w. These comodules need no chosen coaugmentation.

Write $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$ for the categories obtained by inverting maps that are quasi-isomorphisms in every weight, respectively for these modules and comodules. This specifies the weak equivalences; no coderived localization is implicit. Define

$$F(M) = C \widehat{\otimes}_\tau M, \qquad G(N) = A \widehat{\otimes}_\tau N.$$

The first has the coaction induced by Δ_C and the second the left A-action. Writing $\Delta(c) = \sum c_{(1)} \otimes c_{(2)}$ and $\rho(n) = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a\tau(n_{(-1)}) \otimes n_{(0)}.$$

All sums are finite in each weight. The twisting equation, associativity, and coassociativity make both differentials square to zero.

Theorem 8.14 (Koszul module equivalence in positive weights)**.** *These functors induce inverse equivalences*

$$D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C).$$

Their counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The differential formulas show that these are cochain maps, respectively A-linear and C-colinear. The twisting terms cancel in the unit equation; the counit equation uses the module action. Coassociativity and the counit identities give the two triangle identities.

The complex $GF(M)$ is the two-sided normalized bar resolution $A\widehat{\otimes}B(A_{\mathrm{alg}})\widehat{\otimes}M$. For its contraction, use the simplicial boundary convention $\partial = \sum_i (-1)^i d_i$. Multiplying a bar word of length r by $(-1)^r$ translates this convention to the displayed twisting differentials. Writing $\bar{a} = a - \epsilon_A(a)1$, the extra degeneracy is

$$h\big(a_0[a_1|\cdots|a_r]m\big) = 1[\bar{a}_0|a_1|\cdots|a_r]m.$$

The first face returns the original term minus its scalar first factor. The other faces cancel against applying h after the boundary, including the module-action face. The internal differential of M cancels by the suspension sign. Thus $\partial h + h\partial = 1 - i\epsilon_M$, with $i(m) = 1[\,]m$. This is a linear contraction; i need not be A-linear. It preserves total weight, so the counit is a quasi-isomorphism in every weight.

At fixed total weight, filter $FG(N)$ by the weight of its final N-factor. This filtration is finite. The reduced-coaction term lowers that weight and disappears in the associated graded. The associated graded unit is therefore the inclusion into $B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The rightmost extra degeneracy contracts this augmented bar onto k:

$$h\big([a_1|\cdots|a_r]a\big) = (-1)^{r+1}[a_1|\cdots|a_r|\bar{a}]1$$

in the same simplicial convention. The last face gives $1 - i\epsilon$; all other faces cancel in pairs. Consequently the unit is a quasi-isomorphism on each associated graded component, and the finite filtration proves the same for each weight of N.

Both functors preserve weightwise quasi-isomorphisms. Filter by the weight of the final module or comodule component. The twisting terms strictly change that weight in one direction. On associated graded components, the functors tensor complexes of k-vector spaces, which preserves quasi-isomorphisms. The filtration is finite at fixed total weight. Hence F, G descend to the stated localizations, where their unit and counit are isomorphisms. □

This theorem uses an explicit weight decomposition and a lower weight bound on every module and comodule. Derived I-adic completion, with $I = \ker(A \to k)$, instead uses $M \to \lim\limits_{\longleftarrow p} (A/I^p \otimes^{\mathbb{L}}_A M)$. The two completions require a comparison of their filtrations. A coderived category also has a separately specified class of coacyclic complexes. Neither identification follows from the algebra counit alone. For right and left modules, the two-sided bar computes their relative derived tensor product in the same finite weight components. An identification with a geometric derived cotensor product additionally requires the two-sided twisting comparison and its compatibility with the chosen completion and collision maps.

8.10.1 A generator and Morita descent

Let C_I be presentable stable linear categories on the configuration charts, with a compatible family of objects b_I. Assume each b_I is a compact generator. Set

$$A_I = \mathrm{RHom}_{C_I}(b_I, b_I).$$

Composition gives an associative algebra. We use right A_I-modules, so the local Morita functor is $x \mapsto \mathrm{RHom}(b_I, x)$ with action by precomposition.

Theorem 8.15 (Morita equivalence with factorization descent)**.** *Suppose the local categories, generators and algebra objects have specified enriched factorization data, and the local Morita functors commute coherently with the transition functors and disjoint factorization maps. Then the local Morita equivalences assemble to an equivalence of the resulting factorization categories with the corresponding family of module categories.*

Proof. Compact-generator Morita theory identifies C_I with right A_I-modules. The compatibility hypothesis identifies these functors and their inverses on every transition and refinement. Taking descent preserves the equivalence. □

A chosen boundary object need not be a compact generator. Computing its endomorphism algebra supplies only the local algebra; the generator and descent hypotheses decide whether it reconstructs the category.

8.11 Sign calculation for the length-three bar

For homogeneous $a, b, c \in \bar{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}\big[(ab)c - a(bc)\big],$$

which vanishes by associativity.

Chapter 9

Explicit associative algebras

The following examples are algebras in complexes over a field. They make the bar, quadratic dual and exchange relations concrete. Their promotion to geometric factorization objects requires coefficient families and compatible collision maps in addition to the algebra formulas.

9.1 Free and square-zero pairs

Work over a field k in dg vector spaces, with the usual tensor. For a complex M the free augmented algebra is

$$T(M) = k \oplus M \oplus M^{\otimes 2} \oplus \cdots .$$

The unit is the empty word and the augmentation kills positive word length. Assume M is perfect when taking its derived dual.

Theorem 9.1 (Free and square-zero bars)**.** *There is a quasi-isomorphism of coaugmented coalgebras*

$$k \oplus sM \longrightarrow B(T(M)),$$

where sM is primitive. Consequently

$$\mathrm{RHom}_{T(M)}(k,k) \simeq k \oplus M^{\vee}[-1]$$

as an augmented algebra with square-zero reduced product. For the square-zero algebra $A = k \oplus M$, one has

$$B(A) = T^{c}(sM), \qquad \mathrm{RHom}_{A}(k,k) \simeq \prod_{r \geq 0} (M^{\vee}[-1])^{\otimes r}.$$

The last product has the word topology and concatenation multiplication.

Proof. In $B(T(M))$, fix a word of N factors from M and vary its decomposition into nonempty bar blocks. For $N > 1$ the bar differential removes internal cuts. The cut complex is a tensor product of $N - 1$ elementary contractible complexes. Fix the first internal position. Inserting that cut when it is absent, and assigning zero when it is present, contracts the cut complex with its prefix sign. The surviving summands are the empty word and the one-factor word sM. The inclusion of these summands is a coalgebra map and a quasi-isomorphism. Dualizing gives the square-zero product because the surviving reduced coalgebra is primitive. For $A = k \oplus M$ the multiplication part of the bar differential is zero. Dualizing the direct-sum tensor coalgebra gives the product of its tensor duals, with concatenation induced by deconcatenation. □

These are linear coefficient constructions. A geometric version must supply factorization coefficients whose whole bars have the required restriction and collision compatibilities. The displayed ambient direct sum alone does not do so.

9.2 The quantum plane

Let

$$A_q = \Bbbk\langle x, y\rangle/(yx - qxy), \qquad q \in \Bbbk^\times.$$

Give x and y weight one.

Theorem 9.2 (PBW basis of the quantum plane)**.** *The monomials $x^a y^b$, $a, b \geq 0$, form a basis of A_q.*

Proof. Use the rewriting rule $yx \mapsto qxy$. Every reduction decreases the inversion number of a word. The leading word yx has distinct initial and terminal generators, so every overlap decomposes into independent reductions. Bergman's diamond argument therefore gives the ordered monomials as normal forms. □

The quadratic dual is

$$A_q^! = \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Theorem 9.3 (Koszulness)**.** *A_q is Koszul, and its Koszul complex is*

$$0 \to A_q \otimes \Bbbk(\xi\eta) \xrightarrow{d_2} A_q \otimes (\Bbbk\xi \oplus \Bbbk\eta) \xrightarrow{d_1} A_q \to \Bbbk \to 0,$$

with

$$d_1(a \otimes \xi + b \otimes \eta) = ax + by, \quad d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Proof. Direct substitution gives $d_1 d_2 = ayx - qaxy = 0$. The image of d_1 is the augmentation ideal (x, y), so the complex is exact at A_q and has cokernel $\Bbbk$. We now compute $\ker d_1$.

Write

$$u = \sum_{i,j\geq 0} u_{ij} x^i y^j, \qquad v = \sum_{i,j\geq 0} v_{ij} x^i y^j.$$

The relation $y^j x = q^j x y^j$ gives

$$ux + vy = \sum_{a,b\geq 0} \left(q^b u_{a-1,b} + v_{a,b-1}\right) x^a y^b,$$

where coefficients with a negative index are zero. Thus $ux + vy = 0$ implies $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Define $c = \sum_{i,j\geq 0} u_{i,j+1} x^i y^j$. Then $u = cy$ and, using $x^i y^j x = q^j x^{i+1} y^j$,

$$v = -qcx.$$

Hence $(u, v) = d_2(c)$. Finally, $d_2(c) = 0$ implies $cy = 0$; the PBW basis and $q \neq 0$ show that A_q is a domain, so $c = 0$. The complex is exact, and the quadratic Koszul criterion proves that A_q is Koszul. □

Theorem 9.4 (Yoneda algebra of the quantum plane)**.** *Use the convention that a Yoneda product $\alpha\beta$ means first β and then α. The augmentation module has*

$$\operatorname{Ext}^*_{A_q}(\Bbbk, \Bbbk) \cong \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi) = A_q^!$$

as a bigraded algebra, where $|\xi| = |\eta| = 1$ in cohomological degree and each generator retains its internal weight one.

Proof. Apply $\mathrm{Hom}_{A_q}(-, \Bbbk)$ to the minimal Koszul resolution in the preceding theorem. Every matrix entry of d_1 and d_2 lies in the augmentation ideal, so the dual differential vanishes. The cohomology has basis

$$1, \quad \xi, \eta, \quad \xi\eta$$

in cohomological degrees 0, 1, 1, 2. Splicing the two degree-one extensions gives the degree-two class. The syzygy $y \otimes \xi - qx \otimes \eta$ identifies the two splicings by $\xi\eta = -q\eta\xi$, while the absence of degree-two x^2 and y^2 syzygies gives $\xi^2 = \eta^2 = 0$. These relations account for the complete four-dimensional cohomology, and therefore present the Yoneda algebra. □

The algebra, its bar, and its augmentation module are explicit. To realize A_q transversely over a curve one must construct a compatible factorization family of algebras. Tensoring every configuration coefficient by one fixed copy of A_q does not suffice: at two separated supports, factorization would require two copies of that algebra.

9.3 Quiver path algebras

Let Q be a finite quiver and $S = \bigoplus_{i \in Q_0} ke_i$. Assign a dg vector space M_a to each arrow $a : i \to j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i, \qquad T_S(M) = S \oplus M \oplus M \otimes_S M \oplus \cdots .$$

Words are composable paths.

For the quiver $1 \xrightarrow{a} 2$, the path algebra has basis e_1, e_2, a and relation $a = e_2 a e_1$. The simple modules S_1, S_2 satisfy

$$\mathrm{Ext}^1(S_1, S_2) \cong \Bbbk, \qquad \mathrm{Ext}^1(S_i, S_i) = 0, \qquad \mathrm{Ext}^1(S_2, S_1) = 0.$$

The bar differential records concatenation of paths, and the quadratic dual reverses the arrow with cohomological shift.

Theorem 9.5 (Relative path-algebra bar)**.** *For a tensor path algebra $T_S(M)$ over semisimple S,*

$$S \otimes^{\mathbb{L}}_{T_S(M)} S \simeq S \oplus sM$$

as a coaugmented coalgebra in S-bimodules. A quadratic quotient $T_S(M)/(R)$ is Koszul precisely when its relative quadratic complex is exact away from weight zero.

Proof. The last-arrow contraction is S-bilinear and gives the same extra degeneracy as in the free case. The quotient statement follows from the relative twisting-morphism criterion. □

9.4 Weyl and Clifford deformations

Let V be a finite-dimensional vector space over a field of characteristic zero, and let $\omega : V \otimes V \to \Bbbk$ be alternating. The Weyl algebra is

$$W_\hbar(V, \omega) = T(V)[[\hbar]]/\langle uv - vu - \hbar\omega(u, v)\rangle.$$

For a purely odd space and symmetric ω, the same formula with the supercommutator $uv + vu$ gives a Clifford algebra.

Theorem 9.6 (Filtered PBW theorem)**.** *For alternating ω in the even Weyl case, or symmetric ω in the purely odd Clifford case, the associated graded algebra for the word filtration is* $\mathrm{Sym}(V)[[\hbar]]$ *in the Weyl case and* $\bigwedge(V)[[\hbar]]$ *in the Clifford case, interpreted degreewise in the $\hbar$-adic tensor category.*

Proof. Filter by word length. The constant term lowers filtration by two, so the associated graded relation is the commutative or skew-commutative quadratic relation. The cubic ambiguities reduce to the cocycle identity for a central scalar form, which holds because scalars commute. The ordered monomials give the PBW basis. □

Let $R \subset V^{\otimes 2}$ be the span of the homogeneous commutator relations, or supercommutator relations in the odd case. The quadratic-linear-constant presentation has a curved Koszul dual. Its constant term gives the curvature functional

$$\theta : s^2 R \longrightarrow k[[\hbar]][2], \qquad \theta\big(s^2(u \otimes v - v \otimes u)\big) = \hbar\omega(u, v)$$

in the even convention, with the corresponding plus sign for odd generators. Dualizing this functional gives the curvature element m_0 of degree two. A comparison with geometric anomaly curvature requires a map between their deformation complexes.

9.5 The Hall algebra of the A_2 quiver

Let $Q : 1 \to 2$ over $\mathbb{F}_q$. Write S_1, S_2 for the simples and M for the unique non-split extension

$$0 \to S_2 \to M \to S_1 \to 0.$$

Use $[U] * [V] = v^{\langle \dim U, \dim V \rangle} \sum_W F^W_{U,V} [W]$, where $F^W_{U,V}$ counts subobjects isomorphic to V with quotient U, and $\langle \alpha, \beta \rangle = \alpha_1\beta_1 + \alpha_2\beta_2 - \alpha_1\beta_2$. Direct counting gives

$$[S_1] * [S_2] = v^{-1}[S_1 \oplus S_2] + v^{-1}[M], \qquad [S_2] * [S_1] = [S_1 \oplus S_2], \qquad v^2 = q.$$

Putting $E_i = [S_i]$, the root vector is

$$E_{12} = E_1E_2 - v^{-1}E_2E_1 = v^{-1}[M].$$

Theorem 9.7 (Quantum Serre relation)**.** *The Hall generators satisfy*

$$E_1^2E_2 - (v + v^{-1})E_1E_2E_1 + E_2E_1^2 = 0,$$

and the analogous relation with 1 *and* 2 *exchanged.*

Proof. Every representation of dimension vector $(2, 1)$ is a linear map $f : \mathbb{F}_q^2 \to \mathbb{F}_q$ of rank zero or one. Read a word from the top quotient to the bottom quotient. The complete-flag counts are

$\operatorname{rk} f$	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

For rank zero every one-dimensional subspace of $\mathbb{F}_q^2$ is admissible, giving $q + 1$ flags in each word. For rank one, the word 121 forces the unique kernel line, the word 211 is incompatible with the final subrepresentation, and the word 112 allows every line at the remaining stage.

The total Euler-form exponents of the three words are respectively $-1, 0, 1$. Their twisted coefficients are therefore $v^{-1}, 1, v$. On the rank-zero stratum the Serre combination has coefficient

$$(q + 1)\big(v^{-1} - (v + v^{-1}) + v\big) = 0.$$

On the rank-one stratum it has coefficient

$$v^{-1}(q + 1) - (v + v^{-1}) = v^{-1}(v^2 + 1) - (v + v^{-1}) = 0.$$

The two isomorphism classes exhaust dimension vector $(2, 1)$, proving the relation. Vector-space duality, followed by interchanging the vertices, gives the second Serre relation. These are the relations in Ringel's realization of $U_v^+(\mathfrak{sl}_3)$ [43, 80]. □

Chapter 10

Spectral exchange and formal fields

An exchange relation compares two orders of separated spectral labels. Normal forms require its overlap equations. A field realization also requires truncation on each state, a vacuum, translation, and associativity of iterated fields. We construct these structures in that order.

10.1 Labelled exchange and its normal forms

Let k have characteristic zero, let V be finite-dimensional, and work over rational functions in pairwise distinct labelled parameters $u_1, \dots, u_n$, with all denominators of the operators below inverted. Fix an invertible $R(u) \in \mathrm{End}(V^{\otimes 2}) \otimes k(u)$ satisfying

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \tag{10.1}$$

Write P for the tensor flip and $c_{u,v} = PR(u-v) : V_u \otimes V_v \to V_v \otimes V_u$ for the crossing. For exchange in both directions impose

$$R^{21}(-u)R(u) = 1,$$

so $c_{v,u}c_{u,v} = 1$. Write $R(e_i \otimes e_j) = \sum_{k,l} R_{ij}^{kl} e_k \otimes e_l$. The Zamolodchikov–Faddeev relation is

$$a_i(u)a_j(v) = \sum_{k,l} R_{ij}^{kl}(u-v)a_l(v)a_k(u). \tag{10.2}$$

Thus it identifies two labelled tensor words by $c_{u,v}$.

Theorem 10.1 (Normal ordering on the distinct-label sector)**.** *Fix an order of the labels. In the multilinear sector containing each label exactly once, the quotient by these exchanges is canonically isomorphic to the tensor product in that order. In particular a tensor basis of the ordered product is a basis of this sector.*

Proof. Orient each exchange to remove an inversion. The inversion number decreases, so reduction terminates. Exchanges on disjoint positions commute. For three adjacent distinct labels the two possible reductions agree by (10.1). Reverse exchanges impose no further relation because $c_{v,u}$ is the inverse of $c_{u,v}$. Every reduction therefore gives the same linear map to the ordered tensor product. The inclusion of that product into the quotient and the reduction map are inverse: reduction fixes an already ordered tensor, and each reduction step is a defining relation. □

Conversely, an injective ordered-tensor presentation for three distinct labels forces the Yang–Baxter overlap equation; the two reductions represent the same word and must agree as operators. This argument concerns the labelled multilinear sector. Repeated labels and coincident parameters require their own relations. Without inverse exchange the full two-way relation can collapse even at length two. For $V = k$ and $R = 2$, Yang–Baxter holds, but the two exchanges imply $ab = 4ab$; over k this forces every such two-label product to vanish.

10.1.1 *Diagonal exchange*

In a basis e_i of V, choose nonzero scalars or rational functions $q_{ij}(u)$ with $q_{ij}(u)q_{ji}(-u) = 1$ and $q_{ii}(u) = 1$. The operator $R(u)(e_i \otimes e_j) = q_{ij}(u)e_i \otimes e_j$ satisfies Yang–Baxter because its three diagonal factors commute. Thus

$$a_i(u)a_j(v) = q_{ij}(u-v)a_j(v)a_i(u).$$

In the distinct-label sector the preceding theorem applies. For three reversed labels the coefficient is

$$a_i(w)a_j(v)a_k(u) = q_{ij}(w-v)q_{ik}(w-u)q_{jk}(v-u)a_k(u)a_j(v)a_i(w).$$

Both reduced orders cross exactly the same three pairs, proving the formula. For two fixed generators x, y, the relation $yx = qxy$ is the quantum plane already constructed; its complete PBW assertion follows from its separate one-rule reduction proof.

10.1.2 *The rational permutation solution*

For $R(u) = 1 - \hbar P/u$, expansion of both sides of Yang–Baxter has identical linear terms. The quadratic terms agree by the permutation relations and $1/(u(u+v)) + 1/(v(u+v)) = 1/(uv)$. The cubic terms agree because $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$. However,

$$R^{21}(-u)R(u) = \left(1 - \frac{\hbar^2}{u^2}\right)I.$$

The unitary normalization

$$\widetilde{R}(u) = \frac{R(u)}{1-\hbar/u} = \frac{uI - \hbar P}{u - \hbar}$$

satisfies $\widetilde{R}^{21}(-u)\widetilde{R}(u) = I$. Scalar normalization preserves Yang–Baxter.

10.2 The RTT Yangian of $\mathfrak{gl}_2$

Work over $k[\hbar]$, with k of characteristic zero. Put

$$T(u) = 1 + \sum_{r\ge 1} T^{(r)}u^{-r}, \qquad R(u) = 1 - \frac{\hbar P}{u}.$$

The RTT Yangian $Y_\hbar(\mathfrak{gl}_2)$ is generated by the coefficients $t_{ij}^{(r)}$ subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$

Equivalently,

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar\big(t_{kj}(u)t_{il}(v) - t_{kj}(v)t_{il}(u)\big).$$

Proposition 10.2 (Coefficient relations)**.** *Put $t_{ij}^{(0)} = \delta_{ij}$. The RTT identity is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar\big(t_{kj}^{(r)}t_{il}^{(s)} - t_{kj}^{(s)}t_{il}^{(r)}\big), \qquad r, s \ge 0.$$

In particular,

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}),$$

so the first-mode relation has the Rees parameter $\hbar$. At $\hbar = 0$ it is commutative; setting $\hbar = 1$ gives the ordinary $\mathfrak{gl}_2$ relation.

Proof. Clear the denominator $u - v$ in RTT and compare the coefficient of $u^{-r}v^{-s}$. The two terms contributed by multiplication by $u - v$ produce the two commutators on the left, and the two ordered products on the right are the corresponding coefficients of the RTT numerator. Setting $r = 0$ gives the displayed first-mode relation. □

Theorem 10.3 (Hopf structure)**.** *The formulas*

$$\Delta(T(u)) = T(u)\dot{\otimes}T(u), \qquad \varepsilon(T(u)) = 1, \qquad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra structure on the coefficient algebra. Formal matrix identities are taken in its u^{-1}-adic series ring; each individual coefficient is a finite sum. Coefficientwise,

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^{2} \sum_{p=0}^{r} t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}.$$

Proof. Apply the RTT relation in each tensor factor. Matrix multiplication in the auxiliary space gives the RTT relation for $\Delta(T)$. Coassociativity is associativity of matrix multiplication. Expanding the matrix product gives the coefficient formula. The inverse series exists uniquely because $T(u) = 1 + O(u^{-1})$; its defining identities give the antipode axioms. □

The evaluation map

$$\mathrm{ev}_a : T(u) \longmapsto 1 + \frac{\hbar E}{u - a}, \qquad E = (E_{ij}),$$

lands in $U(\mathfrak{gl}_2)[\hbar][[u^{-1}]]$. Its validity follows by substituting $[E_{ij}, E_{kl}] = \delta_{kj}E_{il} - \delta_{il}E_{kj}$ into the coefficient relation. Give $t_{ij}^{(r)}$ degree $r - 1$ and $\hbar$ degree zero. The PBW theorem identifies the associated graded with $U_\hbar(\mathfrak{gl}_2[t])$, where $U_\hbar(\mathfrak{l})$ has relations $xy - yx = \hbar[x, y]_1$. At $\hbar = 1$ this is $U(\mathfrak{gl}_2[t])$. The distinct filtration assigning degree r to $t_{ij}^{(r)}$ has a commutative associated graded. These are the two filtrations in [73, Sections 1.16–1.27]; their distinction fixes the deformation parameter.

This example contains three structures: ordered multiplication in the Yangian, coproduct on tensor products of modules, and exchange by the rational R-matrix.

10.3 RTT COPRODUCTS AND REFLECTION

For a fixed numerical R, the general RTT algebra $\mathcal{T}_R$ is the coefficient algebra of $T(u)$ subject to

$$R_{12}(u - v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u - v). \tag{10.3}$$

Suppose its formal expansion convention makes every coefficient well-defined. Its coproduct and counit are

$$\Delta(t_{ij}(u)) = \sum_a t_{ia}(u) \otimes t_{aj}(u), \qquad \epsilon(t_{ij}(u)) = \delta_{ij}.$$

To verify the relation for the coproduct, write U, V for two copies of T with commuting coefficient algebras. Then

$$RU_1V_1U_2V_2 = RU_1U_2V_1V_2 = U_2U_1RV_1V_2 = U_2U_1V_2V_1R = U_2V_2U_1V_1R.$$

Thus UV satisfies RTT. Matrix multiplication proves coassociativity, and substitution of the identity proves the counit equations. This proves a bialgebra structure; an antipode additionally requires the relevant matrix inverse, as in the Yangian normalization above.

For a constant invertible R satisfying Yang–Baxter, the reflection algebra $\mathcal{K}_R$ is the free algebra on the entries of K modulo

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}. \tag{10.4}$$

This always defines an associative quotient. Its interpretation as boundary transport is a statement about operators satisfying that relation. Indeed, put $B = PR$. Since $PK_2P = K_1$, the reflection equation is equivalent to

$$BK_2BK_2 = K_2BK_2B.$$

Together with Yang–Baxter for the adjacent B_i and commutation at disjoint positions, this is the boundary braid relation. Invertible solutions therefore represent the corresponding boundary braid group; noninvertible solutions give the specified monoid action. This operator coherence proves no PBW theorem for arbitrary generator relations in $\mathcal{K}_R$. For $R = I$ the relation says that all entries of K commute, giving a polynomial algebra and its usual monomial basis.

The RTT coproduct, reflection quotient and ZF labelled sector have different carriers, despite their use of the same numerical R-operator.

10.4 Fields and weak associativity

For a vector space W set

$$\mathcal{E}(W) = \mathrm{Hom}_k(W, W((z))).$$

A field is an endomorphism-valued formal distribution whose action on each state has only finitely many negative powers. The bound may depend on that state. Thus $\mathcal{E}(W)$ generally differs from $\mathrm{End}(W)((z))$, which imposes one uniform bound on the operator coefficients. For example, on $W = \bigoplus_{m\geq 0} ke_m$ the rule $a(z)e_m = z^{-m}e_m$ is a field, but has no uniform lower bound.

Definition 10.4 (Nonlocal vertex algebra)**.** A nonlocal vertex algebra consists of a vector space V, a vacuum $\mathbf{1}$, and a linear map $Y : V \to \mathcal{E}(V)$ such that

$$Y(\mathbf{1}, z) = \mathrm{id}, \qquad Y(a, z)\mathbf{1} \in a + zV[[z]],$$

and for each $a, b, c \in V$ some integer $l \geq 0$ satisfies weak associativity:

$$(z_0 + z_2)^l Y(a, z_0 + z_2)Y(b, z_2)c = (z_0 + z_2)^l Y(Y(a, z_0)b, z_2)c.$$

Substitution and expansion are in the formal-variable convention of [64, Definition 2.3]: powers of $z_0 + z_2$ are expanded in nonnegative powers of the second variable after clearing the indicated factor.

The translation operator is $Ta = \partial_z Y(a, z)\mathbf{1}|_{z=0}$; it satisfies $[T, Y(a, z)] = Y(Ta, z) = \partial_z Y(a, z)$. The vacuum also implies injectivity of the state-field map. Neither fact replaces weak associativity.

Theorem 10.5 (Reconstruction from compatible fields)**.** *Let $U \subset \mathcal{E}(W)$ be quasi-compatible: for every finite ordered list of its fields there is a nonzero polynomial in pairs of variables whose product clears their iterated product into* $\mathrm{Hom}(W, W((z_1, \dots, z_r)))$*. Then U and the identity generate a nonlocal vertex algebra of fields, with W a faithful quasi-module. If U is S-local in Li's sense, the generated algebra is a weak quantum vertex algebra and W is a module.*

Proof. For a compatible pair choose a clearing polynomial $p(z_1, z)$ and define its field product by

$$Y_{\mathcal{E}}(a(z), z_0)b(z) = \iota_{z,z_0} p(z + z_0, z)^{-1} p(z_1, z)a(z_1)b(z)\big|_{z_1=z+z_0}.$$

The result is independent of the clearing polynomial, and compatibility is preserved by its coefficients. Closing under those coefficients and adjoining the identity gives weak associativity by comparing the same cleared triple product in its two nested expansions. These are the constructions and assertions of [64, Theorems 2.17, 2.20 and 5.8]. The last theorem uses S-locality to strengthen the quasi-module identity to the module identity. A canonical Yang–Baxter operator on the generated state space requires the additional nondegeneracy hypothesis of [64, Theorem 5.11 and Corollary 5.13]. □

10.5 The formal deformation parameter

For $V = V_0[[\hbar]]$ and $W = W_0[[\hbar]]$, use

$$V_\hbar((z)) = \varprojlim_M (V/\hbar^M V)((z)), \qquad V \widehat{\otimes} W = (V_0 \otimes W_0)[[\hbar]].$$

The pole bound may now also depend on M. A topological state-field map is $Y : V \widehat{\otimes} V \to V_\hbar((z))$ with vacuum and translation. A braiding has the form $S(z) = I + O(\hbar)$, with completed Laurent coefficients. In the convention used here, S-locality means

$$\begin{aligned} &(z_1 - z_2)^N Y(z_1)(1 \otimes Y(z_2))(S(z_1 - z_2) \otimes 1)(a \otimes b \otimes c) \\ &\quad = (z_1 - z_2)^N Y(z_2)(1 \otimes Y(z_1))(P \otimes 1)(a \otimes b \otimes c) \pmod{\hbar^M}. \end{aligned} \tag{10.5}$$

Here P is the tensor flip (the graded flip in the super version), the Laurent coefficients are expanded in the indicated formal regions, and N can depend on a, b, M but works for every c. Weak associativity is required modulo each $\hbar^M$, with its corresponding state-dependent clearing exponent. These are the quantum vertex axioms of [28, Definitions 3.4, 3.5 and 3.12]. In particular the ordinary local case is $S = I$, since P is already displayed.

A chosen exchange operator may satisfy the stronger exact equations

$$S^{21}(-z)S(z) = I, \qquad S_{12}(z_1 - z_2)S_{13}(z_1 - z_3)S_{23}(z_2 - z_3) = S_{23}(z_2 - z_3)S_{13}(z_1 - z_3)S_{12}(z_1 - z_2).$$

They give coherent exchange on labelled tensor products by the same argument as above. Compatibility of that exchange with a collision product is expressed by the hexagon, in one convention,

$$S(w)(Y(z) \otimes 1) = (Y(z) \otimes 1)S_{23}(w)\iota_{w,z}S_{13}(z + w).$$

This is additional exact data if degeneracy of Y leaves a kernel: associativity alone gives the hexagon only after applying the relevant field map [28, Proposition 3.14]. Conversely, the exact hexagon and braided locality imply weak associativity by [28, Proposition 3.13]. Formal exchange is not, without an analytic continuation theorem, an action by monodromy on actual functions of complex configurations.

For Yangian currents, a scalar factor $f(u)$ with

$$f(u)f(-u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1}$$

makes $f(u)(1 - \hbar P/u)$ unitary; the explicit factor above is one choice. RTT itself does not supply field truncation. For instance an evaluation series $E/(u - a) = \sum_{r \geq 0} a^r E\, u^{-r-1}$, expanded at infinity with

$a \neq 0$, has unbounded negative powers on any state on which E is nonzero. A quantum vertex realization must specify its state space, expansion, vacuum, translation, and weak-associativity or reconstruction theorem. The ϕ-coordinated module construction uses further chosen coordinate data [65].

The same distinction applies to Ran coefficients. Given an actual translation-equivariant coefficient system and a restriction map to fields, the preceding axioms can identify its field realization. Conversely, a nonlocal field algebra does not by its definition provide algebraic coefficients on all configuration charts, collision pullbacks, or an associative algebra for $\otimes^{\mathrm{ch}}$. Those coefficient maps and their descent equations are the required geometric comparison.

10.6 Shuffle multiplication

Let $K(z)$ be rational and set $\mathrm{Sh}_n = k(z_1, \dots, z_n)^{\mathfrak{S}_n}$, with $\mathrm{Sh}_0 = k$. For $f \in \mathrm{Sh}_m$, $g \in \mathrm{Sh}_n$, define

$$f * g = \sum_{\sigma \in \mathrm{Sh}(m,n)} \sigma\left(f(z_1, \dots, z_m)\, g(z_{m+1}, \dots, z_{m+n}) \prod_{i \leq m < j} K(z_i - z_j) \right). \tag{10.6}$$

Here $\mathrm{Sh}(m, n)$ consists of permutations preserving order within each block. Equivalently this is the full permutation sum divided by $m!n!$. The unit is $1 \in \mathrm{Sh}_0$.

Proposition 10.6 (Associative shuffle multiplication)**.** *The product is associative and unital. The same holds on any subspace closed under the product with its specified pole and wheel conditions.*

Proof. Both parenthesizations enumerate each ordered partition of the final labels into three blocks of the prescribed sizes exactly once. Every pair from different blocks contributes the same kernel factor on both sides. An empty block contributes one term and no kernel, proving the unit equation. □

For the additive affine Yangian model, take

$$K(z) = \frac{(z + \epsilon_1)(z + \epsilon_2)(z + \epsilon_3)}{z^3}, \qquad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

In [83, Section 7.3], the small additive shuffle algebra consists of functions $f / \prod_{i \neq j}(z_i - z_j)$ with symmetric polynomial numerator f vanishing when three distinct indices satisfy $z_{i_1} - z_{i_2} = \epsilon_1$ and $z_{i_2} - z_{i_3} = \epsilon_2$. Its product uses the equivalent pair factor $\omega(z_j, z_i) = K(z_i - z_j)$. For generic parameters, meaning that an integral relation $a\epsilon_1 + b\epsilon_2 + c\epsilon_3 = 0$ forces $a = b = c$, [83, Theorem 7.6] identifies it with the positive affine Yangian by sending its current generator e_r to z_1^r. Degreewise factorial rescaling identifies averaged and shuffle-sum normalizations. This cited identification has its specified numerator, wheel and parameter hypotheses; an arbitrary rational shuffle algebra is a larger object.

For two degree-one elements the formula is explicitly

$$z^r * z^s = z_1^r z_2^s K(z_1 - z_2) + z_2^r z_1^s K(z_2 - z_1).$$

For generic nonzero ϵ_a, the numbers $-\epsilon_a$ are zeros of the individual kernel, not poles. The kernel has its pole at zero. The wheel condition is a vanishing condition on a polynomial numerator and cannot be read as a residue at those zeros.

10.6.1 A coproduct is a separate operation

A chamber expansion of a product kernel does not by itself prove an ordinary bialgebra structure. One must give the tensor recipient, the cross factors and any Cartan currents, then verify multiplicativity, the counits, coassociativity and continuity. A double also needs a Hopf pairing with its stated nondegeneracy and completion. A localized or braided target changes these requirements.

A polynomial example exhibits the obstruction already in charge two. Over $R = \mathbb{Q}[h_1, h_2]$, set $h_3 = -h_1 - h_2$, $a = -(h_1^2 + h_1h_2 + h_2^2)$ and $b = -h_1h_2(h_1 + h_2)$. Use even polynomial charge pieces $A_n = R[x_1, \dots, x_n]^{S_n}$ and subset shuffle kernels

$$K_j(v) = v^{2j-1}(v^3 + av + b), \qquad j = 0, 1,$$

with the argument $v = x_{\text{right}} - x_{\text{left}}$. For $j = 0$, pairing terms that exchange a colliding cross-block pair cancels their possible simple-pole residues, so the symmetric product is polynomial. For $j = 1$ the kernel is already polynomial. Thus these are polynomial shuffle algebras. Write $e_m = x^m \in A_1$; the algebra unit is $\mathbf{1}_0 \in A_0$. Direct multiplication gives

$$[e_0, e_3] - 3[e_1, e_2] + a[e_0, e_1] = b(e_0 * e_0).$$

Indeed the commutator of $f, g \in A_1$ is $2b(y - x)^{2j-1}(f(x)g(y) - g(x)f(y))$ and $e_0 * e_0 = 2(y - x)^{2j}((y - x)^2 + a)$. A charge-preserving unital algebra coproduct with charge-zero projection as both counits, and with the ordinary Koszul multiplication on its tensor target, would make every e_m primitive. The left side of the relation would then be primitive, whereas the reduced coproduct of its right side would be $2b\, e_0 \otimes e_0 \neq 0$. This rules out that ordinary cooperation on these polynomial carriers. It does not rule out a construction in a different localized or braided recipient.

Chapter II

Centres, cyclic structures, and cotangent models

Ordered multiplication gives Hochschild operations. A cyclic pairing allows further contractions, while a quantum field theory requires a compatible quantization and pushforward. Finite cotangent models make the algebraic part of these requirements explicit.

II.1 The derived centre and a bulk action

For an associative algebra A in a specified presentable closed symmetric monoidal stable coefficient category, its Hochschild cochains are

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A),$$

with derived tensors and internal Hom in that category. Assume the coefficient tensor preserves the realizations used to define bimodules. The Deligne construction gives an E_2 structure on these cochains [63, 71]. Its brace operations insert cochains into the inputs of other cochains; the resulting operad models the chains of little two-disks. As endomorphisms of the identity bimodule, the centre represents coherent central actions on A.

A physical bulk-to-boundary comparison must supply a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_{\partial})$$

preserving those operations. Locality of a specified bulk action can make it central and hence produce β by the universal property. An equivalence requires a separate completeness theorem for that boundary condition. The centre is an algebraic object even when the physical map is unknown. In a Ran model these constructions also require their restriction and collision compatibilities; an internal Hom in an ambient category does not automatically inherit factorization.

II.2 Cyclic structure and surface sewing

Hochschild chains $CC_*(A)$ retain the differential and the circle action. Their degree-zero homology $\mathrm{HH}_0(A)$ forgets both higher degrees and the coherent circle data needed for cyclic contractions. For a proper dg algebra a d-Calabi–Yau trace is a circle-equivariant map

$$\tau : CC_*(A) \longrightarrow k[-d]$$

whose induced bimodule pairing is nondegenerate. Equivalently the map is defined from the homotopy orbits of the circle action. In a finite cyclic A_∞ model this is represented by a nondegenerate pairing compatible cyclically with every higher multiplication. A functional on $\mathrm{HH}_0(A)$ alone is not that datum.

Theorem 11.1 (Open theory and its closed extension)**.** *Let A be a unital cyclic A_∞ algebra over a characteristic-zero field, with finite-dimensional chain model and a nondegenerate cyclic pairing $A \otimes A \to k[-d]$. It defines an open topological conformal field theory in the cyclic model. Its homotopy-universal open–closed extension has closed state complex quasi-isomorphic to the Hochschild chains of A. The surface operations use the dimension-d determinant local system.*

Proof. The pairing contracts the two flags of an internal ribbon edge; cyclicity makes this contraction independent of a cyclic presentation at a vertex. Boundary strata joining adjacent inputs give the A_∞ identities. These operations give the open theory of [20, Theorem A]. For its universal extension, an annulus with ordered open inputs gives the cyclic tensor word of those inputs. A unit at a nondistinguished position is degenerate, and a collision of consecutive marked points composes the corresponding two factors. Thus its normalized chain differential is precisely the Hochschild differential, including the last-to-first term. This identifies the closed state complex as in [20, Proposition 7.4.3]. □

Properness can provide a finite model, but a chosen strict cyclic pairing and its homotopy compatibility must still be available in the model being used. For a smooth category the corresponding Calabi–Yau datum is instead a negative-cyclic class inducing a nondegenerate dualizing-bimodule comparison. Its direction differs from that of the proper trace. A dualizable linear category with either such structure has framed E_2 Hochschild cochains by the cyclic Deligne theorem [11, Theorem 1.0.1 and Section 4]. The circle-invariant class transports the Hochschild calculus to the framed operations; its nondegeneracy supplies the required equivalence.

Stable-curve sewing is a further action of a modular operad with specified clutching maps. To combine it with ribbon-surface operations one must identify their coefficient carriers and verify the mixed contraction equations. A common scalar trace is insufficient to make these operations agree.

11.3 Matrix traces and the necklace bracket

Let $F = \Bbbk\langle x, y\rangle$ and let $F_{\mathrm{cyc}} = F/[F, F]$ be the quotient by the commutator subspace. Cyclic derivatives are defined by

$$\partial_x^{\mathrm{cyc}}[a_1 \cdots a_n] = \sum_{a_i = x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}$$

and similarly for y. Define

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f \, \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f \, \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}.$$

Theorem 11.2 (Trace realization)**.** *For matrices $X, Y \in \mathfrak{gl}_N$ with $\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}$ and the other coordinate brackets zero,*

$$\{\operatorname{Tr} f(X, Y), \operatorname{Tr} g(X, Y)\} = \operatorname{Tr}\{f, g\}_{\mathrm{neck}}(X, Y).$$

Proof. Differentiate the trace by moving the varied matrix entry cyclically to the first position. This gives the cyclic derivative. Contract the X derivative of one trace with the Y derivative of the other and subtract the reversed contraction. The matrix indices close into the trace of the cyclic product. □

On the locus of simultaneous diagonalization in the Hamiltonian reduction of pairs of matrices with moment map $[X, Y]$, use eigenvalue coordinates with $\{x_i, y_j\} = \delta_{ij}$. The invariant functions are

$$p_{ab} = \operatorname{Tr}(X^a Y^b).$$

Then

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1,b+d-1}.$$

This follows by differentiating $p_{ab} = \sum_i x_i^a y_i^b$. A term with a negative index has zero coefficient and is interpreted as zero. For two eigenvalues, the relative invariants

$$A = (x_1 - x_2)^2, \quad B = (x_1 - x_2)(y_1 - y_2), \quad C = (y_1 - y_2)^2$$

satisfy $B^2 = AC$. Since $\{x_1 - x_2, y_1 - y_2\} = 2$, their brackets are

$$\{A, B\} = 4A, \qquad \{B, C\} = 4C, \qquad \{A, C\} = 8B.$$

Thus $e = A/4, h = -B/2, f = -C/4$ satisfy $\{h, e\} = 2e, \{h, f\} = -2f, \{e, f\} = h$. The same polynomial identities hold on the associated symmetric-product Poisson variety.

11.4 A finite cotangent BF model

Let L be a finite-dimensional dg Lie algebra. Its cotangent BV space is

$$\mathcal{E}_L = L[1] \oplus L^*[-2],$$

with the canonical degree -1 symplectic pairing. Write $A \in L[1]$ and $B \in L^*[-2]$. The cotangent action is

$$S_L(A, B) = \langle B, d_L A + \frac{1}{2}[A, A] \rangle.$$

Theorem 11.3 (Cotangent master equation)**.** *The Hamiltonian vector field of S_L is square-zero, and consequently $\{S_L, S_L\} = 0$.*

Proof. The Hamiltonian vector field is the cotangent lift of the Chevalley vector field, with the suspension signs of the cotangent pairing. Its square on the base vanishes by $d_L^2 = 0$, the derivation identity for d_L, and the Jacobi identity. The cotangent lift preserves commutators of vector fields, so the full Hamiltonian vector field is square-zero. Hence $\{S_L, S_L\}$ is constant; evaluating at the origin gives zero. □

Proposition 11.4 (Coordinate form of the modular term)**.** *Let L be an ordinary finite-dimensional Lie algebra with basis e_a and structure constants $[e_a, e_b] = f_{ab}^c e_c$. Let c^a be the odd coordinates on $L[1]$ and b_a their BV-conjugate coordinates. With*

$$S = \frac{1}{2} b_c f_{ab}^c c^a c^b, \qquad \Delta = -\sum_a \frac{\partial}{\partial c^a} \frac{\partial}{\partial b_a},$$

one has

$$\Delta S = \sum_b \operatorname{Tr}(\mathrm{ad}_{e_b}) c^b.$$

Thus the one-loop class of the finite cotangent theory is exactly the modular character.

Proof. Differentiation first in b_a gives $\frac{1}{2} f_{ij}^a c^i c^j$. Odd differentiation in c^a gives $f_{aj}^a c^j$, and the chosen sign in Δ converts this to $f_{ja}^a c^j = \operatorname{Tr}(\mathrm{ad}_{e_j}) c^j$. This is the displayed formula. □

A finite-dimensional dg Lie algebra is unimodular when the supertrace of every adjoint endomorphism vanishes. Every finite-dimensional nilpotent Lie algebra in degree zero is unimodular because each ad_x is nilpotent.

Theorem 11.5 (Finite-dimensional quantum master equation)**.** *Let L be unimodular and let Δ be the BV Laplacian determined by the translation-invariant Berezinian on $\mathcal{E}_L$. Then*

$$\Delta S_L = 0, \qquad \tfrac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L = 0,$$

and equivalently, in the formal exponential module,

$$\hbar \Delta\big(e^{S_L/\hbar}\big) = 0.$$

The quantum observable differential

$$\delta_{S_L} = \{S_L, -\} + \hbar\Delta$$

satisfies $\delta_{S_L}^2 = 0$.

Proof. The classical master equation gives $\{S_L, S_L\} = 0$. The scalar ΔS_L is the divergence of the Chevalley vector field on $L[1]$. The degree-one linear differential has zero supertrace, so contributes no constant term. Its linear coefficient is the modular character $x \mapsto \mathrm{str}(\mathrm{ad}_x)$, which vanishes by unimodularity. The BV exponential identity

$$\hbar^2 e^{-S_L/\hbar} \Delta(e^{S_L/\hbar}) = \tfrac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L$$

proves the exponential form. The standard BV commutator calculation then gives

$$\delta_{S_L}^2 = \Big\{\tfrac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L, -\Big\} = 0.$$

□

For a finite-dimensional nilpotent Lie algebra $\mathfrak{l}$, a local input on $X \times \mathbb{R}$ is

$$\mathcal{L}_X = \Omega^{0,*}(X) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathfrak{l},$$

with differential $\bar\partial + d_{\mathbb{R}}$ and pointwise bracket. Its cotangent theory uses the compactly supported density dual and the shifts appropriate to the spacetime integration pairing. The classical observables are functions on the full cotangent field complex with the cotangent Chevalley differential, including the dual fields. They are holomorphic along X and topological along the real direction in the corresponding local classical theory.

The finite theorem applies to a finite dg Lie model with its specified cotangent pairing and unimodularity. A linear deformation retract alone may transfer higher L_∞ operations; a quantum pushforward may also produce loop terms. Neither is computed by the cubic finite action without that comparison.

An algebraic interval model can be constructed independently. The enveloping algebra $U(\mathfrak{l})$ quantizes the linear Lie–Poisson space, and its two augmentation modules give

$$k \otimes^{\mathbb{L}}_{U(\mathfrak{l})} k \simeq B(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}).$$

A field-theoretic identification needs the boundary quantization and the map identifying interval sewing with this derived tensor product. A renormalized solution of the local quantum master equation supplies quantum observables in its perturbative domain; the finite-dimensional QME does not supply that solution or the boundary comparison.

11.5 BRST AND ORDERED REDUCTION

Let (A, Q) be an augmented dg ordered algebra whose differential preserves the augmentation. Extend Q to the reduced bar factorwise. Write b for the multiplication bar differential.

Proposition 11.6 (BRST–bar bicomplex)**.** *With the suspension sign convention,*

$$Q_{\mathrm{Bar}} b + b Q_{\mathrm{Bar}} = 0.$$

Hence $D = Q_{\mathrm{Bar}} + b$ satisfies $D^2 = 0$.

Proof. Apply Q to a bar face that multiplies adjacent entries. The derivation formula splits the result into the two terms obtained by applying Q before that face. The totalization sign makes their sum the anticommutator. □

Over a field, the Künneth isomorphism identifies the first page and its first differential as

$$(E_1, d_1) \cong B(H^*(A, Q)).$$

The spectral sequence converges to $H^*(B(A), D)$ when the length filtration is exhaustive and bounded in each total degree, or weightwise under the positive-weight hypotheses above. For an ordered geometric tensor, the same argument requires the corresponding Künneth and convergence statements. A BRST reduction, including a current-algebra Drinfeld–Sokolov construction, fits this bicomplex only after its product, augmentation and derivation properties have been supplied.

11.6 CORNERS AND OPERATION IDENTITIES

Let K be a regular cell complex arising from a manifold with corners. Assign a cochain complex E_σ to each cell and a degree-zero cochain map $\rho_{\sigma,\tau} : E_\sigma \to E_\tau$ to each codimension-one face $\tau \subset \overline{\sigma}$. Form the total complex with the cellular chain degree placed in negative cohomological degrees. Orient the cells and use the usual incidence signs. Its candidate differential is the internal differential plus the signed face maps.

Proposition 11.7 (The corner equation)**.** *This candidate is a differential if and only if, for every codimension-two incidence $\nu \subset \overline{\sigma}$, the equation*

$$\sum_{\sigma \supset \tau \supset \nu} [\sigma : \tau][\tau : \nu]\, \rho_{\tau,\nu} \rho_{\sigma,\tau} = 0$$

holds.

Proof. The internal square is zero. Because every face map is a cochain map, its two mixed terms with the internal differential cancel under the totalization sign. The remaining terms are the two-face composites displayed above, grouped by source and target. This proves necessity and sufficiency. □

For constant coefficients the equation is the cellular identity $\partial^2 = 0$. With operation-valued coefficients, the maps themselves must satisfy it. Associahedra express higher associative identities in this form. Collision compactifications and stable-graph spaces express chiral or modular identities only after their coefficient maps have been supplied. An identity among residues alone is not a construction of the required homotopies.

11.7 Two edge colours

Consider stable graphs with transverse and chiral edges, and coefficients with compatible contraction maps. Give the edges the determinant orientation, with an ordering of colours and the Koszul sign when generators are exchanged. Complete by a graph grading for which each fixed coefficient receives only finitely many contractions. In the associated Feynman-transform construction, separating edges produce brackets $[-,-]_{\perp}$, $[-,-]_{\mathrm{ch}}$ and loops produce $\Delta_{\perp}$, Δ_{ch}.

Proposition 11.8 (The equation for two-colour contractions)**.** *Suppose the chosen coefficient modular operad and its endomorphism pairing satisfy the contraction and symmetry axioms, with the determinant twist. In the resulting completed graph convolution model, a structure element Θ commutes with the Feynman-transform differential exactly when*

$$d\Theta + \tfrac{1}{2}[\Theta,\Theta]_{\perp} + \tfrac{1}{2}[\Theta,\Theta]_{\mathrm{ch}} + \hbar_{\perp}\Delta_{\perp}\Theta + \hbar_{\mathrm{ch}}\Delta_{\mathrm{ch}}\Theta = 0.$$

Here the loop parameters record the corresponding loop contractions and the suspensions are chosen so every summand has the same cohomological degree.

Proof. The transform differential is dual to an edge contraction and inserts that edge on the dual generators. Evaluating a structure map turns a separating edge into a composition of two vertex coefficients and gives the indicated quadratic term; interchanging the two vertices accounts for the factor $1/2$. A loop contracts two flags of one coefficient and gives its linear loop term. For a pair of edges, the coefficient composites agree by the modular axioms, while exchanging their determinant generators changes the sign. This proves cancellation also for one edge of each colour, in particular $\Delta_{\perp}\Delta_{\mathrm{ch}} + \Delta_{\mathrm{ch}}\Delta_{\perp} = 0$. The map from the free graph construction is a chain map exactly when the one-edge formula holds on its generators, which is the displayed equation. This is the edgewise form of the Feynman transform [37]. □

The proposition starts with compatible coefficient contractions. It does not infer them from the existence of the two types of graph edges.

11.8 Typed anomalies

An obstruction to extending a solution is a cohomology class in its deformation complex. The governing datum depends on the construction:

construction	governing object	datum
central extension	local Lie algebra cochains	cocycle
BRST quantization	BRST deformation complex	nilpotence obstruction
BV quantization	local BV cochains	quantum master anomaly
conformal-block family	Atiyah algebra of its determinant line	projective curvature
curved Koszul duality	curved convolution Lie algebra	curvature m_0

A curved differential need not square to zero, so its curvature is not automatically a class in an ordinary cochain complex. When the deformation complexes and obstruction classes are defined, a comparison requires a morphism of those complexes and equality of the transported classes. This retains the geometric origin of each scalar.

11.9 Calabi–Yau categories and moduli

For a dualizable linear category with a specified smooth or proper Calabi–Yau structure, cyclic Deligne gives the framed E_2 structure described above. For a smooth dg category of finite type with a d-Calabi–Yau structure, the derived moduli of objects carries a $(2-d)$-shifted symplectic form under the hypotheses of the moduli construction [10,77]. At an object E, the tangent complex is $\mathrm{RHom}(E,E)[1]$; the Calabi–Yau pairing gives its degree-$(2-d)$ nondegeneracy, and the negative-cyclic lift supplies closedness of the form.

These are two outputs of Calabi–Yau data on specified recipients. A geometric BV theory or a factorization family can yield an ordered chiral algebra when the coefficient comparison and interval-recognition hypotheses are satisfied. Neither output by itself constructs that family or the physical bulk map.

Chapter 12

Hall coefficients and the Igusa comparison

The general constructions become relevant to the Igusa denominator only after their carriers and operations have been compared with the reduced geometric theory. The root presentation supplies an explicit algebraic recipient. The remaining maps must retain the coefficient system, orientations, and prepared states.

12.1 Hall multiplication and geometric coefficients

For a moduli problem with extension stacks, write

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}^{\mathrm{ext}}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

A pull–push product requires a coefficient theory, the relevant pullback along p, a defined pushforward along q, and all shifts, orientations and base-change maps. When these give $q_* p^*$, its expression is

$$a * b = q_* p^* (a \boxtimes b).$$

The two orders of multiplication are compared through full two-step flag stacks. Their identification proves associativity only when the pull–push base-change and orientation identifications agree on those flags.

The quiver constructions of Theorem 4.9 supply positive enveloping comparisons with their stated coefficient hypotheses [26]. Preprojective and surface CoHA comparisons with Yangian halves have their corresponding domains and parameters [29, 81]. To obtain an ordered chiral algebra, multiplication must further extend to a factorization family over the chosen curve or defect parameter. To obtain a double, it must also admit a compatible coproduct and Hopf pairing in the stated tensor and topology. The charge-two example above shows why ordinary cooperation cannot simply be inferred from a shuffle product.

12.2 Two positive root completions

Work over a field k of characteristic zero. Let $\mathfrak{n} = \bigoplus_{\alpha>0} \mathfrak{n}_\alpha$ be a Lie superalgebra with finite-dimensional root spaces and an additive integer height $h(\alpha) > 0$, with only finitely many roots below each height bound. Put

$$\mathfrak{n}_H = \mathfrak{n} / \bigoplus_{h(\alpha)>H} \mathfrak{n}_\alpha.$$

Each quotient is finite-dimensional and nilpotent, since a sufficiently long iterated bracket has height greater than H. There are two enveloping constructions:

$$U_{\mathrm{tower}} = \varprojlim_H U(\mathfrak{n}_H), \qquad \widehat{U}_h(\mathfrak{n}) = \prod_\gamma U(\mathfrak{n})_\gamma.$$

The latter completes by total root height of products. For $\mathfrak{n} = kx$ even of height one, they are respectively $k[x]$ and $k[[x]]$. We therefore keep their topologies separate.

Theorem 12.1 (The two bar comparisons)**.** *Antisymmetrization gives quasi-isomorphisms*

$$\widehat{C}_{*,h}(\mathfrak{n}) \longrightarrow \widehat{B}_{\mathrm{word},h}U(\mathfrak{n}),$$

and

$$\varprojlim_{H} C_*(\mathfrak{n}_H) \longrightarrow \varprojlim_{H} B_{\mathrm{word}}U(\mathfrak{n}_H).$$

In each homological degree the two Chevalley completions agree. The coefficient map from the bar tower to the bar completed by total height is also a quasi-isomorphism. These statements do not identify the two enveloping algebras.

Proof. At fixed root degree, positive height bounds the number of inputs. The PBW filtration reduces antisymmetrization to the super-symmetric Koszul resolution, proving the first assertion coefficientwise. For the second assertion, the quotient maps are surjective on every chain group of both bar and Chevalley complexes and hence on their mapping cones K_H. The degreewise exact Roos sequence

$$0 \to \varprojlim_{H} K_H \to \prod_{H} K_H \xrightarrow{1-\mathrm{shift}} \prod_{H} K_H \to 0$$

proves that the limit cone is acyclic, since its component cones are acyclic and products of vector-space complexes preserve acyclicity. In Chevalley degree r, a monomial with each input of height at most H has total height at most rH; the two height filtrations are cofinal in that degree. The coefficient map composed with the tower antisymmetrization is the first comparison, so it is a quasi-isomorphism as well. These are the constructions of Theorem 3.1 and Proposition 3.2 on the Igusa root algebra. □

For the positive Igusa Lie superalgebra $\mathfrak{n}_{\Delta_5,+}$, the Euler supercharacter is

$$\chi_{\mathrm{root}}\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0}(1 - e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

An intrinsically even root vector gives an exterior Chevalley factor; an odd one gives a symmetric factor. Combining the intrinsic parity and homological Euler signs gives the stated exponent, and proper height makes every coefficient finite. With the root-torus convention $e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m$ and $e^{-\rho} = Q^{1/2}R^{1/2}S^{1/2}$, the denominator theorem gives

$$e^{-\rho}\chi_{\mathrm{root}}\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) = D_5(Z_{\mathrm{root}}), \qquad D_5(2\Omega) = 2^{-6}\Delta_5^{\theta}(2\Omega)$$

in the period convention $Z_{\mathrm{root}} = 2\Omega$ [44, 46]. The change of variable acts on the Weyl factor as well as on the product. This is an ordinary word-bar character identity.

12.3 Currents and finite cotangent models

At every finite height, the current Lie conformal superalgebra is

$$\mathrm{Cur}(\mathfrak{n}_H) = k[\partial] \otimes \mathfrak{n}_H, \qquad [a_\lambda b] = [a, b].$$

Its level-zero vertex envelope V_H has primitive current generators:

$$\Delta(a(-1)\mathbf{1}) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}.$$

The current brackets respect this assignment, so the universal property of the envelope extends it to a vertex-algebra morphism. The quotient maps form an inverse system of vertex algebras. At fixed root degree and mode energy it stabilizes by PBW, as in Proposition 3.4.

The resulting pro-object has fields at every finite height. Its inverse-limit field values naturally lie in $\varprojlim_H V_H((z))$. Identifying them with ordinary fields valued in $(\varprojlim_H V_H)((z))$ would require a uniform pole bound on each completed state pair. These targets need not agree: $\sum_{m\geq 0} x^m z^{-m}$ is Laurent modulo every x^{H+1}, but has no lower Laurent bound over $k[[x]]$. The BD current field, its underlying main-diagonal Ran Lie object, and the associative Ran envelope are the separate constructions of Section 3.2 and Proposition 3.5.

Each finite positive-height Lie superalgebra is unimodular: every homogeneous adjoint operation strictly raises height, so its supertrace vanishes. Its finite cotangent model therefore satisfies the QME above. The independently defined algebraic boundary object $U(\mathfrak{n}_H)$ has interval complex $C_*(\mathfrak{n}_H)$ through the bar comparison. A local holomorphic–topological theory can also be formed from the current local Lie algebra and its density dual, at the classical level. Identifying these three constructions requires boundary quantization and sewing maps. At infinite height it additionally requires compatible quantum pushforwards: dualizing the Lie quotient maps reverses their direction on the cotangent fields. Neither a formal inverse system nor the denominator character supplies these maps or the compact $K3 \times E$ realization.

12.4 Character, bracket, parity, and orientation

An Euler character retains an alternating scalar invariant of a complex. Reconstruction also requires the bracket, parity decomposition and orientation line carried by the intended object.

Theorem 12.2 (Three reconstruction fibres)**.** *The following fibres contain distinct structured objects with the same scalar image.*

(i) *For positive integers a, b, the PBW-character fibre of a graded vector space with basis x, y, z in degrees $a, b, a+b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*

(ii) *The superdimension-zero fibre contains both the zero super vector space and $\mathbb{k}^{1|1}$.*

(iii) *The groupoid of square roots of a line D contains $(L, \phi : L^{\otimes 2} \simeq D)$ and its twists by line bundles χ with a chosen trivialization $\chi^{\otimes 2} \simeq \mathbf{1}$. Nonisomorphic twists can have the same square.*

Thus these scalar images do not determine the bracket, the two parity dimensions, or the square-root datum. A reconstruction must supply enough structure to distinguish the displayed fibres.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $(L \otimes \chi)^{\otimes 2} \cong L^{\otimes 2} \cong D$. On a space with a nontrivial two-torsion line, $\mathbf{1}$ and that line are distinct roots of $\mathbf{1}$. Twisting a root line does not automatically provide a section of the twist; a prescribed section requires its own lift. □

Recognition Theorem 12.3 (Minimal PBW recognition)**.** *Let $\phi : \mathfrak{g} \twoheadrightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.*

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. □

12.5 The geometric comparison

The constructions have three distinct starting points:

input	tensor	bar
transverse associative algebra	$\otimes^{!}$	$\mathrm{Bar}^{\perp}$
associative Ran algebra	$\otimes^{\mathrm{ch}}$	$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$
chiral Lie algebra	$\otimes^{\mathrm{ch}}$	$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$.

The ordinary associative bar and cobar give their tensor reconstruction on the proved positive-weight domain. The enveloping comparison relates an associative bar to Chevalley chains when its PBW hypotheses hold. The interval construction computes a relative derived tensor product. Hochschild cochains supply the algebraic centre, and a Calabi–Yau structure supplies cyclic and framed operations on their own coefficient categories.

For a compact protected realization of the Igusa denominator, these outputs must be connected by actual geometric maps. The first data are a global reduced oriented coefficient system on compact extension flags and a multiplication with its base-change and orientation coherence. The next comparison identifies its primitive algebra with the fixed root presentation through the generating maps and relations, rather than through the character alone. A chiral comparison must retain supported collision operations and the empty unit. A boundary comparison must retain the state, trace domain and sewing maps. A modular comparison must identify the determinant line and its normalized section.

The algebraic root and current constructions provide recipients for these maps. They do not determine a compact state space or a bulk solely from its boundary centre. The complete principal-part filtration, transverse ordering, braid transport and surface contractions remain operations to be compared, each with its own carrier and coherence equations.

Chapter 13

Enveloping bars and planar path sequences

The dimension of a bar chain space and the dimension of its homology are different invariants. For an enveloping algebra, the Chevalley comparison gives a Lie-theoretic model for the homology. This makes it possible to compare Motzkin and Riordan path counts with the ordered bar on a specified carrier.

Bar length and Chevalley degree are homological degrees: H_n denotes degree-n homology. The cohomology of the dual Chevalley complex is written H^n. For finite-dimensional weight pieces their dimensions agree.

13.1 Chevalley comparison

Theorem 13.1 (Bar–Chevalley comparison)**.** *Let k be a field of characteristic zero and let $\mathfrak{g}$ be an ordinary Lie algebra over k. Give $U(\mathfrak{g})$ its augmentation killing $\mathfrak{g}$. There is a natural quasi-isomorphism of ordinary conilpotent differential graded coalgebras*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\perp}\big(U(\mathfrak{g})\big),$$

where both constructions use direct sums and ordinary tensors over k. It preserves the PBW filtration. If $\mathfrak{g} = \bigoplus_{w \geq 1} \mathfrak{g}_w$ and $[\mathfrak{g}_u, \mathfrak{g}_v] \subseteq \mathfrak{g}_{u+v}$, the map is a quasi-isomorphism in each total weight. It extends to the degreewise product over weights, using completed tensor products by total weight. The completed coalgebras are conilpotent in each weight; no global conilpotence is asserted for infinite sums.

Proof. Put $C_n(\mathfrak{g}) = \Lambda^n \mathfrak{g}$ in homological degree n, with differential

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

Its coproduct splits the exterior factors between two ordered sublists, with the sign of the corresponding permutation. The ordinary normalized bar differential joins positions $i, i+1$ with sign $(-1)^{i-1}$. Define

$$\kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

At degree zero this map preserves the unit. At degree two its differential is $[x_1 x_2 - x_2 x_1] = [[x_1, x_2]]$. In every degree, pair permutations whose merged adjacent entries occur in opposite orders. Their difference is the bracket of those entries, with the displayed exterior sign. This proves the chain-map equation. Cutting an antisymmetrized word amounts to choosing an ordered subset and antisymmetrizing its two subwords. The two procedures give the same permutations and signs, proving the coalgebra identity.

Filter $U(\mathfrak{g})$ by PBW degree, the bar by total PBW degree, and $\Lambda^n \mathfrak{g}$ by degree n. PBW identifies the associated graded algebra with $S(\mathfrak{g})$. The associated graded comparison is the antisymmetrization $\Lambda(\mathfrak{g}) \to B(S(\mathfrak{g}))$, with zero differential on its source. To verify that it is a quasi-isomorphism, consider the free Koszul resolution $S(V) \otimes \Lambda(V)$ for $V = \mathfrak{g}$. For a basis (x_i) of V, its differential is $\partial = \sum_i x_i \iota_i$, where ι_i contracts the exterior factor x_i. On polynomial degree p and exterior degree q, with $p + q > 0$, the map

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \, \frac{\partial}{\partial x_i}$$

satisfies $\partial h + h\partial = 1$. Only finitely many summands act on each element, even when V is infinite-dimensional. The identity follows by cancellation of cross terms and the Euler identity in the two degrees. Thus this free resolution has augmentation homology k only. The free normalized bar resolution has the same homology by its extra-degeneracy contraction. Antisymmetrization defines an augmentation-preserving comparison between these free resolutions. The simplicial bar sign convention is related to the displayed bar convention by multiplication by $(-1)^n$ in degree n. Free bases permit a reverse comparison and homotopies to be lifted inductively, since every positive-degree cycle in a resolution is a boundary. Tensoring these homotopies with k proves the asserted quasi-isomorphism on associated graded complexes.

Every cycle in the cone of κ has finite PBW filtration degree. A primitive for its highest associated graded part lowers that degree; iteration terminates because the filtration is bounded below. The cone is acyclic, proving the ordinary assertion. For a positive weight grading, PBW degree is at most total weight. The same argument is therefore finite within each weight. Finally, kernels and images of a degreewise product differential are products of kernels and images, so the componentwise quasi-isomorphisms extend to the stated completion. □

The ordinary comparison does not itself construct a morphism in $\mathrm{FactD}(X)$. For geometric coefficient systems, one must give compatible enveloping and Chevalley constructions on every chart and coherent factorization equivalences for the whole Chevalley complex. On each disjointness locus, with restriction denoted by $j^!$, the required equation is

$$f_{B,I}\, j^! \kappa_I = (\kappa_{I_1} \boxtimes \kappa_{I_2})\, f_{C,I}, \qquad I = I_1 \sqcup I_2.$$

Here $f_{C,I}$ and $f_{B,I}$ are the specified Chevalley and bar factorization maps. The bar map may use the signed shuffle comparison under its geometric functor hypotheses. The Chevalley factorization maps, collision compatibility, refinement coherence, and continuity of the completed maps remain additional geometric data to construct or verify. Internal tensor formulas and PBW alone do not provide them.

Calculation 13.2 (Finite-dimensional simple case)**.** For a finite-dimensional complex simple Lie algebra of rank r with exponents $m_1, \dots, m_r$, the Chevalley cohomology is the exterior algebra on primitive classes of degrees $2m_i + 1$. The corresponding homology Poincaré polynomials are

$$\mathrm{Bar}^{\perp} U(\mathfrak{sl}_2) : 1 + t^3, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{sl}_3) : 1 + t^3 + t^5 + t^8, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{g}_2) : 1 + t^3 + t^{11} + t^{14}.$$

The corresponding Betti sequences for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are $1, 0, 0, 1$ and $1, 0, 0, 1, 0, 1, 0, 0, 1$. For a finite-dimensional Lie algebra, the Chevalley chain dimensions are $\binom{\dim \mathfrak{g}}{n}$.

For an ordinary augmented algebra A over $\mathbb{C}$, let $I = \ker(A \to \mathbb{C})$. Its normalized two-sided bar with trivial coefficients has homological chain spaces

$$B_0(A) = \mathbb{C}, \qquad B_n(A) = I^{\otimes n} \quad (n \geq 1).$$

Here n is bar length, independently of any internal weight or PBW degree. If I has finite dimension d, then $\dim B_n(A) = d^n$; in particular, an augmentation ideal of dimension three gives the chain counts 3^n. In a cohomological convention that places B_n in degree $-n$, its homology is denoted H^{-n} instead of H_n.

For $A = U(\mathfrak{sl}_2)$, PBW gives the basis $e^a h^b f^c$, with $a, b, c \geq 0$, where $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The augmentation ideal has all these basis elements except 1. Consequently $B_n(A)$ is countably infinite-dimensional for every $n \geq 1$. Its total PBW filtration has finite-dimensional associated graded pieces, with generating function

$$\sum_{d \geq 0} \dim \mathrm{gr}^{\mathrm{PBW}}_d B_n(A)\, z^d = \big((1 - z)^{-3} - 1\big)^n.$$

In particular, 3^n is the dimension in the least possible total PBW degree $d = n$. These least-degree dimensions are 3, 9, 27, 81, 243 for $n = 1, 2, 3, 4, 5$; the full bar chain spaces are infinite-dimensional. The quotient of indecomposables is zero: the perfectness of $\mathfrak{sl}_2$ implies $I/I^2 = 0$.

The Chevalley chain model is different: its degree-n space is $\Lambda^n \mathfrak{sl}_2$, of dimensions 1, 3, 3, 1 in degrees 0, 1, 2, 3. With the convention $d(x \wedge y) = [x, y]$, the map $d : \Lambda^2 \mathfrak{sl}_2 \to \mathfrak{sl}_2$ sends

$$e \wedge h \longmapsto -2e, \qquad e \wedge f \longmapsto h, \qquad h \wedge f \longmapsto -2f,$$

and is an isomorphism. The next differential vanishes, since

$$d(e \wedge h \wedge f) = -2e \wedge f - h \wedge h - 2f \wedge e = 0.$$

Together with $d : \mathfrak{sl}_2 \to \mathbb{C}$ equal to zero, this gives Chevalley homology dimensions 1, 0, 0, 1. The ordinary bar–Chevalley comparison identifies these homology groups with those of $U(\mathfrak{sl}_2)$; it does not identify the two chain spaces.

13.2 Positive Witt homology

The positive Witt algebra is

$$L_1 = \bigoplus_{n \geq 1} \mathbb{C} e_n, \qquad [e_i, e_j] = (j - i) e_{i+j}.$$

It is an ordinary positively graded Lie algebra with finite-dimensional weight pieces. These are the positive modes of the centreless Virasoro bracket. Its enveloping bar is an explicit ordinary associative construction; identifying it with the transverse algebra of a particular chiral theory is an additional comparison.

Theorem 13.3 (Pentagonal support)**.** *For $n \geq 1$, the degree-n homology $H_n(\mathrm{Bar}^{\perp} U(L_1))$ is one-dimensional in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$, and zero at every other weight. Also $H_0 = \mathbb{C}$ in weight zero. Its Euler character is*

$$\chi_q\big(\mathrm{Bar}^{\perp} U(L_1)\big) = \prod_{m \geq 1} (1 - q^m). \tag{13.1}$$

Proof. Theorem 13.1 identifies the bar homology with Chevalley homology. In each weight the Chevalley complex is finite-dimensional, so its homology is dual to the continuous cohomology in that weight. Goncharova's main theorem and Example 1.1 compute the whole cohomology: in degree n there are precisely the two basic sequences $(2, 5, \dots, 3n - 1)$ and $(3, 6, \dots, 3n)$ in her exponent indexing [39]. Subtracting one from each exponent gives total weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. This proves both the two nonzero dimensions and the vanishing in all other weights.

Independently, the exterior Chevalley chain complex has Euler product $\prod_{m \geq 1}(1 - q^m)$, because there is one even Lie generator in every positive weight. Its coefficients are finite sums at each weight. Euler's pentagonal identity expands it as $\sum_{r \in \mathbb{Z}} (-1)^r q^{r(3r-1)/2}$, agreeing with the computed homology. This agreement checks the Euler character; the vanishing assertion uses Goncharova's homology theorem, not the Euler identity alone. □

For $n = 1, 2, 3, 4$ the weight pairs are (1, 2), (5, 7), (12, 15) and (22, 26). The corresponding nonzero Euler coefficients at 0, 1, 2, 5, 7, 12, 15, 22, 26 have signs +, −, −, +, +, −, −, +, +.

13.3 Planar path sequences

A Motzkin path is a lattice path from $(0,0)$ to $(n,0)$, staying at nonnegative height, with steps $(1,1)$, $(1,0)$, and $(1,-1)$. Let $M(n)$ count all such paths of length n, and let $R(n)$ count those with no horizontal step at height zero. Both counts include the empty path: $M(0) = R(0) = 1$, while $M(1) = 1$ and $R(1) = 0$. For $n \geq 1$ consider the two sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \ldots, \qquad b_n = R(n+3) = 3, 6, 15, 36, 91, \ldots.$$

Proposition 13.4 (Motzkin and Riordan identities)**.** *For every $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \qquad M(n+1) - M(n) = R(n+2) - R(n).$$

The ordinary generating functions of the two path counts, and hence those of a_n and b_n, belong to $\mathbb{Q}(z, \sqrt{1-2z-3z^2})$.

Proof. Work in the formal power-series ring $\mathbb{Q}[[z]]$, and write $M(z) = \sum_{n\geq 0} M(n)z^n$ and $R(z) = \sum_{n\geq 0} R(n)z^n$. A nonempty Motzkin path either starts with a horizontal step followed by an arbitrary Motzkin path, or has a unique first-return decomposition $UPDQ$, where U and D are the up and down steps, P and Q are arbitrary Motzkin paths, and P is drawn one unit above the horizontal axis. Thus

$$M = 1 + zM + z^2M^2.$$

A nonempty path counted by R has no initial horizontal step. Its first return similarly has the form $UPDQ$, with P arbitrary and Q counted by R. Consequently

$$R = 1 + z^2MR, \qquad R = (1 - z^2M)^{-1}.$$

The inverses here exist because their denominators have constant term one. Using $z^2M^2 = M - 1 - zM$, compute

$$(1 + zM)(1 - z^2M) = 1 + zM - z^2M - z^3M^2 = 1 + z.$$

It follows that $(1+z)R = 1 + zM$. Comparing coefficients of z^{n+1} gives $R(n+1) + R(n) = M(n)$ for all $n \geq 0$. Subtracting consecutive identities gives the difference formula.

Solving the quadratic equation for the formal series with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1-2z-3z^2}}{2z^2}, \qquad R(z) = \frac{1 + zM(z)}{1+z},$$

where the square root has constant term one. The apparent pole in the formula for M is removable as a formal series. Finite index shifts and differences preserve the indicated quadratic function field, proving the last assertion. These identities are consequences solely of the path decompositions. □

Remark 13.5 (Low-degree homology). Theorem 13.3 gives homology dimensions 2, 2, 2 in degrees 1, 2, 3, whereas a_1, a_2, a_3 are 1, 2, 5. These triples agree only in degree two. The degree-one discrepancy already excludes equality of the full homology dimension sequences.

Theorem 13.6 (Enveloping bars and planar path counts)**.** *Let $B_\bullet(A)$ be the ordinary normalized bar over $\mathbb{C}$, with trivial coefficients and homological degree given by bar length. With $a_n = M(n+1) - M(n)$ and $b_n = R(n+3)$ indexed by $n \geq 1$ as above,*

$$\dim H_1B_\bullet(U(\mathfrak{sl}_2)) = 0 \neq b_1 = 3, \qquad \dim H_1B_\bullet(U(L_1)) = 2 \neq a_1 = 1,$$

where $L_1 = \bigoplus_{m\geq 1} \mathbb{C}e_m$ and $[e_i, e_j] = (j-i)e_{i+j}$. Hence neither sequence is the bar homology dimension sequence of the corresponding enveloping algebra. Neither is its full bar chain dimension sequence: in both cases B_n is countably infinite-dimensional for every $n \geq 1$.

Proof. For any augmented algebra A with augmentation ideal I, the bar differential out of $B_1 = I$ is zero and the image from $B_2 = I \otimes I$ is I^2. Therefore

$$H_1 B_\bullet(A) = I/I^2.$$

For an ordinary Lie algebra $\mathfrak{g}$, the inclusion of generators induces an isomorphism

$$\mathfrak{g}/[\mathfrak{g},\mathfrak{g}] \xrightarrow{\sim} I/I^2, \qquad I = \ker(U(\mathfrak{g}) \to \mathbb{C}).$$

Indeed the generators span I/I^2, and $[x, y] = xy - yx$ vanishes there. For injectivity, equip $\mathbb{C} \oplus \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ with the square-zero product on its second summand. The quotient map from $\mathfrak{g}$ to that summand is a Lie map into the commutator Lie algebra of this associative algebra. The universal property of $U(\mathfrak{g})$ extends it to an augmented algebra map, which kills I^2 and provides an inverse on the displayed quotient.

The brackets $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$ show that $\mathfrak{sl}_2$ equals its commutator algebra. For L_1, every nonzero bracket has weight at least three. Conversely, for $m \geq 3$,

$$e_m = \frac{1}{m-2}[e_1, e_{m-1}].$$

Thus $[L_1, L_1] = \bigoplus_{m\geq 3} \mathbb{C}e_m$, and its abelianization has basis the classes of e_1, e_2, in weights one and two. This proves both degree-one assertions without using any higher-degree calculation.

PBW gives a countably infinite basis of the augmentation ideal of either enveloping algebra. In the L_1 case the basis consists of nonempty ordered monomials in the countable set $e_1, e_2, \ldots$, each monomial having finite length. Finite tensor powers of either augmentation ideal are therefore countably infinite-dimensional. The path counts are finite in every degree, so they cannot count these full chain spaces. □

In particular, a complex whose homology dimensions are b_n cannot be quasi-isomorphic to $B_\bullet(U(\mathfrak{sl}_2))$, and one whose homology dimensions are a_n cannot be quasi-isomorphic to $B_\bullet(U(L_1))$, with the stated degree convention. A discrepancy of chain dimensions alone would not imply this conclusion; the degree-one homology discrepancy does.

The weight grading gives a different, finite comparison. An ordered PBW monomial of weight w specifies a partition of w, so for $n \geq 1$,

$$\sum_{w\geq 0} \dim B_n(U(L_1))_w q^w = \left(\prod_{m\geq 1}(1-q^m)^{-1} - 1\right)^n.$$

Each coefficient is finite. Thus a weightwise dimension, a dimension after a specified weight cutoff, and an ungraded dimension are distinct invariants. In a weight completion, dimensions must likewise be interpreted weightwise; the countable algebraic dimensions above refer to the ordinary direct-sum enveloping algebras.

Remark 13.7 (Chiral coefficients and transverse multiplication). The central charge of a Virasoro coefficient specifies no additional transverse multiplication. The free, square-zero, enveloping and quantum-plane examples give ordinary associative bars with different homology. Realizing one beside a fixed chiral coefficient requires an actual coefficient family and compatible operations; tensoring by an arbitrary fixed algebra does not itself preserve factorization. A comparison with an ordered geometric bar must construct that family and a quasi-isomorphism preserving its transverse multiplication.

Chapter 14

Information retained by comparison maps

Dimensions, Euler characters and anomaly coefficients each forget structure. The following examples identify that lost information and the extra maps needed to recover it. The ordinary algebra examples in this chapter are over $k = \mathbb{C}$.

14.1 Alignment and exchange

Proposition 14.1 (The aligned retraction)**.** *In the finite decomposition model, let $\iota : \mathcal{M}^{\mathrm{al}} \to \mathcal{M}$ and $\pi : \mathcal{M} \to \mathcal{M}^{\mathrm{al}}$ be the inclusion and projection. Then $\pi\iota = \mathrm{id}$ and $\iota\pi = e_{\mathrm{al}}$. These maps are inverse on the full object precisely when its complementary summand vanishes. A comparison on the aligned image alone does not specify one on the complement or prove collision descent.*

Proof. On an aligned summand both composites are the identity. On an off-aligned summand $\iota\pi$ is zero. This proves the identities and the criterion for this retraction to be an equivalence. To obtain a mixed distributive law, one must separately verify its unit and iteration diagrams and its compatibility with all coefficient transitions. The retraction identities make no assertion excluding some other comparison on the full object. □

ZF exchange gives concrete coherent crossings on a specified labelled sector. To make it a mixed Ran operation one still needs the coefficient diagram and its collision maps; the spectral word relation does not define those maps.

Theorem 14.2 (Dimensions do not determine exchange)**.** *The dimensions of the kernels of $V^{\otimes n} \to \mathrm{Sym}^n V$ do not determine an R-operator, an associative multiplication or a Hopf structure on an algebra built from V.*

Proof. Take $V = kx \oplus ky$. The polynomial algebra $k[x, y]$ and the quantum plane $A_q = k\langle x, y\rangle/(yx - qxy)$ have the same graded vector-space dimensions, namely $\dim(A_q)_n = n + 1$, by the PBW basis above. For $q \neq 1$, one is commutative and the other is not. If q has infinite multiplicative order, the centre of A_q is k: a monomial commuting with x must have zero y-exponent, and commuting with y then forces zero x-exponent. The two module categories therefore also have different centres.

On $V^{\otimes 2}$ both the identity and the diagonal operator with $q_{12} = q, q_{21} = q^{-1}, q_{11} = q_{22} = 1$ satisfy the exchange equations. The kernel dimensions depend only on V, so cannot select between them. A coproduct or associator requires still further maps and coherence equations. □

14.2 Coefficient operations and the centre

Proposition 14.3 (Topology and coefficient contractions)**.** *The cellular equation $\partial^2 = 0$ does not specify operation-valued maps on the strata of a compactified moduli space. Such maps define a differential only when their signed composites satisfy the corner equation.*

Proof. With constant coefficients the two incident face maps agree and their orientation signs cancel. For coefficient maps the terms are their actual composites; the calculation in the corner proposition gives the necessary and sufficient equation. Their equality is extra data. □

Proposition 14.4 (A boundary centre need not recover a bulk)**.** *The derived centre represents central actions on a specified boundary algebra. Its universal property does not imply that an acting algebra maps to it by an equivalence.*

Proof. For the semisimple algebra $B = k \times k$, take the boundary module $b = (k, 0)$. Its derived endomorphism algebra is k, and the action of the centre $HH^{\circ}(B) = k \times k$ on it is projection to the first factor. The second factor acts trivially. The same boundary algebra occurs for $B = k$, whose centre is k. Thus the boundary algebra alone does not recover the larger centre; b is not a generator of B-modules. A physical bulk-to-centre identification likewise needs its own equivalence theorem. □

14.3 Central charge and anomaly classes

Proposition 14.5 (Equal central charge)**.** *Central charge is not a complete invariant of a vertex operator algebra, even among holomorphic theories.*

Proof. The even unimodular rank-sixteen lattices $E_8 \oplus E_8$ and D_{16}^{+} give holomorphic lattice vertex operator algebras of central charge 16 [31]. Their weight-one Lie algebras are respectively $\mathfrak{e}_8 \oplus \mathfrak{e}_8$ and $\mathfrak{so}_{32}$. These Lie algebras are not isomorphic, so neither are the vertex operator algebras. The same scalar anomaly therefore does not determine the chiral multiplication. □

A representation category, braid transport and conformal blocks require the vertex operations and their representation data. The central charge specifies the coefficient of the Virasoro central extension; it supplies none of those maps.

Proposition 14.6 (The types of anomaly data)**.** *A current-algebra level, Virasoro central charge, determinant-line curvature, BRST obstruction and automorphic weight can be compared only after their coefficient spaces and a comparison map have been specified. Numerical agreement alone identifies none of these structured data.*

Proof. A level scales a current cocycle, and a central charge scales the Virasoro cocycle. A determinant curvature is a two-form representing the line's characteristic class. A BRST obstruction lies in the cohomology of the deformation complex of the BRST differential. An automorphic weight is a character of the relevant structure group. These are different types of objects, not components of one canonically given cohomology class. A construction relating two of them must supply the intervening functor or cochain map and identify its image. □

14.4 Bars and scalar sequences

Proposition 14.7 (A comparison of bar functors)**.** *An identification of* $\mathrm{Bar}^{\perp}(A^{\perp})$ *with* $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}})$ *requires a map relating their coefficient categories, tensor products, augmentations and multiplication operations. Agreement of Euler characters does not give such a map.*

Proof. Both normalized constructions can have an ordinary word-length grading, but their word factors and adjacent-face maps use different tensors and products. A chain comparison must commute with those face maps and with the units; a coalgebra comparison must also preserve the cut maps. The Euler character retains only an alternating sum of dimensions. □

Theorem 14.8 (A scalar sequence is not homology data)**.** *A chain complex, its differential and its grading must be specified before an integer sequence can be identified with bar homology dimensions. Equal chain dimensions, discriminants or growth rates do not determine homology.*

Proof. The two complexes $[k \xrightarrow{0} k]$ and $[k \xrightarrow{1} k]$ have the same chain dimensions and different homology. For $U(\mathfrak{sl}_2)$, Calculation 13.2 gives countably infinite-dimensional full bar chain spaces in positive length. The numbers 3^n count only their least total PBW-degree pieces. Its Chevalley chain dimensions are 1, 3, 3, 1, while its homology dimensions are 1, 0, 0, 1. The quasi-isomorphism identifies homology and not chain spaces. Proposition 13.4 gives path sequences related by an explicit transform; that transform is not a comparison with either bar complex. □

These examples leave positive constructions intact: the CE comparison, path recursions, exchange normal forms and finite cotangent model each have an explicit carrier and proof. A further geometric identification must construct a map preserving the additional operations it claims to recover.

Book IV

Hamiltonian quantization and relative centres

The Poisson bracket on a symplectic plane gives a Lie algebra of Hamiltonians. Its enveloping algebra quantizes functions on the dual Lie algebra; its Hochschild cochains describe the relative derived centre of that quantization. Write $\hbar$ for the formal quantization parameter. The first comparison in this part is therefore between two complexes with the same coefficient parameter: Chevalley cochains with differential $\hbar d_{\mathrm{CE}}$ and relative Hochschild cochains of a Rees enveloping algebra.

Finite pointed jets make this comparison explicit. They also separate three geometric constructions: a degree-one symplectic target for a two-dimensional sigma model, a finite odd cotangent space with a quantum master equation, and the symplectic plane on which Moyal multiplication acts. Their coordinates, degrees, and quantizations will be specified before they are compared.

Two further passages require additional maps. Formal Hamiltonians form an inverse limit, whereas their continuous dual is a direct limit. Matrix Hamiltonians form a necklace Lie algebra, whose abelianization maps onto commutative Hamiltonians. Neither passage supplies the continuous centre comparison by itself. The positive finite constructions, a binary matrix homotopy, and the precise comparison data will give the local models for the global question.

Chapter 15

Hamiltonian jets and cotangent fields

15.1 Hamiltonians and the Poisson bracket

Let $V = \mathbb{C}^2$ with coordinates (z_1, z_2) and symplectic form

$$\omega = \mathrm{d}z_1 \wedge \mathrm{d}z_2.$$

Let $\mathfrak{m} = (z_1, z_2)$. The pointed formal Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[\![z_1, z_2]\!], \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

It consists of Hamiltonians whose vector fields fix the chosen Darboux point. For $p, q \geq 0$, the complete filtration $F^p\mathfrak{h} = \mathfrak{m}^{p+2}$ obeys

$$\{F^p\mathfrak{h}, F^q\mathfrak{h}\} \subset F^{p+q}\mathfrak{h}.$$

Write $R_r = \mathrm{Sym}^r(V^\vee)$ for the homogeneous polynomials of degree r. The bracket obeys

$$\{R_r, R_s\} \subset R_{r+s-2}.$$

For $d \geq 2$ define the finite jet quotient

$$\mathfrak{h}_{\leq d} = \bigoplus_{r=2}^{d} R_r,$$

where every bracket component of degree greater than d is set equal to zero.

Theorem 15.1 (Finite Hamiltonian jet algebra)**.** *The truncated bracket makes $\mathfrak{h}_{\leq d}$ a finite-dimensional Lie algebra. Its dimension is*

$$\dim \mathfrak{h}_{\leq d} = \frac{d(d+3)}{2} - 2.$$

The quadratic part is

$$R_2 \cong \mathfrak{sl}_2,$$

and each R_r is the irreducible $\mathfrak{sl}_2$-module $\mathrm{Sym}^r(\mathbb{C}^2)$.

Proof. The Jacobiator of the polynomial Poisson bracket vanishes before truncation. Projection to degrees at most d preserves this identity because the discarded terms form a Lie ideal. The dimension is

$$\sum_{r=2}^{d} (r+1) = \frac{d(d+3)}{2} - 2.$$

Quadratic Hamiltonians act linearly on V and preserve ω; this identifies R_2 with $\mathfrak{sp}(V) = \mathfrak{sl}_2$. The standard action on homogeneous polynomials gives the final statement. □

For monomials $z^\alpha = z_1^a z_2^b$ and $z^\beta = z_1^c z_2^d$, the bracket is

$$\{z^\alpha, z^\beta\} = \det(\alpha, \beta) z^{\alpha+\beta-\epsilon}, \qquad \epsilon = (1, 1).$$

The coefficient of the Jacobiator on three monomials is

$$\sum_{\text{cyc}} \det(\alpha, \beta) \det(\alpha + \beta - \epsilon, \gamma) = 0.$$

Expanding the second determinant cancels its quadratic terms in pairs. The remaining terms vanish by $\det(\alpha, \beta)\gamma + \det(\beta, \gamma)\alpha + \det(\gamma, \alpha)\beta = 0$ in dimension two. This is the coordinate form of the identity $[\partial_{z_1} \wedge \partial_{z_2}, \partial_{z_1} \wedge \partial_{z_2}]_{\text{Sch}} = 0$.

The affine Hamiltonian Lie algebra $\mathfrak{h}^{\text{aff}} = \mathbb{C}[\![z_1, z_2]\!]/\mathbb{C}$ also contains linear Hamiltonians, which generate translations. It does not have the same finite jet quotients: for $d \geq 2$, the bracket of z_1^{d+1} with z_2 is $(d+1)z_1^d$, so the discarded higher-degree subspace is no longer an ideal.

15.2 Unimodularity

Theorem 15.2 (Unimodularity of every finite jet)**.** *For every $d \geq 2$ and every $x \in \mathfrak{h}_{\leq d}$,*

$$\operatorname{Tr}_{\mathfrak{h}_{\leq d}}(\operatorname{ad}_x) = 0.$$

Proof. A homogeneous Hamiltonian of degree greater than two raises polynomial degree. Its adjoint action is strictly upper triangular in the decomposition $\bigoplus_{r=2}^{d} R_r$, hence has trace zero. A quadratic Hamiltonian belongs to $\mathfrak{sl}_2$. Its action on each finite-dimensional $\mathfrak{sl}_2$-module R_r has trace zero. Linearity gives the result. □

15.3 The degree-one symplectic target

Use cohomological grading, with $(M[k])^i = M^{i+k}$. Put $L = \mathfrak{h}_{\leq d}$ in degree zero and define the semidirect dg Lie algebra

$$\mathfrak{g}_d = L \ltimes L^\vee[-1].$$

The dual summand has degree 1, is abelian, and carries the coadjoint action $(\operatorname{ad}_x^* \lambda)(y) = -\lambda([x, y])$. Thus $\mathfrak{g}_d[1] = L[1] \oplus L^\vee$ is the degree-one symplectic space $T^*[1]L^\vee$. Its functions have even coordinates x_a on $L^\vee$ and odd coordinates c^a of degree 1 in the cotangent direction. These x_a are linear functions corresponding to a basis e_a of L; write $[e_a, e_b] = f^c{}_{ab} e_c$.

Proposition 15.3 (Chevalley–Poisson identification)**.** *There is a canonical isomorphism of commutative dg algebras*

$$C^*_{\text{CE}}(\mathfrak{g}_d) \cong C^*(L, \operatorname{Sym} L) = \big(\operatorname{Sym} L \otimes \Lambda L^\vee, d_{\text{CE}}\big).$$

Under $\operatorname{Sym} L = \mathcal{O}(L^\vee)$*, this is the polynomial polyvector complex for the linear Poisson tensor on $L^\vee$.*

Proof. Chevalley functions are $\operatorname{Sym}(\mathfrak{g}_d^\vee[-1])$. The dual of $L^\vee[-1]$, followed by the shift $[-1]$, is L in degree zero. The other summand contributes $L^\vee[-1]$ in degree 1. The Lie bracket and coadjoint action give

$$d_{\text{CE}} c^c = -\frac{1}{2} f^c{}_{ab} c^a c^b, \qquad d_{\text{CE}} x_a = f^c{}_{ba} c^b x_c. \tag{15.1}$$

The Schouten differential of the linear Poisson tensor has these same values on polynomial coordinates and constant vector fields. Both differentials are derivations, so the identifications agree on the entire algebra. □

In particular, for a polynomial F one has

$$d_{\mathrm{CE}}F = c^a\{x_a, F\}_{\mathrm{lin}}, \qquad \{x_a, x_b\}_{\mathrm{lin}} = f^c{}_{ab}x_c.$$

The identity $d_{\mathrm{CE}}^2 = 0$ is the Poisson Jacobi identity. With the degree--1 Poisson bracket $\{c^a, x_b\} = \delta^a_b$, the Hamiltonian

$$\Theta = \frac{1}{2}x_c f^c{}_{ab}c^a c^b$$

has degree 2 and generates this differential. It is the Hamiltonian of the degree-one symplectic target.

15.4 The finite BV space

The finite BV construction instead uses

$$\mathcal{E}_L = T^*[-1](L[1]) = L[1] \oplus L^\vee[-2].$$

Its linear coordinates are c^a of degree 1 and b_a of degree -2. Choose the odd Poisson bracket $\{b_a, c^b\}_{\mathrm{BV}} = \delta^b_a$ and the constant Berezinian, with left odd derivatives in the Laplacian

$$\Delta = \sum_a \frac{\partial}{\partial b_a}\frac{\partial}{\partial c^a}.$$

The degree-zero cotangent action is

$$S_L = -\frac{1}{2}b_c f^c{}_{ab}c^a c^b.$$

Its Hamiltonian vector field on the c coordinates is the Chevalley differential in (15.1).

Theorem 15.4 (Finite quantum master equation)**.** *For $L = \mathfrak{h}_{\le d}$ the constant Berezinian gives*

$$\frac{1}{2}\{S_L, S_L\}_{\mathrm{BV}} = 0, \qquad \Delta S_L = \sum_b \mathrm{Tr}_L(\mathrm{ad}\, e_b)c^b = 0.$$

Consequently S_L satisfies the finite quantum master equation

$$\frac{1}{2}\{S_L, S_L\}_{\mathrm{BV}} + \hbar\Delta S_L = 0.$$

Proof. The classical term is the cotangent Hamiltonian for the square of the Chevalley differential, hence vanishes by Jacobi. Left differentiation of the two odd variables gives

$$\Delta S_L = -\sum_{a,b} f^a{}_{ab}c^b = \sum_b \mathrm{Tr}_L(\mathrm{ad}\, e_b)c^b.$$

Unimodularity of the finite Hamiltonian jet algebra gives the second vanishing. The chosen Berezinian is translation invariant; multiplying it by a nonzero constant does not change Δ. □

The two cotangent constructions have different degrees and functions. The even variables x_a in the Poisson complex have degree zero; the finite BV antifields b_a have degree -2. The finite master equation is an algebraic statement on $\mathcal{E}_L$. A field theory requires a source manifold, its field complex, and boundary conditions.

15.5 The Poisson sigma model and its boundary product

Let Σ be an oriented disk and use the degree-one symplectic target $T^*[1]L^\vee$. The graded mapping space $\mathrm{Map}(T[1]\Sigma, T^*[1]L^\vee)$ has a degree--1 symplectic form. Its degree-zero classical fields are a map $X : \Sigma \to L^\vee$ and an L-valued one-form η. For the linear Poisson tensor π on $L^\vee$, the classical action is

$$S_\Sigma(X,\eta) = \int_\Sigma \Big(\eta_a \wedge \mathrm{d}X^a + \frac{1}{2}\pi^{ab}(X)\eta_a \wedge \eta_b \Big).$$

Impose $\eta|_{T\partial\Sigma} = 0$. Integration by parts identifies the action with

$$\int_\Sigma \Big\langle X, \mathrm{d}\eta + \frac{1}{2}[\eta,\eta] \Big\rangle.$$

The boundary term vanishes under the stated condition. This is two-dimensional BF theory for L; the full mapping-space fields add the ghosts and antifields required by its gauge symmetry.

For boundary insertions at two ordered points and a fixed value of X at a third point, the formal perturbation expansion gives the Kontsevich graph product. The propagator and the finite counterterm which removes tadpoles are described in [18, Sections 3.7–3.8]. This is a perturbative interpretation of the product. Its associativity and the following algebraic identification use the formality theorem.

Theorem 15.5 (Linear Poisson boundary algebra)**.** *Let $R = \mathbb{C}[\![\hbar]\!]$ and let $\star_L$ be the Kontsevich product for $\hbar\pi$ on* $\mathrm{Sym}\, L[\![\hbar]\!]$*. There is an R-algebra isomorphism*

$$(\mathrm{Sym}\, L[\![\hbar]\!], \star_L) \xrightarrow{\sim} U_\hbar(L) = T(L)[\![\hbar]\!]/\overline{\langle xy - yx - \hbar[x,y]\rangle}.$$

It fixes the linear generators. The associative algebra on the right defines a locally constant factorization algebra on the oriented line, whose product for adjacent intervals is the enveloping multiplication.

Proof. For linear functions the star commutator is exactly $\hbar[x,y]$. The universal property gives the homomorphism from the Rees enveloping algebra to the star algebra. Its reduction modulo $\hbar$ is the identity on $\mathrm{Sym}\, L$. PBW and completeness lift this to an isomorphism; the inverse is the displayed map. This is the linear Poisson identification of [62, Section 8.2]. The usual associative factorization envelope on a line assigns the algebra to an interval and its ordered multiplication to disjoint subintervals. Associativity supplies its gluing maps. □

PBW symmetrization is the underlying R-module isomorphism. It also transports the enveloping product to the Gutt product. The equivalence with the Kontsevich product is an algebra map and contains its own ordering correction. Identifying an analytic or renormalized field-theoretic boundary observable algebra with this algebraic factorization model requires the corresponding observable map.

Chapter 16

Relative centres and affine formality

16.1 The coefficient ring and the Chevalley differential

The Rees relation scales both the Lie bracket and its adjoint action. Its Chevalley differential must retain the same parameter. For example, take the quadratic jet $\mathfrak{l} = \mathfrak{sl}_2$ with $[h, e] = 2e$. Modulo $\hbar$, the enveloping algebra is commutative, so the Hochschild differential of e vanishes. The unscaled Chevalley differential instead satisfies $(d_{\mathrm{CE}} e)(h) = 2e$.

Let $\mathfrak{l}$ be a finite-dimensional complex Lie algebra. For $N \geq 0$, put

$$R_N = \mathbb{C}[\hbar]/(\hbar^{N+1}), \qquad \mathfrak{l}_N = R_N \otimes_{\mathbb{C}} \mathfrak{l}, \qquad [x, y]_N = \hbar[x, y],$$

and

$$A_N = U_{R_N}(\mathfrak{l}_N) = T_{R_N}(\mathfrak{l}_N)/(xy - yx - \hbar[x, y]).$$

Here N denotes the order in $\hbar$, independently of a Chan–Paton rank. Write $\bar{A}_N$ for the augmentation ideal and $S = \mathrm{Sym}_{\mathbb{C}} \mathfrak{l}$. PBW symmetrization is the R_N-module isomorphism

$$\mathrm{sym}_N : S \otimes R_N \longrightarrow A_N, \qquad x_1 \cdots x_r \longmapsto \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

The source has its ordinary symmetric product. Use normalized relative Hochschild cochains

$$C^n_{\mathrm{HH}}(A_N/R_N, A_N) = \mathrm{Hom}_{R_N}(\bar{A}_N^{\otimes_{R_N} n}, A_N)$$

with differential

$$\begin{aligned}(bF)(a_1, \dots, a_{n+1}) = {} & a_1 F(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

Let d_{CE} act on $C^n(\mathfrak{l}, S) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, S)$ by the adjoint Chevalley differential.

Theorem 16.1 (Relative Rees comparison)**.** *Antisymmetrization defines a quasi-isomorphism of R_N-complexes*

$$\mathrm{Alt}_N : C^{\bullet}_{\mathrm{HH}}(A_N/R_N, A_N) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S) \otimes R_N, \hbar d_{\mathrm{CE}}\big),$$

given by

$$(\mathrm{Alt}_N F)(x_1, \dots, x_n) = \mathrm{sym}_N^{-1} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) F(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

These maps commute with coefficient reduction in N.

Proof. Write $A = A_N$, $R = R_N$, and $A^e = A \otimes_R A^{\mathrm{op}}$. In homological degree n, set

$$K_n = A \otimes_R \Lambda_R^n \mathfrak{l}_N \otimes_R A.$$

Its augmentation is multiplication. For $X = x_1 \wedge \cdots \wedge x_n$, let $X_{\widehat{i}}$ and $X_{\widehat{ij}}$ omit the indicated entries without reordering. Define

$$\begin{aligned}\partial(a \otimes X \otimes b) = \sum_i (-1)^{i+1}\big(a x_i \otimes X_{\widehat{i}} \otimes b - a \otimes X_{\widehat{i}} \otimes x_i b\big) \\ + \hbar \sum_{i<j} (-1)^{i+j} a \otimes [x_i, x_j] \wedge X_{\widehat{ij}} \otimes b.\end{aligned}$$

The normalized relative bimodule bar has terms $\mathcal{B}_n = A \otimes_R \bar{A}^{\otimes_R n} \otimes_R A$. Define

$$\iota_n(a \otimes X \otimes b) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a[x_{\sigma(1)}| \cdots |x_{\sigma(n)}]b.$$

The outer bar faces give the first sum in ∂. Pair adjacent transpositions in the internal faces and use $xy - yx = \hbar[x, y]$ to obtain the second sum. For two inputs this reads

$$\begin{aligned}b\big(1[x|y]1 - 1[y|x]1\big) = x[y]1 - y[x]1 + 1[x]y - 1[y]x \\ - \hbar\, 1[[x, y]]1.\end{aligned}$$

Thus $b\iota = \iota\partial$. PBW makes $\mathfrak{l}_N$ a free summand of $\bar{A}$. Since every factorial is invertible in R, ι is injective. The identity $b^2 = 0$ then gives $\partial^2 = 0$.

Give K_n filtration degree equal to its outer PBW degrees plus n. The associated graded is the diagonal Koszul complex of $\mathrm{Sym}_R(\mathfrak{l}_N)$. Its differential uses differences of left and right polynomial generators. A linear change of variables makes these independent polynomial variables. They form a regular sequence even when R has nilpotents. For an augmented cycle, lift a primitive of its highest-degree symbol and subtract its differential. The filtration degree decreases, so this procedure terminates. Hence $K_\bullet \to A$ is a free bimodule resolution. The relative bar is also free over A^e, with its usual unit-insertion contraction. The map ι lifts the identity of A. Projectivity constructs its chain homotopy inverse degree by degree. Applying $\mathrm{Hom}_{A^e}(-, A)$ gives a cochain homotopy equivalence.

The induced differential on $\mathrm{Hom}_R(\Lambda_R^n \mathfrak{l}_N, A)$ is

$$\begin{aligned}(\delta_N \phi)(x_1, \dots, x_{n+1}) = \sum_i (-1)^{i+1} [x_i, \phi(x_1, \dots, \widehat{x_i}, \dots, x_{n+1})]_A \\ + \hbar \sum_{i<j} (-1)^{i+j} \phi([x_i, x_j], x_1, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

Commuting x past the factors of a symmetrized monomial gives

$$[x, \mathrm{sym}_N(f)]_A = \hbar\, \mathrm{sym}_N(\mathrm{ad}_x f).$$

Both sums therefore become $\hbar$ times the ordinary Chevalley differential. Precomposition with ι gives Alt_N. All formulas commute with coefficient reduction. □

16.2 The formal coefficient limit

Put $R = \mathbb{C}[\![\hbar]\!]$ and $A = \varprojlim_N A_N$. PBW identifies its underlying module with $S[\![\hbar]\!]$. Each coefficient of $\hbar$ is a polynomial. Define

$$A^{\widehat{\otimes}_R n} = \varprojlim_N A_N^{\otimes_{R_N} n}, \qquad C^\bullet_{\mathrm{HH,cont}}(A/R, A) = \varprojlim_N C^\bullet_{\mathrm{HH}}(A_N/R_N, A_N).$$

Equivalently, these are continuous R-multilinear normalized cochains on the completed relative tensors.

Corollary 16.2 (Formal relative Rees comparison)**.** *The compatible antisymmetrizations induce a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH,cont}}(A/R, A) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S)[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

Proof. Coefficient reduction is surjective in each cochain degree. For Hochschild cochains, lift outputs on the PBW input basis. The Chevalley exterior input has finite rank, so its outputs also lift. The comparison cones are acyclic and their transition maps are degreewise surjective. The long exact sequence shows that every transition kernel is acyclic. Choose a primitive of a compatible cycle at level zero. Lift it one level and correct its differential by a primitive in the transition kernel. Iteration gives compatible primitives at every order. The inverse-limit cone is therefore acyclic. □

The coefficient ring is part of this result. Absolute $\mathbb{C}$-linear cochains also detect variations of the coefficient parameter. For $\mathfrak{l} = 0$, the continuous absolute derivation $\partial_\hbar$ on R has no relative counterpart. For nonabelian $\mathfrak{l}$, the associated graded for the $\hbar$ filtration has zero differential, whereas d_{CE} need not vanish.

16.3 Linear Poisson formality

For $\mathfrak{l} = \mathfrak{h}_{\le d}$, put $A_d = U_\hbar(\mathfrak{h}_{\le d})$ and

$$\mathcal{P}_d = \big(C^*(\mathfrak{h}_{\le d}, \operatorname{Sym} \mathfrak{h}_{\le d})[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

This is the polynomial polyvector complex on $\mathfrak{h}^*_{\le d}$ with differential $[\hbar\pi_d, -]$ for the linear Poisson tensor π_d. The relative comparison gives $C^{\bullet}_{\mathrm{HH,cont}}(A_d/R, A_d) \to \mathcal{P}_d$ as a map of complexes. Higher operations require a formality map with specified operadic structure.

Fix Kontsevich's formality morphism $\mathcal{U}$ on the affine space $\mathfrak{h}^*_{\le d}$. Its twist by $\hbar\pi_d$ defines a star product $\star_d$ and an L_∞ quasi-isomorphism on the degree-shifted complexes. Let $T_d : (\operatorname{Sym} \mathfrak{h}_{\le d}[\![\hbar]\!], \star_d) \to A_d$ be the algebra identification fixing the linear generators. This identification includes the equivalence between the Kontsevich and PBW products. PBW symmetrization alone identifies the underlying modules. The first Taylor coefficient of the twisted comparison is

$$\Phi_d(v) = (T_d)_* \sum_{r \ge 0} \frac{1}{r!} \mathcal{U}_{r+1}(\underbrace{\hbar\pi_d, \dots, \hbar\pi_d}_{r}, v).$$

It has source $\mathcal{P}_d$ and target $C^{\bullet}_{\mathrm{HH,cont}}(A_d/R, A_d)$. At each order in $\hbar$, polynomial inputs have polynomial outputs. Its reduction modulo $\hbar$ is the Hochschild–Kostant–Rosenberg map, with the normalized antisymmetrization convention. That map is a quasi-isomorphism for the polynomial algebra. The complete $\hbar$ filtration then proves that Φ_d is a quasi-isomorphism. The higher Taylor coefficients supply the L_∞ identities after shifting by $[1]$. These statements use the affine formality construction [62, Sections 6, 8.2, and 8.3].

On invariant polynomials, the standard Duflo normalization gives

$$\mathrm{Duf}_\hbar = \mathrm{sym}_\hbar \circ j^{1/2}(\hbar\partial), \qquad j^{1/2}(x) = \det_{\mathfrak{h}_{\le d}}{}^{1/2}\left(\frac{\sinh(\mathrm{ad}_x/2)}{\mathrm{ad}_x/2}\right).$$

It is an algebra isomorphism from $(\operatorname{Sym} \mathfrak{h}_{\le d})^{\mathfrak{h}_{\le d}}[\![\hbar]\!]$ to $Z(A_d)$, by the Duflo theorem [62, Section 8.3.4]. The square root has constant term one. This assertion concerns invariant degree-zero cohomology. It does not identify antisymmetrization with a homotopy-E_2 map.

16.4 The quadratic Duflo correction

First set the coefficient parameter to 1 in the polynomial Rees algebra. For the resulting ordinary $\mathfrak{sl}_2$ enveloping algebra, take generators

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

The invariant polynomial and the quadratic Casimir are

$$q = 2ef + \frac{1}{2}h^2, \qquad \Omega = ef + fe + \frac{1}{2}h^2.$$

Calculation 16.3 (The zero-point shift)**.** The Duflo map gives

$$\mathrm{Duf}(q) = \Omega + \frac{1}{2}.$$

Let V_n be the irreducible highest-weight-n module, with $n \geq 0$. Then

$$\left(\Omega + \frac{1}{2}\right)\Big|_{V_n} = \frac{(n+1)^2}{2}\,\mathrm{id}\,.$$

Proof. Write $x = ue + vf + wh$. Direct adjoint matrices give $\mathrm{Tr}(\mathrm{ad}_x^2) = 8uv + 8w^2$. Thus the quadratic term of $j^{1/2}(\partial)$ is $(\partial_e\partial_f + \partial_h^2)/6$. Applied to $q = 2ef + h^2/2$, it gives $(2+1)/6 = 1/2$. Higher differential terms annihilate this quadratic polynomial. Symmetrization sends q to Ω. On a highest-weight vector, ef acts by n, fe by zero, and $h^2/2$ by $n^2/2$. Centrality gives the eigenvalue on the entire irreducible module. □

Restoring the Rees parameter gives $\mathrm{Duf}_\hbar(q) = \Omega + \hbar^2/2$. The polynomial calculation permits specialization at 1; arbitrary formal power series do not admit that specialization. Higher wheels assemble into $j^{1/2}$ on higher-degree invariants.

For the quadratic jet, the invariant polynomial $q = 2ef + h^2/2$ is d_{CE}-closed. Its Duflo image is a degree-zero central class. The oscillator quotient below sends that class to $\hbar^2/8$; this scalar follows from the quotient's own Casimir relation.

16.5 Hochschild braces

For Hochschild cochains $F \in C^p(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \ldots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F. For an algebra in degree zero, an insertion has sign $(-1)^{\sum_i (q_i - 1) j_i}$, where j_i is the number of input positions preceding G_i in the expanded expression. The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots. Together with the cup product and unit, these operations give the Hochschild form of the Deligne structure [71, Theorem 1.1]. The operadic construction in [63, Section 7] gives the corresponding structure for A_∞ algebras. Here the relative operations are taken at each coefficient precision and then passed to the complete limit; every fixed-arity insertion is continuous.

The relative Hochschild cochains carry this brace structure. Their canonical coherent evaluation action on A_d gives the closed–open pair $(C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d), A_d)$. The closed algebra is the relative derived centre. A homotopy-E_2 refinement means a morphism or zigzag

$$\widetilde{\Phi}_d : \mathcal{P}_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$$

in complete R-linear homotopy-E_2 algebras, with the chosen Gerstenhaber structure on $\mathcal{P}_d$. A compatible Swiss-cheese refinement also specifies the mixed operations on the pair and their coherence homotopies.

Corollary 16.4 (Open–closed structure with a refinement)**.** *Given these refinements, the pair $(\mathcal{P}_d, A_d)$ has the transported homotopy Swiss-cheese action. Its open product is the enveloping product. The specified closed and mixed Taylor coefficients determine its closed bracket and boundary action.*

Proof. Transport the two-colored operations through the supplied equivalence of complete operadic algebras. The coherence homotopies are part of this equivalence. □

Theorem 16.5 (Brace realization with a refinement)**.** *For $A = A_d$, supply the homotopy-E_2 and mixed refinements of Corollary 16.4. They identify the selected homotopy Gerstenhaber operations on $\mathcal{P}_d$ with the relative Hochschild centre operations. The chosen mixed refinement also identifies their action on the boundary algebra.*

Proof. The relative Hochschild complex has its brace operations. The supplied operadic equivalence preserves their coherent homotopies. An equivalence of complexes alone would not imply this conclusion. □

16.6 The first mixed identities

A relative Hochschild 1-cocycle D of an algebra concentrated in degree zero is an R-linear derivation:

$$D(ab) = D(a)b + aD(b).$$

For two such cocycles D, E, their Gerstenhaber bracket acts by

$$[D, E](a) = D(E(a)) - E(D(a)).$$

For a graded algebra and homogeneous derivations, these formulas become

$$D(ab) = D(a)b + (-1)^{|D||a|}aD(b),$$
$$[D, E](a) = D(E(a)) - (-1)^{|D||E|}E(D(a)).$$

Higher Hochschild cochains act through multilinear evaluations and braces. They need not act by derivations or carry a coproduct defining a diagonal action. A compatible mixed refinement transports these operations, including their homotopies.

Chapter 17

Finite presentations and Moyal multiplication

17.1 Quadratic Hamiltonians

Set

$$e = \frac{1}{2} z_1^2, \qquad f = -\frac{1}{2} z_2^2, \qquad h = -z_1 z_2.$$

A direct differentiation gives

$$\{h, e\} = 2e, \qquad \{h, f\} = -2f, \qquad \{e, f\} = h.$$

Thus $\mathfrak{h}_{\le 2} = R_2$ is $\mathfrak{sl}_2$ with its standard Chevalley basis. Its Rees enveloping algebra is $U_\hbar(\mathfrak{sl}_2)$; the Moyal realization below is a proper associative quotient of this algebra.

17.2 Cubic Hamiltonians

Let

$$v_j = z_1^{3-j} z_2^{j}, \qquad 0 \le j \le 3.$$

Then

$$\{h, v_j\} = (3 - 2j) v_j, \qquad \{e, v_j\} = j v_{j-1}, \qquad \{f, v_j\} = (3 - j) v_{j+1},$$

with the boundary terms understood as zero. Hence R_3 is the irreducible highest-weight-three module.

Proposition 17.1 (The cubic jet algebra)**.** *The degree-three jet algebra is the semidirect product*

$$\mathfrak{h}_{\le 3} \cong \mathfrak{sl}_2 \ltimes \mathrm{Sym}^3(\mathbb{C}^2),$$

where the second summand is abelian.

Proof. The bracket of two cubic polynomials has degree four and vanishes in the degree-three quotient. The bracket with a quadratic polynomial is the displayed $\mathfrak{sl}_2$ action. □

17.3 The first nonlinear layer

Write $w_k = z_1^{4-k} z_2^{k}$, $0 \le k \le 4$. In $\mathfrak{h}_{\le 4}$ the cubic modes satisfy

$$\{v_i, v_j\} = 3(j - i) w_{i+j-1}, \tag{17.1}$$

whenever the index lies between 0 and 4; the coefficient makes every remaining case vanish.

Calculation 17.2 (Two brackets)**.**

$$\{z_1^3, z_2^3\} = 9 z_1^2 z_2^2 = 9 w_2, \qquad \{z_1^2 z_2, z_1 z_2^2\} = 3 z_1^2 z_2^2 = 3 w_2.$$

The two independent cubic pairs reach the same quartic channel with amplitudes in the ratio $3 : 1$.

Proof. Differentiate each monomial in the two symplectic coordinates and substitute in the Poisson formula. The first pair gives $3z_1^2 \cdot 3z_2^2$. The second gives $2z_1z_2 \cdot 2z_1z_2 - z_1^2 \cdot z_2^2 = 3z_1^2z_2^2$. □

Theorem 17.3 (The quartic jet presentation)**.** *The twelve symbols $e, f, h, v_0, \dots, v_3, w_0, \dots, w_4$ present $\mathfrak{h}_{\le 4}$ with the following relations: the $\mathfrak{sl}_2$ relations, the displayed action on v_j,* (17.1)*, and*

$$[h, w_k] = (4-2k)w_k, \qquad [e, w_k] = kw_{k-1}, \quad [f, w_k] = (4-k)w_{k+1},$$
$$[v_i, w_k] = 0, \qquad [w_k, w_l] = 0.$$

Indices outside their indicated ranges contribute zero. The seven symbols $e, f, h, v_0, \dots, v_3$ already generate this Lie algebra.

Proof. Every bracket of two displayed symbols is a linear combination of the twelve symbols. Induction on a Lie word therefore shows that the presented algebra has dimension at most twelve. The polynomial Poisson bracket, truncated above degree four, satisfies every relation. Indeed, a cubic–quartic bracket has degree five and a quartic–quartic bracket has degree six, so both vanish in this quotient. The images of the twelve symbols form the monomial basis of $R_2 \oplus R_3 \oplus R_4$, which has dimension twelve. The resulting surjection is consequently an isomorphism. Finally, (17.1) produces w_0, w_1, w_2, w_3, w_4 from the pairs (v_0, v_1), (v_0, v_2), (v_0, v_3), (v_1, v_3), (v_2, v_3), with nonzero coefficients 3, 6, 9, 6, 3. □

17.4 The Moyal product

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \ge 0} \frac{1}{n!} \left(\frac{\hbar}{2}\right)^n \sum_{r=0}^{n} (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f)(\partial_1^r \partial_2^{n-r} g). \tag{17.2}$$

Every polynomial pair gives a finite sum. Associativity follows directly from the constant Poisson tensor. On a triple tensor product, the operators $P_{ij} = \partial_1^{(i)} \partial_2^{(j)} - \partial_2^{(i)} \partial_1^{(j)}$ commute. Both parenthesizations of the product are therefore

$$m_3 \exp\!\left(\frac{\hbar}{2}(P_{12} + P_{13} + P_{23})\right)(f \otimes g \otimes h),$$

where m_3 multiplies the three polynomial factors.

Theorem 17.4 (Quadratic exactness)**.** *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar\{f, g\}.$$

The quadratic generators therefore define a Lie representation of the Rees-scaled $\mathfrak{sl}_2$. Their associative algebra has an additional Casimir relation.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. □

Proposition 17.5 (The oscillator quotient)**.** *Let $R_N = \mathbb{C}[\hbar]/(\hbar^{N+1})$ and let $W_N = R_N[q, p]$ have the Moyal product for $\{q, p\} = 1$. Let U_N have generators E, F, H with relations*

$$[H, E] = 2\hbar E, \qquad [H, F] = -2\hbar F, \qquad [E, F] = \hbar H.$$

Set $\Omega = EF + FE + H^2/2$. The assignments

$$E \longmapsto e = q^2/2, \qquad F \longmapsto f = -p^2/2, \qquad H \longmapsto h = -qp$$

induce an algebra isomorphism

$$U_N/(\Omega + 3\hbar^2/8) \xrightarrow{\sim} W_N^{(q,p)\mapsto(-q,-p)}.$$

These compatible isomorphisms also hold after $\hbar$-adic completion. If linear Hamiltonians are retained, the generated associative algebra is the full Weyl algebra.

Proof. Quadratic exactness proves the three commutator relations. The associative products are

$$\begin{aligned} e \star f &= -q^2p^2/4 - \hbar qp/2 - \hbar^2/8, \\ f \star e &= -q^2p^2/4 + \hbar qp/2 - \hbar^2/8, \\ h \star h &= q^2p^2 - \hbar^2/4. \end{aligned}$$

Thus Ω maps to $-3\hbar^2/8$. Its commutator with each of E, F, H vanishes by the defining relations. Give these generators PBW degree one. Give an even monomial $q^a p^b$ degree $(a+b)/2$. The leading symbol of $\Omega + 3\hbar^2/8$ is $C = 2EF + H^2/2$. It is a non-zero-divisor in $R_N[E,F,H]$, since $2C$ is monic in H. Centrality and this property give

$$\operatorname{gr}\big(U_N/(\Omega + 3\hbar^2/8)\big) = R_N[E,F,H]/(C).$$

Indeed, every element of the two-sided ideal is a multiple of the central generator. Its leading symbol is the product of the two leading symbols. The quotient has basis $E^a H^b F^c$ with $a, c \geq 0$ and $b = 0, 1$. Their leading symbols map to unit multiples of the distinct monomials $q^{2a+b}p^{2c+b}$, precisely the even monomials. Hence the associated-graded map is an isomorphism. Lifting highest-degree terms proves surjectivity and comparing them proves injectivity. Every element at fixed N has finite polynomial degree. The formulas commute with reduction in N, so their inverse limit gives the formal isomorphism. Finally q, p generate W_N and topologically generate its formal limit. □

The primitive coproduct on U_N does not descend to this quotient. Modulo $\hbar$, it sends the classical relation to

$$\Delta(C) = C \otimes 1 + 1 \otimes C + 2E \otimes F + 2F \otimes E + H \otimes H.$$

The cross-term is nonzero in the tensor square of $\mathbb{C}[E,F,H]/(C)$. For example, evaluation at $(E,F,H) = (1,0,0)$ in the first factor and $(0,1,0)$ in the second gives 2. Thus the oscillator quotient is an associative algebra, with no coproduct inherited from these primitive generators.

Calculation 17.6 (Cubic correction)**.** For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first higher-order correction to this cubic commutator.

Proof. In (17.2), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3(3!) = 3\hbar^3/2$. □

17.5 Weyl generators

With the convention $\{q, p\} = 1$, the symbols satisfy $[q, p]_\star = \hbar$. Their Weyl-ordered differential operators are

$$\widehat{q} = q, \qquad \widehat{p} = -\hbar\partial_q, \qquad [\widehat{q}, \widehat{p}] = \hbar.$$

The abstract Weyl algebra is defined by this commutator relation. Weyl ordering transports its multiplication to the Moyal product. The differential-operator representation realizes these relations over $\mathbb{C}[\hbar]$ and its formal completion. The finite quotient argument above uses the abstract R_N-algebra, so it does not require a faithful differential-operator representation at nilpotent $\hbar$.

Chapter 18

Continuous Hamiltonians and symplectic descent

18.1 Residue duality and the direction of truncation

The finite jet algebras are quotients of $\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[\![z_1, z_2]\!]$. Their duals embed into the continuous dual of $\mathfrak{h}$. Continuous duals use the adic topology and the discrete coefficient field. To see the different directions explicitly, put

$$\rho_{ab} = z_1^{-a-1} z_2^{-b-1}\, \mathrm{d}z_1 \mathrm{d}z_2, \qquad a, b \geq 0.$$

The residue pairing is

$$\operatorname{Res}_0(z_1^i z_2^j \rho_{ab}) = \delta_{ia}\delta_{jb}.$$

Every continuous functional on the power-series ring depends on finitely many Taylor coefficients. Hence its continuous dual is $\bigoplus_{a,b\geq 0} \mathbb{C}\rho_{ab}$, and

$$\mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{a+b\geq 2} \mathbb{C}\rho_{ab} = \varinjlim_d (\mathfrak{h}_{\leq d})^\vee.$$

The two degree-one residue modes pair with the translations in the affine Hamiltonian algebra.

Proposition 18.1 (Coadjoint residue formula)**.** *For $f = z_1^i z_2^j \in \mathfrak{h}$,*

$$\mathrm{ad}^*_f \rho_{ab} = -\big(i(b+1) - j(a+1)\big)\rho_{a-i+1,b-j+1}.$$

A mode with a negative index or total index less than two is interpreted as zero on $\mathfrak{h}$. For a formal f, only finitely many of its monomials contribute to the action on a fixed residue mode.

Proof. By definition, $\langle \mathrm{ad}^*_f \rho_{ab}, g\rangle = -\langle \rho_{ab}, \{f, g\}\rangle$. For $g = z_1^u z_2^v$, the bracket is $(iv - ju) z_1^{i+u-1} z_2^{j+v-1}$. The residue can be nonzero only for $u = a - i + 1$, $v = b - j + 1$; substitution gives the stated coefficient. Nonnegative output indices bound $i \leq a + 1$ and $j \leq b + 1$, which proves finiteness for a power series. □

The continuous semidirect product

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[-1]$$

has its invariant evaluation pairing with values in $\mathbb{C}[-1]$. Give $\mathfrak{h}$ its inverse-limit topology and the residue dual the discrete direct-sum topology. The coadjoint action is continuous by the finite dependence just proved. The finite pairs $\mathfrak{g}_d$ remain valid cotangent dg Lie algebras, but they are not the images of a componentwise Lie quotient of $\mathfrak{g}$.

Proposition 18.2 (Failure of componentwise cotangent truncation)**.** *Let $d \geq 2$. Project $\mathfrak{h}$ to degrees at most d, and project its residue dual by retaining only modes with $a + b \leq d$. Their direct sum is not a Lie morphism $\mathfrak{g} \to \mathfrak{g}_d$.*

Proof. Both z_1^{d+1} and $\rho_{d,1}$ project to zero. Their coadjoint bracket is

$$\mathrm{ad}^*_{z_1^{d+1}} \rho_{d,1} = -2(d+1)\rho_{0,2},$$

which survives the projection. Thus the kernel is not a Lie ideal. The natural dual map of the Hamiltonian quotient is instead the inclusion $(\mathfrak{h}_{\le d})^\vee \hookrightarrow \mathfrak{h}^\vee_{\mathrm{cont}}$. □

18.2 Jet completion and comparison data

The Hamiltonian Lie algebra and its completed Rees enveloping algebra do have the quotient systems

$$\mathfrak{h} = \varprojlim_d \mathfrak{h}_{\le d}, \qquad \widehat{U}_\hbar(\mathfrak{h}) = \varprojlim_d U_\hbar(\mathfrak{h}_{\le d}).$$

This is the jet-quotient topology on the enveloping algebra, together with its $\hbar$ topology. It does not complete arbitrary PBW powers of a fixed finite jet: for example, a bare infinite sum of powers of one cubic generator is not an element of $U_\hbar(\mathfrak{h}_{\le 3})$. A Lie quotient does not by itself define a covariant map between Chevalley cochains with adjoint coefficients. Nor does an algebra quotient define a map between all Hochschild cochains with values in the algebra itself. A cochain must descend through the input quotient and the output quotient. Consequently the coefficient limit of Corollary 16.2 is distinct from this jet limit.

A *compatible jet comparison* consists of the following data. Choose complete R-linear cochain models P_d, H_d with transition maps, levelwise comparisons $P_d \simeq \mathcal{P}_d$ and $H_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$, and compatible quasi-isomorphisms $P_d \to H_d$. Use degreewise surjective transitions and ordinary inverse limits, or use derived inverse limits throughout the following datum. Choose a continuous Hamiltonian polyvector model $\mathcal{P}_\infty$, including its underlying topological space and the differential $[\hbar\pi, -]$ for its linear Poisson tensor. Specify identifications

$$\varprojlim_d P_d \simeq \mathcal{P}_\infty, \qquad \varprojlim_d H_d \simeq C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})),$$

The topology on the right is the stated jet-quotient topology together with the $\hbar$ completion. The identifications must use this same topology. For higher operations, also supply compatible homotopy-E_2 and Swiss-cheese refinements whose operations converge in that topology.

Theorem 18.3 (Continuous comparison from compatible jets)**.** *A compatible jet comparison induces a continuous quasi-isomorphism*

$$\Phi_\infty : \mathcal{P}_\infty \xrightarrow{\sim} C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})).$$

Compatible homotopy-E_2 and Swiss-cheese refinements give the corresponding complete open–closed operations.

Proof. For surjective transitions, the cones of the levelwise maps are acyclic inverse systems with surjective maps in each degree. The primitive-lifting argument of Corollary 16.2 proves exactness of their limit. For a derived inverse limit, use its invariance under levelwise equivalences. Apply the supplied identifications of the two limits. Compatibility and convergence of the operadic maps give the final assertion. □

The finite PBW comparison constructs the relative centre at each fixed jet. A continuous field-theory comparison additionally requires the jet datum and a map from the specified field observables to $\mathcal{P}_\infty$. Universal graph formulas alone do not provide the cochain transition maps.

Corollary 18.4 (Continuous open–closed structure with compatible jets)**.** *Under the jet comparison and mixed refinement hypotheses of Theorem 18.3, the pair* $(\mathcal{P}_\infty, \widehat{U}_\hbar(\mathfrak{h}))$ *has the corresponding complete homotopy Swiss-cheese action. Its closed color is equivalent to the continuous relative derived centre.*

Proof. The supplied finite operations commute with the transition maps and converge in the specified completion. Their limit acts on the two limit objects. The comparison theorem identifies the closed complex with continuous relative Hochschild cochains. □

18.3 Holomorphic deformation-quantization algebroids

The symplectic surface gives a second geometric construction. Let (S, ω) be a complex surface with a nowhere-degenerate closed holomorphic two-form, and let $\pi = \omega^{-1}$. A deformation-quantization algebroid on S is an R-linear stack locally represented by a sheaf of star-product algebras, with

$$f \star g - g \star f = \hbar\{f, g\}_\pi + O(\hbar^2).$$

Local algebra identifications and their inner coherence maps are part of the stack. A global object, which would present the algebroid by one global algebra sheaf, is additional data.

Theorem 18.5 (Symplectic quantization algebroid)**.** *A holomorphic symplectic surface admits a deformation-quantization algebroid* $\mathcal{W}_{S,\hbar}$ *whose classical limit is the trivial algebroid of* $\mathcal{O}_S$*. With the classical trivialization in the symplectic classification, its Fedosov class has the form*

$$\Theta(\mathcal{W}_{S,\hbar}) \in \frac{[\omega]}{\hbar} + H^2(S; \mathbb{C}[\![\hbar]\!]).$$

Changing this class can change the equivalence class of the algebroid.

Proof. Apply the symplectic algebroid existence and classification theorems of [13, Theorems 7.1–7.2 and Section 7.3] to the complex distribution $T^{0,1}S$. Its invariant functions are $\mathcal{O}_S$; the local Dolbeault lemma gives the required exactness along the distribution. The holomorphic symplectic tensor is the nondegenerate Poisson tensor in that theorem. The canonical quantization of the trivial classical gerbe supplies the asserted algebroid. The Fedosov class records the difference from this canonical quantization. □

On a formal Darboux disk the completed local algebra is a formal Weyl algebra. A Fedosov resolution expresses its differential in the form

$$D_F = \nabla + \frac{1}{\hbar}[\gamma, -],$$

where the required fibrewise differential may be included in γ. The square is an inner derivation by the curvature, so central curvature gives $D_F^2 = 0$ while still contributing to the quantization class. Changing that central class is therefore not the same operation as changing a resolution within a fixed class.

A global holomorphic symplectic connection cannot be assumed even on a K3 surface. Its tangent bundle has second Chern number 24. A holomorphic connection would have curvature of type $(2, 0)$, so its second Chern form would have type $(4, 0)$ and vanish on a complex surface, a contradiction. Local holomorphic constructions, or Dolbeault resolutions with their descent data, are consequently essential. Covariant formality for smooth manifolds [30] uses a chosen connection; equivariance there requires an invariant connection.

Holomorphic Lie algebroid formality also gives a sheafwise L_∞ comparison between polyvectors and polydifferential operators, together with the compatible comparison of their differential-form and Hochschild-chain modules [15, Theorem 5.2 and Corollary 5.3]. For the tangent Lie algebroid, this supplies holomorphic formality without a global holomorphic connection. The additional E_2 structure and descent for a specified quantization class are the data in the following comparison.

18.4 Centre descent with a fixed formal Poisson class

All cochains in this section are continuous and relative to $R = \mathbb{C}[\![\hbar]\!]$. Let $\{U_i\}$ be a Darboux cover and choose a formal Poisson tensor

$$\Pi_i = \hbar\pi_i + \hbar^2\pi_{i,2} + \cdots, \qquad [\Pi_i, \Pi_i]_{\mathrm{Sch}} = 0,$$

whose quantization represents the chosen local algebroid. Write PV_{U_i} for holomorphic polyvector fields, placing $\Lambda^k T_{U_i}$ in degree k. Fix homotopy-E_2 structures and local equivalences

$$\widetilde{\Phi}_i : (\mathrm{PV}_{U_i}[\![\hbar]\!], [\Pi_i, -]) \simeq C^{\bullet}_{\mathrm{HH,cont}}(\mathcal{W}_{U_i,\hbar}/R, \mathcal{W}_{U_i,\hbar}).$$

Require compatible equivalences on overlaps and their coherent homotopies on all higher intersections, both for the source complexes and for these maps. Denote the resulting sheaf of source algebras by $\mathcal{P}_{S,\Pi}$ and the algebroid's relative Hochschild cochain sheaf by $\mathcal{C}_{S,\hbar}$.

Theorem 18.6 (Global centre comparison with coherent descent)**.** *The specified data induce a sheafwise homotopy-E_2 equivalence*

$$\mathcal{P}_{S,\Pi} \simeq \mathcal{C}_{S,\hbar}$$

and an equivalence of their derived global sections. If the chosen continuous Morita theory identifies $Z^{\mathrm{der}}_{\mathrm{Mor},R}(\mathrm{Perf}(\mathcal{W}_{S,\hbar}))$ with $\mathbb{R}\Gamma(S, \mathcal{C}_{S,\hbar})$, this gives

$$Z^{\mathrm{der}}_{\mathrm{Mor},R}(\mathrm{Perf}(\mathcal{W}_{S,\hbar})) \simeq \mathbb{R}\Gamma(S, \mathcal{P}_{S,\Pi}).$$

Proof. The overlap maps and their higher homotopies define a map of descent objects. Its restriction to each U_i is an equivalence, so it is an equivalence of sheaves. Derived global sections preserve this equivalence. Applying the stated Morita identification gives the last formula. □

When the chosen formal Poisson class is represented globally by $\Pi = \hbar\pi$ with the indicated untwisted descent, the source is $(\mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -])$. An arbitrary quantization period does not specify this representative. Likewise, equivariance under holomorphic symplectomorphisms requires equivariant comparison and descent data; local formality alone does not supply them.

18.5 Topological factorization with holomorphic coefficients

Give $\mathbb{R}^2$ its standard framing. A complete homotopy-E_2 algebra internal to sheaves on S has a locally constant factorization envelope on $\mathbb{R}^2$ valued in those sheaves, using the chosen completed tensor product. Applied to the preceding equivalence, this gives

$$\mathrm{Fact}_{\mathsf{E}_2}(\mathcal{P}_{S,\Pi}) \simeq \mathrm{Fact}_{\mathsf{E}_2}(\mathcal{C}_{S,\hbar}).$$

The assertion follows by applying the factorization envelope to an equivalence of its coefficient algebras. It retains the relative centre action on the corresponding local perfect-module categories.

The holomorphic variable here is a sheaf variable: restriction to smaller open sets is contravariant. A factorization algebra on $\mathbb{R}^2 \times S$ also requires covariant operations for disjoint holomorphic open sets and the relevant descent condition. Such operations require a geometric construction in the coefficient direction. Once supplied compatibly on both sides, they promote the displayed equivalence to that geometric factorization comparison. They are not part of the definition of an E_2-algebra sheaf.

18.6 Classical mixed fields and the quantization question

There is a local classical BF construction with the desired mixed directions. On a holomorphic symplectic chart $U \subset S$, let

$$\mathcal{D} = \Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(U), \qquad D = \mathrm{d}_{\mathbb{R}^2} + \bar{\partial}.$$

Here the tensor is the completed projective tensor of smooth-form spaces. The differential acts by the graded tensor rule. For a finite-dimensional Lie algebra L, take fields

$$A \in \mathcal{D} \otimes L[1], \qquad B \in \mathcal{D} \otimes L^\vee[2].$$

The two de Rham and two Dolbeault degrees make the integration degree four. The shifts above therefore give the BV pairing degree -1. With compact support, the degree-zero action is

$$S_{\mathrm{BF}}(A, B) = \int_{\mathbb{R}^2 \times U} \omega \wedge \left\langle B, DA + \frac{1}{2}[A, A] \right\rangle.$$

Its classical master equation follows from $D^2 = 0$, the derivation property of D, Jacobi, and integration by parts. For $L = \mathfrak{h}_{\leq d}$ this is a finite coefficient model. A formal Hamiltonian field theory must additionally specify its continuous coefficient complex and the gluing of formal coordinate charts.

Renormalized quantum observables require compatible gauge fixing and counterterms satisfying the field-theoretic quantum master equation. The finite identity $\Delta S_L = 0$ controls its stated finite model; it does not establish the local field-theoretic quantum master equation. Even after quantization, identifying the observable algebra with the relative centre requires a map from those observables to the selected Hamiltonian or symplectic polyvector complex, with the factorization operations and descent respected.

Chapter 19

Derived matrix equations and necklace actions

19.1 The derived commuting fibre

For a positive integer N, let $X = (X_{ij})$ and $Y = (Y_{ij})$ be two generic $N \times N$ matrices. Define the commutative dg algebra

$$\mathcal{K}_N = \mathbb{C}[X_{ij}, Y_{ij}] \otimes \Lambda(\psi_{ij}), \qquad |X_{ij}| = |Y_{ij}| = 0, \quad |\psi_{ij}| = -1,$$

with differential

$$QX = QY = 0, \qquad Q\psi = [X, Y].$$

Here ψ is a matrix with anticommuting entries, and products of such matrices use the ordinary matrix rule with graded coefficients. The algebra $\mathcal{K}_N$ is the function algebra of the derived zero fibre of the moment map

$$\mu : \mathfrak{gl}_N^2 \longrightarrow \mathfrak{gl}_N, \qquad (X, Y) \longmapsto [X, Y].$$

Indeed, the Koszul resolution of the origin in the target, pulled back along μ, computes the derived tensor product defining this fibre. Its zeroth cohomology is the coordinate algebra of the ordinary commuting scheme. Its negative cohomology retains the failure of the commuting equations to form a regular sequence.

For example, $\operatorname{Tr} \psi$ is a closed element of degree -1 because $\operatorname{Tr}[X, Y] = 0$. It is not exact: setting $X = Y = 0$ annihilates every boundary but leaves the nonzero exterior element $\operatorname{Tr} \psi$. At rank one this becomes

$$\mathcal{K}_1 = \mathbb{C}[x, y] \otimes \Lambda(\psi), \qquad Q = 0.$$

Thus even the simplest derived commuting fibre has an odd direction in addition to its ordinary affine plane.

Simultaneous conjugation acts on $\mathcal{K}_N$. Two quotients of this action will be distinguished:

$$[\operatorname{Spec} \mathcal{K}_N / GL_N], \qquad \mathcal{B}_N = C^*_{\mathrm{CE}}(\mathfrak{gl}_N, \mathcal{K}_N).$$

The first is the derived group quotient; the second computes Chevalley cochains for the infinitesimal action. They need not have the same function complex. At rank one conjugation is trivial, and rational $\mathbb{G}_m$ invariants are exact, so the group quotient has function complex $\mathcal{K}_1$. In contrast,

$$\mathcal{B}_1 = \mathcal{K}_1 \otimes \Lambda(c), \qquad |c| = 1,$$

with zero differential. This additional Chevalley ghost records Lie algebra cohomology.

19.2 Cyclic words and the moment map

Let $F = \mathbb{C}\langle x, y\rangle$ and let $F_{\mathrm{cyc}} = F/[F, F]$, where $[F, F]$ is the linear span of associative commutators. For a word $a_1 \cdots a_r$, its cyclic derivative in x is

$$\partial_x^{\mathrm{cyc}}[a_1 \cdots a_r] = \sum_{a_i = x} a_{i+1} \cdots a_r a_1 \cdots a_{i-1}.$$

Define $\partial_y^{\mathrm{cyc}}$ similarly. These derivatives are well defined on cyclic words and take values in the free algebra. Put

$$\delta_f(x) = -\partial_y^{\mathrm{cyc}} f, \qquad \delta_f(y) = \partial_x^{\mathrm{cyc}} f,$$

and extend by the derivation rule.

Theorem 19.1 (Necklace Lie algebra and action)**.** *The bracket*

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f \, \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f \, \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}$$

makes F_{cyc} a Lie algebra, with the constants central. The map $f \mapsto \delta_f$ is a Lie morphism, factors through $F_{\mathrm{cyc}}/\mathbb{C}$, and satisfies

$$\delta_f([x, y]) = 0.$$

Proof. Consider the bilinear map $F \otimes F \to F \otimes F$ determined by the generator values $\{\{x, y\}\} = 1 \otimes 1$, $\{\{y, x\}\} = -1 \otimes 1$, with the other generator brackets zero, the outer Leibniz rule in the second argument, and skew-symmetry with interchange of the two output factors. This is a double bracket. Its double Jacobiator is the cyclic sum of nested brackets, with the outer bracket acting in the first output factor. It vanishes on generator triples, since the inner brackets are constant. The derivation property in each argument then gives its vanishing on all words. Its induced bracket on cyclic words is the displayed formula. The double Jacobi identity gives both the Lie identity and $[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}}$; see [84, Section 2.4 and Theorem 6.3.1]. The cyclic derivative identity

$$[x, \partial_x^{\mathrm{cyc}} f] + [y, \partial_y^{\mathrm{cyc}} f] = 0$$

follows by cancellation of successive cyclic cuts of each word. Substitution into the derivative of $[x, y]$ proves the moment-map identity. Constant cyclic words have zero derivatives. □

Proposition 19.2 (Action on the derived commuting fibre)**.** *Matrix evaluation of δ_f on X, Y, together with $\delta_f \psi = 0$, defines an action of $F_{\mathrm{cyc}}/\mathbb{C}$ on $\mathcal{K}_N$ by dg derivations. It is equivariant for conjugation and hence acts on $\mathcal{B}_N$ as well.*

Proof. The moment-map identity gives $\delta_f(Q\psi) = \delta_f[X, Y] = 0 = Q(\delta_f \psi)$. The commutator with Q also vanishes on X, Y, so it vanishes on the whole algebra. Matrix evaluation preserves the free-algebra derivation and Lie identities. All expressions are equivariant polynomials in the matrices; extending the derivation by zero on Chevalley ghosts therefore gives the action on $\mathcal{B}_N$. □

The ordinary matrix space carries the symplectic form $\mathrm{Tr}(\mathrm{d}X \wedge \mathrm{d}Y)$. For $f \in F_{\mathrm{cyc}}$ define $J_N(f) = \mathrm{Tr}\, f(X, Y)$. Constants are retained here: $J_N(1) = N$.

Theorem 19.3 (Trace moment map)**.** *With this symplectic convention,*

$$\{J_N(f), J_N(g)\}_{\mathfrak{gl}_N^2} = J_N(\{f, g\}_{\mathrm{neck}}).$$

Proof. Cyclic differentiation gives

$$\mathrm{d}J_N(f) = \mathrm{Tr}\big((\partial_x^{\mathrm{cyc}} f)(X, Y)\mathrm{d}X + (\partial_y^{\mathrm{cyc}} f)(X, Y)\mathrm{d}Y\big).$$

Contraction with the inverse trace symplectic form gives the necklace formula. Equivalently, the matrix-entry brackets are $\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}$, as in [84, Propositions 7.5.1–7.5.2]. □

19.3 Formal completion and the enveloping action

Let $F_{\mathrm{cyc},r}$ be the cyclic words of length r and put

$$\mathfrak{n} = \prod_{r \geq 2} F_{\mathrm{cyc},r}.$$

Its bracket sends lengths r, s to length $r + s - 2$. Complete $\mathcal{K}_N$ by the weights

$$\operatorname{wt} X_{ij} = \operatorname{wt} Y_{ij} = 1, \qquad \operatorname{wt} \psi_{ij} = 2.$$

The differential preserves weight. For a Hamiltonian of length r, its derivation raises weight by $r - 2$ on the even generators and annihilates ψ. The sum of these derivations therefore converges on the weight completion $\widehat{\mathcal{K}}_N$. Chevalley ghosts have weight zero, giving the corresponding completion $\widehat{\mathcal{B}}_N$.

Let $\mathfrak{n}_{\leq d} = \bigoplus_{2 \leq r \leq d} F_{\mathrm{cyc},r}$ with the projected bracket, and define $\widehat{U}_\hbar(\mathfrak{n}) = \varprojlim_d U_\hbar(\mathfrak{n}_{\leq d})$. The continuous Lie action gives a Rees enveloping action

$$\widehat{U}_\hbar(\mathfrak{n}) \longrightarrow \mathrm{End}_{\mathrm{cont}}(\widehat{\mathcal{B}}_N)[\![\hbar]\!], \qquad f \longmapsto \hbar \partial_f.$$

The enveloping completion uses these length quotients and the coefficient parameter. A Hamiltonian of length greater than d acts trivially on the quotient by matrix weights at least d. Thus the finite actions are compatible on every bounded matrix weight quotient, and their inverse limit defines the displayed map. The Rees relation follows directly from

$$[\hbar \partial_f, \hbar \partial_g] = \hbar^2 \partial_{\{f,g\}_{\mathrm{neck}}}.$$

At any fixed output weight only finitely many Hamiltonian lengths contribute. The topology on endomorphisms is the inverse-limit topology of these weight quotients. No completion by arbitrary PBW powers in a fixed finite length quotient is imposed.

19.4 The commutative Hamiltonian lift

Abelianization of words gives a continuous surjective Lie map

$$\mathfrak{n} \longrightarrow \mathfrak{h}.$$

Here x maps to z_1 and y to z_2. A linear section is cyclic symmetrization: average the words with a prescribed number of x's and y's and pass to their cyclic class. Denote it by σ. It need not be a Lie section.

Proposition 19.4 (The first symmetrization defect). *For $f = x^3$ and $g = y^3$,*

$$\begin{aligned} D &= \{\sigma f, \sigma g\}_{\mathrm{neck}} - \sigma\{f, g\} \\ &= 3\big([xxyy] - [xyxy]\big) = -\frac{3}{2}\big[[x, y]^2\big]_{\mathrm{cyc}} \neq 0. \end{aligned}$$

On the derived commuting fibre, the corresponding derivation defect is null-homotopic. A degree--1 derivation giving the homotopy is

$$H(X) = -3[X, \psi], \qquad H(Y) = -3[Y, \psi], \qquad H(\psi) = 3\psi^2.$$

It satisfies $QH + HQ = \partial_D$ and extends equivariantly to $\mathcal{B}_N$.

Proof. There are six words with two copies of each generator. Four have cyclic class $[xxyy]$ and two have cyclic class $[xyxy]$, so

$$\sigma(x^2y^2) = \frac{2}{3}[xxyy] + \frac{1}{3}[xyxy].$$

The necklace bracket of $[x^3]$ and $[y^3]$ is $9[xxyy]$, whereas the commutative bracket is $9x^2y^2$. Their difference is the first displayed expression for D. Expansion of $[x, y]^2$ gives $2[xyxy] - 2[xxyy]$ after cyclic reduction. The two necklaces are distinct basis classes, proving nonvanishing. The derivation defect is also nonzero at rank two: for $X = \mathrm{diag}(1, 0)$ and $Y = E_{12}$ one has $[X, [X, Y]] = E_{12}$, hence $\delta_D(X) = -3E_{12}$.

Put $\mu = [X, Y]$. Cyclic differentiation of D gives

$$\delta_D(X) = -3[X, \mu], \qquad \delta_D(Y) = -3[Y, \mu], \qquad \delta_D(\psi) = 0.$$

The first two identities for $QH + HQ$ follow from $Q\psi = \mu$. On the odd generator, the graded derivation rule gives

$$QH(\psi) = 3(\mu\psi - \psi\mu), \qquad HQ(\psi) = -3(\mu\psi - \psi\mu),$$

so these terms cancel. Equality on generators gives equality on the whole free graded algebra and hence after matrix evaluation. The formulas are conjugation equivariant, so extending H by zero on the Chevalley ghosts preserves the same homotopy identity. □

This homotopy resolves one binary defect. An action of all commutative formal Hamiltonians on the derived matrix object requires higher homotopies satisfying the L_∞ identities and converging in the specified topology. Constructing such a lift, or a different strict Lie lift, remains the comparison problem. The necklace action is already a complete positive construction; the failed strictness of σ is not an obstruction to every possible Hamiltonian action.

19.5 The first two ordinary trace models

Passing to ordinary rank-one functions gives $\mathbb{C}[x, y]$ and $J_1(f) = f(x, y)$ after abelianization. The necklace bracket becomes the usual Poisson bracket. Pointed Hamiltonians act by vector fields fixing the origin; linear Hamiltonians generate translations. The odd coordinate of $\mathcal{K}_1$ and the Chevalley ghost of $\mathcal{B}_1$ belong to their larger derived models.

For two-by-two matrices, Cayley–Hamilton gives

$$X^2 - (\mathrm{Tr}\, X)X + (\det X)I = 0$$

and the analogous identity for Y. These reduce a consecutive power X^2 or Y^2 in a word after adjoining the indicated trace and determinant coefficients. Commuting complex matrices admit a common invariant flag and hence can be simultaneously triangularized. Their traces are therefore computed from the diagonal pairs (x_i, y_i), even when the matrices are not simultaneously diagonalizable. Write $p_{ab} = \sum_{i=1}^{2} x_i^a y_i^b$. Then

$$p_{20} = p_{10}^2 - 2\det X, \qquad p_{02} = p_{01}^2 - 2\det Y,$$

$$p_{11}^2 - p_{20}p_{02} = -(x_1y_2 - x_2y_1)^2.$$

The first two identities are Cayley–Hamilton trace identities; the third follows by expanding the two diagonal sums. These are identities of ordinary trace functions. They do not identify the full derived quotient with its eigenvalue model.

Chapter 20

Stable traces and the centre comparison

20.1 Graded stabilization

The eigenvalue model of rank N has coordinate ring

$$\mathcal{T}_N = \mathbb{C}[x_1, y_1, \dots, x_N, y_N]^{S_N}.$$

Give x_i, y_i weight one. For $a, b \geq 0$ with $a + b > 0$, put

$$p_{ab}^{(N)} = \sum_{i=1}^{N} x_i^a \, y_i^b.$$

Adding the zero pair defines restriction maps $\mathcal{T}_{N+1} \to \mathcal{T}_N$ preserving this weight. Define the graded stable algebra and its weight completion by

$$\mathcal{T}_{\mathrm{gr}} = \bigoplus_{w \geq 0} \varprojlim_N (\mathcal{T}_N)_w, \qquad \widehat{\mathcal{T}}_{\mathrm{wt}} = \prod_{w \geq 0} (\mathcal{T}_{\mathrm{gr}})_w.$$

Theorem 20.1 (Stable multisymmetric algebra)**.** *For each weight w, restriction stabilizes once $N \geq w$. There is a graded algebra isomorphism*

$$\mathcal{T}_{\mathrm{gr}} \cong \mathbb{C}[\, p_{ab} \mid a, b \geq 0,\ a + b > 0 \,], \qquad \mathrm{wt}\, p_{ab} = a + b.$$

Its weight completion is $\widehat{\mathcal{T}}_{\mathrm{wt}}$.

Proof. A monomial orbit sum is indexed by a multiset of nonzero exponent pairs $\alpha_1, \dots, \alpha_l$, placed in distinct slots among $1, \dots, N$. At total weight w one has $l \leq w$. For $N \geq w$ all such multisets occur, and zero-pair restriction identifies their orbit-sum bases. This proves stabilization.

Expand $p_{\alpha_1} \cdots p_{\alpha_l}$ by assigning a slot to each factor. The assignments with distinct slots give the corresponding orbit sum with coefficient $\prod_\alpha m_\alpha!$, where m_α is the multiplicity of α in the multiset. All other assignments merge two or more exponent pairs and have fewer occupied slots. Induction on l therefore gives a triangular change of basis between the power-sum monomials and the orbit sums. The diagonal coefficients are nonzero over $\mathbb{C}$. In each weight the two bases are consequently equivalent, proving generation and algebraic independence of the stable power sums. Taking the product over weights gives the stated completion. □

This definition distinguishes graded stabilization from an inverse limit of ungraded polynomial rings. For example, $\sum_{r \geq 0} p_{10}^r$ is a valid weight-completed series but does not restrict to a polynomial at rank one. It is therefore not an element of $\varprojlim_N \mathcal{T}_N$ with uncompleted terms. Finite-rank relations and the formal topology must both be specified before taking a stable limit.

20.2 The rank parameter and the pointed Poisson algebra

On $(\mathbb{C}^2)^N$ use $\{x_i, y_j\} = \delta_{ij}$. The invariant bracket is

$$\{p_{ab}^{(N)}, p_{cd}^{(N)}\} = (ad - bc)\, p_{a+c-1,b+d-1}^{(N)}, \qquad p_{00}^{(N)} = N.$$

It follows by differentiating the two sums; terms with distinct indices have zero bracket. A negative output index occurs only with zero coefficient. In particular, $\{p_{10}^{(N)}, p_{01}^{(N)}\} = N$. Adding the zero pair is therefore not Poisson on the full trace algebra.

There are two compatible ways to retain this formula. First, introduce a central symbol ν and define

$$\widetilde{\mathcal{T}} = \mathbb{C}[\nu, p_{ab} \mid a + b > 0], \qquad p_{00} = \nu,$$

with the displayed bracket on generators and $\{\nu, -\} = 0$. This is the symmetric Poisson algebra of the Lie algebra of polynomial Hamiltonians including its constant. Jacobi follows from the polynomial Poisson bracket. Specialization $\nu = N$ and $p_{ab} \mapsto p_{ab}^{(N)}$ is a Poisson map for every rank.

Second, use the pointed subalgebra generated by p_{ab} with $a + b \geq 2$. The bracket of two such generators again has degree at least two, so it never produces the rank parameter. Zero-pair restriction is Poisson on this subalgebra. Its weight completion is also Poisson: the bracket lowers weight by two, and only finitely many homogeneous input weights can contribute to a fixed output weight. This gives the stable pointed Hamiltonian trace algebra.

20.3 The two-brane cone

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \qquad v = y_1 - y_2, \qquad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 20.2 (The A_1 Kirillov–Kostant cone)**.** *With* $\{u, v\} = 2$,

$$\{H, E\} = 4E, \qquad \{H, F\} = -4F, \qquad \{E, F\} = 2H.$$

The generators $E/2$, $F/2$, $H/2$ *satisfy the standard* $\mathfrak{sl}_2$ *Lie–Poisson brackets. Their quadratic relation defines its nilpotent cone.*

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

20.4 Reduced cyclic traces and stable Lie homology

A trace of the unit records the representation dimension. For $B = \mathbb{C}$, one has $\mathrm{Tr}_N(1) = N$, so ordinary traces cannot define a rank-compatible map on all of $HC_0(B)$. A scalar-reduced trace removes this dependence after choosing an augmentation.

Let B be a unital associative $\mathbb{C}$-algebra. The affine representation scheme $\mathrm{Rep}_N B$ represents unital algebra maps $B \to \mathrm{Mat}_N(D)$ for commutative $\mathbb{C}$-algebras D. Put

$$\mathcal{R}_N = \mathcal{O}(\mathrm{Rep}_N B)^{GL_N}, \qquad \mathfrak{c}_B = HC_0(B) = B/[B,B],$$

where $[B,B]$ denotes the linear span of commutators. The ordinary trace is the linear map

$$\mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N, \qquad \mathrm{Tr}_N([b])(\rho) = \mathrm{Tr}(\rho(b)).$$

Cyclicity makes it independent of the representative, and conjugation invariance puts its image in $\mathcal{R}_N$.

For Lie and Poisson compatibility, specify a Lie bracket on $\mathfrak{c}_B$, a Poisson bracket on each $\mathcal{R}_N$, and the identities

$$\{\mathrm{Tr}_N \beta, \mathrm{Tr}_N \gamma\}_N = \mathrm{Tr}_N [\beta,\gamma]_{\mathrm{cyc}} \qquad (\beta,\gamma \in \mathfrak{c}_B). \tag{20.1}$$

A Hamiltonian comparison also specifies a Lie map $j : \mathfrak{h} \to \mathfrak{c}_B$ for this bracket. These are additional data on B. The preceding necklace bracket and trace moment map supply the bracket and trace data for $B = \mathbb{C}\langle x, y\rangle$ with its canonical symplectic bracket. The map j remains an additional comparison. Here j is an algebraic Lie map. A continuous formal comparison also requires a topology on the cyclic recipient and continuity of j and the trace maps. An arbitrary unital algebra does not carry these structures by definition.

For rank stabilization, choose an augmentation $\epsilon : B \to \mathbb{C}$. It defines the rank-stabilization morphism

$$s_N : \mathrm{Rep}_N B \longrightarrow \mathrm{Rep}_{N+1} B, \qquad \rho \longmapsto \rho \oplus \epsilon.$$

Restriction gives $s_N^* : \mathcal{R}_{N+1} \to \mathcal{R}_N$. Define the reduced cyclic space and trace by

$$\bar{\mathfrak{c}}_B = B/(\mathbb{C}1 + [B,B]), \qquad t_N(\bar{b})(\rho) = \mathrm{Tr}(\rho(b)) - N\epsilon(b).$$

Let $\mathfrak{m}_N \subset \mathcal{R}_N$ be the ideal of functions vanishing at the representation $b \mapsto \epsilon(b) I_N$. The completed invariant rings and stable target are

$$\widehat{\mathcal{R}}_N = \varprojlim_{r \geq 1} \mathcal{R}_N/\mathfrak{m}_N^r, \qquad \mathcal{R}_\infty^\wedge = \varprojlim_N \widehat{\mathcal{R}}_N.$$

Use the inverse-limit topologies. Write J for the augmentation ideal in $\mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B$ and put

$$\widehat{\mathrm{Sym}}\bar{\mathfrak{c}}_B = \varprojlim_{r \geq 1} \mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B / J^r.$$

The stable matrix Lie algebra is $\mathfrak{gl}_\infty B = \varinjlim_N \mathfrak{gl}_N B$ under block-zero Lie embeddings. For its homology, use trivial coefficients $\mathbb{C}$. Let $V_B = \bigoplus_{n \geq 1} HC_{n-1}(B)$, with $HC_{n-1}(B)$ placed in homological degree n. The notation $\mathrm{Sym}_{\mathrm{gr}} V_B$ denotes the free graded-commutative algebra, with the Koszul sign rule.

Theorem 20.3 (Cyclic traces and stable Lie homology)**.** *Assume the compatible jet comparison and its homotopy Gerstenhaber refinement. Write $\mathcal{A} = \widehat{U}_\hbar(\mathfrak{h})$. For the augmented algebra and completion above, there are maps*

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^*_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A}), \qquad \mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N,$$

$$\widehat{\mathrm{Sym}}\bar{\mathfrak{c}}_B \longrightarrow \mathcal{R}_\infty^\wedge, \qquad \mathrm{Sym}_{\mathrm{gr}} V_B \xrightarrow{\sim} H_*^{\mathrm{CE}}(\mathfrak{gl}_\infty B; \mathbb{C}).$$

The first preserves the selected homotopy Gerstenhaber structure. The second is linear and, under the bracket data (20.1), is a Lie map. Under those data, its composite with j preserves the Hamiltonian bracket. The third is a continuous homomorphism of commutative algebras, and the fourth is an isomorphism of graded Hopf algebras. These maps are natural for morphisms preserving the stated data.

Proof. The first arrow is the chosen comparison of Theorem 18.3. Its operadic refinement gives the asserted Gerstenhaber compatibility. The trace formula above constructs the second arrow. Equation (20.1) supplies its Lie compatibility, and the Lie property of j gives the composite assertion.

The augmentation annihilates commutators. Both terms of t_N vanish on a commutator, and their values on a scalar multiple of the unit cancel. Thus t_N is well defined on $\bar{\mathfrak{c}}_B$. At the trivial representation its value is zero, so its image lies in $\mathfrak{m}_N$. Moreover,

$$\begin{aligned}(s_N^* t_{N+1})(\bar{b})(\rho) &= \operatorname{Tr}(\rho(b)) + \epsilon(b) - (N+1)\epsilon(b)\\ &= t_N(\bar{b})(\rho).\end{aligned}$$

The point $N\epsilon$ maps to $(N+1)\epsilon$, so $s_N^*(\mathfrak{m}_{N+1}^r) \subset \mathfrak{m}_N^r$. Hence restriction defines the transition maps between the completed rings. The symmetric-algebra extension of t_N sends J^r into $\mathfrak{m}_N^r$. It therefore induces a ring homomorphism on each corresponding quotient. Taking the inverse limit in r gives a continuous map to $\widehat{\mathcal{R}}_N$. The displayed stabilization identity makes these maps compatible in N. Their inverse limit is the third arrow.

The diagonal Lie map $\mathfrak{gl}_\infty B \to \mathfrak{gl}_\infty B \oplus \mathfrak{gl}_\infty B$ induces the homology coproduct. Block sum in the opposite direction induces the homology product, using the Künneth isomorphism over $\mathbb{C}$. The Loday–Quillen–Tsygan theorem [66, Theorem 6.2] identifies primitive homology in degree n with $HC_{n-1}(B)$. The stable homology is a connected graded-commutative and cocommutative Hopf algebra over $\mathbb{C}$. The characteristic-zero structure theorem therefore identifies it with the free graded-commutative Hopf algebra on its primitives, giving the fourth arrow. The trace formulas, the augmentation-preserving rank maps, and stable Lie homology are natural. Naturality of the first arrow requires preservation of the selected jet comparison. □

Scalar reduction need not preserve the supplied Poisson bracket. For example, for $B = \mathbb{C}[x, y]$ with $\{x, y\} = 1$ and $\epsilon(x) = \epsilon(y) = 0$, traces on N points satisfy

$$\{\operatorname{Tr}_N x, \operatorname{Tr}_N y\} = N, \qquad t_N(\bar{1}) = 0.$$

A reduced Poisson comparison requires the cyclic bracket to descend through the scalar quotient and to satisfy the reduced trace identity. For example, centrality of the unit and vanishing of ϵ on the selected cyclic brackets give these properties: subtracting $N\epsilon(b)$ does not change the bracket of two trace functions, and then $t_N([b, c]) = \operatorname{Tr}_N[b, c]$. A restriction to a Hamiltonian subalgebra may supply these conditions. Only multiplicativity and continuity are asserted for the third arrow. For $B = \mathbb{C}$, the reduced cyclic space is zero and that arrow is the identity on $\mathbb{C}$.

20.5 The local centre and its modules

Assume a compatible jet comparison and a complete homotopy-E_2 refinement as in Theorem 18.3. Put

$$\mathcal{A}_{\mathrm{bulk}}^{\mathrm{alg}} = \operatorname{Fact}_{\mathsf{E}_2}(\mathcal{P}_\infty).$$

The factorization envelope uses the standard framing of the two topological directions. The formal holomorphic coefficient remains at the Darboux point. Let C_∂ be the category of perfect complete $\mathcal{A}$-modules in the chosen continuous R-linear Morita theory. Assume its centre is computed by $C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A})$.

Theorem 20.4 (Local derived-centre comparison)**.** *These data give an equivalence of complete homotopy-E_2 factorization algebras*

$$\mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} \xrightarrow{\sim} \mathrm{Fact}_{\mathsf{E}_2}\Big(Z^{\mathrm{der}}_{\mathrm{Mor},R}(C_\partial)\Big).$$

On a topological disk the coefficient map is the selected refinement of Φ_∞. Every perfect complete boundary module has the induced derived centre action.

Proof. The compatible jet theorem gives the complete homotopy-E_2 comparison. The factorization envelope preserves this equivalence. The assumed continuous Morita model identifies its target with the centre. A centre element is a derived natural endomorphism of the identity functor, so evaluation gives the action on each boundary module. □

More generally, a complete module M receives the same action whenever the continuous Morita theory supplies the completed derived tensor and $\mathcal{A}\widehat{\otimes}^{\mathbf{L}}_{\mathcal{A}}M \simeq M$. Apply $-\widehat{\otimes}^{\mathbf{L}}_{\mathcal{A}}M$ to the derived bimodule endomorphisms of $\mathcal{A}$ which define Hochschild cochains.

Corollary 20.5 (Global K3 comparison with descent data)**.** *Let S be a K3 surface with holomorphic symplectic form. Suppose the local comparisons extend, with the coherent descent data specified above, to a sheafwise equivalence with $C_{S,\hbar}$, and that the stated continuous Morita identification holds. Suppose further that the selected cyclic and mixed refinements extend and satisfy descent in the same completed category. Then derived global sections give the corresponding global centre, cyclic, and open–closed comparisons.*

Proof. Derived global sections preserve equivalences of sheaves. Apply them to the supplied sheafwise maps and their coherence homotopies. □

20.6 The remaining comparison maps

At a fixed Hamiltonian jet, PBW antisymmetrization proves the relative Rees comparison, including both Chevalley terms scaled by $\hbar$. Affine formality supplies a comparison with its selected L_∞ structure, and Duflo identifies the degree-zero invariant algebras. A homotopy-E_2 or mixed refinement is specified when those operations are used. The oscillator algebra is a separate associative quotient; its Casimir relation obstructs descent of the primitive coproduct.

For the Hamiltonian inverse limit, the compatible cochain systems of Theorem 18.3 are the required datum. With its operadic refinement and continuous Morita model, the centre acts on every supplied complete boundary module M through

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R,\mathcal{A}) \longrightarrow \mathrm{RHom}_{\mathcal{A}}(M,M).$$

The final arrow evaluates a derived natural endomorphism at M. For the matrix object $\widehat{\mathcal{B}}_N$, the necklace action has been constructed. A coherent lift of commutative Hamiltonians would provide the additional action needed to compare it with a module over the Hamiltonian boundary algebra. The explicit binary homotopy gives one equation in this problem.

The symplectic surface has its own quantization algebroid and its fixed-class descent question. Relating its centre to Hamiltonian Lie cochains requires a comparison between those carriers. Relating either centre to scalar traces requires an actual map to the stated cyclic or representation-invariant target; surjectivity is a further assertion. The reduced trace homomorphism and the stable Lie-homology theorem construct their stated maps without making that additional identification.

Calabi–Yau Quantum Groups

Extensions of geometric objects define Hall products. Tensor products of representations require coproducts; exchange requires intertwiners. A trace on a specified state space then gives an amplitude. We construct these operations in categorical, surface and root models, and determine the maps needed to join them to fields on a compact Calabi–Yau threefold.

Chapter 21

Calabi–Yau structures and local observables

21.1 Calabi–Yau categories

Let $\Bbbk$ be a field of characteristic zero and let C be a small stable $\Bbbk$-linear category. Smoothness means that its diagonal bimodule is perfect; properness means that every morphism complex is perfect over $\Bbbk$. These finiteness conditions lead to two cyclic duality data. Write $CC_*(C)$ for Hochschild chains, with their circle action, and $C^!$ for the inverse-dualizing bimodule.

A smooth, or left, d-Calabi–Yau structure is a negative-cyclic class

$$\theta_{\mathrm{CY}} : \Bbbk[d] \longrightarrow CC_*(C)^{hS^1}$$

whose Hochschild image induces an equivalence $C^![d] \simeq C$. A proper, or right, structure is a circle-equivariant trace

$$CC_*(C) \longrightarrow \Bbbk[-d]$$

whose composition pairing on morphisms is nondegenerate. For a smooth proper category we use compatible left and right structures. A scalar trace without its circle-equivariant lift is insufficient for this datum.

For an object E, the tangent complex of the derived moduli stack is

$$T_E\mathcal{M}_C \simeq \mathrm{RHom}_C(E,E)[1].$$

Composition and the trace give its pairing.

Theorem 21.1 (The two universal categorical outputs)**.** *Let C be smooth and proper with the compatible cyclic structures above. Its Hochschild cochains carry a framed E_2-algebra structure. If C is of finite type and the moduli of objects satisfies the representability hypotheses of the shifted-symplectic moduli theorem, then $\mathcal{M}_C$ is $(2-d)$-shifted symplectic.*

Proof. Cyclic Deligne applies to the nondegenerate circle-equivariant trace and refines the Hochschild brace structure [11]. The moduli pairing at E is

$$\mathrm{RHom}(E,E)[1]^{\otimes 2} \xrightarrow{\circ} \mathrm{RHom}(E,E)[2] \xrightarrow{\mathrm{Tr}} \Bbbk[2-d].$$

Calabi–Yau duality makes it nondegenerate. The negative-cyclic lift provides the closed form, globally on the representable moduli problem [10, 77]. □

The shift of this form depends on d. The Hochschild operations remain framed E_2 operations. Neither output alone specifies a local field theory or a factorization algebra on a chosen manifold.

21.2 Local BV theories and factorization observables

21.2.1 Cyclic elliptic local Lie algebras

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local pairing inducing the degree--1 BV form on $\mathcal{L}[1]$, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\mathrm{Obs}^{\mathrm{cl}}(U) = C^*_{\mathrm{CE,cont}}(\mathcal{L}(U)).$$

Here $\mathcal{L}(U)[1]$ is the complex of unrestricted smooth fields. We use the differentiable pro-cochain model and tensor products of the elliptic observable construction [21, 22]. Continuous linear functionals are compactly supported distributions. The underlying cochain product is defined in this coefficient category; its Poisson bracket is defined on the regular observable subcomplex on which the contractions exist. An arbitrary pair of distributional functionals need not admit such a contraction. For $U \subset V$, restriction of fields $\mathcal{L}(V) \to \mathcal{L}(U)$ induces the covariant map on observables. For pairwise disjoint $U_i \subset V$, restrict to $\bigoplus_i \mathcal{L}(U_i)$ and multiply the pulled-back cochains. These are the prefactorization maps.

Theorem 21.2 (Factorization observables)**.** *An elliptic classical BV theory produces a factorization algebra* $\mathrm{Obs}^{\mathrm{cl}}$*. A renormalized quantization satisfying the quantum master equation produces a factorization algebra* $\mathrm{Obs}^{\mathrm{q}}$ *over* $\Bbbk[[\hbar]]$.

Proof. Restriction is a morphism of local L_∞ algebras, so its contravariant continuous Chevalley map commutes with the differential. For disjoint opens, the completed cochains of the finite direct sum are the completed tensor product of the cochains. This constructs the displayed maps and their composition identities. The elliptic descent theorem in this completed coefficient category supplies Weiss descent [21, 22]. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [21, 22]. □

21.2.2 Boundary and defect restriction

Let a theory on $C \times \mathbb{R}$ be holomorphic on the curve and locally constant on the oriented real line. A boundary or defect has its own observable category; a normal half-line by itself does not equip one fixed endomorphism complex with curve coefficients.

Choose instead a family of boundary objects b_I on the curve's configuration spaces, with diagonal transition maps and disjoint-locus factorization equivalences. Suppose its derived endomorphism objects and their compositions respect those maps. They define a coefficient object A_b with local values $\mathrm{RHom}(b_I, b_I)$.

Theorem 21.3 (Ordered boundary algebra)**.** *Under these coefficient and composition hypotheses,*

$$A_b \in \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(C), \otimes^!).$$

If each b_I is a compact generator and the local module equivalences preserve the specified coefficient transitions and factorization, they descend to an equivalence with A_b-modules in these coefficients.

Proof. Composition and the identity give the internal associative operations. Their compatibility with the coefficient maps is precisely the condition that these operations are morphisms in $\mathrm{Fact}_D(C)$. Compact-generator Morita theory gives the local module equivalences; their assumed descent compatibility glues them. □

21.3 Three curve-level algebraic outputs

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 21.4 (Curve-level faces)**.** A transverse face is

$$A^{\perp} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(C), \otimes^!).$$

An associative chiral face is

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$, and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$.

Theorem 21.5 (Source separation)**.** *The factorizing boundary construction gives the transverse face. A multiplication for $\star^{\mathrm{ch}}$ gives the associative chiral face, and its commutator gives the Lie face. In a specified stable symmetric monoidal model, let $U_{\star^{\mathrm{ch}}}(\mathfrak{g})$ denote the associative enveloping algebra of a Lie object. If the PBW and completed bar–Chevalley comparison hold in that model, then*

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(U_{\star^{\mathrm{ch}}}(\mathfrak{g})) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}).$$

A mixed construction requires a distributive law compatible with all diagonal and factorization transitions.

Proof. The commutator is operadic restriction. The displayed comparison is the enveloping bar–Chevalley map in the named monoidal category, with the PBW filtration and convergence hypotheses just stated. A coherent distributive law composes the two associative operations as in Volume I. □

The associative envelope $U_{\star^{\mathrm{ch}}}$ is specified in this category; it is not identified here with the Beilinson–Drinfeld chiral envelope. Neither the dimension of the source nor an ordinary vector-space calculation supplies the missing geometric coefficient maps.

21.4 Holomorphic Chern–Simons theory

Let X be a complex threefold with holomorphic volume form Ω_X, and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega^{0,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

Local integrals are evaluated on compactly supported variations. The action is

$$S_{\mathrm{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 21.6 (Classical hCS factorization algebra)**.** *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X.*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial}A + \frac{1}{2}[A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. □

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization solves the quantum master equation order by order, trivializing its obstruction classes compatibly with restriction and factorization [23].

21.5 Pushforward to a curve

Let $p : X \to C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathbb{F}$ on X, define

$$(p_*\mathbb{F})(U) = \mathbb{F}(p^{-1}U).$$

Theorem 21.7 (Factorization pushforward)**.** *The assignment $p_*\mathbb{F}$ is a factorization algebra on C. Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathbb{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathbb{F})(U) \simeq \int_{F_p} \mathbb{F}|_{F_p \times U},$$

compatible on overlaps. To obtain algebraic factorization D-modules, require in addition regular-holonomic algebraic models on every configuration space and compatible algebraic models for all structure maps.

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. The additional algebraic models give the last realization. Riemann–Hilbert can compare specified constructible and regular-holonomic models; it does not algebraize an arbitrary holomorphic factorization algebra. □

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

21.6 The derived centre and the physical bulk

For an ordered boundary algebra A, define

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

The tensor and derived Hom are internal to the chosen closed monoidal coefficient category, with admissible derived tensor products. The Hochschild brace operations give its internal E_2 structure. A coherent bulk action in a compatible coefficient model gives

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A).$$

Definition 21.8 (Open–closed completeness)**.** A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase ‘‘bulk equals centre’’ into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

Chapter 22

Hall algebras and symplectic surfaces

22.1 Extension correspondences

Let $\mathcal{M} = \coprod_{\gamma\in\Gamma} \mathcal{M}_\gamma$ be a moduli stack of objects in a fixed exact category or heart of a stable category. Let $\mathcal{M}^{\mathrm{ext}}_{\alpha,\beta}$ classify extensions

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The correspondence is

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}^{\mathrm{ext}}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Theorem 22.1 (Hall associativity)**.** *Assume a monoidal coefficient theory with the required Künneth maps, pullbacks, pushforwards and coefficient comparisons, satisfying base change for the two-step filtration stack. Denote its resulting pull–push by $q_* p^*$. Then*

$$a * b = q_* p^* (a \boxtimes b)$$

defines an associative product.

Proof. The two parenthesizations are the two projections from the stack of filtrations $0 \subset E_1 \subset E_2 \subset E_3$. Base change identifies both composites with pullback to this stack followed by pushforward to E_3. □

For critical CoHAs, the coefficient theory includes vanishing cycles and a coherent orientation. For symplectic CoHAs, it includes the selected equivariant Borel–Moore theory. Completion is chosen so that each output degree receives a convergent sum.

22.2 Orientation data and the threefold Hall product

An orientation is a square root of the determinant line of the three-Calabi–Yau moduli problem. Strong orientation also specifies its multiplicative isomorphisms on extension correspondences, compatible with triple flags and the zero object. Joyce–Upmeier construct natural orientations for the compact threefold moduli problems in their theorem; the strong enhancement is a further condition [54, 61].

Let Φ_γ be the Donaldson–Thomas perverse sheaf on $\mathcal{M}_\gamma$. Define the shift $\kappa(\alpha,\beta)$ by the hyperbolic comparison

$$(\Phi_\alpha \boxtimes \Phi_\beta)[\kappa(\alpha,\beta)] \simeq R p_* q^! \Phi_{\alpha+\beta}.$$

This fixes its sign convention. The index obeys

$$\kappa(\alpha,\beta) + \kappa(\alpha+\beta,\gamma) = \kappa(\beta,\gamma) + \kappa(\alpha,\beta+\gamma), \qquad \kappa(0,\alpha) = \kappa(\alpha,0) = 0.$$

Thus the tensor

$$(V \otimes_\kappa W)_\gamma = \bigoplus_{\alpha+\beta=\gamma} (V_\alpha \otimes W_\beta)[\kappa(\alpha,\beta)]$$

has the induced associativity constraint.

Theorem 22.2 (Threefold Hall algebra from strong orientation)**.** *Assume the three-Calabi–Yau moduli stack satisfies the geometric hypotheses of* [61, Theorem 8.6 and Corollary 8.8]*: its classical stack is quasi-separated Artin with affine stabilizers, it is* Θ*-reductive, and the zero object is closed. Strong orientation defines a unital associative product on*

$$H = \bigoplus_{\gamma} H^{\bullet}(\mathcal{M}_{\gamma}, \Phi_{\gamma}), \qquad H \otimes_{\kappa} H \longrightarrow H.$$

Proof. The hyperbolic comparison has adjoint $p^{*}(\Phi_{\alpha} \boxtimes \Phi_{\beta})[\kappa] \to q^{!}\Phi_{\alpha+\beta}$. Properness of q on the chosen filtration component allows its counit to define a map on ordinary cohomology. The triple-flag comparison and strong orientation identify the two threefold products. The zero-endpoint correspondences give the unit [61, Proposition 8.29]. □

Definition 22.3 (Reduced $K3 \times E$ Hall datum)**.** A reduced datum additionally specifies the elliptic translation quotient, reduced critical coefficient complexes, their pull–push maps, and orientation comparisons on all flags. A chosen support convention and charge completion must make these maps defined and continuous. A full quantum-group realization further requires the Hopf and pairing data described below.

The theorem uses ordinary critical cohomology. Neither an E-equivariant Borel–Moore product nor descent of reduced coefficients follows from strong orientation alone.

22.3 The geometric E-equivariant Hall source

Let S be a projective K3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\leq 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Let Γ_X be its numerical charge group and Γ_X^{+} the monoid of charges of coherent sheaves. The degree of the projected one-cycle is additive:

$$m : \Gamma_X \longrightarrow \mathbb{Z}, \qquad m(\Gamma_X^{+}) \subseteq \mathbb{Z}_{\geq 0}.$$

For $\gamma \in \Gamma_X^{+}$, let $\mathcal{M}_{\gamma}$ be the moduli stack of objects of charge γ. Translation on the elliptic factor gives an action of E on every $\mathcal{M}_{\gamma}$ and on every extension stack.

Oberdieck and Shen construct the E-equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E,\mathrm{mot}} = \widehat{\bigoplus}_{\gamma \in \Gamma_X^{+}} K_0^{E}(\mathrm{St}/\mathcal{M}_{\gamma}),$$

where only effective charges occur. Fix an admissible filtration for which each coefficient of a product receives finitely many contributions. It must control Euler characteristic and zero-dimensional charges as well as curve class; a curve-degree filtration alone does not do this.

Theorem 22.4 (Geometric reduced Hall algebra)**.** *The extension correspondence defines the associative equivariant motivic Hall product. Let* H_{reg} *be the Oberdieck–Shen regular subalgebra and let* $H_{\mathrm{sc}} = H_{\mathrm{reg}}/(\mathbb{L} - 1)H_{\mathrm{reg}}$ *be its semiclassical quotient, where* $\mathbb{L} = [\mathbb{A}^1]$*, with the coefficient localization of that construction. Reduced integration gives a multiplicative map*

$$I_X^{E} : \widehat{H}_{\mathrm{sc}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X]}[\epsilon]/(\epsilon^2)$$

on every compatible admissible completion.

Proof. Translation preserves exact sequences and numerical charge. The two-step flag correspondence therefore proves equivariant associativity. The reduced integration theorem is [76, Theorem 5]. For an equivariant variety $Z \to \mathcal{M}_\gamma$ its coefficient is

$$I_X^E[Z \to \mathcal{M}_\gamma] = \big(\chi(Z^E, \nu) - \chi((Z - Z^E)/E, \nu)\epsilon\big)e^\gamma,$$

with the pulled-back Behrend function ν and weighted orbifold Euler characteristic of the moving quotient. For sheaves supported in dimension at most one on the threefold, Riemann–Roch gives $\chi(F, G) = 0$: both Chern characters begin in degree four, so their product has no degree-six component. The target's Euler twist is thus trivial. The theorem is multiplicative on H_{sc}, and continuity extends it to the selected completion. □

Its composite with $H_{\mathrm{reg}} \to H_{\mathrm{sc}}$ will also be denoted I_X^E. It is not a map from every stack class with arbitrary denominators. The sign of the moving term is part of the reduced convention.

22.4 Local and wrapped geometry inside the Hall algebra

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\mathrm{loc}} = \widehat{\bigoplus}_{m(\gamma)=0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma), \qquad \mathcal{H}_X^{\mathrm{wr}} = \widehat{\bigoplus}_{m(\gamma)>0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma).$$

A pure one-dimensional support of degree zero projects to a finite subset of E. A support of positive degree projects onto E. The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem 22.5 (Geometric local–wrapped decomposition)**.** *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\mathrm{loc}}$ is a subalgebra, $\mathcal{H}_X^{\mathrm{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E,\mathrm{mot}} \cong \mathcal{H}_X^{\mathrm{loc}} \ltimes \mathcal{H}_X^{\mathrm{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. On its regular domain, reduced integration preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E. □

This is the geometric local–wrapped grading. The root construction has a separate grading of the same form. Comparing them requires a primitive map that preserves this degree, the brackets and the coefficients.

22.5 Cohomological refinement and the Gram character

A cohomological refinement starts with a specified reduced equivariant coefficient theory. Suppose it supplies complexes $\Phi_\gamma^{\mathrm{red}}$ on the chosen quotient or equivariant moduli stacks and admissible pull–push maps on extension and flag correspondences. In a Borel–Moore model write

$$\mathcal{H}_X^{E,\mathrm{coh}} = \widehat{\bigoplus}_{\gamma \in \Gamma_X^+} H_*^{\mathrm{BM},E}(\mathcal{M}_\gamma, \Phi_\gamma^{\mathrm{red}}),$$

including the degree shifts of that model.

Theorem 22.6 (Cohomological lift from compatible coefficients)**.** *If these maps satisfy the oriented base-change and triple-flag identities, they define an associative product on $\mathcal{H}_X^{E,\mathrm{coh}}$. Elliptic degree gives a subalgebra in degree zero and a closed ideal in positive degree.*

Proof. The supplied coefficient identities make both threefold products the same map through the flag correspondence. Additivity of numerical charge gives the degree decomposition. The underlying correspondences are the motivic ones; a realization between their coefficient theories is further data. □

Use the Mukai lattice $\widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$, with

$$((r,c,s),(r',c',s'))_{\mathrm{Muk}} = c\cdot c' - rs' - r's.$$

For the dyonic charge lattice

$$\Gamma_{\mathrm{phys}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z}),$$

write

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

To compare sheaf charges with this lattice, choose an additive map

$$c : \Gamma_X \longrightarrow \Gamma_{\mathrm{phys}}.$$

Put $B_\Pi(a,b) = \Pi(a+b) - \Pi(a) - \Pi(b)$ and give $\widetilde{\Gamma}_{\mathrm{phys}} = \Gamma_{\mathrm{phys}} \times \mathbb{Z}^3$ the group law

$$(a,u)(b,v) = (a+b, u+v+B_\Pi(a,b)).$$

Then $j_\Pi(a) = (a,\Pi(a))$ is a homomorphism. Assume $j_\Pi c$ preserves the selected charge completions. On the regular subalgebra $\mathcal{H}_{\mathrm{reg}}$ of the reduced integration theorem, define

$$I^E_{X,\Pi,c} = (j_\Pi c)_* I^E_X : \mathcal{H}_{\mathrm{reg}} \longrightarrow \widehat{\mathbb{Q}[\widetilde{\Gamma}_{\mathrm{phys}}]}[\epsilon]/(\epsilon^2).$$

Write $I^E_{X,\Pi}$ for this map when c has been fixed.

Proposition 22.7 (Polarized reduced character)**.** *The map $I^E_{X,\Pi,c}$ is multiplicative. Its coefficients are the reduced invariants supplied by I^E_X, regraded by $j_\Pi c$. If $g \in 1 + F^1H$ is group-like in a complete filtered Hall bialgebra and the logarithm converges, then* $\log g$ *is primitive.*

Proof. Expanding $\Pi(a+b+d)$ gives

$$B_\Pi(a,b) + B_\Pi(a+b,d) = B_\Pi(b,d) + B_\Pi(a,b+d).$$

This proves associativity of the group law. The definition gives $j_\Pi(a+b) = j_\Pi(a)\,j_\Pi(b)$. Composing its group-algebra map with reduced integration proves multiplicativity on $\mathcal{H}_{\mathrm{reg}}$. For the last assertion, the two factors in $\Delta g = (g\widehat{\otimes}1)(1\widehat{\otimes}g)$ commute. Taking the convergent logarithm gives $\Delta \log g = \log g \widehat{\otimes} 1 + 1 \widehat{\otimes} \log g$. □

The primitive reduced Gromov–Witten theorem supplies the separate character $-\chi_{10}^{-1}$ in classes primitive in the K3 factor [75]. Comparing this character with the Hall character requires the same charge map, sector, and normalization. A primitive root framing must also preserve the extension bracket and parity of the chosen BPS states.

22.6 BPS primitives

A coproduct defines primitives by

$$\operatorname{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

A BPS algebra can also be constructed geometrically before such a coproduct is chosen. The distinction matters for the full CoHA.

Theorem 22.8 (Two-Calabi–Yau BPS algebra)**.** *In the setting of* [26]*, assume the extension stack is proper over the object stack, the moduli stack has the required three-term resolution of derived Hom, its good moduli space is separated with finite-type components, direct sum on that space is finite, and the simple-object category has the specified right two-Calabi–Yau duality. With the associative unital Hall product and the prescribed sign twist, the BPS algebra is the positive enveloping algebra of the constructed generalized Kac–Moody Lie algebra.*

Proof. The simple BPS sheaves supply its generators. The extension product, local quiver comparison and Serre relations give a morphism from their generalized Kac–Moody envelope. The PBW and support decomposition in [26, Theorems 1.1–1.2] identify this morphism with the BPS algebra. □

This theorem applies to the semistable K3 examples satisfying those hypotheses. The full CoHA has a PBW description involving $\mathfrak{g}_{\mathrm{BPS}} \otimes H^*(B\mathbb{G}_m)$; that description is an isomorphism of the stated graded sheaves or vector spaces. It does not identify the full Hall multiplication with multiplication in $U(\mathfrak{g}_{\mathrm{BPS}})$.

22.7 The A_2 Hall algebra

For the quiver $1 \to 2$ over $\mathbb{F}_q$, put $v^2 = q$ and let $E_i = [S_i]$ be the simple-object classes. Use the Hall twist $v^{\langle \alpha, \beta \rangle}$, where $\langle (a,b),(c,d) \rangle = ac + bd - ad$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2,1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

After the Hall twists $v^{-1}, 1, v$, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [43, 80].

22.8 The tripled Jordan quiver and $\mathbb{C}^3$

For the one-vertex quiver with loops x, y, z, take the potential $W = \operatorname{Tr} x[y,z]$. Its Jacobi algebra is $\mathbb{C}[x,y,z]$. Dimensional reduction compares its critical CoHA with the CoHA of zero-dimensional sheaves on $\mathbb{A}^2$.

Theorem 22.9 (Affine-Yangian positive half)**.** *Let $T = (\mathbb{C}^*)^2$ act with weights t_1, t_2 on the plane and weight $-t_1 - t_2$ on the third loop. Over $H_T^*(\mathrm{pt}, \mathbb{Q}) = \mathbb{Q}[t_1, t_2]$, the surface CoHA is isomorphic to the strictly positive affine Yangian of $\mathfrak{gl}_1$ in the presentation of* [24]*. The isomorphism sends its current generator e_i to the corresponding one-point class $\alpha_1^{(i)}$.*

Proof. Dimensional reduction identifies the Hall products. Davison proves spherical generation and the current relations over the polynomial parameter ring, then uses the affine BPS description and PBW to prove the isomorphism. The shuffle comparison gives an additional localized presentation with $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$ [24, 27, 81, 83]. □

No parameter localization is needed for the stated polynomial-ring isomorphism. Nonequivariantly the algebra is $U(\mathrm{gr}_F W^+_{1+\infty})$; the one-parameter version is the corresponding Rees envelope. Here

$$W^+_{1+\infty} = \mathrm{span}_{\mathbb{Q}}\{z^m D^n : m > 0,\ n \geq 0\}, \qquad D = z\frac{d}{dz},$$

and

$$[z^m D^a, z^n D^b] = z^{m+n}\big((D+n)^a D^b - D^a (D+m)^b\big).$$

The order filtration is $F_i = \mathrm{span}\{z^m D^a : a \leq (i+2)/2\}$. The full affine Yangian has separate negative and Cartan currents. A vacuum representation yields a vertex algebra only after a vacuum, translation operator, fields and a reconstruction theorem have been specified; an unspecified evaluation homomorphism does not provide them.

22.9 Surface CoHAs and ADE Yangians

Let $S_\Gamma \to \mathbb{C}^2/\Gamma$ be the minimal resolution for a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$, and let C be its exceptional curve. Use the diagonal torus action commuting with Γ.

Theorem 22.10 (Curve-supported surface CoHA)**.** *The equivariant CoHA of coherent sheaves set-theoretically supported on C, including their zero-dimensional subobjects, has the completed nonstandard affine ADE Yangian presentation of* [29]. *Its completion uses quasi-compact and slope bounds; its Yangian transition maps are the truncated braid maps between quotients of the standard negative half.*

Proof. The reflection equivalences and Hecke correspondences identify the finite slope pieces. Their compatibility with the truncated braid maps gives the inverse-system comparison. The multiplication and continuity theorems in [29] give the completed isomorphism. □

An ADE configuration on a K3 surface suggests a local application. It requires an identification of the supported categories, coefficient theories and equivariant geometry. The presence of the intersection matrix alone does not supply that identification or a global torus action.

22.10 Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [12]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. In models with a specified boundary construction, its geometric data determine the relevant shift and truncation.

The BFN construction supplies a second geometric source of Yangians. Loop rotation supplies the quantization parameter and convolution supplies the product. A coproduct requires an additional construction. The surface CoHA uses sheaf extensions. A comparison between the two is a model-specific equivalence.

22.11 The global zero-dimensional surface algebra

Let S be a projective K3 surface, and let $\mathcal{M}_S(n)$ classify zero-dimensional sheaves of length n. In the cohomological grading and rational Borel–Moore theory of [72], put

$$H_S^{\circ} = \bigoplus_{n \geq 0} H_*^{\mathrm{BM}}(\mathcal{M}_S(n), \mathbb{Q}).$$

Choose a hyperbolic plane with basis e, f in $H^2(S, \mathbb{Z})$, where $e^2 = f^2 = 0$ and $e \cdot f = 1$. The class $q = e - 12f$ satisfies $q^2 = -24[\mathrm{pt}] = c_1(S)^2 - c_2(S)$. This is a cohomology class; no Hodge-type condition on q is required.

Define $\mathfrak{w}_S^+$ on the symbols $z^m D^n \lambda$, linear in $\lambda \in H^*(S, \mathbb{Q})$, with $m > 0$, $n \geq 0$, by

$$[z^m D^n \lambda, z^a D^b \mu] = z^{m+a} \frac{(D + aq)^n D^b - D^n (D + mq)^b}{q} \lambda\mu.$$

The quotient means the polynomial obtained before specializing the formal parameter to q. Its numerator is divisible by that parameter; the nilpotent class q is never inverted. To see Jacobi, first impose $Dz = z(D + t)$ over $\mathbb{Q}[t]$. Its commutator divided by t is a polynomial Lie bracket, so specialization $t \mapsto q$ preserves Jacobi.

Write $W^+(S)$ for the positive surface current algebra. It is generated by $T_n(\lambda)$, $n \geq 0$, linearly in $\lambda \in H^*(S, \mathbb{Q})$, with coefficient migration

$$[T_m(\lambda\mu), T_n(\nu)] = [T_m(\lambda), T_n(\mu\nu)],$$

the current relation

$$\begin{aligned} 0 = {} & [T_m(\lambda), T_{n+3}(\mu)] - 3[T_{m+1}(\lambda), T_{n+2}(\mu)] \\ & + 3[T_{m+2}(\lambda), T_{n+1}(\mu)] - [T_{m+3}(\lambda), T_n(\mu)] \\ & - [T_m(\lambda), T_{n+1}(q^2\mu)] + [T_{m+1}(\lambda), T_n(q^2\mu)], \end{aligned}$$

and the cubic relation

$$\sum_{\sigma \in S_3} \sigma \cdot [T_{m_3}(\lambda_3), [T_{m_2}(\lambda_2), T_{m_1+1}(\lambda_1)]] = 0.$$

Here the indices are nonnegative and σ permutes the three pairs (m_i, λ_i).

Theorem 22.11 (Global surface CoHA)**.** *There is an algebra isomorphism*

$$H_S^{\circ} \simeq W^+(S) \simeq U(\mathfrak{w}_S^+).$$

The length-one current class indexed by $D^n\lambda$ corresponds to $zD^n\lambda$.

Proof. The surface CoHA theorem identifies H_S° with the positive deformed W-algebra [72, Theorem B]. For $c_1 = 0$ and $c_1^2 - c_2 = q^2$, the current relations identify this algebra with $U(\mathfrak{w}_S^+)$ [72, Theorem 3.5]. The proof constructs the current homomorphism; surjectivity and the proved PBW dimensions make it an isomorphism. □

22.12 A global surface coproduct

Let Γ be the lattice of Mukai vectors of coherent sheaves on S, where $v(F) = \mathrm{ch}(F)\sqrt{\mathrm{td}(S)}$. Let χ be the surface Euler pairing, the negative of the Mukai pairing above. For the stack $\mathcal{M}_S(\gamma)$ of objects

of vector γ, put $v_\gamma = -\chi(\gamma,\gamma)$, its virtual dimension. Work in rational pro-complexes obtained by restriction to quasi-compact open substacks:

$$A_\gamma = R\Gamma^{\mathrm{BM}}_{\mathrm{pro}}(\mathcal{M}_S(\gamma),\mathbb{Q})[-v_\gamma].$$

Write $A = (A_\gamma)_\gamma$ for this charge-graded pro-object. There are two charge tensors. On total charge γ they are

$$(A\widehat{\otimes}'_\Gamma B)_\gamma = \bigoplus_{\alpha+\beta=\gamma} A_\alpha \widehat{\otimes} B_\beta,$$
$$(A\widehat{\otimes}_\Gamma B)_\gamma = \prod_{\alpha+\beta=\gamma} A_\alpha \widehat{\otimes} B_\beta.$$

Here each tensor on the right is the tensor of the specified pro-objects. In addition to the cochain Koszul sign, the symmetry has the factor $(-1)^{\chi(\alpha,\beta)}$.

The coefficient construction uses oriented shifted Lagrangian classes

$$[L]_{\mathrm{Lag}} : f_1^*\Phi_1 \longrightarrow f_2^!\Phi_2[-\operatorname{vdim} L]$$

for a correspondence $X_1 \xleftarrow{f_1} L \xrightarrow{f_2} X_2$. We require their unit, composition, external-product and étale-base-change identities, together with dimensional reduction for shifted cotangent stacks. These are the functorial coefficient hypotheses used in [25, Theorem 2.4 and Section 6].

Theorem 22.12 (Eigenvalue coproduct for a symplectic surface)**.** *With this coefficient package, the surface Hall construction gives*

$$m : A\widehat{\otimes}'_\Gamma A \longrightarrow A, \qquad \Delta : A \longrightarrow A\widehat{\otimes}_\Gamma A.$$

The coproduct is coassociative and cocommutative for the Euler-twisted symmetry. It is compatible with the Hall product in the completed tensor domains of [25, Proposition 6.4]*. The resulting cohomology is a topological graded bialgebra.*

Proof. Enhance a sheaf to (F,ϕ) with $\phi \in \mathrm{End}(F)$. For an open $U \subset \mathbb{C}$, restrict to endomorphisms whose eigenvalues lie in U. Dimensional reduction identifies the critical cohomology over a nonempty convex U with A. For two disjoint such opens, generalized eigenspaces split the sheaf canonically. Oriented Thom–Sebastiani identifies the corresponding critical complexes with the product-completed tensor of their two factors.

Restriction from a convex open containing both gives Δ. Three disjoint opens give the same restriction in either order, proving coassociativity. Interchanging the two summands changes the orientation by $(-1)^\chi$, which gives the stated cocommutativity. The zero-sheaf component gives the counit.

For multiplicativity, split each term of an extension into its two generalized-eigenvalue parts. Off-diagonal extensions between disjoint spectra vanish. The resulting square of extension correspondences identifies the two product–coproduct composites. The composition and external-product identities for the shifted Lagrangian classes make their coefficient maps agree; the middle interchange has the Euler and Koszul signs. Properness of the flag maps gives the locally finite extension of multiplication on this completed target [25, Propositions 6.1 and 6.4]. □

For this argument the object stack is quasi-smooth, locally of finite type with affine diagonal, closed under direct sums and summands, contains the zero object as an open and closed component, and is Θ-reductive. The pro-system handles its non-quasi-compactness. Multiplication is not asserted on an arbitrary product over all charge decompositions.

Let $H_S = \bigoplus_\gamma H^*(A_\gamma)$ after topological realization. Fix a polarization and a reduced Hilbert polynomial p, and write $H_S^{\mathrm{ss}}(p)$ for the corresponding semistable Hall algebra. The semistable charge cone

is strictly convex and its cohomology has finite graded pieces. Let $\mathfrak{g}_S$ and $\mathfrak{g}_S^{\mathrm{ss}}(p)$ be the primitives for the two coproducts. In this same symmetry,

$$U(\mathfrak{g}_S^{\mathrm{ss}}(p)) \simeq H_S^{\mathrm{ss}}(p), \qquad U(\mathfrak{g}_S) \hookrightarrow H_S$$

with dense image in the second map. Restriction identifies the primitives of each semistable charge with the corresponding global primitives; their graded spaces are the affinized BPS cohomology, including the $H^*(B\mathbb{G}_m)$ factor and the normalization of dimensional reduction [25, Theorem 6.8].

The semistable assertion follows from connected cocommutative reconstruction. For the global assertion, ordered Harder–Narasimhan products give PBW tensors of semistable primitive envelopes. An iterated coproduct followed by semistable restriction detects the first nonzero Harder–Narasimhan term, proving injectivity. The same triangular argument approximates every class on each quasi-compact truncation, proving density. Purity and the strict-nilpotent-support comparison identify the primitive spaces in this argument.

This constructs a global K3 coproduct on the stated surface coefficients. Its comparison with reduced critical coefficients on $K3 \times E$ is a separate problem. Its pro-tensors and Euler symmetry also differ from the fixed-parameter polynomial shuffle carrier whose ordinary Koszul cooperation fails in Volume I.

Chapter 23

Amalgams, doubles and tensor representations

23.1 The integral Hall–Borcherds amalgam

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra and let $\mathfrak{n}_{\Delta_s,+}$ be the positive Igusa Borcherds Lie superalgebra. Choose a finite-dimensional abelian Cartan algebra $\mathfrak{h}_\Pi$ and additive weight maps from both charge groups to $\mathfrak{h}_\Pi^*$. For a homogeneous x_γ, the action is $[h, x_\gamma] = \lambda(\gamma)(h)x_\gamma$. Additivity of λ makes this action a derivation. Form

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \qquad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta_s,+}.$$

The two chosen weights must agree along any primitive comparison. The polarized group law makes j_Π additive, but its coordinate $\Pi(a)$ is not an additive scalar weight.

For example, let $[x, y] = z$ in the Heisenberg Lie algebra, with charges 1, 1, 2. An additive grading derivation has eigenvalues 1, 1, 2. The quadratic prescription $D(x) = x$, $D(y) = y$, $D(z) = 4z$ gives

$$D[x, y] = 4z, \qquad [Dx, y] + [x, Dy] = 2z.$$

It therefore cannot define a Cartan action. The polarized character and the additive Cartan action remain separate constructions.

Fix a nonnegative filtration whose positive parts have strictly positive integer height and finite-dimensional pieces. Keep the Cartan in degree zero, with a specified compatible coefficient topology. Complete the ordinary amalgam by this filtration; all mapping properties below are in the resulting category of continuous filtered maps.

Definition 23.1 (Integral protected Lie algebra)**.** The integral Hall–Borcherds algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X\boxtimes\Delta} = \mathfrak{b}_X \widehat{\amalg}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 23.2 (Hall–Borcherds amalgamation)**.** *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_X, \qquad r_\Delta : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X\boxtimes\Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The two factor inclusions are split injections. Their Lie brackets and enveloping products are retained. A representation or character of either factor pulls back along its retraction; this does not identify the two factors' characters or construct a compact determinant map.

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by the chosen positive-height filtration. Finite-dimensional graded pieces and positivity make each fixed-height Lie calculation finite. Degree-zero Cartan brackets reduce to the prescribed weight action. The separated inverse limit has the stated continuous mapping property. The projection $\mathfrak{b}_X \to \mathfrak{h}_\Pi \to \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. □

Definition 23.3 (Comparison ideal)**.** A primitive comparison $\varphi : \mathfrak{n}_{\Delta_s,+} \to \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta_s,+} \rangle} \subset \mathfrak{b}_{X \boxtimes \Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X \simeq \Delta, \varphi} = \mathfrak{b}_{X \boxtimes \Delta} / \mathcal{J}_\varphi.$$

Proposition 23.4 (Graph quotient of a primitive comparison)**.** *Let $\varphi : \mathfrak{b}_\Delta \to \mathfrak{b}_X$ be a continuous Lie morphism restricting to the identity on $\mathfrak{h}_\Pi$. Use a completion in which the amalgam and closed quotient have their stated universal properties. Then*

$$\mathfrak{b}_{X \boxtimes \Delta} / \mathcal{J}_\varphi \cong \mathfrak{b}_X.$$

The two factor maps become id *and φ. If φ is an isomorphism, the quotient is also isomorphic to $\mathfrak{b}_\Delta$.*

Proof. The maps id and φ agree on $\mathfrak{h}_\Pi$. They give $q_\varphi : \mathfrak{b}_{X \boxtimes \Delta} \to \mathfrak{b}_X$, which kills the generators of $\mathcal{J}_\varphi$. The inclusion of $\mathfrak{b}_X$ gives a map s in the other direction. The composite $q_\varphi s$ is the identity. The other composite is the identity on the geometric factor. On the Borcherds factor, the defining relation identifies x with $s\varphi(x)$, so it is also the identity there. The two factor images topologically generate the quotient. Continuity proves that the maps are inverse on the completion. □

The graph quotient implements a specified primitive comparison. Constructing that comparison requires geometric simple states whose extension brackets satisfy the Borcherds relations. The quotient formula alone produces neither those states nor their orientation data.

23.2 Coproducts, pairings, and doubles

A finite double shows which structures enter before completion. Let G be a finite group and let $D(G)$ have basis $\delta_g h$, $g, h \in G$. Write e for the group identity and set

$$(\delta_g h)(\delta_x k) = \begin{cases} \delta_g(hk), & g = hxh^{-1}, \\ 0, & \text{otherwise}, \end{cases} \qquad 1 = \sum_g \delta_g e.$$

Its coalgebra and antipode are

$$\Delta(\delta_g h) = \sum_{ab=g} \delta_a h \otimes \delta_b h, \quad \epsilon(\delta_g h) = \delta_{g,e}, \quad S(\delta_g h) = \delta_{h^{-1}g^{-1}h} h^{-1}.$$

Put $1h = \sum_x \delta_x h$.

Theorem 23.5 (Finite group double)**.** *These formulas make $D(G)$ a Hopf algebra with universal matrix*

$$R = \sum_g \delta_g e \otimes 1g, \qquad R^{-1} = \sum_g \delta_g e \otimes 1g^{-1}.$$

It satisfies

$$R\Delta(a) = \Delta^{\mathrm{op}}(a)R, \quad (\Delta \otimes \mathrm{id})R = R_{13}R_{23}, \quad (\mathrm{id} \otimes \Delta)R = R_{13}R_{12}.$$

Proof. A triple product is nonzero precisely when $g = hxh^{-1}$ and $x = kyk^{-1}$; either order then gives $\delta_g(hkl)$. This proves associativity. Both iterated coproducts sum over $abc = g$. Conjugation respects multiplication in G, so Δ preserves products. The two counit identities follow by setting one factor of $ab = g$ equal to e.

In $m(S \otimes \mathrm{id})\Delta(\delta_g h)$ a term survives only when $b = a^{-1}$, hence only when $g = e$. In that case the sum is $\sum_a \delta_{h^{-1}a^{-1}h} e = 1$. The other antipode identity is the same calculation. Orthogonality of the δ_g gives the displayed inverse of R.

Write R in its function and group factors. Moving a group element past a function conjugates its index, which gives $R\Delta(\delta_g h) = \Delta^{\mathrm{op}}(\delta_g h)R$. For the first hexagon, both sides sum the terms with indices a, b, ab. For the second, orthogonality in the first tensor factor forces the same index g in both remaining group factors. These are the stated two equations. Applying either counit to R gives 1. □

The conjugation and the inverse in S are essential. Already for $G = S_3$, erasing conjugation leaves the tensor-product algebra $\mathbb{Q}^G \otimes \mathbb{Q}G$, but the displayed R no longer intertwines its coproduct. Erasing g^{-1} in S destroys the convolution identity.

For a finite-dimensional Hopf algebra H with invertible antipode, the convention of [34, Section 7.14] pairs H with $H^{*\mathrm{cop}}$. The opposite coproduct and antipode enter the cross-relations; the dual bases give the canonical tensor. This gives a full positive model without any convergence assumption.

Definition 23.6 (Completed Hall double)**.** Choose positive and negative Hopf Borel halves with invertible antipodes, their common Cartan identifications, and a Hopf pairing with the specified opposite-coproduct convention. Require a separated triangular multiplication in the chosen completed tensor category, with underlying module

$$\widehat{D}(H^+) = H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+.$$

Here the notation includes the chosen negative and Cartan data. Its canonical tensor must be supplied by locally perfect dual pieces or by an explicit continuous coevaluation, and must converge on the selected modules. The Cartan reduction must be a Hopf quotient. This datum defines the completed Hall double.

Theorem 23.7 (Double construction)**.** *Suppose the finite dual pieces or finite quotients in this datum carry their compatible ordinary doubles, cross-relations and canonical tensors. Their separated limit is a complete Hopf double. On the module domain where its canonical tensor converges, it is quasitriangular and hence defines braided tensor products.*

Proof. At each finite stage the Hopf pairing and the opposite dual coproduct give the ordinary Drinfeld double. Compatibility makes every associativity, antipode and quasitriangular equation an equation of inverse systems. Passing to the separated limit preserves those equations in the specified tensor domains. The coevaluation hypothesis supplies the limiting canonical tensor; its inverse and the two hexagons are inherited from the finite stages. A compatible Cartan Hopf quotient preserves these identities [32, 34, 35]. □

Nondegeneracy of a pairing alone need not produce a coevaluation or a convergent sum of dual bases. A general infinite Hall construction must establish those conditions directly when no finite-double system is available.

Theorem 23.8 (Quantum Hall–Borcherds amalgam)**.** *Let H_X and H_Δ be complete Hopf algebras containing the same complete Cartan Hopf algebra H°, with continuous Hopf retractions onto H°. Form their algebraic amalgamated free product and choose a separated multiplicative filtration for which the two coproducts and antipodes extend continuously to the completed tensor products. Their completed amalgamated free product*

$$H_{X\boxtimes\Delta} = H_X \mathbin{\widehat{*}_{H^\circ}} H_\Delta$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents the pairs of filtered continuous Hopf morphisms whose induced map is continuous for this filtration, and has Hopf retractions to the factors when those retractions preserve the filtration. A universal R-matrix requires an invertible element in its completed tensor square satisfying intertwining with both generating factors, the two hexagons and the counit normalizations. The two factor R-matrices do not by themselves supply it.

Proof. The ordinary amalgam is the free product modulo the relations identifying the Cartan copies; its chosen separated completion exists as their filtered inverse limit. The two coproducts agree on H°, so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the filtration-compatible factor retractions onto H°. All extensions are continuous by the filtration hypothesis. The additional quasitriangular equations must hold on both generating factors and in the chosen tensor completion. □

A retraction from a full quantum group to its unrestricted Cartan is itself a substantial assumption. For $U_q(\mathfrak{sl}_2)$ over $\mathbb{Q}(q)$, killing E, F in $[E, F] = (K - K^{-1})/(q - q^{-1})$ forces $K^2 = 1$. The usual positive Borel projection therefore does not extend to that Cartan retraction of the full quantum group.

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

23.3 Ordered chiral bialgebras

A tensor product of representations requires compatible actions on threefold products and the unit. Retain the factorization coefficient category $(\mathrm{Fact}_D(C), \otimes^!)$ on the smooth complex curve C. Fix a category $\widehat{\mathcal{V}}_C$ of complete filtered objects with specified presentations $F \simeq \varprojlim_r F_r$, where $F_r \in \mathrm{Fact}_D(C)$. Use derived limits in the fixed Ran enhancement and define

$$F \mathbin{\widehat{\otimes}^!} G = \varprojlim_{r,s} (F_r \otimes^! G_s).$$

Assume these limits remain factorizing and give $\widehat{\mathcal{V}}_C$ a symmetric monoidal structure with unit $\mathbf{1}$. Its associativity, symmetry, and unit equivalences must preserve the presentations and the Ran structure. Morphisms are continuous maps compatible with these presentations. These are hypotheses on the chosen completion.

Definition 23.9 (Ordered chiral bialgebra)**.** An ordered chiral bialgebra is a counital E_1-coalgebra object in complete ordered chiral algebras:

$$A \in \mathrm{CoAlg}_{\mathsf{E}_1}\Big(\mathrm{Alg}_{\mathsf{E}_1}(\widehat{\mathcal{V}}_C, \widehat{\otimes}^!)\Big).$$

It has continuous multiplication m and unit η, and continuous morphisms of unital ordered chiral algebras

$$\Delta : A \longrightarrow A \widehat{\otimes}^! A, \qquad \epsilon : A \longrightarrow \mathbf{1}.$$

The counit is the chosen augmentation. Write τ for the coefficient symmetry, including Koszul signs. Suppressing ambient associativity and unit equivalences, compatibility reads

$$\Delta m = (m \widehat{\otimes}^! m)(\mathrm{id} \widehat{\otimes}^! \tau \widehat{\otimes}^! \mathrm{id})(\Delta \widehat{\otimes}^! \Delta),$$
$$\Delta\eta = \eta \widehat{\otimes}^! \eta, \qquad \epsilon m = \epsilon \widehat{\otimes}^! \epsilon, \qquad \epsilon\eta = \mathrm{id}_{\mathbf{1}}.$$

Coassociativity and the counit identities read

$$(\Delta \widehat{\otimes}^! \mathrm{id})\Delta = (\mathrm{id} \widehat{\otimes}^! \Delta)\Delta,$$
$$(\epsilon \widehat{\otimes}^! \mathrm{id})\Delta = \mathrm{id}_A = (\mathrm{id} \widehat{\otimes}^! \epsilon)\Delta.$$

For a homotopy-coherent presentation, these identities include the higher coherences of the displayed coalgebra object. Every operation and coherence preserves diagonal transitions and disjoint-locus factorization maps. An antipode is a continuous convolution inverse $S : A \to A$ of id_A:

$$m(S \widehat{\otimes}^! \mathrm{id})\Delta = \eta\epsilon = m(\mathrm{id} \widehat{\otimes}^! S)\Delta.$$

A bialgebra with an antipode is a Hopf object in this coefficient category.

A continuous left A-module is an object $M \in \widehat{\mathcal{V}}_C$ with a unital E_1 action $\rho_M : A \widehat{\otimes}^! M \to M$. Its action and coherences are morphisms in these factorization coefficients. Write $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ for their category.

Theorem 23.10 (Representation fusion)**.** *An ordered chiral bialgebra makes* $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ *monoidal. Its product is* $M \widehat{\otimes}^! N$ *with action*

$$\begin{aligned} A \widehat{\otimes}^! (M \widehat{\otimes}^! N) &\xrightarrow{\Delta \widehat{\otimes}^! \mathrm{id}} (A \widehat{\otimes}^! A) \widehat{\otimes}^! (M \widehat{\otimes}^! N) \\ &\xrightarrow{\mathrm{id} \widehat{\otimes}^! \tau_{A,M} \widehat{\otimes}^! \mathrm{id}} (A \widehat{\otimes}^! M) \widehat{\otimes}^! (A \widehat{\otimes}^! N) \\ &\xrightarrow{\rho_M \widehat{\otimes}^! \rho_N} M \widehat{\otimes}^! N. \end{aligned}$$

The unit is $\mathbf{1}$ *with action induced by* ϵ*. The ambient associator and unit equivalences are module equivalences.*

Suppose a continuous universal R*-matrix is also given. It is an invertible morphism* $R : \mathbf{1} \to A \widehat{\otimes}^! A$ *for the tensor-product algebra, satisfying*

$$R\Delta(a) = \Delta^{\mathrm{op}}(a)R, \qquad \Delta^{\mathrm{op}} = \tau_{A,A}\Delta,$$
$$(\Delta \widehat{\otimes}^! \mathrm{id})(R) = R_{13}R_{23}, \qquad (\mathrm{id} \widehat{\otimes}^! \Delta)(R) = R_{13}R_{12},$$
$$(\epsilon \widehat{\otimes}^! \mathrm{id})(R) = \eta = (\mathrm{id} \widehat{\otimes}^! \epsilon)(R).$$

The first equation is an identity of morphisms from A to its completed tensor square. Leg notation uses the unit and symmetry of $\widehat{\mathcal{V}}_C$. In a homotopy-coherent presentation, require compatible lifts of these identities. Then $c_{M,N} = \tau_{M,N} \circ R_{M,N}$ is a braiding, where $R_{M,N}$ acts through the external tensor-product module action.

Proof. The symmetry combines the independent module actions into an $A\widehat{\otimes}^{!}A$-action. Restriction along the unital algebra map Δ gives the displayed action. For three modules, the two actions use $(\Delta\widehat{\otimes}^{!}\mathrm{id})\Delta$ and $(\mathrm{id}\,\widehat{\otimes}^{!}\Delta)\Delta$. Coassociativity makes the ambient associator A-linear. The counit identities make the ambient unit equivalences A-linear. The coalgebra coherences give the pentagon and triangle coherences. The completion hypotheses keep these maps continuous and factorizing. Quasitriangularity makes $c_{M,N}$ an intertwiner and invertibility makes it an equivalence. The two coproduct identities for R give the hexagons. Its counit identities give compatibility with the unit. □

The tensor in this construction is the completed transverse tensor $\otimes^{!}$ on factorization coefficients. A chiral-convolution or Hall coproduct requires its own construction and a comparison in the specified category. A spectral version additionally requires a difference coordinate, a Laurent-series completion, and compatible domains for iterated expansions. Its identities must hold in those domains before defining module actions.

Example 23.11 (A counital algebra map without coassociativity)**.** In ordinary complex vector spaces, let $A = \mathbb{C}[x]$ and put

$$\Delta(x) = x \otimes 1 + 1 \otimes x + x \otimes x^2, \qquad \epsilon(x) = 0.$$

Polynomial substitution extends these assignments to unital algebra maps. Writing $F(u, v) = u + v + uv^2$, the equations $F(0, v) = v$ and $F(u, 0) = u$ prove both counit identities. But

$$\begin{aligned} F(F(u,v),w) - F(u,F(v,w)) &= -uv^2w^4 - uv^2w^2 - 2uvw^3 - 2uvw \\ &= -uvw(1+w^2)(vw+2) \neq 0. \end{aligned}$$

On three regular modules, the two actions differ already on $1 \otimes 1 \otimes 1$. Thus the canonical tensor associator is not A-linear. This ordinary example shows why multiplicativity and the counit identities must be supplemented by coassociativity. It makes no assertion about a geometric realization in factorization coefficients.

23.4 Tensor reconstruction and geometric kernels

Let $\mathscr{C}$ be an essentially small rigid linear monoidal category and let $\omega : \mathscr{C} \to \mathrm{Vect}_{\mathrm{fd}}(\mathbb{Q})$ be a $\mathbb{Q}$-linear strong monoidal functor. A choice of geometric kernels specifies the morphisms of $\mathscr{C}$; it is part of the input. Choose a small skeleton and define

$$H = \left(\bigoplus_{M \in \mathscr{C}} \omega(M)^* \otimes \omega(M) \right) \Big/ \left\langle [\lambda \circ \omega(f) \otimes v]_M - [\lambda \otimes \omega(f)v]_N \right\rangle,$$

where $f : M \to N$, $v \in \omega(M)$, and $\lambda \in \omega(N)^*$.

Proposition 23.12 (Matrix-coefficient coalgebra)**.** *For a basis (e_i) with dual basis (e^i), the formulas*

$$\Delta[\lambda \otimes v]_M = \sum_i [\lambda \otimes e_i]_M \otimes [e^i \otimes v]_M, \qquad \epsilon[\lambda \otimes v]_M = \lambda(v)$$

are basis-independent and make H a coalgebra. Tensor product gives its multiplication, and duals give its antipode. Each $\omega(M)$ is a comodule by

$$v \longmapsto \sum_i e_i \otimes [e^i \otimes v]_M .$$

Every selected kernel is a comodule morphism.

Proof. The identity tensor $\sum_i e_i \otimes e^i$ is basis-independent. Inserting it twice proves coassociativity; evaluation proves both counit identities. The defining relations express naturality under f and are preserved by these formulas. The strong monoidal constraints identify the coefficient of $M \otimes N$ with the product of coefficients of M and N. Their pentagon and unit identities give the bialgebra axioms. The evaluation and coevaluation identities for duals give the antipode identities. The same relations prove that $\omega(f)$ intertwines the coactions. □

The linear dual of H represents linear natural endomorphisms of ω. Tensor-compatible natural automorphisms are the corresponding multiplicative, unit-preserving families, not all elements of this dual. An equivalence with a comodule category requires the additional exactness and reconstruction hypotheses stated in the Tannaka theorem below. An action algebra generated by selected kernels retains those operators; this coend encodes the natural coefficient coactions that make the selected morphisms intertwiners.

23.5 Tannaka reconstruction

Let $\mathcal{L}$ be an essentially small rigid $\Bbbk$-linear abelian monoidal category with finite-dimensional morphisms, finite-length objects and exact tensor product. Let

$$\omega : \mathcal{L} \longrightarrow \mathrm{Vect}_{\mathrm{fd}}$$

be faithful, exact and strong monoidal. Form its matrix-coefficient Hopf algebra H by the preceding coend construction. The underlying vector space of H may be infinite-dimensional.

Extend ω to $\overline{\omega} : \mathrm{Ind}(\mathcal{L}) \to \mathrm{Vect}$. Assume it has a right adjoint R, is comonadic, and that the canonical coend map identifies its comonad with $- \otimes H$. Require also that objects with finite-dimensional $\overline{\omega}$ are exactly the objects of $\mathcal{L}$.

Theorem 23.13 (Rigid Tannaka reconstruction)**.** *Under these hypotheses the comparison is a monoidal equivalence*

$$\mathcal{L} \simeq \mathrm{Comod}_{\mathrm{fd}}(H).$$

A braiding on $\mathcal{L}$ gives the corresponding coquasitriangular form on H.

Proof. The coend formulas give each $\omega(M)$ its universal coaction. Tensor products and duals give the Hopf operations. Comonadicity identifies $\mathrm{Ind}(\mathcal{L})$ with coalgebras for $\overline{\omega}R \simeq - \otimes H$. Restricting to finite-dimensional underlying spaces gives the displayed equivalence. The two hexagons transport to the two multiplicativity equations of the coquasitriangular form. □

The comonad lives on Vect; an infinite-dimensional H does not define an endofunctor $- \otimes H$ of $\mathrm{Vect}_{\mathrm{fd}}$. A spectral version additionally needs a parameter base, meromorphic coefficient domains and compatible expansions. Ordinary rigid reconstruction does not supply those data.

23.6 From positive half to full quantum group

A local or compact positive algebra enters a full quantum group through Hopf Borel halves, a continuous pairing, Cartan reduction and the tensor completion of the double construction. On the modules where the canonical tensor acts, it gives braided tensor products. For a topologically free Hopf deformation $H_\hbar$ of $U(\mathfrak{g})$, the first antisymmetric term

$$\delta(x) = \hbar^{-1}(\Delta - \Delta^{\mathrm{op}})(\widetilde{x}) \pmod{\hbar}$$

defines its semiclassical cobracket on primitives, when the stated lift and topological freeness conditions hold.

The affine Yangian of $\mathfrak{gl}_1$ has independently presented negative and Cartan currents. The exceptional-curve algebra has its specified nonstandard completed positive presentation. The root Borcherds construction in Volume III has a paired Borel realization with its own topology. None of these statements identifies an arbitrary reduced compact Hall source with a full quantum group.

23.7 KZ and rational exchange

For a simple complex Lie algebra $\mathfrak{g}$, fix an invariant nondegenerate form and let Ω be its inverse tensor. On a tensor product of modules, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i}\sum_{i<j}\Omega_{ij}\, d\log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Over formal $\mathbb{C}[[\hbar]]$, the Drinfeld–Kohno comparison identifies the corresponding categories of deformations of finite-dimensional modules after a choice of associator, with $q = e^{\hbar/2}$ in this normalization [34, Section 8.27.7]. A specialization requires its own parameter and convergence hypotheses.

For the standard module of $\mathfrak{gl}_N$, rational difference exchange uses the Yangian matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

where $P(v \otimes w) = w \otimes v$. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

23.8 The centre and the double

The Hochschild centre $Z_{\mathrm{ord}}(A)$ controls central operations on the ordered algebra in its specified coefficient category. To form a Drinfeld centre of modules, first specify a monoidal module category. For an ordered chiral bialgebra satisfying Definition 23.9, use

$$\mathcal{M}_A = \mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$$

with the monoidal product of Theorem 23.10. An object of $Z(\mathcal{M}_A)$ is a module M together with natural equivalences

$$c_{M,N} : M \widehat{\otimes}^{!} N \xrightarrow{\sim} N \widehat{\otimes}^{!} M \qquad (N \in \mathcal{M}_A)$$

satisfying the half-braiding hexagon and unit identities, with their coherences. This categorical centre is braided. A comparison with the Hochschild centre must specify a realization functor, the source and

target of the comparison, and the operations it preserves. Dualizability alone does not supply such a comparison. Derived loop-space descriptions apply in the geometric categories and under the hypotheses of their respective centre theorems [7].

The centre and the double answer complementary questions. The centre classifies central actions in the chosen category. The double reconstructs a quantum group from paired positive and negative halves. A field-theory comparison requires explicit realizations of the relevant line and bulk operations and a map compatible with both constructions.

Chapter 24

Geometric actions on K3 moduli

24.1 The Mukai lattice

For a K3 surface S, the Mukai lattice is

$$\widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$$

with pairing

$$((r,c,s),(r',c',s')) = c\cdot c' - rs' - r's.$$

It is even unimodular of signature $(4,20)$ and rank 24. The Heisenberg commutator below uses the Poincaré pairing $\int_S \alpha\beta$, which has different signs on $H^0 \oplus H^4$. These two pairings are not identified without that sign change.

24.2 The Nakajima Heisenberg algebra

Let

$$\mathbb{H}_S = \bigoplus_{n\geq 0} H^*(S^{[n]}).$$

Use rational cohomology. For $m \in \mathbb{Z}\setminus\{0\}$ and $\alpha \in H^*(S)$, normalize the Nakajima operators $p_m(\alpha)$ so positive modes annihilate the vacuum and the following commutator holds.

Theorem 24.1 (K3 Heisenberg action)**.** *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m\,\delta_{m+n,0}\int_S \alpha\cup\beta\ \mathrm{id}\,.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^*(S^{[0]})$.

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha\beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [74]. □

This is a nonabelian central extension: opposite modes have the displayed scalar commutator. Each creation half is abelian. Its Fock representation is a global K3 construction; a larger action must contain it through an actual homomorphism.

24.3 The orthogonal LLV algebra

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 24.2 (Orthogonal LLV symmetries)**.** *Over* $\mathbb{R}$*, for a K3 surface* S*, the LLV algebra acting on* $H^*(S,\mathbb{R})$ *is*

$$\mathfrak{g}_{\mathrm{LLV}}(S) \cong \mathfrak{so}\big(\widetilde{H}(S), (-,-)_{\mathrm{Muk}}\big) \cong \mathfrak{so}(4,20).$$

For an irreducible hyperkähler manifold M *of* $K3^{[n]}$ *type with* $n \geq 2$*, one has* $b_2(M) = 23$ *and*

$$\mathfrak{g}_{\mathrm{LLV}}(M) \cong \mathfrak{so}(4,21).$$

Smooth compact hyperkähler moduli spaces of stable objects on S *carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the K3 lattice.*

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S, the extension has signature $(4,20)$. For $K3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3,20)$, so adjoining the hyperbolic plane gives signature $(4,21)$ [68, 85]. □

Over $\mathbb{Q}$ the orthogonal Lie algebra is specified by its rational quadratic form, rather than by a real signature alone. The LLV algebra is a classical cohomological symmetry. A Yangian realization of the full orthogonal action requires a current algebra, its action on the stated cohomology, and a compatible spectral coproduct.

24.4 Local ADE Yangians

An ADE configuration of (-2)-curves on S gives a root sublattice of $H^2(S,\mathbb{Z})$. An identification of the supported categories and equivariant coefficient systems with the resolution model gives its completed positive affine ADE Yangian. Such an identification is additional geometry, beyond the intersection matrix. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus the ADE models provide local nonabelian Yangian sectors when these geometric comparisons are given. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

24.5 Global and local K3 algebras

The global zero-dimensional CoHA is $W^+(S)$, and it admits the associative double $W^{(c)}(S)$ of [72]. This algebra has positive and negative current halves and a commutative Cartan part, with a linear triangular decomposition

$$W^-(S) \otimes W^0(S) \otimes W^+(S) \simeq W^{(c)}(S).$$

At an integer central parameter $c = r$ the algebra is defined over $\mathbb{Q}$. Its positive currents $T_n^+(\lambda)$ have bidegree $(1, 2n-2+\deg\lambda)$; negative currents have first degree -1. The algebra is defined by the current presentation, including its mixed relation, in [72, Definition 3.10].

The Heisenberg embedding has

$$[q_m(\lambda), q_n(\mu)] = m\delta_{m+n,0}\, c \int_S \lambda\mu.$$

For $c_1(S) = 0$, the colored Virasoro central form is $\eta(\lambda,\mu) = c\int_S c_2(S)\lambda\mu$; in the convention of [72, Proposition 3.20],

$$[L_m(\lambda), L_n(\mu)] = (n-m)L_{m+n}(\lambda\mu) - \frac{n^3-n}{12}\delta_{m+n,0}\eta(\lambda,\mu).$$

This is the opposite of the usual Virasoro sign convention. On a K3, $\eta(1,1) = 24c$.

A two-sided Hecke pattern is a family of moduli spaces, open in the specified sheaf stacks, stable under the elementary modifications used for both creation and annihilation. With the regularity and properness hypotheses of [72, Proposition 6.8], a rank-r pattern carries an action of $W^{(r)}(S)$ on its tautological classes. The virtual-class formulation gives the corresponding action without the regularity assumption in that theorem. The Hilbert-scheme case recovers the Nakajima operators. Separately, the universal Fock-type module of [72, Proposition 4.8] is faithful for $r > 0$. It is neither irreducible nor a highest-weight module; the geometric Hilbert-scheme representation can factor through a central quotient of the double.

These are global associative algebra and representation results. They do not assert a Hopf double or a universal R-matrix for $W^{(c)}(S)$, nor an embedding of the full LLV algebra into it. The following comparisons retain their separate carriers:

Structure	Carrier	Operation
Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation, annihilation
LLV	$H^*(S)$ or $H^*(M)$ for hyperkähler M	Lefschetz commutators
Global $W^{(r)}(S)$	specified two-sided Hecke patterns	positive and negative currents
Local ADE Yangian	equivariant exceptional-curve moduli	completed Hecke currents
Affinized BPS algebra	surface Hall pro-cohomology	Euler-twisted coproduct

Maps between these actions must compare their state spaces, coefficients and mixed commutators.

24.6 K3 CORRESPONDENCES ON FIXED COEFFICIENTS

Fix rational cohomology and a collection $\mathscr{M}_S$ of smooth projective moduli spaces on a projective K3 surface S. Include the Hilbert schemes $S^{[n]}$ and close the collection under finite products. For $M, N \in \mathscr{M}_S$, define the graded kernel space

$$K(M,N) = H^*(M \times N, \mathbb{Q})[2\dim_{\mathbb{C}} M].$$

A kernel of degree r acts from $H^j(M)$ to $H^{j+r}(N)$ by

$$T_\alpha(v) = (p_N)_*(p_M^* v \cup \alpha).$$

The shift in $K(M,N)$ accounts for integration over M. All pushforwards in this construction are proper.

Proposition 24.3 (Kernel composition)**.** *The product*

$$\beta \circ \alpha = (p_{13})_*(p_{12}^*\alpha \cup p_{23}^*\beta)$$

defines an associative graded category with diagonal identities. The kernel action identifies $K(M,N)$ with $\mathrm{Hom}^*(H^*(M), H^*(N))$.

Proof. The Künneth isomorphism and Poincaré duality identify $H^*(M \times N)[2\dim M]$ with $H^*(M)^\vee \otimes H^*(N)$. Under this identification, T_α is the associated linear map. Integration over the middle factor contracts its dual pair, so $T_{\beta\circ\alpha} = T_\beta T_\alpha$. Faithfulness then proves associativity and the diagonal identity. External products give tensor products with their Koszul signs. □

A geometric symmetry is specified by selected kernels in this category. Nakajima incidence kernels connect different Hilbert schemes. Lefschetz multiplication and its adjoint act on each individual cohomology space. Using every cohomology kernel would instead give all linear maps; the selected geometric kernels carry the representation-theoretic content.

24.7 The algebra generated by geometric actions

Put $V_S = \bigoplus_{M \in \mathscr{M}_S} H^*(M, \mathbb{Q})$. Choose a set of homogeneous operators T_λ on V_S, each sending a vector with finite support to a vector with finite support. The Nakajima operators on the Hilbert-scheme summand satisfy this condition: each changes the point number by a fixed integer. Individual Lefschetz operators, extended by zero on other summands, also satisfy it. Let F be the free unital graded algebra on symbols t_λ.

Proposition 24.4 (Geometric action algebra)**.** *The map*

$$\rho : F \longrightarrow \mathrm{End}_{\mathbb{Q}}(V_S), \qquad t_\lambda \longmapsto T_\lambda$$

defines a faithful action of $A_S = F/\ker\rho$. For every algebra B, maps $A_S \to B$ are exactly maps $F \to B$ that kill $\ker\rho$*. Products and commutators of the selected kernels are their convolution products and commutators on the relevant moduli spaces.*

Proof. Finite words in the selected operators preserve finite support, so ρ is defined by composition. The quotient acts faithfully by definition of its kernel. The factorization property is the universal property of an algebra quotient. The convolution assertion follows from Proposition 24.3. □

An equivariant ADE construction uses a separate coefficient field, for example $\mathrm{Frac}\, H^*_T(\mathrm{pt}, \mathbb{Q})$, and its prescribed localization and proper pushforwards. Joining it to A_S requires maps between the actual state spaces that intertwine those operators and their coefficients. A completion of A_S requires a specified topology and continuity of every generator. The indefinite Mukai norm alone supplies neither condition. The desired common K3 quantum algebra must construct these local-to-global actions and their mixed convolution relations.

24.8 The three K3 brackets

The Heisenberg bracket is the excess intersection of Nakajima incidence kernels. The LLV bracket is the commutator of Lefschetz kernels and their adjoints. Write $\mathrm{Corr}_{K3}(S)$ for the rational kernel category constructed above. An ADE Yangian initially lives in its equivariant coefficient category. When an actual coefficient and state comparison puts its operators on a common carrier, mixed commutators are differences of the corresponding triple pushforwards, with Koszul signs. The action algebra retains these operators and relations. The coend additionally requires the specified rigid monoidal category and fibre functor.

Chapter 25

Protected roots, cyclic operations and characters

25.1 Finite quivers and Calabi–Yau completions

For a finite quiver Q with zero potential, let $\Gamma(Q, 0)$ be its path-length-completed Ginzburg algebra. Path products are read from left to right: if $a : i \to j$ and $b : j \to k$, then $ab : i \to k$. For each arrow $a : i \to j$, add $a^* : j \to i$ of degree -1. At each vertex add a loop t_i of degree -2. Original arrows have degree zero, and

$$da = da^* = 0, \qquad dt_i = e_i\Big(\sum_a [a, a^*]\Big)e_i.$$

Proposition 25.1 (Restriction to a full subquiver)**.** *If Q_0 is a full subquiver, retaining its generators and killing all other vertices, arrows, reverse arrows, and loops defines a continuous dg-algebra projection*

$$\Gamma(Q, 0) \longrightarrow \Gamma(Q_0, 0).$$

The inclusion of retained graded generators need not commute with the differential.

Proof. The projection respects source and target idempotents and extends multiplicatively. At a retained vertex, every term involving a discarded arrow vanishes. The other terms are precisely dt_i in the smaller algebra. All arrow differentials vanish, so the projection is a chain map. It preserves the path-length filtration and is continuous. For the final assertion, start with one vertex i and no arrows, where $dt_i = 0$. Add a vertex j and an arrow $a : i \to j$. In the larger algebra $dt_i = aa^*$, a nonzero length-two path. Thus the graded inclusion is not a dg map. □

The algebraic Ginzburg algebra for this finite quiver is homologically smooth and three-Calabi–Yau [57, Theorem 6.3]. Its cyclic class comes from the potential; zero potential satisfies the negative-cyclic lift condition [58]. The path-length-completed algebra is topologically homologically smooth and three-Calabi–Yau in the pseudocompact bimodule category [59, Theorem A.17]. Its enveloping tensor product and inverse-dualizing complex are continuous. This identifies the finite completed model without replacing those tensors by ordinary algebraic ones. The displayed projections specify an inverse system for compatible zero-potential windows. A Calabi–Yau structure on a completed limit requires its own nondegeneracy and continuity argument. Comparisons with Hall categories and quantized Borcherds algebras must also preserve orientations, root relations, and the paired Hopf data.

25.2 The protected input

The K3 elliptic genus is $2\phi_{0,1}$. Use the normalized denominator $D_5 := D_5$ of Volume III. The Borcherds half determines this weight-five form and its Borcherds Lie superalgebra $\mathfrak{g}_\Delta$. Fix its positive root spaces with their parity and proper height. The positive algebra and its root Fock space are defined from this Lie superalgebra, independently of a compact Hall realization.

For each finite zero-potential quiver window $Q_{\Delta,H}$, form the Ginzburg algebra $\Gamma_{\Delta,H}$. Restriction to full subquivers gives the projections of Proposition 25.1. Write Γ_Δ for this specified inverse system. A nondegenerate continuous Calabi–Yau structure on a completed limit is additional data. An ordinary Hall category $\mathcal{B}_\Delta$ must carry its chosen quiver coefficients and extension correspondences. Its identification with a root presentation requires the corresponding Hall theorem on that precise datum.

Definition 25.2 (Root-datum realizations)**.** A protected root-datum realization records

$$\mathfrak{P}(K_3 \times E) = (\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \widehat{U}_q(\mathfrak{g}_\Delta), \mathbb{A}_Z).$$

Here $\mathcal{H}_{\mathrm{ip}} = \bigoplus_{\alpha>0}(\mathfrak{g}_\Delta)_\alpha$ with its parity. The current-factorization, paired quantum-group, and Fredholm entries retain their own coefficient categories, completions, and defining hypotheses. Maps among these entries are specified realization maps, not consequences of using the same root labels.

The enveloping and Chevalley constructions are functorial under the primitive maps of Theorem 25.7. The root Fock trace is computed from the actual graded root spaces. A Fredholm realization adds a family of operators on a specified domain and the convergence and orientation data identifying its Weyl-shifted Pfaffian with the denominator. These are the analytic inputs to the square-root comparison below.

Definition 25.3 (Integral protected realization)**.** Let $\mathfrak{p}^+_X$ be a BPS primitive Lie algebra supplied by a selected reduced cohomological Hall realization, with an actual BPS comparison or coproduct. The integral protected realization is

$$\mathfrak{P}_{\mathrm{int}}(X, \Delta) = \left(\mathcal{H}^{E,\mathrm{mot}}_X, \mathfrak{P}(K_3 \times E), \mathfrak{b}_{X\boxtimes\Delta}, r_X, r_\Delta, I^E_{X,\Pi}\right).$$

The motivic component and its regular reduced character are defined as above; the root realizations retain their independent hypotheses. The additional BPS or coproduct construction supplies $\mathfrak{p}^+_X$; compatible additive Cartan weights then give the amalgamated component.

Proposition 25.4 (Primitive graph comparison)**.** *Suppose the two primitive Borels, their common Cartan action, and their completed amalgam are supplied as above. Their Cartan retractions induce retractions from the amalgam to each factor. For each specified primitive map φ preserving Cartan action, its graph quotient is canonically the geometric Borel.*

Proof. The identity on one Borel and the Cartan projection of the other agree on the common Cartan, so the amalgam gives the retraction. The quotient statement is Proposition 23.4. □

The motivic integration map and primitive comparison are different maps. Strong orientation gives the coefficient coherence of a cohomological Hall construction under its stated geometric hypotheses. Extracting a primitive Lie algebra additionally requires the appropriate BPS or coproduct construction.

25.3 Cyclic open-string operations

Let A be a finite-dimensional, uncurved cyclic A_∞ algebra over $\mathbb{C}$, with a nondegenerate pairing of degree -3. Use the cohomological suspension $s : A \to A[1]$ of degree -1. The suspended operations $b_n : (A[1])^{\otimes n} \to A[1]$ have degree one. Our sign convention is the coderivation convention $b^2 = 0$ on the tensor coalgebra, with the Koszul rule for moving homogeneous inputs. Write ω for the resulting degree--1 symplectic form on $A[1]$. Fix the Hamiltonian convention $\iota_{X_F}\omega = dF$ and $[X_F, X_G] = X_{\{F,G\}}$.

For the universal degree-zero coordinate x on $A[1]$, put

$$S(x) = \sum_{n\geq 1} \frac{1}{n+1}\omega(x, b_n(x^{\otimes n})).$$

Cyclic invariance is read with the suspended Koszul signs. All functions are formal power series at the origin. An equivalent unsuspended convention writes this as $\frac{1}{2}\langle a, m_1 a\rangle + \sum_{n\geq 2} \frac{1}{n+1}\langle a, m_n(a^{\otimes n})\rangle$.

Proposition 25.5 (Cyclic master equation)**.** *The action satisfies* $\{S, S\} = 0$.

Proof. Cyclicity makes the $n+1$ derivatives of the arity-$n+1$ term equal, with the suspended Koszul signs. Thus $X_S(x) = \sum_{n\geq 1} b_n(x^{\otimes n})$. The coderivation identity implies $X_S^2 = 0$. The Hamiltonian identity gives $X_{\{S,S\}} = 2X_S^2 = 0$. Nondegeneracy of ω makes $\{S, S\}$ constant. Uncurvedness removes b_0, so S starts in order two and its bracket has no constant term. The constant is therefore zero. □

A finite-height Ginzburg realization requires such a cyclic model on the selected proper objects before this scalar action is applied. A Hall realization additionally requires a moduli problem with critical coefficient complexes. For an extension $A_0 \to F \to B_0$, put $L(A_0, B_0) = \det R\,\mathrm{Hom}(A_0, B_0)$. Three-Calabi–Yau duality gives

$$R\,\mathrm{Hom}(B_0, A_0) \simeq R\,\mathrm{Hom}(A_0, B_0)^\vee[-3].$$

Applying the determinant functor to the two derived Hom triangles gives

$$\det R\,\mathrm{Hom}(F, F) \cong \det R\,\mathrm{Hom}(A_0, A_0) \otimes \det R\,\mathrm{Hom}(B_0, B_0) \otimes L(A_0, B_0)^{\otimes 2}.$$

Indeed, the four subquotients are the two diagonal Hom complexes and the two cross complexes. Dualization and the odd shift cancel on their determinant lines, so the two cross determinants coincide. Orientation lines require square-root isomorphisms along this formula, compatible on triple flags and with any quotient descent. The cyclic master equation does not select these isomorphisms.

25.4 An algebraic open–closed comparison

A finite-dimensional ordinary Lie algebra $\mathfrak{l}$ gives a precise coefficient model for an enveloping-algebra centre. Let $U = U(\mathfrak{l})$ over $\mathbb{C}$. The adjoint module U^{ad} has action $x \cdot u = xu - ux$. Its Chevalley cochains are

$$C^n(\mathfrak{l}, U^{\mathrm{ad}}) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, U),$$

with differential

$$\begin{aligned}(df)(x_1, \dots, x_{n+1}) = &\sum_i (-1)^{i+1}[x_i, f(x_1, \widehat{x_i}, \dots, x_{n+1})] \\ &+ \sum_{i<j} (-1)^{i+j} f([x_i, x_j], x_1, \widehat{x_i}, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

The second line places the bracket first and retains the order of the remaining arguments.

Theorem 25.6 (Hochschild–Chevalley comparison)**.** *Restriction followed by antisymmetrization defines a quasi-isomorphism*

$$C^\bullet_{\mathrm{HH}}(U, U) \longrightarrow C^\bullet(\mathfrak{l}, U^{\mathrm{ad}}), \qquad \phi \longmapsto \left[(x_1, \dots, x_n) \mapsto \sum_{\sigma\in S_n} \mathrm{sgn}(\sigma)\phi(x_{\sigma(1)}, \dots, x_{\sigma(n)})\right].$$

This is a comparison of ordinary cochain complexes.

Proof. The Chevalley bimodule resolution has terms $K_n = U \otimes \Lambda^n \mathfrak{l} \otimes U$. Its differential has the left-minus-right multiplication terms and the bracket terms with the signs dual to the displayed differential. Filter K_n by total PBW degree, counting each exterior generator as degree one. The multiplication terms preserve this degree and bracket terms lower it. The associated graded is the Koszul resolution of the diagonal in $\mathrm{Sym}(\mathfrak{l}) \otimes \mathrm{Sym}(\mathfrak{l})$, with regular sequence $x_i \otimes 1 - 1 \otimes x_i$. That Koszul complex is exact away from degree zero. The filtration is bounded below and exhaustive, and each element has finite filtration degree, so lifting a leading term proves exactness of $K_\bullet$ by induction. Its augmentation has image U. Thus $K_\bullet$ is a free bimodule resolution.

Antisymmetrization gives a map from $K_\bullet$ to the ordinary bar resolution, lifting the identity of U. Adjacent products in its bar differential pair to commutators; $xy - yx = [x, y]$ gives exactly the bracket terms of $K_\bullet$. The outer products give its left-minus-right terms. A comparison between projective resolutions lifting the identity is a homotopy equivalence. Applying $\mathrm{Hom}_{U \otimes U^{\mathrm{op}}}(-, U)$ gives the stated formula and quasi-isomorphism. □

This is the classical Cartan–Eilenberg comparison. For a finite Lie superalgebra, the same construction uses the super-exterior powers and permutation Koszul signs. Odd generators produce symmetric exterior directions, so the resolution can be unbounded; its algebraic direct sums and cochain products remain part of the convention. The PBW and super-Koszul proof applies with those conventions.

A physical mixed holomorphic–topological realization requires a map from its independently defined bulk observables to this cochain object. The theorem specifies the adjoint coefficients and the actual algebraic comparison. An E_2 refinement, a current-factorization realization, and compatibility with root-height transitions require maps preserving their additional operations. A quotient of Lie algebras alone does not provide a covariant map of Hochschild centres.

25.5 Primitive comparisons and their algebraic extensions

Let $\mathfrak{n}_\Delta$ be the positive Borcherds Lie superalgebra. Choose homogeneous parity-preserving images of its simple generators in a charge-graded Lie superalgebra $\mathfrak{p}$. Assume these images satisfy every defining relation.

Theorem 25.7 (Extension of a primitive comparison)**.** *The chosen images determine a unique Lie-superalgebra map $f : \mathfrak{n}_\Delta \to \mathfrak{p}$. It induces*

$$U(f)(x_1 \cdots x_r) = f(x_1) \cdots f(x_r), \qquad C_*(f) = \mathrm{Sym}^c(f[1]).$$

These are respectively an enveloping-algebra map and a Chevalley chain-coalgebra map. Filtration-preserving maps extend to the specified inverse-limit completions. If f is surjective and both algebras have finite charge components and proper positive height, equality of their parity-refined PBW characters implies that f and $U(f)$ are isomorphisms.

Proof. The defining presentation gives f. Its tensor-algebra extension carries $xy - (-1)^{|x||y|} yx - [x, y]$ to the corresponding relation. It therefore descends to $U(f)$. The shifted symmetric-coalgebra map commutes with the Chevalley differential because f preserves brackets. Compatible maps on filtration quotients give maps on their inverse limits. Surjectivity of f gives surjectivity of $U(f)$. Character equality makes each finite charge component an isomorphism, separately in both parities. The PBW embeddings then imply injectivity of f. □

At level zero the current Lie conformal algebra is $\mathbb{C}[\partial] \otimes \mathfrak{n}_\Delta$ with $[x_\lambda y] = [x, y]$ on constant generators. The map $\partial^j x \mapsto \partial^j f(x)$ preserves this bracket and sesquilinearity. A nonzero central level additionally requires preservation of the specified invariant pairing. Hall convolution, Calabi–Yau structures, quantum coproducts, and determinant lines require their own comparison data.

25.6 The exact scalar and first-order observables

Set

$$\mathrm{Den}_\Delta(Z) = D_5(2Z), \qquad \jmath(Z) = Z/2, \qquad \jmath^*\,\mathrm{Den}_\Delta = D_5.$$

The one-particle root spaces give

$$\mathrm{sch}\,\mathrm{Fock}(\mathfrak{g}_{\Delta_5+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{-\dim \mathfrak{g}_{\Delta_5\alpha}}.$$

Write $P(Z) = \prod_{\alpha>0}(1-e^{-\alpha(Z)})^{\dim \mathfrak{g}_{\Delta_5\alpha}}$. The normalized root operator gives, on its analytic domain,

$$\mathrm{sPf}_F(\mathbb{A}_Z) = P(Z), \qquad e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}^{\mathrm{red}}_{K_3\times E} = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

Theorem 25.8 (Protected square-root theorem)**.** *Assume the stated root-space Fredholm comparison with its orientation, Weyl shift, and convergence domain. Within that realization, the denominator section* Den_Δ *is simultaneously:*

(i) *the Weyl denominator of* $\mathfrak{g}_{\Delta_5}$*;*

(ii) *the Weyl-shifted Euler character of the bar–Chevalley complex;*

(iii) *the Weyl-shifted oriented Fredholm Pfaffian of* $\mathbb{A}_Z$*;*

(iv) *the Weyl-shifted PBW determinant of the parity-refined root superspace* $\mathcal{H}_{\mathrm{rp}}$*;*

(v) *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree. Across Weyl chambers they preserve transport when the corresponding determinant-line isomorphisms and orientation choices are included in the datum.

Proof. The denominator and bar–Chevalley theorems give $e^{-\rho}P$. The Fredholm theorem gives P, and multiplication by the same Weyl factor gives the third identification. PBW for the Borcherds Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $\jmath^*\,\mathrm{Den}_\Delta = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. □

25.7 Physical meaning

The root-datum realizations retain several distinct constructions. The finite Ginzburg algebras and their projections form an inverse system. A completed three-Calabi–Yau sector requires a continuous nondegenerate Calabi–Yau structure and compatible cyclic models on the selected proper objects. The quiver Hall category retains its chosen coefficients and extension correspondences. A Hall theorem on that datum must identify its root presentation. The Borcherds root superspace defines the root-graded Fock space and its supercharacter. Its identification with physical one-particle states requires a parity-preserving graded comparison intertwining the primitive actions.

A realization on the elliptic curve must construct the current object with its local and wrapped supports, collision operations, and comparison maps. The quantum Borcherds algebra, with its specified pairing and completion, acts as a line-operator symmetry only through a representation preserving fusion and exchange. The root operator gives the Fredholm product on its specified convergence domain. A

physical determinant interpretation requires an operator comparison and an orientation-preserving identification of the determinant lines and sections. The Weyl-normalized doubled Fock character gives the inverse scalar denominator. An operator trace additionally requires its state space and trace domain.

Simple-root states satisfying the defining relations give a primitive Lie map. The explicit enveloping and Chevalley extensions preserve products and brackets. A compact string realization must additionally construct its brane functor, Hall orientation maps, paired quantum coproduct, and trace-line comparison.

25.8 Borcherds products as characters

Let $\mathfrak{n}_+$ have finite-dimensional charge spaces and a proper positive height. The alternating supercharacter of its Chevalley chains, using homological sign and intrinsic parity, is

$$\chi_s C_*(\mathfrak{n}_+) = \prod_{\alpha>0} (1 - e^{-\alpha})^{\mathrm{sdim}(\mathfrak{n}_+)_\alpha}.$$

A Borcherds product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

25.9 The universal Borcherds weight formula

For a weight-zero weak Jacobi form of positive integral index satisfying the integrality hypotheses of the multiplicative Borcherds lift,

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borcherds lift has weight

$$\mathrm{wt}(\mathrm{Borch}(\varphi)) = \frac{c(0,0)}{2}.$$

For the K3 half-index, $c(0,0) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

25.10 The Gritsenko–Cléry atlas

The eight diagonal-divisor forms are

$$\Delta_5, \Delta_2, \Delta_1, \Delta_{1/2}, \nabla_3, \nabla_2, \nabla_{3/2}, Q_1$$

with weights

$$5, 2, 1, \frac{1}{2}, 3, 2, \frac{3}{2}, 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [19]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

25.11 Scalarization hierarchy

Once the displayed comparison maps are constructed, one can pass through the following recipients:

$$\begin{aligned}\text{factorization or Hall object} &\longrightarrow \text{primitive Lie algebra}\\ &\longrightarrow \text{PBW character} \longrightarrow \text{scalar partition function}.\end{aligned}$$

Primitive extraction requires a coproduct or a BPS comparison; the last arrow requires a state, trace and normalization. These are not automatic arrows for an arbitrary factorization algebra. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

Chapter 26

Cotangent theories on K3 times E

26.1 Holomorphic cotangent BF theory on $K3 \times E$

Let S be a K3 surface with holomorphic symplectic form σ_S, let E be an elliptic curve with holomorphic form $\mathrm{d}z$, and put

$$X = S \times E, \qquad \Omega_X = \sigma_S \wedge \mathrm{d}z.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ of the positive Igusa Borcherds algebra, take fields

$$A \in \Omega^{0,\bullet}(X, \mathfrak{n}_{\Delta_5,+}{}^{(H)})[1], \qquad B \in \Omega^{0,\bullet}(X, (\mathfrak{n}_{\Delta_5,+}{}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A,B) = \int_X \Omega_X \wedge \langle B, \bar\partial A + \frac{1}{2}[A,A]\rangle. \tag{26.1}$$

The two shifts are both $[1]$ in complex dimension three. After integration against Ω_X, evaluation has degree -1 on these fields. This is the holomorphic cotangent theory of the dg Lie superalgebra of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$-valued Dolbeault forms.

Theorem 26.1 (Classical compact-background source)**.** *The action* (26.1) *satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X.*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ together with $\bar\partial^2 = 0$ and the derivation property of $\bar\partial$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. □

26.2 Loop weights and the quantum master equation

Every interaction vertex in (26.1) has one B-leg and two A-legs. The propagator pairs an A-leg with a B-leg.

Lemma 26.2 (Loop bound)**.** *A connected graph with at least one interaction vertex and the stated propagator has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B-half-edges. Every internal edge consumes one of them, hence $E \le V$. The loop number is $E - V + 1 \le 1$. □

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero supertrace.

Proposition 26.3 (Vanishing of one-loop Lie weights)**.** *For every finite positive-height quotient, the Lie weight of each one-loop graph with at least one interaction vertex vanishes.*

Proof. A one-loop graph has $E = V$. Every B-leg is consumed by an internal edge, so its external legs are A-legs. Its Lie weight is a cyclic supertrace of adjoint operators, or the dual supertrace with the corresponding Koszul signs. Each external label has positive root height. The product therefore strictly raises height on the finite-dimensional adjoint representation and has zero supertrace. The same argument applies to the dual representation. □

This vanishing removes the one-loop Lie coefficients in a perturbative construction with the stated propagator and vertex types. A renormalized quantum factorization algebra still requires admissible analytic kernels, local extensions, the quantum master equation, and compatible height transitions on the chosen observable spaces. These are the remaining quantum realization data. A quotient of positive Lie algebras induces an inclusion on linear duals. Thus the A- and B-field directions do not automatically give a same-direction tower of cotangent theories or quantum observables.

26.3 Classical reduction along K3

Choose a Kähler metric on S and normalize a closed harmonic $(0,2)$-form η by $\int_S \sigma_S \wedge \eta = 1$. Let $K = \mathbb{C}[\eta]/(\eta^2)$ with $|\eta| = 2$ and zero differential. There is a unital dg-algebra map

$$i : K \longrightarrow \Omega^{0,*}(S), \qquad \eta \longmapsto \eta.$$

Its image is closed under multiplication because $(0,4)$-forms vanish. The Dolbeault cohomology of a K3 surface is one-dimensional in degrees zero and two and zero in degree one, so i is a quasi-isomorphism. It preserves the trace defined by integration against σ_S.

Let $\mathfrak{l}$ be a finite-dimensional Lie superalgebra, and write $\mathfrak{d} = \mathfrak{l} \ltimes \mathfrak{l}^*$ for its cotangent Lie superalgebra, with its invariant graded evaluation pairing. For an open $U \subset E$, tensor with $\Omega^{0,*}(U)$, using the completed smooth tensor product in the S direction. The field comparison is

$$K \otimes \Omega^{0,*}(U) \otimes \mathfrak{d} \xrightarrow{i \otimes \mathrm{id}} \Omega^{0,*}(S \times U) \otimes \mathfrak{d}.$$

The bracket is multiplication of forms followed by the Lie bracket, with Koszul signs.

Proposition 26.4 (Classical field comparison)**.** *The displayed map is a trace-preserving dg Lie quasi-isomorphism, compatible with restriction in U. Its induced minimal operations in the K3 factor have no higher products.*

Proof. Multiplicativity of i proves bracket compatibility. Hodge projection p and $h = \bar\partial^* G$ on the compact surface satisfy $pi = \mathrm{id}$ and $\bar\partial h + h\bar\partial = \mathrm{id} - ip$. Tensoring these continuous operators with forms on U gives the same contraction, with the tensor-product signs. Thus $i \otimes \mathrm{id}$ is a quasi-isomorphism. Restriction acts only on the U factor and commutes with it. The normalized trace on K is the restriction of the Dolbeault trace. Choose the contraction with $hi = 0$. In every higher transferred tree, the first product of inputs from $i(K)$ remains in $i(K)$ and the next internal homotopy kills it. Thus every higher transferred product vanishes. □

The coefficient algebra K retains both the degree-zero and degree-two K3 modes. The proposition compares classical field complexes and their brackets. Constructing an equivalence of quantum observable factorization algebras requires compatible quantization, support conventions, and the induced maps on observable completions. The ordinary Hochschild comparison of Theorem 25.6 is a separate algebraic calculation, not that quantum field map.

26.4 Protected states and the Igusa trace

The formal root Fock object is

$$\widehat{\mathrm{Sym}}(\mathcal{H}_{\mathrm{ip},\bar{\mathrm{o}}})\widehat{\otimes}\widehat{\Lambda}(\mathcal{H}_{\mathrm{ip},\bar{\mathrm{i}}}), \qquad \mathcal{H}_{\mathrm{ip}} = \bigoplus_{\alpha>0} \mathfrak{g}_{\Delta_5\,\alpha},$$

completed in proper positive height. Its character is a formal product. For an operator trace, give each finite root space a Hilbert norm, take their Hilbert direct sum, and form the bosonic and fermionic Hilbert Fock spaces. Put $q_\alpha = e^{-\alpha(Z)}$ and let e^{-H_Z} act diagonally on one-particle states.

Theorem 26.5 (Protected Hamiltonian trace)**.** *Assume*

$$\sum_{\alpha>0} \dim(\mathfrak{g}_{\Delta_5\,\alpha})|q_\alpha| < \infty, \qquad \sup_{\alpha>0} |q_\alpha| < 1.$$

Then its second quantization is trace class on the Hilbert Fock space, and its supertrace is

$$\mathrm{str}(e^{-H_Z}) = \prod_{\alpha>0} (1 - q_\alpha)^{-\operatorname{sdim} \mathfrak{g}_{\Delta_5\,\alpha}} = P(Z)^{-1}.$$

The all-parity estimates of Volume III give a nonempty domain satisfying these hypotheses. With $\mathrm{Den}_\Delta = e^{-\rho} P$*, the primitive reduced scalar is*

$$-\big(\jmath^* \mathrm{Den}_\Delta\big)^{-2} = -\chi_{10}^{-1}, \qquad \jmath(Z) = Z/2,$$

in the Oberdieck–Pixton normalization and primitive K3 curve sector.

Proof. The absolute trace of each even oscillator is $(1 - |q_\alpha|)^{-1}$, while that of each odd oscillator is $1 + |q_\alpha|$. The two hypotheses make the product of these absolute traces finite, by the logarithm bounds on a disk of radius less than one. Thus tensoring over the root basis gives a trace-class operator. The supertrace replaces the odd factor by $1 - q_\alpha$, giving P^{-1}. The denominator identity is $\mathrm{Den}_\Delta = e^{-\rho} P$. Its half-coordinate pullback is D_5, whose square is χ_{10}; the primitive reduced curve-counting theorem supplies the final minus sign [75]. □

This analytic trace is formed on the Hilbert completion just specified. The formal product completion is a different state object. Identifying either with a compact physical state space requires a representation and trace comparison.

26.5 The compact-source comparison

The compact threefold provides the classical cotangent field theory and the dg Lie comparison of Proposition 26.4. The root datum separately gives the Borcherds algebra, its enveloping algebra, and the root-graded Fock space. The latter has the trace computed above. Proposition 26.3 supplies the finite one-loop Lie cancellation, while Theorem 25.6 computes the ordinary algebraic centre model with adjoint coefficients.

Problem 26.6 (Geometric compact comparison)**.** Construct objects E_t representing the simple branes and maps on their derived morphism complexes that preserve units, composition, and higher composition homotopies. Realize these maps on extension stacks with the reduced critical coefficients and orientation isomorphisms on triple flags. Construct compatible tensor and unit constraints on the resulting boundary functor. Compare the chosen quantum bulk action with the independently defined boundary centre, including support, collisions, and completion. Finally construct a trace and a map of its value line to the automorphic line in the same charge normalization.

A primitive map satisfying the root relations gives the enveloping and Chevalley maps of Theorem 25.7. A strong monoidal equivalence transports categorical centres. Neither construction determines the brane functor or its cyclic trace. These are the additional maps required for compact geometric reconstruction.

26.6 Holomorphic–topological boundary theories

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A factorizing boundary family with compatible composition gives the ordered algebra constructed above. Local constancy and coherent operations in two real directions give an E_2 structure when topological factorization recognition applies. A meromorphic line category also needs its holomorphic parameter and expansion data. A coherent bulk action then factors through the stated centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}).$$

For $\mathfrak{l} = \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, it computes the finite-height root Chevalley character. Identifying it with boundary observables requires the boundary construction; the automorphic denominator also carries its Weyl factor.

26.7 Six-dimensional hCS on $K3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{0,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS}. Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S.

A protected reduction to a chiral algebra consists of a deformation retract of the K3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a factorizing boundary or defect family with the coefficient comparisons above, and an admissible quantization solving the quantum master equation. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S\times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

26.8 Omega backgrounds

Let a torus act on the local geometry and suppose the cochain model has operators Q of degree 1 and ι_V of degree -1, commuting with $\mathcal{L}_V$ and satisfying

$$Q^2 = 0, \qquad [Q, \iota_V] = \mathcal{L}_V, \qquad \iota_V^2 = 0.$$

For an even parameter u of cohomological degree two, put

$$Q_\Omega = Q + u\iota_V, \qquad Q_\Omega^2 = u\mathcal{L}_V.$$

It is a differential on the invariant subcomplex. A general holomorphic differential does not carry these contraction operators without additional data.

Equivariant localization in suitable toric or ADE models supplies deformation parameters. A spectral R-matrix still requires its correspondences and tensor identities. A compact K3 has no nonzero holomorphic vector field: its symplectic form identifies such fields with holomorphic one-forms, and $H^0(S, \Omega^1_S) = 0$. Consequently a localized model reaches a compact generic K3 only through a specified deformation family and coefficient comparison, with monodromy and wall-crossing compatibility.

Chapter 27

Anomalies and reconstruction

27.1 Anomalies as typed classes

The constructions have the following obstruction recipients:

Construction	Recipient
hCS quantization	local BV cochains
Critical CoHA orientation	Picard and gerbe theory of the critical stack
Hall coproduct	specified coefficient maps on flag correspondences
Projective conformal blocks	Atiyah algebra of the determinant line
Automorphic multiplier	character group of the modular cover

Some recipients are complexes, others are Picard groupoids, gerbes or character groups. A comparison must first supply maps in the appropriate categories; when an additive deformation-complex model exists, those maps must identify the transported obstruction classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

27.2 The foundational theorem

Definition 27.1 (Complete Calabi–Yau quantum-group datum)**.** A complete comparison datum specifies independent categorical, field, boundary, Hall, Hopf, and trace objects together with:

(i) PTVV-representable moduli for the smooth proper Calabi–Yau category;

(ii) a renormalized BV quantization satisfying the quantum master equation and a factorizing family of local compact generators b_I;

(iii) a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;

(iv) a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;

(v) paired negative and Cartan halves, a completed coproduct, a nondegenerate continuous Hopf pairing with its opposite-dual convention, a continuous coevaluation or locally perfect dual pieces, and a separated triangular completion with compatible Cartan reduction;

(vi) compatible open–closed and Hall–boundary morphisms;

(vii) a parity-refined primitive comparison for every automorphic identification.

Theorem 27.2 (Calabi–Yau quantum-group assembly)**.** *Fix a complete comparison datum as above. In particular it includes:*

(i) *a smooth proper category with compatible cyclic Calabi–Yau data and the representable moduli-of-objects hypotheses stated above;*

(ii) *a quantized cyclic elliptic BV theory and the compatible factorizing boundary or defect family;*

(iii) *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*

(iv) *paired Hopf Borel halves with antipodes, compatible Cartan data, a Hopf pairing, and the completed-double and coevaluation hypotheses;*

(v) *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

(i) *framed* E_2 *Hochschild cochains and the shifted-symplectic moduli structure;*

(ii) *the three curve-level algebraic faces and their three bars;*

(iii) *the positive Hall algebra and, when a BPS comparison or coproduct is supplied, its specified primitive Lie algebra;*

(iv) *the completed quantum double and braided representation category;*

(v) *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*

(vi) *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [10, 11, 77]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [61]. The completed-double hypotheses give the double and its canonical R-matrix on the stated module domain. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. □

The local quiver, equivariant surface, K3 correspondence, and Igusa Borcherds constructions have the carriers stated above. A geometric $K3 \times E$ comparison requires the primitive map and determinant-line map of Problem 26.6.

27.3 Landscape of established realizations

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2/\Gamma}$	$Y_\infty^+(\widehat{\mathfrak{g}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
$K3$	BPS envelope; $W^+(S)$ for zero-dimensional sheaves	global $W^{(r)}(S)$ actions; surface bialgebra in its pro-coefficients	K3 Hall characters
$K3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_5})$ and the root-current model and classical BF theory	$-\chi_{10}{}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; a paired full double additionally requires all the Hopf, coevaluation and topology hypotheses stated above. The K3 row includes the global BPS algebra, the zero-dimensional W-algebra and its associative double actions, and the pro-cohomological surface coproduct with its functorial coefficient hypotheses. The final row records

independently constructed root, character, and classical field objects. Their geometric Hall realization requires the comparison in Problem 26.6.

27.4 Constructed comparisons and their geometric extensions

The Hall flag argument constructs associativity once its coefficient and orientation maps are defined. The tensor-action formula constructs representation fusion from a coassociative counital coproduct. The matrix-coefficient formulas construct the coend of a supplied fibre functor. The primitive presentation gives enveloping and Chevalley maps. The classical cyclic and K3 field calculations retain their actual differentials, traces, and brackets.

The quantum field realization requires admissible quantization and observable completions. The compact Hall realization requires the brane and orientation maps of Problem 26.6. The Fredholm and automorphic comparisons require their analytic domains and value-line isomorphisms. These constructions preserve the full geometric reconstruction problem beyond its finite algebraic and classical calculations.

27.5 Actions and comparison maps

For a specified boundary algebra A_b, a coherent bulk action is a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_b).$$

The target is the derived endomorphism object of the diagonal bimodule in the chosen factorization coefficient category. A Hall construction instead starts with

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Critical coefficients, orientations, and admissible pull–push maps give its product. Framed moduli supply a representation after their action correspondences and triple-flag identities have been constructed. A paired Hopf structure gives $D(H^+, H^\circ, H^-)$. Its coproduct defines tensor products of continuous modules, and a convergent universal R-matrix gives exchange.

A strong monoidal equivalence between this module category and a boundary category transports categorical centres. It does not identify that categorical centre with the object $Z_{\mathrm{ord}}(A_b)$ without a realization theorem. A trace on a specified representation then gives an amplitude. Its comparison with an automorphic section requires a map of value lines with the same normalization.

27.6 Comparison data for Calabi–Yau constructions

Comparisons among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products require the following structures:

(i) for a smooth proper d-Calabi–Yau category with compatible nondegenerate left and right cyclic structures, the categorical outputs are framed E_2 Hochschild cochains and, when the moduli stack of objects is representable in the PTVV setting, $(2-d)$-shifted symplectic moduli; the shift d does not change the Hochschild operadic level to E_n;

(ii) a geometric BV theory gives factorization observables, and a factorizing boundary family with coherent composition gives a transverse ordered chiral algebra; associative chiral convolution is recorded as a separate supplied structure;

(iii) pushforward to a curve uses an actual map $p : X \to C$; formal cycle-specialization notation alone specifies no pushforward;

(iv) extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;

(v) Hopf Borel halves, opposite dual convention, Cartan reduction, pairing, coevaluation and topology define the stated double;

(vi) K3 Heisenberg, LLV, local ADE Yangians, the global surface W-algebra and affinized BPS algebra have their stated carriers and coefficient systems; constructed homomorphisms join their actions;

(vii) the graph of the $K3 \times E$ Gram map becomes multiplicative in the polarized central extension;

(viii) point-local and wrapped sectors form a hybrid factorization problem;

(ix) the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and a compact Hall identification requires the primitive and trace-line comparisons;

(x) anomaly identities are transported through explicit maps between their complexes, Picard objects or character groups.

The categorical, geometric, Hall, and automorphic operations thus have separate carriers and comparison maps.

27.7 Typed data table

Object	Minimal data	Primary operation
CY category	compatible smooth/proper cyclic data	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^!$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\mathrm{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull–push product
Bialgebra	compatible product and coassociative counital coproduct	fusion of representations
Double	Hopf halves, duality, Cartan reduction and convergent coevaluation	universal R-matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

27.8 Low-degree formulas

Several small formulas distinguish the operations in this construction. The K3 half-index begins

$$\phi_{0,1}(\tau, z) = r^{-1} + 10 + r + O(q),$$

and its higher Fourier rows are obtained by the exact theta division through q^3 in Volume III. The constant coefficient 10 gives weight 5. The Gram polarization satisfies

$$B_\Pi(a, b) + B_\Pi(a + b, d) = B_\Pi(b, d) + B_\Pi(a, b + d),$$

which proves the extension law used above. The rational exchange matrix $1 - \hbar P/u$ satisfies the three-factor Yang–Baxter equation; its normalization for two-sided unitarity is separate, as in Volume I. The

two A_2 flag rows give the quantum Serre identity. Finally the type-II Gram matrix in the basis r_1, r_2, r_3 and its simple reflections are

$$G = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}, \qquad s_i(v) = v - (v, r_i) r_i .$$

Since $(r_i, r_i) = 2$, direct expansion gives $s_i^2 = 1$ and $(s_i v, s_i w) = (v, w)$. Thus these reflections preserve the lattice form; they do not identify it with the additive Cartan weights.

Book V

Boundary operations and three-dimensional gravity

Chapter 28

Interfaces and associative duality

A boundary comparison must preserve products, interfaces and their higher identities. Their deformation equations admit a local BV construction when a local field realization is supplied. Three-dimensional Cartan gravity provides a concrete classical theory for comparison; quantum reconstruction also involves measures, boundary conditions, states and global sectors.

28.1 Three products and one interface

Holomorphic collisions and transverse composition have different coefficient objects. To compare them, one needs maps that put both actions on a common interface. The ordinary two-sided bar describes the algebraic relation these maps must preserve.

Let X be a smooth complex curve. Fix a six-functor enhancement of $\mathcal{D}_X = D\text{-}\operatorname{Mod}(\operatorname{Ran} X)$, with pointwise tensor $\otimes^!$ and chiral convolution $\star^{\mathrm{ch}}$. A module for one tensor is not thereby a module for the other. Choose a recipient for the interface, coefficient realization functors into that recipient, and their comparison maps.

Definition 28.1 (Complete mixed boundary datum)**.** A complete mixed datum consists of a chiral Lie algebra L, an associative algebra A^{ch} for $\star^{\mathrm{ch}}$, a transverse E_1 algebra $A^{\perp}$ in the specified factorization coefficient category, and an interface K. It includes a Lie map $\iota : L \to (A^{\mathrm{ch}})_{\mathsf{Lie}}$ and actions on K of the two realized associative algebras. The actions commute coherently: their comparison homotopy and all its higher identities are part of the datum, denoted by λ. Every operation is continuous for the declared filtration. Augmentations, when required, are additional specified maps to the appropriate tensor units. They do not identify those units with a conformal vacuum or a chosen physical state. Write $\mathbb{A} = (L, A^{\mathrm{ch}}, A^{\perp}, K, \iota, \lambda)$.

The three native bars retain their recipients:

$$\operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(L), \qquad \operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}), \qquad \operatorname{Bar}^{\perp}(A^{\perp}).$$

A fourth bar describes the interface after coefficient realization. The following strict algebraic model fixes its order and signs.

Construction 28.2 (Two-sided interface bar)**.** Let A, B be augmented associative algebras over a field k, concentrated in degree zero, and let K be a left A, right B bimodule. Put $\bar{A} = \ker \epsilon_A$, $\bar{B} = \ker \epsilon_B$. The term of bidegree (p, q) is

$$C_{p,q} = (s\bar{A})^{\otimes p} \otimes K \otimes (s\bar{B})^{\otimes q}, \qquad |sa| = -1.$$

There is one left block, one interface factor, and one right block. For an element $m \in K$, on $[a_1|\cdots|a_p]m[b_1|\cdots|b_q]$, let d_L multiply adjacent a_i, a_{i+1} with sign $(-1)^{i-1}$ and act by a_p on m with

sign $(-1)^{p-1}$. Let d_R first act by b_1 on m with sign $+1$, and multiply b_i, b_{i+1} with sign $(-1)^i$. Empty sums vanish. Set

$$b|_{C_{p,q}} = d_L + (-1)^p d_R.$$

The direct-sum total complex computes $k \otimes^{\mathbb{L}}_A K \otimes^{\mathbb{L}}_B k$. For dg algebras, add the internal differential using the tensor Koszul signs. For positive weights with a lower bound on K, take the degreewise product over weights; each fixed weight has only finitely many bar lengths.

Theorem 28.3 (Interface differential). *The displayed differential squares to zero. Its identities are associativity of A, B, their module axioms on K, and commutation of the two actions.*

Proof. Two disjoint contractions in d_L^2 have opposite signs; two overlapping ones agree by associativity or the left module axiom. Thus $d_L^2 = 0$. The same argument gives $d_R^2 = 0$. The unsuspended left and right contractions commute, including at K, where the identity is $(am)b = a(mb)$. Applying d_L lowers p by one, so the two cross terms in $(d_L + (-1)^p d_R)^2$ have opposite signs. Finally, the augmented one-sided free bar resolutions of k on the two sides give the displayed derived tensor product. Their tensor product is a resolution because k is a field. Positive weights make the same proof finite at every weight. □

In a homotopy-coherent interface, the binary operations alone need not square to zero. The homotopy λ and the higher operations contribute further terms. A supplied coloured bar resolution packages their identities into a square-zero coderivation. Promoting this algebraic construction to $\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)$ requires the actual collision, restriction and coefficient maps in the chosen geometric recipient.

28.2 Low-arity mixed signs

The shortest word involving both actions is $[x]m[y]$. Its boundary is

$$b([x]m[y]) = (xm)[y] - [x](my), \qquad b^2([x]m[y]) = (xm)y - x(my) = 0.$$

The sign is the orientation of the two contraction edges. More generally,

$$\det(E_{\mathrm{ch}} \oplus E_\perp) = \det(E_{\mathrm{ch}}) \otimes \det(E_\perp);$$

interchanging two odd contraction operators changes this orientation. For $A = B = K = k[x]/(x^3)$, omitting the cross sign in the total differential leaves a nonzero multiple $2x^2$ on $[x]1[x]$ in characteristic zero. This is already a failure of the strict bimodule bar, before geometric collision terms enter.

28.3 Compatible modules and the mixed centre

An operation on a boundary theory must act on boundary-changing morphisms as well as on objects. Natural transformations retain this compatibility, and derived natural transformations retain its higher homotopies.

Assume the chosen recipient gives a presentable enriched stable category of compatible modules and the enriched endomorphism object used below exists. A compatible module is a triple $(M^{\mathrm{ch}}, M^\perp, \kappa)$ with an A^{ch} action, an $A^\perp$ action, and an interface comparison over K. Let $\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}$ be their stable category.

Definition 28.4 (Mixed derived centre). The mixed derived centre is

$$Z_{\mathrm{mix}}(\mathbb{A}) = \mathrm{End}\Big(\mathrm{id}_{\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}}\Big).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 28.5 (Universal interior action)**.** *$Z_{\mathrm{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every action already defined as a coherent natural endomorphism of the identity determines a point of this centre. The space of its factorizations through the representing endomorphism object is contractible. Morita-equivalent complete mixed data have equivalent centres.*

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Vertical composition and composition of endofunctors give the coherent interchange operations on endomorphisms of the identity, hence its E_2 structure. In a Hochschild cochain model these are the brace operations. A supplied coherent natural action defines an endomorphism of the identity. A physical bulk insertion determines such an action only after its action on modules, morphisms, and higher compositions has been constructed. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. □

28.4 Relative induction from a chiral algebra

Let $j : \mathcal{P}_{\mathrm{ch}} \to \mathcal{P}_{\mathrm{mix}}$ be a specified morphism of coloured operads in a monoidal category in which the relevant homotopy colimits exist and tensor product preserves them. A chiral algebra V is here an algebra for the source operad, with its chosen unit relations. Relative operadic composition defines

$$j_! V = \mathcal{P}_{\mathrm{mix}} \circ^{\mathbb{L}}_{\mathcal{P}_{\mathrm{ch}}} V.$$

It can be computed by the geometric realization of the relative operadic bar. This is the algebraic collar $\mathrm{Coll}(V)$ for the specified operad map.

Theorem 28.6 (Relative induction and completion)**.** *Relative induction is left adjoint to restriction:*

$$\mathrm{Map}(j_! V, B) \simeq \mathrm{Map}(V, j^* B).$$

If the selected category of complete algebras has a completion left adjoint to inclusion, then $\widehat{\mathrm{Coll}}(V) = \widehat{j_! V}$ has the same mapping property for complete targets. In particular, this holds for the positive weight completion with weight-preserving maps and compatible finite-weight quotients used in the ordinary bar construction.

Proof. Mapping the relative bar realization into B gives the totalization of maps out of its terms. Its compatibilities say exactly that the map from V respects the source-operad action. This proves the first adjunction. Compose it with the completion adjunction for the second. In the positive weight case a compatible map on every weight extends uniquely continuously to the product, and conversely a weight-preserving continuous map restricts to those maps. The equivalence is an equivalence of mapping spaces, including their homotopies. □

A regular interface and its distributive law still have to be constructed in the geometric coefficient category. The associative envelope $U_\star(L)$ of a Lie object for a specified monoidal product is also a separate construction from the Beilinson–Drinfeld chiral envelope. An operadic induction identifies them only through a supplied comparison.

28.5 The Koszul–Morita square

A change from modules to comodules must transport the interface as well as each algebra. The mixed bar supplies the comparison object. Exchange then asks for motion in an additional topological direction, while BRST reduction asks whether the differential respects every transported operation.

Let P be a dualizable boundary state for $A^\perp$. Its boundary dual is

$$A^!_P = \mathrm{RHom}_{A^\perp}(P, P).$$

The ordinary bar construction gives explicit twisted tensor functors. Their extension to the mixed datum must also transport the interface and its geometric operations.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w\geq 0} A_w$ be an augmented associative algebra in cohomological degree zero. Assume $A_0 = k$, augmentation projects onto A_0, and multiplication adds weights. Set $A = \prod_{w\geq 0} A_w$ and $C = \prod_{w\geq 0} B(A_{\mathrm{alg}})_w$, with degreewise products and the weight topology. The bar suspension lowers cohomological degree by one, with $b_2[a|b] = [ab]$. The degree-one twisting map $\tau : C \to A$ sends $[a]$ to a and vanishes on the empty word and longer words.

A module or comodule has the form $\prod_{w\geq w_0} M_w$, with differential preserving weight, for some integer w_0. Actions and coactions preserve weight. Comodule coactions are counital and coassociative, and use the tensor product

$$(C\widehat{\otimes}M)_w = \bigoplus_{u+v=w} B(A_{\mathrm{alg}})_u \otimes M_v.$$

The same convention defines $A\widehat{\otimes}M$. These sums are finite in each weight. Repeated reduced coactions vanish in weight w after more than $w - w_0$ steps. Invert quasi-isomorphisms in each weight to obtain $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$.

Proposition 28.7 (Ordinary twisted tensor equivalence)**.** *The functors*

$$F(M) = C\widehat{\otimes}_\tau M, \qquad G(N) = A\widehat{\otimes}_\tau N$$

induce inverse equivalences $D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C)$. Writing $\Delta c = \sum c_{(1)} \otimes c_{(2)}$ and $\rho n = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a\tau(n_{(-1)}) \otimes n_{(0)}.$$

The counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The twisting equation makes both differentials square to zero. The action, coaction, and counit identities show that the displayed maps are chain maps and satisfy the triangle identities. For $GF(M)$, the extra degeneracy of the two-sided bar inserts $1[\bar{a}_0|a_1|\cdots|a_r]m$ into $a_0[a_1|\cdots|a_r]m$, where $\bar{a}_0 = a_0 - \epsilon_A(a_0)1$. With simplicial boundary $\sum_i (-1)^i d_i$, it contracts the augmented bar. Multiplying length-r words by $(-1)^r$ gives the displayed differential convention. This contraction preserves weight, so the counit is a weightwise quasi-isomorphism. For $FG(N)$, filter each weight by the weight of the final N-factor. The reduced coaction lowers that weight. The associated graded unit is the contracted right-sided bar $k \otimes N_v \to B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The filtration is finite in each total weight, so the unit is also a quasi-isomorphism. The same finite filtrations show that F, G preserve weightwise quasi-isomorphisms, since tensoring complexes over k preserves them. Thus the unit and counit give the asserted inverse equivalences. □

For a right module R and a left module L with the same lower-weight bounds, the two-sided bar has underlying weight-completed complex

$$\prod_n \bigoplus_{u+v_1+\cdots+v_r+w=n} R_u \otimes s\bar{A}_{v_1} \otimes \cdots \otimes s\bar{A}_{v_r} \otimes L_w, \qquad v_i > 0,$$

where the sum also runs over $r \geq 0$ and $\bar{A} = \ker(A_{\mathrm{alg}} \to k)$. Internal differentials and the two endpoint actions and adjacent products form its bar differential. Its realization computes $R \otimes^{\mathbb{L}}_A L$ in these

weight-graded categories. Comparing it with a derived cotensor product requires a two-sided twisting comparison on the chosen comodule category and completion. Weight completion here is not identified with derived augmentation-adic completion.

Proposition 28.8 (Centre transport under a mixed equivalence)**.** *Suppose an equivalence between the compatible mixed module categories is supplied, with coherent compatibility with all four bar objects and the interface operations. It identifies the derived endomorphism objects of their identity functors.*

Proof. Conjugate a natural endomorphism by the equivalence and a quasi-inverse. The unit and counit homotopies give inverse maps on derived endomorphism objects and preserve composition and its coherent operations. □

Constructing such a mixed equivalence requires the restriction, collision, interface, and distributive-law comparisons for the actual geometric coefficients. Its unit and counit must respect those maps and the chosen inverse limits. A diagonal bimodule comparison must produce an actual compatible bicomodule before tensor products can be identified with geometric cotensor products. Transport of mixed Hochschild chains and traces further requires their cyclic comparison and the trace domains used in their definition. These are additional mixed Koszul–Morita constructions. The ordinary weight-graded equivalence supplies its stated one-colour functors, without identifying the other mixed carriers or their physical traces.

28.6 BRST reduction

Let Q be a square-zero derivation of all the supplied mixed operations, including their homotopies. Extend it factorwise to the bar resolution. The total bar operator b then anticommutes with Q.

Theorem 28.9 (BRST–bar bicomplex)**.** *The total differential $D = Q + b$ squares to zero. Suppose its filtration is finite in each weight, or satisfies a stated strong-convergence criterion. The associated spectral sequence has*

$$E_1 = H_Q(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)), \qquad E_1 \Longrightarrow H_D(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)).$$

Over a field, when the relevant tensor and completion functors preserve the needed quasi-isomorphisms, (E_1, d_1) is the bar complex of the Q-cohomology operations. If this cohomology lies in one row, the spectral sequence collapses after this differential, and the target is the cohomology *of that reduced bar complex.*

Proof. The derivation identity gives $Qb + bQ = 0$; the bar identities give $b^2 = 0$. Filter the resulting bicomplex so that Q is its degree-zero differential. Künneth and the specified limit condition identify the first page and its remaining bar differential. A finite weightwise filtration converges; in the one-row case all later differentials have zero target or source. Without that concentration the later differentials remain part of the comparison. □

Chapter 29

Deformations, local fields, and quantization

29.1 The centre-extended deformation algebra

Varying a boundary product changes its compatibility with the interface and the central action. The deformation complex must therefore include all these operations. A local field realization is further data needed to turn their deformation equations into a BV action.

Fix a coloured operad $\mathcal{P}_{\mathrm{mix}}$ presenting the supplied operations and central action. Choose a conilpotent dg cooperad C and a cofibrant resolution $\Omega C \to \mathcal{P}_{\mathrm{mix}}$; for example, use the bar cooperad when its bar–cobar resolution is available. Fixed units and prescribed nullary operations are held relative. Write $\bar{C}$ for the reduced part. The convolution complex

$$\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}} = \mathrm{Hom}_{\mathbb{S}}\big(\bar{C}, \mathrm{End}_{\mathbb{A}}\big)$$

uses the tensor recipient of this operadic presentation. Complete it in a filtration for which the insertion operations converge and the deformation ideal is pronilpotent. It contains the chiral operation, both associative products, the interface, and the central action.

Theorem 29.1 (Complete deformation L_∞ algebra)**.** *The convolution complex has a canonical complete L_∞ structure. Its Maurer–Cartan mapping space describes algebras over ΩC, with the fixed relative data. These are the chosen homotopy-coherent mixed structures together with their central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the chosen cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_∞ identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\mathrm{mix}}$ to $\mathrm{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. □

29.1.1 Field equations and Bianchi identities

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_\alpha x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 29.2 (Universal Bianchi identity)**.** *For every α,*

$$d_\alpha \mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_\alpha^2 = 0$.

Proof. Both identities are the exponential form of the L_∞ Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^{α} and $e^{\alpha} x$. □

29.2 Shifted cotangent transgression

Let $\mathfrak{l}$ be a complete elliptic L_∞ algebra on a three-manifold M. Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathfrak{l}[1] = \mathfrak{l}[1] \oplus \mathfrak{l}^![-2].$$

Here $\mathfrak{l}^!$ is the continuous density-valued dual. The pairing is integrated with compact support or boundary conditions permitting integration by parts. For example, for the de Rham local algebra in three dimensions, $\mathfrak{l}^! \simeq \Omega^\bullet(M, \mathfrak{g}^*)[3]$; both superfields then have de Rham shift $[1]$. This is different from the finite-dimensional cotangent of a Lie algebra, whose dual term has shift $[-2]$. The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem 29.3 (Universal BV envelope)**.** *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \qquad d^!_\alpha \beta = 0.$$

The linearization, including its gauge directions, is the tangent complex of the shifted cotangent moduli problem. At the zero cotangent vector over a Maurer–Cartan point α, it splits as

$$T_\alpha \operatorname{MC}(\mathfrak{l}) \oplus L_\alpha \operatorname{MC}(\mathfrak{l})[-1].$$

At a nonzero β, the derivative of $d^!_\alpha \beta$ also contains the Hessian term in the α direction.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. □

Definition 29.4 (Local realization and chiral BV envelope)**.** A local realization of the complete mixed datum consists of an elliptic local L_∞ algebra $\mathfrak{l}^{\mathrm{full}}_{\mathbb{A}}$ on $M = X^{\mathrm{an}} \times \mathbb{R}$, a specified continuous filtered L_∞ quasi-isomorphism

$$\Gamma(M, \mathfrak{l}^{\mathrm{full}}_{\mathbb{A}}) \longrightarrow \mathrm{Def}^{\mathrm{full}}_{\mathbb{A}},$$

and the local pairing needed for cotangent transgression. Here Γ denotes the chosen complex of global sections with its declared topology. The comparison includes compatibility with the specified boundary operations. The dual fields use the density-valued continuous dual, with compact support or boundary conditions that make the integrated pairing and integration by parts defined.

For a supplied local realization, define

$$\mathbf{BVEnv}^{\mathrm{full}}_3(\mathbb{A}) = T^*[-1] \operatorname{MC}(\mathfrak{l}^{\mathrm{full}}_{\mathbb{A}}).$$

Its action is the cotangent action displayed above. Constructing a local realization from an abstract mixed datum remains an additional existence problem.

29.3 Intrinsic algebraic quantization

An invariant volume gives a finite-dimensional BV operator. To use this construction on an inverse system of charts, its operators must also commute with the transition maps.

Let $\mathfrak{X}$ first be a smooth graded formal chart, with a homological vector field Q and a nonvanishing Berezinian μ invariant under Q. Use a coordinate or filtration completion on which the operators below are continuous. Functions on its shifted cotangent are completed polyvectors:

$$\mathcal{O}(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p\geq 0} \Gamma(\mathfrak{X}, \wedge^p T_{\mathfrak{X}})[p].$$

The volume defines divergence and hence the operator Δ_μ on polyvectors, with its signs fixed by the following bracket convention.

Theorem 29.5 (Invariant-volume BV complex)**.** *The degree-one operator Δ_μ has order two and satisfies*

$$\Delta_\mu^2 = 0, \qquad [d_{\mathfrak{X}}, \Delta_\mu] = 0,$$

$$\Delta_\mu(fg) = \Delta_\mu(f)g + (-1)^{|f|} f\Delta_\mu(g) + (-1)^{|f|}\{f, g\}.$$

Thus $(\widehat{\mathrm{PV}}(\mathfrak{X})[\![\hbar]\!], d_{\mathfrak{X}} + \hbar\Delta_\mu)$ is a complex.

Proof. In coordinates with the selected volume, divergence is the graded sum of paired derivatives in the base and cotangent variables. The graded commutation relations give its square zero. Equivalently, divergence on vector fields is a Lie cocycle; this identity extends to the square of its second-order extension. Invariance of μ under Q gives the commutator with $d_{\mathfrak{X}}$. Expanding the paired derivatives on a product gives the displayed bracket. These identities extend to the chosen completion by continuity; see also the finite BV construction in [82]. □

For an ordinary smooth m-dimensional variety and a nowhere-zero top form, contraction gives the precise comparison

$$\mathrm{PV}^p \longrightarrow \Omega^{m-p}, \qquad (\mathrm{PV}, \Delta_\mu) \simeq (\Omega^\bullet[m], d_{\mathrm{dR}}),$$

with contraction signs chosen compatibly. Polyvector degree is reversed, and the shift $[m]$ makes both sides have degree $-p$. The parameter $\hbar$ gives the corresponding Rees differential. On a graded or derived chart, a Berezinian is not an ordinary top differential form: for one odd coordinate θ, the powers $(d\theta)^n$ persist for all n. A de Rham description therefore needs its own recipient, such as integral forms, and a comparison map with its shifts and completion.

Definition 29.6 (Quantum complex on a chart system)**.** Suppose finite smooth graded formal charts of the localized deformation problem carry these invariant volumes, and are linked by specified maps of BV complexes. If the operators are continuous on the separated inverse limit, denote that limit complex by $\mathbf{QHdg}_3^{\mathrm{full}}(\mathbb{A})$. Existence of compatible orientations alone does not construct the transition maps or a comparison with a derived de Rham complex.

Theorem 29.7 (Quantum master equation)**.** *For a degree-zero formal action $S = S_0 + \hbar S_1 + \cdots$, the equation*

$$dS + \tfrac{1}{2}\{S, S\} + \hbar\Delta_\mu S = 0$$

is equivalent to $(d + \hbar\Delta_\mu)e^{S/\hbar} = 0$ on the formal WKB module generated by $e^{S/\hbar}$, or in any explicitly specified algebra where the exponential and these operations converge.

Proof. The second-order product identity gives

$$(d + \hbar\Delta_\mu)e^{S/\hbar} = \hbar^{-1}\big(dS + \tfrac{1}{2}\{S, S\} + \hbar\Delta_\mu S\big)e^{S/\hbar}.$$

This is an identity of formal WKB symbols. A general exponential with $S_0 \neq 0$ need not be an element of the original $\hbar$-adic algebra. □

29.4 Perturbative factorization realization

The algebraic quantum differential does not specify a propagator or a space of singular kernels. Fix a local realization, elliptic gauge-fixing data, support and boundary conditions, and a renormalization prescription. A heat-kernel propagator is defined away from the diagonal. Extending its graph integrals to collisions requires counterterms and compatibility with the boundary conditions.

Definition 29.8 (Factorization realization data)**.** A factorization realization consists of renormalized local observable complexes, continuous disjoint-insertion maps, and coherent homotopies for their compositions. It includes extensions of graph weights to the required collision strata, their Stokes identities, and a quantum master equation in the permitted local functional complex. Weiss descent is a further required property of these actual complexes. A boundary realization also includes a comparison with the supplied mixed datum, preserving its two products, Lie operation, interface, and mixed law.

Under these hypotheses the disjoint-insertion maps and composition homotopies define the stated factorization object. The boundary comparison identifies its boundary operations with those of $\mathbb{A}$. The construction problem is to produce these data from the physical fields and action. A graphwise identity supplies the corresponding composition equation, whereas descent and boundary identification require their own maps and proofs.

Chapter 30

Exchange and projective boundary families

30.1 Planar exchange and Tannaka reconstruction

A second topological direction can give exchange paths for defects. An actual line category must provide their braiding and its homotopies. In the finite algebraic setting, let $\mathscr{L}$ be an essentially small rigid braided k-linear tensor category with a faithful strong monoidal functor $\omega : \mathscr{L} \to \mathrm{Vect}_k^{\mathrm{fd}}$. Form the coend

$$H = \int^{M \in \mathscr{L}} \omega(M)^* \otimes \omega(M)$$

in all vector spaces. Write $[f \otimes v]_M$ for a matrix coefficient.

Theorem 30.1 (Matrix-coefficient Hopf algebra)**.** *The coend is a coquasitriangular Hopf algebra. Its coalgebra structure is*

$$\Delta[f \otimes v]_M = \sum_i [f \otimes e_i]_M \otimes [e^i \otimes v]_M, \qquad \epsilon[f \otimes v]_M = f(v).$$

Tensor product gives multiplication; duality gives the antipode. Its linear dual acts on each finite fibre. Reconstruction of the entire line category as finite comodules requires the usual exactness and reconstruction hypotheses in addition to faithfulness.

Proof. The coevaluation tensor $\sum_i e_i \otimes e^i$ is basis-independent, so the displayed coproduct is well-defined and coassociative. Evaluation gives the counit. The monoidal constraint identifies the product of coefficients of M, N with a coefficient of $M \otimes N$; functoriality makes this descend through the coend relations. The tensor unit gives its unit, and the duality identities give the antipode equations. Finally, evaluating the braiding on two fibres defines the coquasitriangular pairing. Naturality and the two hexagons are exactly its compatibility with multiplication and coproduct. These are the finite reconstruction operations of Volume IV. □

For a meromorphic spectral family, specify expansion regions, tensor completions and pairings before passing to continuous duals. An actual completed coevaluation may turn the pairing into an R-matrix. Naturality on a chosen generating fibre gives an RTT *relation*. A full RTT presentation further needs a generation theorem and a proof that these relations give its entire kernel.

30.2 Projective connections and the Virasoro tangent complex

A projective connection describes the affine transformation law of a stress tensor. It is distinct from an action on every operation of a mixed boundary theory.

Fix the convention

$$\delta_\epsilon p = \mathcal{D}_{p,c}(\epsilon) = \epsilon\, \partial p + 2(\partial \epsilon)\, p + \frac{c}{12} \partial^3 \epsilon.$$

On a coordinate disc, the third-derivative term is a Lie algebra cocycle: with $L_\epsilon f = \epsilon f' + 2\epsilon' f$,

$$L_\epsilon(\eta''') - L_\eta(\epsilon''') = (\epsilon\eta' - \eta\epsilon')'''.$$

Global projective connections are glued by the corresponding Schwarzian coordinate law. Denote their quotient by coordinate changes by $\mathcal{P}_c(X)$. Its tangent complex is

$$\Omega^{0,\bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{0,\bullet}(X, K_X^{\otimes 2}),$$

with vector fields in quotient-tangent degree -1 and projective variations in degree zero; totalize with the Dolbeault degree.

Proposition 30.2 (A projectively covariant mixed family)**.** *Suppose the coordinate-change action lifts coherently to the chiral and transverse operations, the interface and all their identities. It then defines a family of mixed structures over the projective quotient. Its relative and projective tangent algebras form an extension. A compatible splitting identifies that extension with the corresponding homotopy semidirect product.*

Proof. The equivariant structure descends the family through the coordinate-change groupoid. Differentiation gives the tangent extension; a splitting supplies its action and higher coherence operations. Conversely those operations give the semidirect description. A map $\mathrm{Vir}_c \to A^{\mathrm{ch}}$ supplies the chiral conformal action, but the other actions in the hypothesis must also be constructed. □

30.3 Einstein completion of a conformal pair

Let A_L, A_R be conformal mixed boundary data equipped with the coherent projective families just described and specified maps to the same projective quotient $\mathcal{P}_c(X)$. Their homotopy fibre product is a relative deformation problem. Equality of the two central parameters is necessary for this matching convention, but does not supply these maps or the coupled action.

Definition 30.3 (Candidate Einstein BV completion)**.** Suppose the relative algebra and a continuous L_∞ action of the Cartan algebra have been supplied, together with their local realization. The associated cotangent candidate is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1]\,\mathrm{MC}\big(\mathfrak{g}_{\mathrm{grav}} \ltimes_{\mathrm{Ham}} \mathfrak{m}_{A_L,A_R}\big).$$

Here $\mathfrak{g}_{\mathrm{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L,A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 30.4 (Cotangent action for a supplied coupled algebra)**.** *Suppose projective matching and a continuous L_∞ action have constructed the displayed complete semidirect product. Suppose also that its local realization and integrated dual pairing are defined. Its shifted cotangent has action*

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle$$

and satisfies the classical master equation. Its Bianchi identity is the L_∞ identity of the coupled algebra.

Proof. The supplied semidirect product has a square-zero homological vector field. Its cotangent lift is Hamiltonian for the displayed action and remains square-zero. This proves the classical master equation and the Bianchi identity by the preceding cotangent construction. □

The gravitational summand of this cotangent action is the cotangent theory of Cartan deformations. It introduces independent dual fields. To identify it with Einstein BV theory requires a comparison preserving the BV pairing and the action, including boundary terms, on the invertible-coframe sector. The flatness equations alone do not provide that comparison. A physical coupled completion must also construct the matter action and its equivariant moment map. The intended Einstein reconstruction therefore retains these field and action comparisons as existence problems.

Chapter 31

Cartan gravity and its boundary currents

31.1 Three-dimensional gravity as a flat Cartan theory

Let M be an oriented three-manifold and $\ell > 0$, with $\Lambda = -\ell^{-2}$. Use $\eta = \mathrm{diag}(-1, 1, 1)$ and $\epsilon_{012} = 1$, raising internal indices with η. The Cartan algebra $\mathfrak{g}_\Lambda$ has basis J_a, P_a, $a = 0, 1, 2$, with

$$[J_a, J_b] = \epsilon_{ab}{}^{c} J_c, \qquad [J_a, P_b] = \epsilon_{ab}{}^{c} P_c, \qquad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^{c} J_c.$$

The invariant pairing is $\langle J_a, P_b \rangle = \eta_{ab}$ and vanishes on the two diagonal summands. A Cartan connection is

$$\mathcal{A} = \omega^a J_a + e^a P_a.$$

Put $R^a = d\omega^a + \tfrac{1}{2} \epsilon^a{}_{bc} \omega^b \wedge \omega^c$ and $T^a = de^a + \epsilon^a{}_{bc} \omega^b \wedge e^c$. Its curvature is

$$F_{\mathcal{A}} = \Big(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \Big) J_a + T^a P_a.$$

Theorem 31.1 (Cartan–Einstein equivalence)**.** *On the locus where e is invertible, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \qquad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \qquad g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu.$$

The Chern–Simons action for $\mathcal{A}$ equals the Einstein–Hilbert action with cosmological constant, up to its boundary term and normalization.

Proof. The P component of the flatness equation is the torsion equation. Invertibility of e gives the Levi–Civita spin connection. The J component states that the curvature two-form has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Expanding the Chern–Simons three-form gives the first-order gravitational action with the coefficients computed below [1, 86]. On a flat connection, $\mathcal{L}_\zeta \mathcal{A} = d_{\mathcal{A}}(\iota_\zeta \mathcal{A})$; invertibility of e therefore identifies infinitesimal translations with diffeomorphisms modulo Lorentz transformations, on shell. □

31.2 The Chern–Simons normalization

Normalize the fundamental trace on $\mathfrak{sl}_2$ by $\mathrm{tr}(J_a J_b) = \eta_{ab}/2$. Define

$$\mathrm{CS}_{\mathrm{tr}}(A) = \mathrm{tr}\Big(A \wedge dA + \tfrac{2}{3} A \wedge A \wedge A \Big).$$

For $A^{\pm} = \omega \pm \ell^{-1} e$, direct expansion gives

$$\mathrm{CS}_{\mathrm{tr}}(A^+) - \mathrm{CS}_{\mathrm{tr}}(A^-) = \frac{4}{\ell} \mathrm{tr}(e \wedge R_\omega) + \frac{4}{3\ell^3} \mathrm{tr}(e \wedge e \wedge e) - \frac{2}{\ell} d\,\mathrm{tr}(\omega \wedge e).$$

Since $R^a = \frac{1}{2}\epsilon^a{}_{bc}R^{bc}$, its bulk integral is

$$\frac{k}{4\pi}\int_M \left(\mathrm{CS}_{\mathrm{tr}}(A^+) - \mathrm{CS}_{\mathrm{tr}}(A^-)\right)$$
$$= \frac{k}{4\pi\ell}\int_M \epsilon_{abc}e^a \wedge \left(R^{bc} + \frac{1}{3\ell^2}e^b \wedge e^c\right) + \text{boundary term}.$$

Thus the Einstein normalization $1/(16\pi G)$ gives $k = \ell/(4G)$. With the full Cartan pairing $\langle J_a, P_b\rangle = \eta_{ab}$, the corresponding single Cartan Chern–Simons three-form is the displayed epsilon-form, up to an exact form, and has coefficient $1/(16\pi G)$. The Cartan pairing and the fundamental trace are different invariant forms; their action coefficients must be compared through this expansion [1, 86].

31.3 Four equivalent linear complexes

Fix an invertible constant-curvature solution (e, ω). Its local deformations have four presentations: the flat adjoint de Rham complex, the Einstein–Cartan complex, the metric Einstein complex, and the Calabi complex resolving the sheaf of local Killing fields.

Theorem 31.2 (Local comparison of gravitational complexes)**.** *On a geodesically convex open set, these complexes are quasi-isomorphic. The first comparison is the splitting of the flat Cartan differential into translation and Lorentz components. Eliminating the algebraic torsion variable and Lorentz gauge pair gives the metric complex. In dimension three its curvature and Bianchi terms identify algebraically with the Calabi terms. Each complex resolves the same local Killing sheaf.*

Proof. The Cartan splitting is an isomorphism of complexes, with the formulas below. For invertible e, the map

$$v^a \longmapsto \epsilon^a{}_{bc}v^b \wedge e^c$$

from spin-connection variations to torsion variations is an isomorphism of rank-nine bundles. The torsion equation therefore eliminates v. The antisymmetric coframe variation and the Lorentz gauge parameter form another algebraic contractible pair. Contracting these pairs in the complex leaves the metric differential and its Bianchi identity. In three dimensions the algebraic curvature module is isomorphic to the symmetric two-tensor module by contraction and trace reversal; linearizing at constant curvature gives the corresponding identification of equations.

For local exactness, the flat adjoint complex has the Poincaré homotopy. Its parallel degree-zero sections are the Killing field and its skew first derivative, identified by the coframe. The Calabi complex has a compatible Killing-transport resolution and local homotopies on constant-curvature opens [60, Sec. 2 and Cor. 10]. These yield the same resolution of the Killing sheaf. Local exactness here uses this differential resolution, not merely exactness of its principal symbol. □

31.4 The four complexes in coordinates

The flat adjoint complex is

$$\Omega^0(M, \mathfrak{g}_\Lambda) \xrightarrow{d_A} \Omega^1(M, \mathfrak{g}_\Lambda) \xrightarrow{d_A} \Omega^2(M, \mathfrak{g}_\Lambda) \xrightarrow{d_A} \Omega^3(M, \mathfrak{g}_\Lambda).$$

Its degrees are gauge parameters, fluctuations, linearized equations and Bianchi identities. Write a parameter as (ξ^a, λ^a) and a fluctuation as $(u^a, v^a) = (\delta e^a, \delta\omega^a)$. The first two maps are

$$u^a = D_\omega\xi^a + \epsilon^a{}_{bc}e^b\lambda^c, \qquad v^a = D_\omega\lambda^a + \ell^{-2}\epsilon^a{}_{bc}e^b\xi^c,$$
$$\delta T^a = D_\omega u^a + \epsilon^a{}_{bc}v^b \wedge e^c, \quad \delta F^a = D_\omega v^a + \ell^{-2}\epsilon^a{}_{bc}e^b \wedge u^c.$$

The final map is the linearized Cartan Bianchi identity.

Proposition 31.3 (Cartan chain identity)**.** *The composite of the first two maps is zero on the torsion-free constant-curvature background.*

Proof. Apply D_ω^2 to ξ, λ, substitute $R^a = -\epsilon^a{}_{bc} e^b \wedge e^c/(2\ell^2)$ and $D_\omega e = 0$. The curvature terms cancel the explicit ℓ^{-2} terms. The remaining coefficients vanish by the Jacobi identity. Equivalently, $d_{\mathcal{A}}^2 = [F_{\mathcal{A}}, -] = 0$. □

Modulo Lorentz rotation, the metric fluctuation and its gauge variation are

$$h_{\mu\nu} = 2\eta_{ab} e^a_{(\mu} u^b_{\nu)}, \qquad K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu.$$

Use $\mathrm{Ric}(g) = 2\Lambda g$, $R(g) = 6\Lambda$ and $h = g^{\mu\nu} h_{\mu\nu}$ for its trace. An unambiguous form of the linearized Einstein operator is the Palatini formula

$$\begin{aligned} C^\rho{}_{\mu\nu} &= \tfrac{1}{2} g^{\rho\sigma}(\nabla_\mu h_{\nu\sigma} + \nabla_\nu h_{\mu\sigma} - \nabla_\sigma h_{\mu\nu}), \\ \dot{\mathrm{Ric}}_{\mu\nu} &= \nabla_\rho C^\rho{}_{\mu\nu} - \nabla_\nu C^\rho{}_{\mu\rho}, \\ \dot{R} &= g^{\mu\nu} \dot{\mathrm{Ric}}_{\mu\nu} - 2\Lambda h, \\ (E_\Lambda h)_{\mu\nu} &= \dot{\mathrm{Ric}}_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} \dot{R} - 2\Lambda h_{\mu\nu}. \end{aligned}$$

Naturality and the contracted Bianchi identity give $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$. For a constant rescaling $h = c\,g$, one has $C = 0$, $\dot{R} = -6\Lambda c$, and $E_\Lambda(c\,g) = \Lambda c\, g$.

Proposition 31.4 (Metric reduction on the torsion constraint)**.** *Solving $\delta T = 0$ uniquely for v and substituting in $\delta F = 0$ gives $E_\Lambda h = 0$. On this constrained space, the kernel of $(u, v) \mapsto h$ is the infinitesimal Lorentz action.*

Proof. The invertible algebraic map in the preceding theorem determines v from the first derivative of u. The symmetric part of u gives $h/2$; its antisymmetric part is a Lorentz rotation. Substitution differentiates the Levi–Civita curvature, giving the displayed Palatini operator. A Lorentz rotation determines both its coframe variation and its associated spin-connection variation. An arbitrary v in the unconstrained space would not be detected by h, which is why the torsion constraint is needed. □

For the Calabi comparison, define the constant-curvature defect

$$C(g)_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \Lambda(g_{\mu\rho} g_{\nu\sigma} - g_{\mu\sigma} g_{\nu\rho}).$$

The second Calabi operator is $\dot{C}_g$, and the complex begins

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2 T^*M) \xrightarrow{\dot{C}_g} \Gamma(\mathcal{R}) \xrightarrow{B_g} \Gamma(\mathcal{B}).$$

Here $\mathcal{R}$ is the algebraic curvature bundle and $\mathcal{B}$ the corresponding Bianchi bundle. Differentiating the natural identity $C(g) = 0$ gives $\dot{C}_g K = 0$. The correction by the variation of $\Lambda g \wedge g$ is essential. In dimension three, contraction and trace reversal identify this complex with the metric presentation above.

31.5 Brown–Henneaux and the projective boundary

Take asymptotically AdS boundary conditions for which the Chern–Simons charges are differentiable and finite. With the preceding fundamental trace and boundary orientation, each chiral sector has level $k = \ell/(4G)$. Brown–Henneaux boundary conditions yield the classical central charge [14]

$$c = 6k = \frac{3\ell}{2G}.$$

The local reduction makes its projective term explicit. Choose generators $[L_i, L_j] = (i-j)L_{i+j}$, with

$$L_1 = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \quad L_0 = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad L_{-1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

In the gauge $a_z = L_1 + t(z)L_{-1}$, set

$$\lambda_\epsilon = \epsilon L_1 - \epsilon' L_0 + (t\epsilon + \tfrac{1}{2}\epsilon'')L_{-1}.$$

Matrix multiplication gives

$$\partial\lambda_\epsilon + [a_z, \lambda_\epsilon] = (\epsilon t' + 2\epsilon' t + \tfrac{1}{2}\epsilon''')L_{-1}.$$

Normalize the projective variable by $p = kt$ and the charge by $Q_\epsilon = (2\pi)^{-1}\int \epsilon\, p$ with the selected orientation. Then

$$\delta_\epsilon p = \epsilon p' + 2\epsilon' p + \frac{k}{2}\epsilon''' = \mathcal{D}_{p,6k}(\epsilon).$$

This is a comparison of actual classical boundary charges with the projective transformation law. Identifying another conformal mixed datum with these charges still requires its boundary map.

31.6 Boundary BFV theory and the Virasoro moment map

Variation of a Chern–Simons action with an oriented boundary gives the boundary symplectic form

$$\Omega_\partial = \frac{k}{4\pi}\int_{\partial M} \operatorname{tr}(\delta A \wedge \delta A).$$

A choice of polarization and boundary term specifies the allowed fields and differentiable gauge charges. On a spatial boundary circle, these charges have the affine Poisson central extension; in a compatible sign convention its cocycle is

$$\frac{k}{2\pi}\int_{S^1} \operatorname{tr}(\lambda\, d\eta).$$

The highest-weight reduction just computed gives the classical Virasoro algebra. Neither this symplectic form alone nor an arbitrary boundary condition defines its quantization.

For the algebraic quantum comparison, let $V^\kappa(\mathfrak{sl}_2)$ be the affine vertex algebra with long-root squared length two and $\kappa \neq -2$. Principal quantum Drinfeld–Sokolov reduction, including its ghost and improvement terms, gives a Virasoro algebra of central charge

$$c_{\mathrm{DS}}(\kappa) = 1 - \frac{6(\kappa+1)^2}{\kappa+2} = 13 - 6(\kappa+2) - \frac{6}{\kappa+2}.$$

This formula is the principal rank-one reduction of [2, Example 5.22]. The affine quantum parameter κ and the classical gravitational parameter k have different definitions. Relating them requires the chosen quantization, real form and level convention. Substituting $\kappa = k$ does not give the classical identity $c = 6k$.

31.7 Higher-spin completion

There is a concrete extension of the Chern–Simons construction: choose $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ with its principal $\mathfrak{sl}_2$ embeddings, action levels and asymptotically AdS highest-weight boundary conditions. If V_m denotes the irreducible $\mathfrak{sl}_2$ module of highest weight m, then

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^{N} V_{2s-2}.$$

Theorem 31.5 (Principal spectrum and boundary reduction)**.** *In this specified Chern–Simons model, the frame-like fields have one integer spin s for every $2 \le s \le N$. Principal classical Drinfeld–Sokolov reduction gives one boundary generating current in each of the same conformal weights, with the classical W_N Poisson algebra.*

Proof. The principal embedding decomposes the adjoint as displayed. The connection component in V_{2s-2} is the frame-like spin-s field. Highest-weight gauge leaves one lowest-weight coefficient in each summand; the residual gauge transformations give the principal reduced Poisson algebra. These are the bulk and boundary constructions of [16]; the infinite higher-spin analogue is treated in [49]. □

This theory has no local propagating higher-spin modes in three dimensions, but has nontrivial boundary charges and global sectors. An independently given pair of W_N boundary algebras does not by itself provide its bulk action, states or allowed fillings. A reconstruction must map that pair to the actual boundary reduction and prove the required equivalence.

31.8 The principal embedding count

The exponents of $\mathfrak{sl}_N$ are $1, 2, \dots, N-1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m+1) = (N-1)(N+1) = N^2 - 1.$$

Writing $s = m + 1$ gives one field at every spin $s = 2, \dots, N$.

Chapter 32

Matter, determinants, and gravitational states

32.1 Matter stress tensor and the Einstein moment map

A projective transformation law alone does not make a matter action Hamiltonian. A Hamiltonian coupling needs a symplectic or Poisson matter space, an action, and an equivariant moment map. Its elementary BRST model exhibits the extra ghost terms explicitly.

Let the finite-dimensional ordinary Lie algebra $\mathfrak{h}$ act Hamiltonianly on an ordinary Poisson algebra B over a characteristic-zero field, with generators μ_a and $\{\mu_a, \mu_b\} = f_{ab}{}^c \mu_c$. Adjoin odd variables c^a, b_a of degrees 1, -1, respectively, with the degree-zero Poisson bracket $\{c^a, b_b\} = \delta^a_b$; all other new generator brackets vanish. Put

$$\Omega = c^a \mu_a - \tfrac{1}{2} f_{ab}{}^c c^a c^b b_c .$$

Proposition 32.1 (Hamiltonian BRST charge)**.** *The charge satisfies* $\{\Omega, \Omega\} = 0$*, so* $Q_{\mathrm{BRST}} = \{\Omega, -\}$ *is square-zero. Without the equivariance condition, its square contains the moment-map defect.*

Proof. The term quadratic in ghosts is

$$\tfrac{1}{2} c^a c^b \big(\{\mu_a, \mu_b\} - f_{ab}{}^c \mu_c \big)$$

in $\frac{1}{2}\{\Omega, \Omega\}$. The remaining term is cubic in ghosts and linear in b, with the Jacobi identity as coefficient. Both vanish. The graded Jacobi identity then gives $Q^2_{\mathrm{BRST}} = 0$. For $\{q, p\} = 1$, $\mu_h = qp$, $\mu_e = p$ and $[h, e] = e$, the charge is $c_h q p + c_e p - c_h c_e b_e$. Deleting its cubic ghost term leaves $\{\Omega, \Omega\} = 2 c_h c_e p \neq 0$. □

A local field theory further needs its matter action, ghosts, antifields and higher couplings. A supplied local equivariant deformation algebra has the shifted cotangent BV action already constructed. Identification with an Einstein–matter action requires the action and pairing comparison; the isolated linear term $\langle h, \mu_{\mathrm{matt}} \rangle$ does not replace these data.

The Ward identity also records the matter equations. Define $T^{\mu\nu}$ and E_i by

$$\delta S_{\mathrm{matt}} = \int_M \Big(\tfrac{1}{2} T^{\mu\nu} \delta g_{\mu\nu} + E_i \delta \phi^i \Big) \operatorname{vol}_g .$$

For scalar matter fields, diffeomorphism invariance and compactly supported ξ give

$$0 = \int_M \xi^\nu (-\nabla^\mu T_{\mu\nu} + E_i \partial_\nu \phi^i) \operatorname{vol}_g ,$$
$$\nabla^\mu T_{\mu\nu} = E_i \partial_\nu \phi^i .$$

Thus conservation holds on shell. Other tensor fields contribute the corresponding Euler–Lagrange terms from their Lie derivatives. In a BV action this is the component of the master equation linear in the diffeomorphism ghost, with the equations and antifield terms retained.

32.2 One-loop gravity and analytic torsion

Consider first a closed oriented Riemannian three-manifold and an acyclic unitary flat coefficient bundle E. Fix elliptic gauge fixing and the normalization of the Gaussian measure. If Δ_q is its de Rham Laplacian on q-forms, use the torsion convention

$$\tau_{\mathrm{RS}}(M,E)=\prod_{q=0}^{3}(\det{}'\Delta_q)^{(-1)^{q+1}q/2}.$$

The prime omits zero modes; acyclicity removes their integration here.

Proposition 32.2 (Acyclic Chern–Simons Gaussian)**.** *For a unitary flat adjoint bundle in this setting, the absolute one-loop factor is*

$$|Z_{1\text{-}loop}(\mathcal{A})|=\tau_{\mathrm{RS}}(M,E)^{1/2}.$$

The regularized phase is determined by the relevant eta invariant together with the framing convention.

Proof. Gauge fixing separates exact one-forms from their orthogonal complement. The bosonic and ghost determinants give $(\det{}'\Delta_0)^{3/4}(\det{}'\Delta_1)^{-1/4}$. Hodge duality in dimension three identifies Δ_3 with Δ_0 and Δ_2 with Δ_1, so the displayed torsion is $(\det{}'\Delta_0)^{3/2}(\det{}'\Delta_1)^{-1/2}$. The Gaussian magnitude is its square root. Spectral asymmetry of the first-order kinetic operator gives the phase [79, 87]. □

The gravitational real form and a noncompact AdS manifold require their own integration contour, boundary domains and regularization. The local quasi-isomorphisms of gravitational complexes identify cohomological determinant lines, but do not by themselves identify analytic norms or Gaussian measures. Already the acyclic two-term Euclidean complex $\mathbb{R}\xrightarrow{c}\mathbb{R}$ has torsion $|c|$ in this convention, although it is quasi-isomorphic to zero (take $c\neq 0$). Equality of one-loop factors therefore requires the corresponding norm and measure comparison as well as orientation.

32.3 BTZ, Cardy, and the area law

Let a modular, unitary two-dimensional theory have left and right central charges c_L, c_R and a discrete spectrum with a suitable vacuum. Write $E_L = L_0 - c_L/24$ and $E_R = \bar{L}_0 - c_R/24$. In a regime where the modular transform is vacuum-dominated, the leading Cardy entropy is

$$S_{\mathrm{Cardy}}=2\pi\sqrt{c_L E_L/6}+2\pi\sqrt{c_R E_R/6}.$$

This is an asymptotic statement about a specified spectrum. The usual fixed-central-charge high-energy regime and a simultaneous large-central-charge limit are different limits; the latter needs appropriate spectral conditions, for example the sparse-light-spectrum hypotheses of [48].

For the BTZ charges, take $G, \ell > 0$, $r_+ \geq |r_-|$, and

$$E_L=\frac{(r_+ + r_-)^2}{16G\ell},\qquad E_R=\frac{(r_+ - r_-)^2}{16G\ell},\qquad c_L=c_R=\frac{3\ell}{2G}.$$

Proposition 32.3 (BTZ substitution)**.** *The Cardy expression evaluated on these charges equals $2\pi r_+/(4G)$, the Bekenstein–Hawking area expression. It computes the leading statistical entropy only when the boundary spectrum and energy range justify the Cardy asymptotic.*

Proof. The two terms become $\pi(r_+ + r_-)/(4G)$ and $\pi(r_+ - r_-)/(4G)$; the sign condition on the radii fixes both square roots. Their sum is $2\pi r_+/(4G)$. The BTZ metric and its charges give the geometric area expression [5]; modular high-energy asymptotics supply the statistical comparison [17]. □

32.4 Higher-spin holonomy and boundary charges

Holonomy is the ordered exponential of a connection. It contains global conjugacy data and is different from the polynomial invariant of its local connection coefficient. For example, if a constant traceless 2×2 matrix satisfies $a^2 = qI$ around a circle of length L, then

$$\operatorname{tr}(a^2) = 2q, \qquad \operatorname{tr}(\mathrm{Hol}^s) = 2\cosh(sL\sqrt{q}) = 2 + s^2L^2q + O(q^2).$$

The latter is a power series in q, including at $q = 0$.

Proposition 32.4 (Constant-sector reconstruction from a logarithm)**.** *For a constant semisimple $\mathfrak{sl}_N$ connection, a chosen local branch of the logarithm of its holonomy recovers the conjugacy class of the connection coefficient, subject to the traceless condition. On a regular principal Drinfeld–Sokolov slice, its invariant polynomials then recover the constant boundary current coordinates with their fixed normalizations.*

Proof. Diagonalize the coefficient. Holonomy exponentiates its eigenvalues times L; a local logarithm branch inverts this map, while conjugacy retains the unordered eigenvalues. The elementary symmetric polynomials of those eigenvalues are the characteristic-polynomial coefficients. The regular principal slice gives coordinates for these invariants. For nonconstant connections path ordering intervenes, and a branch of global holonomy does not recover all local current modes. □

The W_N brackets come from the actual boundary Hamiltonian reduction. They are not obtained by identifying every holonomy trace with a current zero mode. Smooth Euclidean saddles impose the appropriate contractible-cycle holonomy condition in the chosen global gauge group and spin structure; the remaining charges and allowed saddles still require these global data.

Chapter 33

Anomalies and the Igusa boundary problem

33.1 Four native anomaly classes

The relevant anomaly data have different recipients. A local quantum BV obstruction is a degree-one class in the local deformation complex at the chosen perturbative order. A determinant line has curvature and possibly flat holonomy. A framing anomaly is encoded by the appropriate transgressed characteristic or invertible-theory datum. A curved bar construction has an arity-zero term satisfying its curved identities. A comparison must map both the class and any proposed primitive between these recipients.

Proposition 33.1 (Central curvature extension)**.** *Let* $(\mathfrak{l}, d, [-,-], \kappa)$ *be a curved dg Lie algebra with central* $\kappa \in \mathfrak{l}^2$*,* $d\kappa = 0$*, so* $d^2 = [\kappa, -] = 0$*. Adjoin a central generator* u *of degree one with* $du = \kappa$*. Then* κ *is exact in the extension, and twisting by* $-u$ *gives zero curvature. The extension respects a pronilpotent completion whenever* u*,* d *and the twist are admissible for that filtration.*

Proof. The differential extends because $d^2 u = d\kappa = 0$ and κ is central. The twisted curvature is

$$\kappa + d(-u) + \tfrac{1}{2}[-u, -u] = 0.$$

The twisted differential remains d because u is central. □

A geometric cancellation is a further statement. If an anomaly is represented by an actual invertible field theory $\mathcal{A}$, tensoring with its inverse cancels that full anomaly. Adjoining u does not construct such an inverse theory or its coupling. In particular, canceling a curvature form leaves open any flat holonomy of the anomaly line.

33.2 Hall and Calabi–Yau sources

Let C be a Calabi–Yau category with a Hall moduli problem. The actual Hall constructions of Volume IV specify coefficient complexes, orientations, admissible pull–push maps and flag identities. With these data the extension correspondence gives H^+. A boundary chiral algebra requires a factorization pushforward compatible with configurations and collisions. Relative induction, when defined, still needs its interface and geometric comparison maps to supply a complete mixed datum.

Proposition 33.2 (Transport under a supplied mixed Morita equivalence)**.** *Suppose a Hall construction supplies a complete mixed boundary datum and a Morita equivalence of its compatible module category with that of* $\mathbb{A}$*. The equivalence identifies their derived centres. A comparison of local deformation algebras, BV pairings, and actions is additional data for a BV comparison. Quantum line and amplitude comparisons additionally require the monoidal and cyclic structures used to define them.*

Proof. Conjugating natural transformations by the equivalence and its inverse identifies the endomorphisms of the two identity functors. The unit and counit homotopies provide the inverse comparison and preserve composition. This proves the centre comparison. It does not construct the additional local, symplectic, or cyclic maps. □

An algebra map alone does not induce the required forward centre map. For example, the homomorphism $\mathbb{C}[x] \to M_2(\mathbb{C})$ with $x \mapsto \operatorname{diag}(0,1)$ sends the central element x to a noncentral matrix. Its commutator with the matrix unit E_{12} is $-E_{12}$. Thus even the degree-zero condition fails for a centre map extending the given algebra map. Hall multiplication maps must be supplemented by an actual coherent module comparison before centre transport can be applied.

For $X = K3 \times E$, suppose a geometric Hall source supplies the complete charge-graded BPS Lie superalgebra and Cartan action required in Volume IV. Its semidirect product with the common Cartan defines $\mathfrak{b}_X$. Volume IV then constructs the completed Lie amalgam with the Igusa Borcherds Borel $\mathfrak{b}_\Delta$. Applying the completed enveloping functor gives the cocommutative Hopf algebra

$$H_{X\boxtimes\Delta} = \widehat{U}(\mathfrak{b}_X)\widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)}\widehat{U}(\mathfrak{b}_\Delta).$$

Under the continuous extension and retraction hypotheses of Volume IV, the two Hopf retractions preserve the geometric and automorphic sectors. A quantum amalgam also requires compatible Hopf structures, continuous coproducts and actual retractions over the common Cartan algebra. A pairing or a Hall product alone does not provide them.

33.2.1 Mixed realization of the amalgam

The Hopf retractions above are maps of their stated algebraic objects. A mixed realization must construct chiral and transverse operations, an interface, and a compatible distributive law. To realize the retractions, it must then supply maps preserving those operations and the chosen completions.

A BV retraction requires further maps between the local deformation algebras, together with compatible pairings and actions. The counterexample to forward centre functoriality prevents deducing these maps from the Hopf retractions alone. The intended geometric–automorphic theory asks for this mixed realization and for a primitive comparison quotient on its actual boundary operators. Its interaction ideal is the kernel of that constructed quotient, rather than an ideal determined by equality of scalar characters.

33.3 Igusa root states and the mixed-boundary comparison

Here D_5 is the monic Igusa form and $\chi_{10} = D_5^2$. The Gritsenko–Nikulin presentation defines the root Lie superalgebra $\mathfrak{g}_{\Delta_5}$ and its positive subalgebra $\mathfrak{n}_+$. Let $d_\alpha = \operatorname{sdim}(\mathfrak{g}_{\Delta_5})_\alpha$, and fix the positive cone, its proper height, and Weyl vector ρ. Put

$$P = \prod_{\alpha>0}(1-e^{-\alpha})^{d_\alpha}, \qquad \operatorname{Den}_\Delta = e^{-\rho}P = D_5(2Z), \qquad \jmath(Z) = Z/2.$$

The products are first interpreted in the positive-height completion.

Theorem 33.3 (Root character and reduced scalar)**.** *The root Fock space* $\operatorname{Sym}(\mathfrak{n}_+)$*, formed with super symmetry, has supercharacter* P^{-1}*. Consequently*

$$-\jmath^*\big(e^{2\rho}\operatorname{sch}(\operatorname{Sym}(\mathfrak{n}_+)^{\otimes 2})\big) = -(\jmath^*\operatorname{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

For a nonsingular projective K3 surface S and an elliptic curve E, this scalar also has an enumerative interpretation. Let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant in class (β_h, d), where β_h is nonzero and primitive with square $2h-2$. Disconnected domains have no collapsed connected components. Use the incidence insertion $\beta_h^\vee \otimes [\text{point}]$, with $\langle \beta_h, \beta_h^\vee\rangle = 1$. Then the series

$$\mathcal{Z}(u,q,s) = \sum_{g,h,d\geq 0} N_{g,h,d}\,u^{2g-2}q^{h-1}s^{d-1}$$

is the Laurent expansion of $-\chi_{10}(r,q,s)^{-1}$ *under* $r = e^{iu}$*, by* [75, Theorem 1].

Proof. An even root vector contributes $(1-e^{-\alpha})^{-1}$ to the supercharacter of its symmetric algebra. An odd root vector contributes $1-e^{-\alpha}$, because its symmetric algebra in super vector spaces is an exterior algebra. Multiplying the two contributions gives P^{-1}. Proper height makes every coefficient calculation finite. The Weyl factor gives $e^{2\rho}P^{-2} = \mathrm{Den}_{\Delta}^{-2}$. Pullback by $\jmath$ gives $D_5(Z)^{-2} = \chi_{10}(Z)^{-1}$. Oberdieck–Pixton, Theorem 1, identifies the separately defined reduced numerical generating series with its negative [75]. This last theorem is an enumerative input; the character calculation supplies the root side. □

A primitive realization starts with parity-preserving root classes in a charge-graded Lie superalgebra $\mathfrak{p}$ satisfying the defining relations. Use the same positive charge cone and height filtration on the target, and require the classes to preserve charge. The free Lie presentation gives an even charge-preserving map $f : \mathfrak{n}_+ \to \mathfrak{p}$.

Applying f to tensor words constructs $U(f)$. With trivial coefficients, use Chevalley chains $C_*^{\mathrm{CE}}(\mathfrak{a}) = \mathrm{Sym}(\mathfrak{a}[1])$, where the homological suspension raises degree by one and the differential contracts pairs by the bracket. Applying the suspension of f to every factor gives a chain map because f preserves brackets.

For the level-zero Lie conformal current, define $\mathrm{Cur}(\mathfrak{a}) = \mathbb{C}[\partial]\otimes\mathfrak{a}$, with ∂ even. Its generator bracket is $[x_\lambda y] = [x,y]$. Extend it by $[\partial x_\lambda y] = -\lambda[x_\lambda y]$ and $[x_\lambda \partial y] = (\partial+\lambda)[x_\lambda y]$. The formula $\partial^j x \mapsto \partial^j f(x)$ defines $\mathrm{Cur}(f)$. Charge preservation makes these maps compatible with the specified proper-height quotients and their inverse limits.

A complete mixed boundary realization requires further geometric maps. The compact source consists of sheaves and extension flags on a common elliptic torsor. Its critical realization must supply reduced coefficient complexes, orientation cross-line maps, unit and flag coherences, and the admissible pull–push operations. A chiral comparison must preserve actual support and moving-insertion operations. A quantum-double comparison must preserve coproducts, pairings and the domain of exchange operators. The finite Calabi–Yau quiver completions require their specified dg maps; ordinary full-subquiver inclusions do not in general give those maps.

The mixed datum to which a BV envelope is applied must therefore include these operations and compatible maps between them. Identifying the primitive algebra of that envelope requires an actual map of its deformation complex to the root complex. A determinant comparison requires a skew perfect family, a square-root line and section, and an isomorphism to the automorphic line preserving transport and normalization. An operator-trace interpretation further requires its state space and trace domain. Constructing this datum and these comparisons is the Igusa mixed-boundary problem. The root character identity specifies the scalar that their final evaluation must recover.

Chapter 34

Descent of coherent operations

34.1 The carrier lattice

Different operations live on different geometric domains. A descent construction records their incidence categories and the maps between them; a list of domain names alone does not determine these categories. To describe the support filtration, take the finite set

$$\mathfrak{C} = \{\mathrm{diag}, \mathrm{ch}, \bot, \mathrm{int}, \mathrm{ex}, \mathrm{cyc}, \mathrm{mod}, \mathrm{BV}, \mathrm{Ein}\}.$$

These names stand for diagonal, chiral, transverse, interface, exchange, cyclic, modular, BV and Einstein operations. Supply an incidence ∞-category $\mathcal{I}$ of the actual operations, faces and higher relations, together with its support subcategories $\mathcal{I}_S$ and an admissible coherent diagram $F : \mathcal{I} \to \mathrm{Spaces}$. Set

$$\mathrm{Str}_S(\mathbb{A}) = \lim_{\mathcal{I}_S} F.$$

Definition 34.1 (Relation-excisive pair and cover)**.** The pair S, T is relation-excisive when its incidence diagram presents

$$\mathcal{I}_{S\cup T} \simeq \mathcal{I}_S \coprod_{\mathcal{I}_{S\cap T}}^{h} \mathcal{I}_T$$

and the structure diagram has the indicated compatible restrictions. A finite cover $U = \bigcup_i S_i$ is relation-complete when the corresponding Čech homotopy colimit of all nonempty intersections presents $\mathcal{I}_U$, with its structure diagram. These are hypotheses about the full categories and their coherent relations. Covering the names of the operations, or checking only individual cells, does not suffice.

Theorem 34.2 (Carrier descent)**.** *For a relation-excisive pair, the square*

$$\begin{array}{ccc} \mathrm{Str}_{S\cup T}(\mathbb{A}) & \longrightarrow & \mathrm{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \mathrm{Str}_T(\mathbb{A}) & \longrightarrow & \mathrm{Str}_{S\cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. For a relation-complete cover,

$$\mathrm{Str}_U(\mathbb{A}) \simeq \operatorname*{holim}_{\varnothing \neq J \subseteq I} \mathrm{Str}_{\bigcap_{j\in J} S_j}(\mathbb{A}).$$

Restriction makes $S \mapsto \mathrm{Str}_S(\mathbb{A})$ a contravariant functor on the support power set.

Proof. The space of cones to $F|_{\mathcal{I}_U}$ is its limit. A presentation of the indexing category and diagram by the stated homotopy colimit identifies such a cone with compatible cones on the pieces, including all higher matching homotopies. This is exactly the displayed homotopy limit. Restriction of cones along inclusions composes coherently, giving functoriality and compatibility under refinements of the cover. □

A support filtration of the deformation complex needs an additional chain-level hypothesis. Here is a form in which its associated graded is explicit. Suppose a cellular model D has graded pieces E_S and

$$D = \bigoplus_{S \subseteq \mathfrak{C}} E_S, \qquad d(E_S) \subseteq \bigoplus_{T \subseteq S} E_T.$$

Put $\mathrm{Def}_{\leq S} = \bigoplus_{T \subseteq S} E_T$ and assume the cellular inclusions compute their homotopy latching objects. Define

$$\mathrm{Def}_{=S} = \mathrm{cofib}\Big(\operatorname*{hocolim}_{T \subsetneq S} \mathrm{Def}_{\leq T} \longrightarrow \mathrm{Def}_{\leq S} \Big).$$

Then $\mathrm{Def}_{=S}$ is E_S with the support-preserving part of d. The latching condition holds, for example, for the displayed split cellular inclusions of complexes over a field.

Theorem 34.3 (Support spectral sequence)**.** *Under these hypotheses, $F_p D = \bigoplus_{|S| \leq p} E_S$ is a finite filtration by subcomplexes, with a strongly convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{|S|=p} H^{p+q}(\mathrm{Def}_{=S}) \Longrightarrow H^{p+q}(D).$$

The first differential is induced by the part of d lowering support cardinality by one, with its cellular signs.

Proof. The triangular condition on d makes every $F_p D$ a subcomplex. Its associated graded is the direct sum of E_S of cardinality p with their support-preserving differentials. For each proper support T, the index category of proper subsets containing T has initial object T; the split latching colimit therefore removes exactly the proper-support pieces. This identifies the associated graded with the displayed cofibres. The spectral sequence of the finite filtration now gives the result and strong convergence. □

A first-page class can be killed by a later differential. For instance, put a degree-zero generator at support $\{a, b\}$ and a degree-one generator at support $\{a\}$, and let the differential map the first isomorphically to the second. The first page has both classes, whereas the total complex is acyclic. A general convolution cochain differential can instead enlarge support; it needs a different invariant filtration or a comparison with such a cellular model before this theorem applies.

34.2 Sixteen comparisons and their hypotheses

The constructions above give the following comparisons, each with its specified carrier and hypotheses.

(i) An operation has its actual geometric coefficient domain.

(ii) A diagonal model can retain principal parts when its filtration and residue or restriction maps are supplied.

(iii) An ordered interval carries transverse fusion with its composition maps.

(iv) Motion in a plane gives exchange when the line category and its braiding have been constructed.

(v) A circle gives a cyclic trace only on the stated cyclic object and domain.

(vi) Stable-curve sewing requires its gluing maps and orientation data.

(vii) The transverse, associative-chiral and chiral-Lie bars have different tensors and units.

(viii) An enveloping comparison uses its actual PBW theorem and functor; the associative monoidal envelope and the chiral envelope retain their types.

(ix) The interface bar records both actions and their higher compatibility.

(x) The derived centre represents coherent natural interior actions on the supplied module category.

(xi) A cooperadic resolution gives the tangent algebra of the full mixed structure with its fixed relative operations.

(xii) Its Maurer–Cartan curvature gives the corresponding field equation after a local realization.

(xiii) Its L_∞ identities give the Bianchi identity.

(xiv) A local dual pairing gives the classical shifted cotangent BV action.

(xv) An invariant volume gives the finite BV complex; a de Rham description or inverse limit needs the additional comparisons just specified.

(xvi) An Einstein realization requires projective matching, coupled local fields, and a comparison preserving the physical action and BV pairing.

These statements allow several formulations of the theory. Equivalences between formulations require maps preserving the listed operations.

34.3 Coherent actions and gravitational reconstruction

A supplied mixed boundary datum defines compatible modules and their derived natural endomorphisms. A specified coloured operad organizes deformations of that datum together with its central action. These are constructions relative to the full operations and their coherence maps. The centre alone does not recover the datum from which the deformation algebra is formed.

A supplied local realization of the deformation algebra admits shifted cotangent transgression. With the invariant volumes and compatible BV transition maps specified above, the finite charts also give the algebraic quantum complex. The physical bulk is defined independently by its fields, action, gauge symmetry, boundary conditions, and quantization. Reconstruction requires a coherent action map from its observables to an independently defined geometric centre, followed by an equivalence theorem on the stated sector.

For gravity, the invertible coframe gives the classical Cartan–Einstein correspondence. The cotangent construction supplies a formal comparison theory with extra dual fields. Its identification with gravitational BV fields requires an action-preserving and pairing-preserving map. Quantum boundary currents, a continuum measure, global holonomy sectors, allowed fillings, and compatible states remain necessary for the quantum and global comparison.

Hall and Calabi–Yau geometry can supply boundary operations when their correspondences, coefficients, orientations, and comparison maps are constructed. Exchange additionally uses a monoidal category and a fibre functor. Cyclic pairings and sewing determine amplitudes only on their stated domains. These comparisons are the mathematical steps by which local boundary operations could reconstruct physical and geometric theories.

Book VI

The Igusa comparisons

Chapter 35

From the denominator to geometric states

35.1 The denominator and the reduced counting series

The Jacobi form $\phi_{0,1}$ supplies the automorphic input. The type-II Lorentzian lattice, positive roots, Weyl vector, and Gritsenko–Nikulin presentation supply the denominator algebra. These data give

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z), \qquad \Delta_5^{\theta} = 64D_5, \qquad \chi_{10} = {D_5}^2.$$

On the root torus, Chevalley chains of $\mathfrak{n}_{\Delta_5,+}$ have Euler supercharacter

$$P = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_{\Delta_5}\alpha}, \qquad e^{-\rho} P = \operatorname{den}(\mathfrak{g}_{\Delta_5}).$$

PBW gives $\operatorname{sch}\operatorname{Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. For $\jmath(Z) = Z/2$,

$$-\jmath^*\Big(e^{2\rho}\operatorname{sch}\big(\operatorname{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big)\Big) = -{\chi_{10}}^{-1}.$$

The primitive reduced curve-counting theorem identifies its numerical generating series with this scalar, in the specified normalization and expansion domain.

35.2 The maps needed for compact realization

The scalar comparison retains signed dimensions. A compact realization begins with reduced coefficient complexes on moduli of objects of $S \times E$. It requires their common-torsor extension and flag correspondences and coherent orientation maps. The Hall product acts on the resulting state complex. Its primitive classes must satisfy the Igusa presentation. Theorem 3.6 constructs the primitive comparison map once these classes and relations are given. It also proves the parity-refined PBW criterion for this map to be an isomorphism.

The coproduct and pairing must be compared before identifying quantum doubles. A chiral realization preserves position, support, and collision operations. A Pfaffian realization requires a family of skew complexes and an isomorphism of its square-root line with the pullback of $\mathcal{L}^5 \otimes \nu_5$. This isomorphism must preserve the section and its transport. An operator realization also specifies its Hilbert or cochain carrier and the domain on which its normalized trace converges.

The diagonal Fredholm construction gives an analytic model on the minimal virtual root superspace with signed multiplicities. Its Berezinian is the root product P on the stated absolute-convergence chamber. Multiplication by the Weyl monomial $e^{-\rho}$ gives the denominator; the half-coordinate pullback $\jmath$ then gives D_5. The virtual parity encodes the signed exponent, whereas the root Lie superalgebra retains the two parity dimensions and bracket. Relating this model to compact states requires an intertwiner and the line comparison above. The full construction problem is to produce compatible maps on geometric states, Hall operations, chiral operations, and square-root lines. Their scalar evaluation must recover the established Igusa identity.

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